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Method for depositing boron and gallium containing silicon germanium layers

Abstract

Methods and devices for epitaxially growing boron- and gallium-doped silicon germanium layers. The layers may be used, for example, as a p-type source and/or drain regions in field effect transistors.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
D30036	12/1898	Rhind	N/A	N/A
D31889	12/1898	Gill	N/A	N/A
D56051	12/1919	Cohn	N/A	N/A
2059480	12/1935	Obermaier	N/A	N/A
2161626	12/1938	Loughner et al.	N/A	N/A
2240163	12/1940	Pick	N/A	N/A
2266416	12/1940	Duclos	N/A	N/A
2280778	12/1941	Andersen	N/A	N/A
D142841	12/1944	D'Algodt	N/A	N/A
2410420	12/1945	Bennett	N/A	N/A
2441253	12/1947	Sarver	N/A	N/A
2480557	12/1948	Cummins	N/A	N/A
2563931	12/1950	Harrison	N/A	N/A
2660061	12/1952	Lewis	N/A	N/A
2745640	12/1955	Cushman	N/A	N/A
2847320	12/1957	Bulloff	N/A	N/A
2990045	12/1960	Root	N/A	N/A
3038951	12/1961	Mead	N/A	N/A
3089507	12/1962	Drake et al.	N/A	N/A
3094396	12/1962	Flugge et al.	N/A	N/A
3197682	12/1964	Klass et al.	N/A	N/A
3232437	12/1965	Hultgren	N/A	N/A
3263502	12/1965	Springfield	N/A	N/A
3332286	12/1966	Strong	N/A	N/A
3410349	12/1967	Troutman	N/A	N/A
3420622	12/1968	Donges et al.	N/A	N/A
3588192	12/1970	Drutchas et al.	N/A	N/A
3634740	12/1971	Stevko	N/A	N/A
3647387	12/1971	Benson	N/A	N/A
3647716	12/1971	Koches	N/A	N/A
3713899	12/1972	Sebestyen	N/A	N/A
3718429	12/1972	Williamson	N/A	N/A
3796182	12/1973	Rosler	N/A	N/A

3814128	12/1973	Grantham	N/A	N/A
3833492	12/1973	Bollyky	N/A	N/A
3854443	12/1973	Baerg	N/A	N/A
3862397	12/1974	Anderson et al.	N/A	N/A
3867205	12/1974	Schley	N/A	N/A
3885504	12/1974	Baermann	N/A	N/A
3887790	12/1974	Ferguson	N/A	N/A
3904371	12/1974	Neti	N/A	N/A
3913058	12/1974	Nishio et al.	N/A	N/A
3913617	12/1974	van Laar	N/A	N/A
3916270	12/1974	Wachtler et al.	N/A	N/A
3947685	12/1975	Meinel	N/A	N/A
3960559	12/1975	Suzuki	N/A	N/A
3962004	12/1975	Sonneborn	N/A	N/A
3983401	12/1975	Livesay	N/A	N/A
3997638	12/1975	Manning et al.	N/A	N/A
4048110	12/1976	Vanderspurt	N/A	N/A
4054071	12/1976	Patejak	N/A	N/A
4058430	12/1976	Suntola et al.	N/A	N/A
4079944	12/1977	Durley et al.	N/A	N/A
4093491	12/1977	Whelpton et al.	N/A	N/A
4099041	12/1977	Berkman et al.	N/A	N/A
D249341	12/1977	Mertz	N/A	N/A
4126027	12/1977	Smith et al.	N/A	N/A
4134425	12/1978	Gussefeld et al.	N/A	N/A
4145699	12/1978	Hu et al.	N/A	N/A
4149237	12/1978	Freitas	N/A	N/A
4152760	12/1978	Freitas et al.	N/A	N/A
4157751	12/1978	Grundken et al.	N/A	N/A
4164959	12/1978	Wurzburger	N/A	N/A
4176630	12/1978	Elmer	N/A	N/A
4179530	12/1978	Koppl et al.	N/A	N/A
4181330	12/1979	Kojima	N/A	N/A
4184188	12/1979	Briglia	N/A	N/A
4194536	12/1979	Stine et al.	N/A	N/A
4217463	12/1979	Swearingen	N/A	N/A
4229064	12/1979	Vetter et al.	N/A	N/A
4234449	12/1979	Wolson et al.	N/A	N/A
4241000	12/1979	McCauley et al.	N/A	N/A
4314763	12/1981	Steigmeier et al.	N/A	N/A
4322592	12/1981	Martin	N/A	N/A
4324611	12/1981	Vogel et al.	N/A	N/A
4333735	12/1981	Hardy	N/A	N/A
4355912	12/1981	Haak	N/A	N/A
4384918	12/1982	Abe	N/A	N/A
4389973	12/1982	Suntola et al.	N/A	N/A
D269850	12/1982	Preisler et al.	N/A	N/A
4393013	12/1982	McMenamin	N/A	N/A
4401507	12/1982	Engle	N/A	N/A
4412133	12/1982	Eckes et al.	N/A	N/A

4413022	12/1982	Suntola et al.	N/A	N/A
4414492	12/1982	Hanlet	N/A	N/A
4436674	12/1983	McMenamin	N/A	N/A
4444990	12/1983	Villar	N/A	N/A
D274122	12/1983	Stahel et al.	N/A	N/A
4454370	12/1983	Voznick	N/A	N/A
4455193	12/1983	Jeuch et al.	N/A	N/A
4465716	12/1983	Baber et al.	N/A	N/A
4466766	12/1983	Geren et al.	N/A	N/A
4479831	12/1983	Sadow	N/A	N/A
4480284	12/1983	Tojo et al.	N/A	N/A
4481300	12/1983	Hartnett et al.	N/A	N/A
4484061	12/1983	Zelinka et al.	N/A	N/A
4488506	12/1983	Heinecke et al.	N/A	N/A
4495024	12/1984	Bok	N/A	N/A
4496828	12/1984	Kusmierz et al.	N/A	N/A
4499354	12/1984	Hill et al.	N/A	N/A
4502094	12/1984	Lewin et al.	N/A	N/A
4504439	12/1984	Elter et al.	N/A	N/A
4512113	12/1984	Budinger	N/A	N/A
4512841	12/1984	Cunningham et al.	N/A	N/A
4520116	12/1984	Gentilman et al.	N/A	N/A
4520421	12/1984	Sakitani et al.	N/A	N/A
4527005	12/1984	McKelvey et al.	N/A	N/A
4534816	12/1984	Chen et al.	N/A	N/A
4535628	12/1984	Hope	N/A	N/A
4537001	12/1984	Uppstrom	N/A	N/A
4548688	12/1984	Mathews	N/A	N/A
4551192	12/1984	Di Milia et al.	N/A	N/A
4554611	12/1984	Lewin	N/A	N/A
4560590	12/1984	Bok	N/A	N/A
4570328	12/1985	Price et al.	N/A	N/A
4575408	12/1985	Bok	N/A	N/A
4575636	12/1985	Caprari	N/A	N/A
4578560	12/1985	Tanaka et al.	N/A	N/A
4579080	12/1985	Martin et al.	N/A	N/A
4579378	12/1985	Snyders	N/A	N/A
4579623	12/1985	Suzuki et al.	N/A	N/A
4581520	12/1985	Vu et al.	N/A	N/A
4590326	12/1985	Woldy	N/A	N/A
4611966	12/1985	Johnson	N/A	N/A
4620998	12/1985	Lalvani	N/A	N/A
4622918	12/1985	Bok	N/A	N/A
4624728	12/1985	Bithell et al.	N/A	N/A
D288556	12/1986	Wallgren	N/A	N/A
4653541	12/1986	Oehlschlaeger et al.	N/A	N/A
4654226	12/1986	Jackson et al.	N/A	N/A
4655592	12/1986	Allemand	N/A	N/A
4662987	12/1986	Bok	N/A	N/A
4664769	12/1986	Cuomo et al.	N/A	N/A

4670126	12/1986	Messer et al.	N/A	N/A
4681134	12/1986	Paris	N/A	N/A
4693211	12/1986	Ogami et al.	N/A	N/A
4700089	12/1986	Fujii et al.	N/A	N/A
4707815	12/1986	Yamasaki	N/A	N/A
4717461	12/1987	Strahl et al.	N/A	N/A
4718637	12/1987	Contin	N/A	N/A
4720362	12/1987	Gentilman et al.	N/A	N/A
4720407	12/1987	Sculke	N/A	N/A
4721533	12/1987	Phillippi et al.	N/A	N/A
4721534	12/1987	Phillippi et al.	N/A	N/A
4722298	12/1987	Rubin et al.	N/A	N/A
4724272	12/1987	Raniere et al.	N/A	N/A
4725204	12/1987	Powell	N/A	N/A
4735259	12/1987	Vincent	N/A	N/A
4738618	12/1987	Massey et al.	N/A	N/A
4738748	12/1987	Kisa	N/A	N/A
4747367	12/1987	Posa	N/A	N/A
4749416	12/1987	Greenspan	N/A	N/A
4750520	12/1987	Heim et al.	N/A	N/A
4753192	12/1987	Goldsmith et al.	N/A	N/A
4753856	12/1987	Haluska et al.	N/A	N/A
4756794	12/1987	Yoder	N/A	N/A
4764076	12/1987	Layman et al.	N/A	N/A
4770590	12/1987	Hugues et al.	N/A	N/A
4771015	12/1987	Kanai	N/A	N/A
4775281	12/1987	Prentakis	N/A	N/A
4776744	12/1987	Stonestreet et al.	N/A	N/A
4780169	12/1987	Stark et al.	N/A	N/A
4781511	12/1987	Harada et al.	N/A	N/A
4789294	12/1987	Sato et al.	N/A	N/A
4790258	12/1987	Drage et al.	N/A	N/A
4802441	12/1988	Waugh	N/A	N/A
4804086	12/1988	Grohrock	N/A	N/A
4808387	12/1988	Datta et al.	N/A	N/A
4812201	12/1988	Sakai et al.	N/A	N/A
4812217	12/1988	George et al.	N/A	N/A
4821674	12/1988	deBoer et al.	N/A	N/A
4827430	12/1988	Aid et al.	N/A	N/A
4828224	12/1988	Crabb et al.	N/A	N/A
4830515	12/1988	Cortes	N/A	N/A
4837113	12/1988	Luttmer et al.	N/A	N/A
4837185	12/1988	Yau et al.	N/A	N/A
4854263	12/1988	Chang et al.	N/A	N/A
4854266	12/1988	Simson et al.	N/A	N/A
4857137	12/1988	Tachi et al.	N/A	N/A
4857382	12/1988	Liu et al.	N/A	N/A
4858557	12/1988	Pozzetti et al.	N/A	N/A
4863374	12/1988	Vukovich	N/A	N/A
4867629	12/1988	Iwasawa et al.	N/A	N/A

4871523	12/1988	Datta et al.	N/A	N/A
4874273	12/1988	Tokisue et al.	N/A	N/A
4880982	12/1988	Hoksaas	N/A	N/A
4882199	12/1988	Sadoway et al.	N/A	N/A
4886162	12/1988	Ambrogio	N/A	N/A
4916091	12/1989	Freeman et al.	N/A	N/A
4917556	12/1989	Stark et al.	N/A	N/A
4920918	12/1989	Adams et al.	N/A	N/A
4925388	12/1989	Iseki et al.	N/A	N/A
4931135	12/1989	Horiuchi et al.	N/A	N/A
4934831	12/1989	Volbrecht	N/A	N/A
4938815	12/1989	McNeilly	N/A	N/A
D309702	12/1989	Hall	N/A	N/A
4949671	12/1989	Davis et al.	N/A	N/A
4949848	12/1989	Kos	N/A	N/A
4950624	12/1989	Inuzima et al.	N/A	N/A
4956538	12/1989	Moslehi	N/A	N/A
4958061	12/1989	Wakabayashi et al.	N/A	N/A
D311126	12/1989	Crowley	N/A	N/A
4962063	12/1989	Maydan et al.	N/A	N/A
4963506	12/1989	Liaw et al.	N/A	N/A
4976996	12/1989	Monkowski et al.	N/A	N/A
4978567	12/1989	Miller	N/A	N/A
4984904	12/1990	Nakano et al.	N/A	N/A
4985114	12/1990	Okudaira	N/A	N/A
4986215	12/1990	Yamada	N/A	N/A
4987102	12/1990	Nguyen et al.	N/A	N/A
4987856	12/1990	Hey	N/A	N/A
4989992	12/1990	Piai	N/A	N/A
4991614	12/1990	Hammel	N/A	N/A
5002632	12/1990	Loewenstein et al.	N/A	N/A
5013691	12/1990	Lory et al.	N/A	N/A
5022961	12/1990	Izumi et al.	N/A	N/A
5027746	12/1990	Frijlink	N/A	N/A
5028366	12/1990	Harakal et al.	N/A	N/A
D320148	12/1990	Andrews	N/A	N/A
5049029	12/1990	Mitsui et al.	N/A	N/A
5053247	12/1990	Moore	N/A	N/A
5057436	12/1990	Ball	N/A	N/A
5060322	12/1990	Delepine	N/A	N/A
5061083	12/1990	Grimm et al.	N/A	N/A
5062386	12/1990	Christensen	N/A	N/A
5064337	12/1990	Asakawa et al.	N/A	N/A
5065698	12/1990	Koike	N/A	N/A
5067437	12/1990	Watanabe et al.	N/A	N/A
5069591	12/1990	Kinoshita	N/A	N/A
5071258	12/1990	Usher et al.	N/A	N/A
5074017	12/1990	Toya et al.	N/A	N/A
5082517	12/1991	Moslehi	N/A	N/A
5084126	12/1991	McKee	N/A	N/A

5088444	12/1991	Ohmine et al.	N/A	N/A
5097890	12/1991	Nakao	N/A	N/A
5098638	12/1991	Sawada	N/A	N/A
5098865	12/1991	Machado	N/A	N/A
5104514	12/1991	Quartarone	N/A	N/A
5107170	12/1991	Ishikawa et al.	N/A	N/A
5108192	12/1991	Mailliet et al.	N/A	N/A
5110407	12/1991	Ono et al.	N/A	N/A
5114683	12/1991	Hirase	N/A	N/A
5116018	12/1991	Friemoth et al.	N/A	N/A
5117121	12/1991	Watanabe et al.	N/A	N/A
D327534	12/1991	Manville	N/A	N/A
5119760	12/1991	McMillan et al.	N/A	N/A
5124272	12/1991	Saito et al.	N/A	N/A
5125358	12/1991	Ueda et al.	N/A	N/A
5125710	12/1991	Gianelo	N/A	N/A
5130003	12/1991	Conrad	N/A	N/A
5134965	12/1991	Tokuda et al.	N/A	N/A
5137286	12/1991	Whitford	N/A	N/A
5151296	12/1991	Tokunaga	N/A	N/A
5154301	12/1991	Kos	N/A	N/A
5158128	12/1991	Inoue et al.	N/A	N/A
D330900	12/1991	Wakegijig	N/A	N/A
5167716	12/1991	Boitnott et al.	N/A	N/A
5167761	12/1991	Westendorp et al.	N/A	N/A
5174881	12/1991	Iwasaki et al.	N/A	N/A
5176451	12/1992	Sasada	N/A	N/A
5178639	12/1992	Nishi	N/A	N/A
5178682	12/1992	Tsukamoto et al.	N/A	N/A
5180273	12/1992	Sakaya et al.	N/A	N/A
5180435	12/1992	Markunas et al.	N/A	N/A
5181779	12/1992	Shia et al.	N/A	N/A
5182232	12/1992	Chhabra et al.	N/A	N/A
5183511	12/1992	Yamazaki et al.	N/A	N/A
5186120	12/1992	Ohnishi et al.	N/A	N/A
D333606	12/1992	Kanemitsu et al.	N/A	N/A
5192717	12/1992	Kawakami	N/A	N/A
5193912	12/1992	Saunders	N/A	N/A
5193969	12/1992	Rush et al.	N/A	N/A
5194401	12/1992	Adams et al.	N/A	N/A
5199603	12/1992	Prescott	N/A	N/A
5208961	12/1992	Lajoie	N/A	N/A
5213650	12/1992	Wang et al.	N/A	N/A
5219226	12/1992	James	N/A	N/A
5221556	12/1992	Hawkins et al.	N/A	N/A
5225366	12/1992	Yoder	N/A	N/A
5226383	12/1992	Bhat	N/A	N/A
5226713	12/1992	Matsumura	N/A	N/A
5226967	12/1992	Chen et al.	N/A	N/A
5228114	12/1992	Suzuki	N/A	N/A

5231062	12/1992	Mathers et al.	N/A	N/A
5232508	12/1992	Arena et al.	N/A	N/A
5234526	12/1992	Chen et al.	N/A	N/A
5242501	12/1992	McDiarmid	N/A	N/A
5242539	12/1992	Kumihashi et al.	N/A	N/A
5243195	12/1992	Nishi	N/A	N/A
5243202	12/1992	Mori et al.	N/A	N/A
5246218	12/1992	Yap et al.	N/A	N/A
5246500	12/1992	Samata et al.	N/A	N/A
5249960	12/1992	Monoe	N/A	N/A
5250092	12/1992	Nakano	N/A	N/A
5252133	12/1992	Miyazaki et al.	N/A	N/A
5252134	12/1992	Stauffer	N/A	N/A
5259881	12/1992	Edwards et al.	N/A	N/A
5261167	12/1992	Sakata	N/A	N/A
5266526	12/1992	Aoyama	N/A	N/A
5268989	12/1992	Moslehi et al.	N/A	N/A
5271967	12/1992	Kramer et al.	N/A	N/A
5273609	12/1992	Moslehi	N/A	N/A
5277932	12/1993	Spencer	N/A	N/A
5278494	12/1993	Obigane	N/A	N/A
5279886	12/1993	Kawai et al.	N/A	N/A
5279986	12/1993	Maloney et al.	N/A	N/A
5280894	12/1993	Witcraft et al.	N/A	N/A
5281274	12/1993	Yoder	N/A	N/A
5284519	12/1993	Gadgil	N/A	N/A
5288684	12/1993	Yamazaki et al.	N/A	N/A
5294572	12/1993	Granneman et al.	N/A	N/A
5294778	12/1993	Carman et al.	N/A	N/A
5295777	12/1993	Hodos	N/A	N/A
5298089	12/1993	Bowe et al.	N/A	N/A
5305417	12/1993	Najm et al.	N/A	N/A
5306666	12/1993	Izumi	N/A	N/A
5306946	12/1993	Yamamoto	N/A	N/A
5308650	12/1993	Krummel et al.	N/A	N/A
5308788	12/1993	Fitch et al.	N/A	N/A
5310410	12/1993	Begin et al.	N/A	N/A
5310456	12/1993	Kadomura	N/A	N/A
5312245	12/1993	Brannen et al.	N/A	N/A
5313061	12/1993	Drew et al.	N/A	N/A
5314538	12/1993	Maeda et al.	N/A	N/A
5314570	12/1993	Ikegaya et al.	N/A	N/A
5315092	12/1993	Takahashi et al.	N/A	N/A
5320218	12/1993	Yamashita et al.	N/A	N/A
5326427	12/1993	Jerbic	N/A	N/A
5328360	12/1993	Yokokawa	N/A	N/A
5328810	12/1993	Lowrey et al.	N/A	N/A
5335309	12/1993	Fujii et al.	N/A	N/A
5336327	12/1993	Lee	N/A	N/A
5338362	12/1993	Imahashi	N/A	N/A

5340261	12/1993	Oosawa et al.	N/A	N/A
5346961	12/1993	Shaw et al.	N/A	N/A
5348774	12/1993	Golecki et al.	N/A	N/A
5350480	12/1993	Gray	N/A	N/A
5354433	12/1993	Granneman et al.	N/A	N/A
5354580	12/1993	Goela et al.	N/A	N/A
5356478	12/1993	Chen et al.	N/A	N/A
5356672	12/1993	Schmitt et al.	N/A	N/A
5360269	12/1993	Ogawa et al.	N/A	N/A
5362328	12/1993	Gardiner et al.	N/A	N/A
5364667	12/1993	Rhieu	N/A	N/A
D353452	12/1993	Groenhoff	N/A	N/A
5374315	12/1993	Deboer et al.	N/A	N/A
D354898	12/1994	Nagy	N/A	N/A
5378501	12/1994	Foster et al.	N/A	N/A
5380367	12/1994	Bertone	N/A	N/A
5382311	12/1994	Ishikawa et al.	N/A	N/A
5387265	12/1994	Kakizaki et al.	N/A	N/A
5388945	12/1994	Garric et al.	N/A	N/A
5393577	12/1994	Uesugi et al.	N/A	N/A
5397395	12/1994	Sano et al.	N/A	N/A
5403630	12/1994	Matsui et al.	N/A	N/A
5404082	12/1994	Hernandez et al.	N/A	N/A
5407449	12/1994	Zinger	N/A	N/A
5407867	12/1994	Iwasaki et al.	N/A	N/A
5413813	12/1994	Cruse et al.	N/A	N/A
5414221	12/1994	Gardner	N/A	N/A
5415753	12/1994	Hurwitt et al.	N/A	N/A
5418382	12/1994	Blackwood et al.	N/A	N/A
5421893	12/1994	Perlov	N/A	N/A
5422139	12/1994	Fischer	N/A	N/A
5423942	12/1994	Robbins et al.	N/A	N/A
5426137	12/1994	Allen	N/A	N/A
5427824	12/1994	Inushima et al.	N/A	N/A
5430011	12/1994	Tanaka et al.	N/A	N/A
5431734	12/1994	Chapple-Sokol et al.	N/A	N/A
5443646	12/1994	Yamada et al.	N/A	N/A
5443648	12/1994	Ohkase	N/A	N/A
5443686	12/1994	Jones et al.	N/A	N/A
5444217	12/1994	Moore	N/A	N/A
5447294	12/1994	Sakata et al.	N/A	N/A
5453124	12/1994	Moslehi et al.	N/A	N/A
D363464	12/1994	Fukasawa	N/A	N/A
5456207	12/1994	Gedridge et al.	N/A	N/A
5456757	12/1994	Aruga et al.	N/A	N/A
5461214	12/1994	Peck et al.	N/A	N/A
5462899	12/1994	Ikeda	N/A	N/A
5463176	12/1994	Eckert	N/A	N/A
5464313	12/1994	Ohsawa	N/A	N/A
5474410	12/1994	Ozawa et al.	N/A	N/A

5474612	12/1994	Sato et al.	N/A	N/A
5478429	12/1994	Komino et al.	N/A	N/A
5480488	12/1995	Bittner et al.	N/A	N/A
5480818	12/1995	Matsumoto et al.	N/A	N/A
5482559	12/1995	Imai et al.	N/A	N/A
5484484	12/1995	Yamaga et al.	N/A	N/A
5494439	12/1995	Goldstein et al.	N/A	N/A
5494494	12/1995	Mizuno et al.	N/A	N/A
5496408	12/1995	Motoda et al.	N/A	N/A
5501740	12/1995	Besen et al.	N/A	N/A
5503875	12/1995	Imai et al.	N/A	N/A
5504042	12/1995	Cho et al.	N/A	N/A
5510277	12/1995	Cunningham et al.	N/A	N/A
5512102	12/1995	Yamazaki	N/A	N/A
5514439	12/1995	Sibley	N/A	N/A
5518549	12/1995	Hellwig	N/A	N/A
5518780	12/1995	Tamor et al.	N/A	N/A
5519234	12/1995	Paz de Araujo	N/A	N/A
5520743	12/1995	Takahashi et al.	N/A	N/A
5523616	12/1995	Yasuhide	N/A	N/A
5527111	12/1995	Lysen et al.	N/A	N/A
5527417	12/1995	Iida et al.	N/A	N/A
5531218	12/1995	Krebs	N/A	N/A
5531835	12/1995	Fodor et al.	N/A	N/A
5537311	12/1995	Stevens	N/A	N/A
5540059	12/1995	Yokokawa	N/A	N/A
5540821	12/1995	Tepman	N/A	N/A
5540898	12/1995	Davidson	N/A	N/A
5554557	12/1995	Koh	N/A	N/A
5556275	12/1995	Sakata et al.	N/A	N/A
5558717	12/1995	Zhao et al.	N/A	N/A
5559046	12/1995	Oishi et al.	N/A	N/A
5562383	12/1995	Iwai et al.	N/A	N/A
5562947	12/1995	White et al.	N/A	N/A
5562952	12/1995	Nakahigashi et al.	N/A	N/A
5565038	12/1995	Ashley	N/A	N/A
5569402	12/1995	Meisser et al.	N/A	N/A
5574247	12/1995	Nishitani et al.	N/A	N/A
5576629	12/1995	Turner	N/A	N/A
5577331	12/1995	Suzuki	N/A	N/A
5583736	12/1995	Anderson et al.	N/A	N/A
5584936	12/1995	Pickering et al.	N/A	N/A
5584963	12/1995	Takahashi	N/A	N/A
5586585	12/1995	Bonora et al.	N/A	N/A
5589002	12/1995	Su	N/A	N/A
5589110	12/1995	Motoda et al.	N/A	N/A
5595606	12/1996	Fujikawa et al.	N/A	N/A
5601641	12/1996	Stephens	N/A	N/A
5602060	12/1996	Kobayashi et al.	N/A	N/A
5604410	12/1996	Vollkommer et al.	N/A	N/A

5611448	12/1996	Chen	N/A	N/A
5616264	12/1996	Nishi et al.	N/A	N/A
5616947	12/1996	Tamura	N/A	N/A
5621982	12/1996	Yamashita	N/A	N/A
5632919	12/1996	MacCracken et al.	N/A	N/A
5637153	12/1996	Niino et al.	N/A	N/A
D380527	12/1996	Velez	N/A	N/A
5645646	12/1996	Beinglass et al.	N/A	N/A
5650013	12/1996	Yamazaki	N/A	N/A
5650351	12/1996	Wu	N/A	N/A
5653807	12/1996	Crumbaker	N/A	N/A
5656093	12/1996	Burkhart et al.	N/A	N/A
5661263	12/1996	Salvaggio	N/A	N/A
5662470	12/1996	Huussen et al.	N/A	N/A
5663899	12/1996	Zvonar et al.	N/A	N/A
5665608	12/1996	Chapple-Sokol et al.	N/A	N/A
5667592	12/1996	Boitnott et al.	N/A	N/A
5670786	12/1996	Meyer et al.	N/A	N/A
5679215	12/1996	Barnes et al.	N/A	N/A
5681779	12/1996	Pasch et al.	N/A	N/A
D386076	12/1996	Moore	N/A	N/A
5683517	12/1996	Shan	N/A	N/A
5683561	12/1996	Hollars et al.	N/A	N/A
5685912	12/1996	Nishizaka	N/A	N/A
5685914	12/1996	Hills et al.	N/A	N/A
5690742	12/1996	Ogata et al.	N/A	N/A
5695567	12/1996	Kordina	N/A	N/A
5697706	12/1996	Ciaravino et al.	N/A	N/A
5698036	12/1996	Ishii et al.	N/A	N/A
5700729	12/1996	Lee et al.	N/A	N/A
5708825	12/1997	Sotomayor	N/A	N/A
5709745	12/1997	Larkin et al.	N/A	N/A
5711811	12/1997	Suntola et al.	N/A	N/A
5716133	12/1997	Hosokawa et al.	N/A	N/A
5718574	12/1997	Shimazu	N/A	N/A
5720927	12/1997	Cripe et al.	N/A	N/A
D392855	12/1997	Pillow	N/A	N/A
5724748	12/1997	Brooks	N/A	N/A
5728223	12/1997	Murakarni et al.	N/A	N/A
5728425	12/1997	Ebe et al.	N/A	N/A
5730801	12/1997	Tepman et al.	N/A	N/A
5730802	12/1997	Ishizumi et al.	N/A	N/A
5732597	12/1997	Devenyi	N/A	N/A
5732744	12/1997	Barr et al.	N/A	N/A
5732957	12/1997	Yu	N/A	N/A
5736314	12/1997	Hayes et al.	N/A	N/A
5753835	12/1997	Gustin	N/A	N/A
5754390	12/1997	Sandhu et al.	N/A	N/A
5759281	12/1997	Gurary et al.	N/A	N/A
5761328	12/1997	Solberg et al.	N/A	N/A

5766365	12/1997	Umutoy et al.	N/A	N/A
5768125	12/1997	Zinger et al.	N/A	N/A
5769952	12/1997	Komino	N/A	N/A
5775889	12/1997	Kobayashi et al.	N/A	N/A
5777838	12/1997	Tamagawa et al.	N/A	N/A
5779203	12/1997	Edlinger	N/A	N/A
5781693	12/1997	Ballance et al.	N/A	N/A
5782979	12/1997	Kaneno	N/A	N/A
5790750	12/1997	Anderson	N/A	N/A
5791782	12/1997	Wooten et al.	N/A	N/A
5792272	12/1997	Van Os et al.	N/A	N/A
5796074	12/1997	Edelstein et al.	N/A	N/A
5801104	12/1997	Schuegraf et al.	N/A	N/A
5801945	12/1997	Comer	N/A	N/A
5804505	12/1997	Yamada et al.	N/A	N/A
5806980	12/1997	Berrian	N/A	N/A
5813851	12/1997	Nakao	N/A	N/A
5818716	12/1997	Chin et al.	N/A	N/A
5819092	12/1997	Ferguson et al.	N/A	N/A
5819434	12/1997	Herchen et al.	N/A	N/A
5820366	12/1997	Lee	N/A	N/A
5820685	12/1997	Kurihira et al.	N/A	N/A
5820686	12/1997	Moore	N/A	N/A
5826129	12/1997	Hasebe et al.	N/A	N/A
5827420	12/1997	Shirazi et al.	N/A	N/A
5827435	12/1997	Samukawa	N/A	N/A
5827757	12/1997	Robinson et al.	N/A	N/A
5836483	12/1997	Disel	N/A	N/A
5837058	12/1997	Chen et al.	N/A	N/A
5837320	12/1997	Hampden-Smith et al.	N/A	N/A
5844683	12/1997	Pavloski et al.	N/A	N/A
5846332	12/1997	Zhao et al.	N/A	N/A
5851293	12/1997	Lane et al.	N/A	N/A
5851294	12/1997	Young et al.	N/A	N/A
5851299	12/1997	Cheng et al.	N/A	N/A
5852445	12/1997	Yoshikawa et al.	N/A	N/A
5852879	12/1997	Schumaier	N/A	N/A
5853484	12/1997	Jeong	N/A	N/A
D403949	12/1998	Nakamura	N/A	N/A
D404370	12/1998	Kimura	N/A	N/A
D404372	12/1998	Ishii	N/A	N/A
5855680	12/1998	Soininen et al.	N/A	N/A
5855681	12/1998	Maydan et al.	N/A	N/A
5855687	12/1998	DuBois et al.	N/A	N/A
5855726	12/1998	Soraoka et al.	N/A	N/A
5857777	12/1998	Schuh	N/A	N/A
5861233	12/1998	Sekine et al.	N/A	N/A
5862302	12/1998	Okase	N/A	N/A
5863123	12/1998	Lee et al.	N/A	N/A
5865205	12/1998	Wilmer	N/A	N/A

5866795	12/1998	Wang et al.	N/A	N/A
5871586	12/1998	Crawley et al.	N/A	N/A
5872065	12/1998	Sivaramakrishnan	N/A	N/A
5873942	12/1998	Park	N/A	N/A
5877095	12/1998	Tamura et al.	N/A	N/A
5879128	12/1998	Tietz et al.	N/A	N/A
5879459	12/1998	Gadgil et al.	N/A	N/A
5880980	12/1998	Rothacher et al.	N/A	N/A
5882165	12/1998	Maydan et al.	N/A	N/A
5884640	12/1998	Fishkin et al.	N/A	N/A
5888304	12/1998	Umotoy et al.	N/A	N/A
5891251	12/1998	MacLeish et al.	N/A	N/A
5893741	12/1998	Huang	N/A	N/A
5897348	12/1998	Wu	N/A	N/A
5897378	12/1998	Eriguchi	N/A	N/A
5897379	12/1998	Ulrich et al.	N/A	N/A
5897710	12/1998	Sato et al.	N/A	N/A
D409894	12/1998	McClurg	N/A	N/A
5904170	12/1998	Harvey et al.	N/A	N/A
D411516	12/1998	Imafuku et al.	N/A	N/A
5908672	12/1998	Ryu	N/A	N/A
5915200	12/1998	Tokumasu et al.	N/A	N/A
5915562	12/1998	Nyseth et al.	N/A	N/A
5916365	12/1998	Sherman	N/A	N/A
D412270	12/1998	Fredrickson	N/A	N/A
5920798	12/1998	Higuchi et al.	N/A	N/A
5928426	12/1998	Aitchison	N/A	N/A
D412512	12/1998	Boisvert	N/A	N/A
5937142	12/1998	Moslehi et al.	N/A	N/A
5937323	12/1998	Orczyk et al.	N/A	N/A
5939886	12/1998	Turner et al.	N/A	N/A
5947718	12/1998	Weaver	N/A	N/A
5950327	12/1998	Peterson et al.	N/A	N/A
5950925	12/1998	Fukunaga et al.	N/A	N/A
5954375	12/1998	Trickle et al.	N/A	N/A
5961775	12/1998	Fujimura	N/A	N/A
5968275	12/1998	Lee et al.	N/A	N/A
5970621	12/1998	Bazydola	N/A	N/A
5972196	12/1998	Murphy et al.	N/A	N/A
5975492	12/1998	Brenes	N/A	N/A
5976973	12/1998	Ohira et al.	N/A	N/A
5979506	12/1998	Aarseth	N/A	N/A
5982931	12/1998	Ishimaru	N/A	N/A
5984391	12/1998	Vanderpot et al.	N/A	N/A
5987480	12/1998	Donohue et al.	N/A	N/A
5989342	12/1998	Ikeda et al.	N/A	N/A
5992453	12/1998	Zimmer	N/A	N/A
5997588	12/1998	Goodwin	N/A	N/A
5997768	12/1998	Scully	N/A	N/A
5998870	12/1998	Lee et al.	N/A	N/A

6000732	12/1998	Scheler et al.	N/A	N/A
6001183	12/1998	Gurary et al.	N/A	N/A
6001267	12/1998	Van Os et al.	N/A	N/A
6004204	12/1998	Luxton et al.	N/A	N/A
D419652	12/1999	Hall et al.	N/A	N/A
6013553	12/1999	Wallace	N/A	N/A
6013920	12/1999	Gordon et al.	N/A	N/A
6014677	12/1999	Hayashi et al.	N/A	N/A
6015459	12/1999	Jamison et al.	N/A	N/A
6015465	12/1999	Kholodenko et al.	N/A	N/A
6017779	12/1999	Miyasaka	N/A	N/A
6017818	12/1999	Lu	N/A	N/A
6022180	12/1999	Motoyama et al.	N/A	N/A
6022802	12/1999	Jang	N/A	N/A
6024799	12/1999	Chen et al.	N/A	N/A
6025117	12/1999	Nakano et al.	N/A	N/A
6027163	12/1999	Longenecker et al.	N/A	N/A
6029602	12/1999	Bhatnagar	N/A	N/A
6030900	12/1999	Grassl et al.	N/A	N/A
6033215	12/1999	Ohsawa	N/A	N/A
6035101	12/1999	Sajoto et al.	N/A	N/A
6035804	12/1999	Arami et al.	N/A	N/A
6039809	12/1999	Toyama et al.	N/A	N/A
6042652	12/1999	Hyun	N/A	N/A
6044860	12/1999	Neu	N/A	N/A
6045260	12/1999	Schwartz et al.	N/A	N/A
6048154	12/1999	Wytman	N/A	N/A
6050506	12/1999	Guo et al.	N/A	N/A
6053982	12/1999	Halpin et al.	N/A	N/A
6053983	12/1999	Saeki et al.	N/A	N/A
6054013	12/1999	Collins et al.	N/A	N/A
6054678	12/1999	Miyazaki	N/A	N/A
6060691	12/1999	Minami et al.	N/A	N/A
6060721	12/1999	Huang	N/A	N/A
6063196	12/1999	Li et al.	N/A	N/A
6066204	12/1999	Haven	N/A	N/A
6068441	12/1999	Raaijmakers et al.	N/A	N/A
6071572	12/1999	Mosely et al.	N/A	N/A
6072163	12/1999	Armstrong et al.	N/A	N/A
6073973	12/1999	Boscaljon et al.	N/A	N/A
6074154	12/1999	Ueda et al.	N/A	N/A
6074443	12/1999	Venkatesh	N/A	N/A
6074514	12/1999	Bjorkman et al.	N/A	N/A
6077027	12/1999	Kawamura et al.	N/A	N/A
6079356	12/1999	Umotoy et al.	N/A	N/A
6079927	12/1999	Muka	N/A	N/A
6083321	12/1999	Lei et al.	N/A	N/A
6086677	12/1999	Umotoy et al.	N/A	N/A
6090212	12/1999	Mahawili	N/A	N/A
6090442	12/1999	Klaus et al.	N/A	N/A

6090659	12/1999	Laibowitz et al.	N/A	N/A
6091062	12/1999	Pfahnl et al.	N/A	N/A
6093252	12/1999	Wengert et al.	N/A	N/A
6093253	12/1999	Lofgren	N/A	N/A
6093611	12/1999	Gardner et al.	N/A	N/A
6095083	12/1999	Rice et al.	N/A	N/A
6096133	12/1999	Yuuki et al.	N/A	N/A
6096267	12/1999	Kishkovich	N/A	N/A
6099302	12/1999	Hong et al.	N/A	N/A
6099649	12/1999	Schmitt et al.	N/A	N/A
6099651	12/1999	Sajoto et al.	N/A	N/A
6102565	12/1999	Kita et al.	N/A	N/A
6104002	12/1999	Hirose et al.	N/A	N/A
6104011	12/1999	Juliano	N/A	N/A
6104401	12/1999	Parsons	N/A	N/A
6106625	12/1999	Koai et al.	N/A	N/A
6106678	12/1999	Shufflebotham	N/A	N/A
6110531	12/1999	Paz de Araujo et al.	N/A	N/A
6113703	12/1999	Anderson et al.	N/A	N/A
6119710	12/1999	Brown	N/A	N/A
6121061	12/1999	Van Bilsen et al.	N/A	N/A
6121158	12/1999	Benchikha et al.	N/A	N/A
6122036	12/1999	Yamasaki et al.	N/A	N/A
6124600	12/1999	Moroishi et al.	N/A	N/A
6125789	12/1999	Gupta et al.	N/A	N/A
6126744	12/1999	Hawkins et al.	N/A	N/A
6126848	12/1999	Li et al.	N/A	N/A
6127249	12/1999	Hu	N/A	N/A
6129044	12/1999	Zhao et al.	N/A	N/A
6129546	12/1999	Sada	N/A	N/A
6132207	12/1999	Stoutjesdijk	N/A	N/A
6134807	12/1999	Komino	N/A	N/A
6135460	12/1999	Wise et al.	N/A	N/A
6137240	12/1999	Bogdan	N/A	N/A
6139239	12/1999	Snijders	N/A	N/A
6139983	12/1999	Ohashi et al.	N/A	N/A
6140252	12/1999	Cho et al.	N/A	N/A
6143079	12/1999	Halpin	N/A	N/A
6143082	12/1999	McInerney et al.	N/A	N/A
6143129	12/1999	Savas et al.	N/A	N/A
6146463	12/1999	Yudovsky et al.	N/A	N/A
6148761	12/1999	Majewski et al.	N/A	N/A
6151446	12/1999	Hunter et al.	N/A	N/A
6152070	12/1999	Fairbairn et al.	N/A	N/A
6152669	12/1999	Morita et al.	N/A	N/A
6156151	12/1999	Komino et al.	N/A	N/A
6158941	12/1999	Muka et al.	N/A	N/A
6159301	12/1999	Sato et al.	N/A	N/A
6160244	12/1999	Ohashi	N/A	N/A
6161500	12/1999	Kopacz et al.	N/A	N/A

6162323	12/1999	Koshimizu	N/A	N/A
6164894	12/1999	Cheng	N/A	N/A
6174809	12/2000	Kang et al.	N/A	N/A
6176929	12/2000	Fukunaga et al.	N/A	N/A
6177688	12/2000	Linthicum et al.	N/A	N/A
6178918	12/2000	Van Os et al.	N/A	N/A
6179955	12/2000	Shin et al.	N/A	N/A
6180979	12/2000	Hofman et al.	N/A	N/A
6182603	12/2000	Shang et al.	N/A	N/A
6183564	12/2000	Reynolds et al.	N/A	N/A
6183565	12/2000	Granneman et al.	N/A	N/A
6187672	12/2000	Zhao et al.	N/A	N/A
6187691	12/2000	Fukuda	N/A	N/A
6189482	12/2000	Zhao et al.	N/A	N/A
6190037	12/2000	Das et al.	N/A	N/A
6190113	12/2000	Bui et al.	N/A	N/A
6190634	12/2000	Lieber et al.	N/A	N/A
6191399	12/2000	Van Bilsen	N/A	N/A
6194037	12/2000	Terasaki et al.	N/A	N/A
6200897	12/2000	Wang et al.	N/A	N/A
6201999	12/2000	Jevtic	N/A	N/A
6203613	12/2000	Gates et al.	N/A	N/A
6203618	12/2000	Hashizume et al.	N/A	N/A
6207932	12/2000	Yoo	N/A	N/A
6207936	12/2000	de Waard	N/A	N/A
6209221	12/2000	Beulens	N/A	N/A
6210485	12/2000	Zhao et al.	N/A	N/A
6212789	12/2000	Kato	N/A	N/A
6213708	12/2000	Allen	N/A	N/A
6214122	12/2000	Thompson	N/A	N/A
6214717	12/2000	Yan et al.	N/A	N/A
6217658	12/2000	Orczyk et al.	N/A	N/A
6217662	12/2000	Kong et al.	N/A	N/A
6218288	12/2000	Li et al.	N/A	N/A
6224679	12/2000	Sasaki et al.	N/A	N/A
6225020	12/2000	Jung et al.	N/A	N/A
6225602	12/2000	Buijze et al.	N/A	N/A
6225745	12/2000	Srivastava	N/A	N/A
6230650	12/2000	Yamazaki	N/A	N/A
6231290	12/2000	Kikuchi et al.	N/A	N/A
6235121	12/2000	Honma et al.	N/A	N/A
6235858	12/2000	Swarup et al.	N/A	N/A
6238734	12/2000	Senzaki et al.	N/A	N/A
6239402	12/2000	Araki et al.	N/A	N/A
6239715	12/2000	Belton	N/A	N/A
6240875	12/2000	Wijck et al.	N/A	N/A
6241822	12/2000	Ide	N/A	N/A
6242359	12/2000	Misra	N/A	N/A
6243654	12/2000	Johnson et al.	N/A	N/A
6245647	12/2000	Akiyama et al.	N/A	N/A

6245665	12/2000	Yokoyama	N/A	N/A
6247245	12/2000	Ishii	N/A	N/A
6250250	12/2000	Maishev et al.	N/A	N/A
6255221	12/2000	Hudson et al.	N/A	N/A
6257758	12/2000	Culbertson	N/A	N/A
6261648	12/2000	Akiba et al.	N/A	N/A
6264467	12/2000	Lue et al.	N/A	N/A
6265113	12/2000	Rosasco et al.	N/A	N/A
6265311	12/2000	Hautala et al.	N/A	N/A
6270572	12/2000	Kim et al.	N/A	N/A
6271148	12/2000	Kao	N/A	N/A
6271320	12/2000	Keller et al.	N/A	N/A
6274496	12/2000	Leusink et al.	N/A	N/A
6274878	12/2000	Li et al.	N/A	N/A
6281098	12/2000	Wang	N/A	N/A
6281141	12/2000	Das et al.	N/A	N/A
6283692	12/2000	Perlov et al.	N/A	N/A
6284050	12/2000	Shi et al.	N/A	N/A
6284149	12/2000	Li et al.	N/A	N/A
6287965	12/2000	Kang et al.	N/A	N/A
6287988	12/2000	Nagamine et al.	N/A	N/A
6293700	12/2000	Lund et al.	N/A	N/A
D449873	12/2000	Bronson	N/A	N/A
6296710	12/2000	Allen et al.	N/A	N/A
6296711	12/2000	Loan et al.	N/A	N/A
6296909	12/2000	Spitsberg	N/A	N/A
6297539	12/2000	Ma et al.	N/A	N/A
6299133	12/2000	Waragai et al.	N/A	N/A
6302964	12/2000	Umotoy et al.	N/A	N/A
6303523	12/2000	Cheung	N/A	N/A
6305898	12/2000	Yamagishi et al.	N/A	N/A
6311016	12/2000	Yanagawa et al.	N/A	N/A
6312525	12/2000	Bright et al.	N/A	N/A
6315512	12/2000	Tabrizi et al.	N/A	N/A
6316162	12/2000	Jung et al.	N/A	N/A
6316371	12/2000	Oosterlaken et al.	N/A	N/A
6320320	12/2000	Bailey et al.	N/A	N/A
6321680	12/2000	Cook et al.	N/A	N/A
6321780	12/2000	Iwabuchi	N/A	N/A
D451893	12/2000	Robson	N/A	N/A
D452220	12/2000	Robson	N/A	N/A
6325858	12/2000	Wengert	N/A	N/A
6326322	12/2000	Kim et al.	N/A	N/A
6326597	12/2000	Lubomirsky et al.	N/A	N/A
6328561	12/2000	Hasper et al.	N/A	N/A
6328864	12/2000	Ishizawa et al.	N/A	N/A
6329297	12/2000	Balish	N/A	N/A
6333275	12/2000	Mayer et al.	N/A	N/A
6335049	12/2001	Basceri	N/A	N/A
6335240	12/2001	Kim et al.	N/A	N/A

6335293	12/2001	Luo et al.	N/A	N/A
6342427	12/2001	Choi et al.	N/A	N/A
6343239	12/2001	Toda et al.	N/A	N/A
6344084	12/2001	Koinuma et al.	N/A	N/A
6344232	12/2001	Jones et al.	N/A	N/A
6346419	12/2001	Ryerson et al.	N/A	N/A
6347636	12/2001	Xia	N/A	N/A
6350391	12/2001	Livshits et al.	N/A	N/A
6352049	12/2001	Yin et al.	N/A	N/A
6352945	12/2001	Matsuki	N/A	N/A
6357984	12/2001	Zinger et al.	N/A	N/A
6363294	12/2001	Coronel et al.	N/A	N/A
D455024	12/2001	Mimick et al.	N/A	N/A
6367410	12/2001	Leahey et al.	N/A	N/A
6368773	12/2001	Jung et al.	N/A	N/A
6368987	12/2001	Kopacz et al.	N/A	N/A
6370796	12/2001	Zucker	N/A	N/A
6372583	12/2001	Tyagi	N/A	N/A
6374831	12/2001	Chandran	N/A	N/A
6375312	12/2001	Ikeda et al.	N/A	N/A
6375749	12/2001	Boydston et al.	N/A	N/A
6375750	12/2001	Van Os et al.	N/A	N/A
6379466	12/2001	Sahin et al.	N/A	N/A
D457609	12/2001	Piano	N/A	N/A
6383300	12/2001	Saito et al.	N/A	N/A
6383566	12/2001	Zagdoun	N/A	N/A
6383955	12/2001	Matsuki	N/A	N/A
6387207	12/2001	Janakiraman	N/A	N/A
6387823	12/2001	Sonderman et al.	N/A	N/A
6387827	12/2001	Mertens et al.	N/A	N/A
6390753	12/2001	de Ridder et al.	N/A	N/A
6390754	12/2001	Yamaga et al.	N/A	N/A
6391803	12/2001	Kim et al.	N/A	N/A
6395650	12/2001	Callegari et al.	N/A	N/A
6398184	12/2001	Sowada et al.	N/A	N/A
6399022	12/2001	Schuler et al.	N/A	N/A
6402806	12/2001	Schmitt et al.	N/A	N/A
6410433	12/2001	Hautala et al.	N/A	N/A
6410459	12/2001	Blalock et al.	N/A	N/A
6410463	12/2001	Matsuki	N/A	N/A
6413321	12/2001	Kim et al.	N/A	N/A
6413583	12/2001	Moghadam et al.	N/A	N/A
6420279	12/2001	Ono et al.	N/A	N/A
6423949	12/2001	Chen et al.	N/A	N/A
D461233	12/2001	Whalen	N/A	N/A
D461882	12/2001	Piano	N/A	N/A
6428859	12/2001	Chiang et al.	N/A	N/A
6429465	12/2001	Yagi et al.	N/A	N/A
6432206	12/2001	Tolt	N/A	N/A
6432255	12/2001	Sun et al.	N/A	N/A

6432479	12/2001	Chang et al.	N/A	N/A
6432849	12/2001	Endo et al.	N/A	N/A
6433298	12/2001	Ishii	N/A	N/A
6435798	12/2001	Satoh	N/A	N/A
6435865	12/2001	Tseng et al.	N/A	N/A
6436819	12/2001	Zhang	N/A	N/A
6437444	12/2001	Andideh	N/A	N/A
6438502	12/2001	Awtrey	N/A	N/A
6439822	12/2001	Kimura et al.	N/A	N/A
6440261	12/2001	Tepman et al.	N/A	N/A
6441350	12/2001	Stoddard et al.	N/A	N/A
6445574	12/2001	Saw et al.	N/A	N/A
6446573	12/2001	Hirayama et al.	N/A	N/A
6447232	12/2001	Davis et al.	N/A	N/A
6447651	12/2001	Ishikawa et al.	N/A	N/A
6447937	12/2001	Murakawa et al.	N/A	N/A
6448192	12/2001	Kaushik	N/A	N/A
6450117	12/2001	Muruges h et al.	N/A	N/A
6450757	12/2001	Saeki	N/A	N/A
6451713	12/2001	Tay et al.	N/A	N/A
6452017	12/2001	Uhlenbrock et al.	N/A	N/A
6454860	12/2001	Metzner et al.	N/A	N/A
6454909	12/2001	Matsuse et al.	N/A	N/A
6455098	12/2001	Tran et al.	N/A	N/A
6455225	12/2001	Kong et al.	N/A	N/A
6455445	12/2001	Matsuki	N/A	N/A
6460482	12/2001	Kuibira et al.	N/A	N/A
6461435	12/2001	Littau et al.	N/A	N/A
6461436	12/2001	Campbell et al.	N/A	N/A
6461439	12/2001	Granneman et al.	N/A	N/A
6462310	12/2001	Ratliff et al.	N/A	N/A
6464825	12/2001	Shinozaki	N/A	N/A
6468924	12/2001	Lee	N/A	N/A
6471779	12/2001	Nishio et al.	N/A	N/A
6472266	12/2001	Yu et al.	N/A	N/A
6474987	12/2001	Huang et al.	N/A	N/A
6475276	12/2001	Elers et al.	N/A	N/A
6475902	12/2001	Hausmann et al.	N/A	N/A
6475930	12/2001	Junker et al.	N/A	N/A
6478872	12/2001	Chae et al.	N/A	N/A
6481945	12/2001	Hasper et al.	N/A	N/A
6482331	12/2001	Lu et al.	N/A	N/A
6482663	12/2001	Buckland	N/A	N/A
6483989	12/2001	Okada et al.	N/A	N/A
6488774	12/2001	Horie et al.	N/A	N/A
6490493	12/2001	Dharnipragada	N/A	N/A
6492625	12/2001	Boguslavskiy et al.	N/A	N/A
6494065	12/2001	Babbitt	N/A	N/A
6494998	12/2001	Brcka	N/A	N/A
6496819	12/2001	Bello et al.	N/A	N/A

6497734	12/2001	Barber et al.	N/A	N/A
6497767	12/2001	Okase et al.	N/A	N/A
6498091	12/2001	Chen et al.	N/A	N/A
6499533	12/2001	Yamada	N/A	N/A
6500487	12/2001	Holst et al.	N/A	N/A
6502530	12/2002	Turlot et al.	N/A	N/A
6503079	12/2002	Kogano et al.	N/A	N/A
6503330	12/2002	Sneh et al.	N/A	N/A
6503365	12/2002	Kim et al.	N/A	N/A
6503562	12/2002	Saito et al.	N/A	N/A
6503826	12/2002	Oda	N/A	N/A
6506009	12/2002	Nulman et al.	N/A	N/A
6506253	12/2002	Sakuma	N/A	N/A
6507410	12/2002	Robertson et al.	N/A	N/A
6511539	12/2002	Raaijmakers	N/A	N/A
6514313	12/2002	Spiegelman	N/A	N/A
6514666	12/2002	Choi et al.	N/A	N/A
6521295	12/2002	Remington	N/A	N/A
6521547	12/2002	Chang et al.	N/A	N/A
6527884	12/2002	Takakuwa et al.	N/A	N/A
6528171	12/2002	Endler et al.	N/A	N/A
6528430	12/2002	Kwan	N/A	N/A
6528752	12/2002	Ishii et al.	N/A	N/A
6528767	12/2002	Bagley et al.	N/A	N/A
6530994	12/2002	Mahawili	N/A	N/A
6531193	12/2002	Fonash et al.	N/A	N/A
6531412	12/2002	Conti et al.	N/A	N/A
6534133	12/2002	Kaloyeros et al.	N/A	N/A
6534395	12/2002	Werkhoven et al.	N/A	N/A
6536950	12/2002	Green	N/A	N/A
6539891	12/2002	Kang et al.	N/A	N/A
6540469	12/2002	Matsunaga et al.	N/A	N/A
6540838	12/2002	Sneh et al.	N/A	N/A
6544906	12/2002	Rotondaro et al.	N/A	N/A
6552209	12/2002	Lei et al.	N/A	N/A
6558517	12/2002	Basceri	N/A	N/A
6558755	12/2002	Berry et al.	N/A	N/A
6559026	12/2002	Rossmann et al.	N/A	N/A
6562094	12/2002	Denker et al.	N/A	N/A
6565763	12/2002	Asakawa et al.	N/A	N/A
6566278	12/2002	Harvey et al.	N/A	N/A
6569239	12/2002	Arai et al.	N/A	N/A
6569971	12/2002	Roh et al.	N/A	N/A
6573030	12/2002	Fairbairn et al.	N/A	N/A
6574644	12/2002	Hsu et al.	N/A	N/A
6576062	12/2002	Matsuse	N/A	N/A
6576064	12/2002	Griffiths et al.	N/A	N/A
6576300	12/2002	Berry et al.	N/A	N/A
6576564	12/2002	Agarwal	N/A	N/A
6578589	12/2002	Mayusumi	N/A	N/A

6579833	12/2002	McNallan et al.	N/A	N/A
6580050	12/2002	Miller et al.	N/A	N/A
6582174	12/2002	Hayashi	N/A	N/A
6583048	12/2002	Vincent et al.	N/A	N/A
6583572	12/2002	Veltrop et al.	N/A	N/A
6587108	12/2002	Guerlain et al.	N/A	N/A
6589352	12/2002	Yudovsky et al.	N/A	N/A
6589707	12/2002	Lee et al.	N/A	N/A
6589868	12/2002	Rossmann	N/A	N/A
6590251	12/2002	Kang et al.	N/A	N/A
6594550	12/2002	Okrah	N/A	N/A
6596398	12/2002	Russo et al.	N/A	N/A
6596653	12/2002	Tan	N/A	N/A
6598559	12/2002	Vellore et al.	N/A	N/A
6602806	12/2002	Xia et al.	N/A	N/A
6607602	12/2002	Granneman et al.	N/A	N/A
6607868	12/2002	Choi	N/A	N/A
6607948	12/2002	Sugiyama et al.	N/A	N/A
6608745	12/2002	Tsuruta et al.	N/A	N/A
6610375	12/2002	Akiba et al.	N/A	N/A
6613685	12/2002	Granneman et al.	N/A	N/A
6616986	12/2002	Sherman	N/A	N/A
6617253	12/2002	Chu et al.	N/A	N/A
6620251	12/2002	Kitano	N/A	N/A
6623799	12/2002	Lee et al.	N/A	N/A
6624064	12/2002	Sahin	N/A	N/A
6627268	12/2002	Fair et al.	N/A	N/A
6627503	12/2002	Ma et al.	N/A	N/A
6630030	12/2002	Suntola et al.	N/A	N/A
6632478	12/2002	Gaillard et al.	N/A	N/A
6633364	12/2002	Hayashi	N/A	N/A
6635115	12/2002	Fairbairn et al.	N/A	N/A
6635117	12/2002	Kinnard et al.	N/A	N/A
6635578	12/2002	Xu et al.	N/A	N/A
6638839	12/2002	Deng et al.	N/A	N/A
6640145	12/2002	Hoffberg et al.	N/A	N/A
6645304	12/2002	Yamaguchi	N/A	N/A
6648974	12/2002	Ogliari et al.	N/A	N/A
6649921	12/2002	Cekic et al.	N/A	N/A
6652924	12/2002	Sherman	N/A	N/A
6656281	12/2002	Ueda	N/A	N/A
6656282	12/2002	Kim et al.	N/A	N/A
6658933	12/2002	Allegre et al.	N/A	N/A
6659111	12/2002	Mouri et al.	N/A	N/A
6660662	12/2002	Ishikawa et al.	N/A	N/A
6662817	12/2002	Yamagishi	N/A	N/A
6663332	12/2002	Sluijk et al.	N/A	N/A
6673196	12/2003	Oyabu	N/A	N/A
6676290	12/2003	Lu	N/A	N/A
6676759	12/2003	Takagi	N/A	N/A

6679194	12/2003	Ham et al.	N/A	N/A
6682971	12/2003	Tsuneda et al.	N/A	N/A
6682973	12/2003	Paton et al.	N/A	N/A
6683274	12/2003	Kwon et al.	N/A	N/A
D486891	12/2003	Cronce	N/A	N/A
6684659	12/2003	Tanaka et al.	N/A	N/A
6684719	12/2003	Gehner et al.	N/A	N/A
6686281	12/2003	Yamazaki et al.	N/A	N/A
6688784	12/2003	Templeton	N/A	N/A
6689220	12/2003	Nguyen	N/A	N/A
6692575	12/2003	Omstead et al.	N/A	N/A
6692576	12/2003	Halpin et al.	N/A	N/A
6692903	12/2003	Chen et al.	N/A	N/A
6696367	12/2003	Aggarwal	N/A	N/A
6699003	12/2003	Saeki	N/A	N/A
6699399	12/2003	Qian et al.	N/A	N/A
6700089	12/2003	Hirooka	N/A	N/A
6709989	12/2003	Ramdani et al.	N/A	N/A
6710364	12/2003	Guldi et al.	N/A	N/A
6710857	12/2003	Kondo	N/A	N/A
6712949	12/2003	Gopal	N/A	N/A
6713824	12/2003	Mikata	N/A	N/A
6716477	12/2003	Komiyama et al.	N/A	N/A
6716571	12/2003	Gabriel	N/A	N/A
6719499	12/2003	Kuznetsov et al.	N/A	N/A
6720260	12/2003	Fair et al.	N/A	N/A
6720262	12/2003	Koh et al.	N/A	N/A
6720531	12/2003	Jacobson et al.	N/A	N/A
6722837	12/2003	Inui	N/A	N/A
6723642	12/2003	Lim et al.	N/A	N/A
6730614	12/2003	Lim et al.	N/A	N/A
6732006	12/2003	Haanstra et al.	N/A	N/A
6734090	12/2003	Agarwala et al.	N/A	N/A
6734631	12/2003	Juestel et al.	N/A	N/A
6737716	12/2003	Matsuo et al.	N/A	N/A
6740853	12/2003	Johnson et al.	N/A	N/A
6743475	12/2003	Skarp et al.	N/A	N/A
6743738	12/2003	Todd et al.	N/A	N/A
6745095	12/2003	Ben-Dov	N/A	N/A
6746240	12/2003	de Ridder et al.	N/A	N/A
6746308	12/2003	Bode et al.	N/A	N/A
6749671	12/2003	Holst et al.	N/A	N/A
6753507	12/2003	Fure et al.	N/A	N/A
6755221	12/2003	Jeong et al.	N/A	N/A
6756085	12/2003	Waldfried	N/A	N/A
6756293	12/2003	Li et al.	N/A	N/A
6756318	12/2003	Nguyen et al.	N/A	N/A
6759098	12/2003	Han	N/A	N/A
6760981	12/2003	Leap	N/A	N/A
6766545	12/2003	Hodges	N/A	N/A

6767447	12/2003	Uno et al.	N/A	N/A
D494552	12/2003	Tezuka et al.	N/A	N/A
6776849	12/2003	Aggarwal et al.	N/A	N/A
6780704	12/2003	Raaijmakers et al.	N/A	N/A
6783875	12/2003	Yamada et al.	N/A	N/A
6784033	12/2003	Yamazaki	N/A	N/A
6784108	12/2003	Donohoe et al.	N/A	N/A
D496008	12/2003	Takahashi et al.	N/A	N/A
6786997	12/2003	Yamazaki	N/A	N/A
D497536	12/2003	Fujiwara	N/A	N/A
D497977	12/2003	Engelbrektsson	N/A	N/A
6811960	12/2003	Lee et al.	N/A	N/A
6812157	12/2003	Gadgil	N/A	N/A
6815350	12/2003	Kim et al.	N/A	N/A
6815352	12/2003	Tamura et al.	N/A	N/A
6818566	12/2003	Leeson et al.	N/A	N/A
6818864	12/2003	Ptak	N/A	N/A
6820570	12/2003	Kilpela et al.	N/A	N/A
6821889	12/2003	Elers et al.	N/A	N/A
6821910	12/2003	Adomaitis et al.	N/A	N/A
6824665	12/2003	Shelnut et al.	N/A	N/A
6825106	12/2003	Gao et al.	N/A	N/A
6825134	12/2003	Law et al.	N/A	N/A
D499620	12/2003	Horner-Richardson et al.	N/A	N/A
6827789	12/2003	Lee et al.	N/A	N/A
6828235	12/2003	Takano	N/A	N/A
6831004	12/2003	Byun	N/A	N/A
6833024	12/2003	Holst et al.	N/A	N/A
6835039	12/2003	Van Den Berg	N/A	N/A
6838122	12/2004	Basceri et al.	N/A	N/A
6841201	12/2004	Shanov et al.	N/A	N/A
6843202	12/2004	Kusuda	N/A	N/A
6843858	12/2004	Rossman	N/A	N/A
6846146	12/2004	Inui	N/A	N/A
6846515	12/2004	Vrtis	N/A	N/A
6846742	12/2004	Rossman	N/A	N/A
6847014	12/2004	Benjamin et al.	N/A	N/A
6849241	12/2004	Dauelsberg et al.	N/A	N/A
6849857	12/2004	Ichiki et al.	N/A	N/A
6854580	12/2004	Braford	N/A	N/A
6858524	12/2004	Haukka et al.	N/A	N/A
6858547	12/2004	Metzner	N/A	N/A
6861642	12/2004	Ichiki et al.	N/A	N/A
6863019	12/2004	Shamouilian	N/A	N/A
6863281	12/2004	Endou et al.	N/A	N/A
6864041	12/2004	Brown	N/A	N/A
6867086	12/2004	Chen et al.	N/A	N/A
6867153	12/2004	Tokunaga	N/A	N/A
6867859	12/2004	Powell	N/A	N/A

6872258	12/2004	Park et al.	N/A	N/A
6872259	12/2004	Strang	N/A	N/A
D504142	12/2004	Horner-Richardson et al.	N/A	N/A
6874247	12/2004	Hsu	N/A	N/A
6874480	12/2004	Ismailov	N/A	N/A
6875477	12/2004	Trickett et al.	N/A	N/A
6875677	12/2004	Conley, Jr. et al.	N/A	N/A
6876017	12/2004	Goodner	N/A	N/A
6876191	12/2004	de Ridder	N/A	N/A
6878206	12/2004	Tzu et al.	N/A	N/A
6878402	12/2004	Chiang et al.	N/A	N/A
6883733	12/2004	Lind	N/A	N/A
6884066	12/2004	Nguyen et al.	N/A	N/A
6884295	12/2004	Ishii	N/A	N/A
6884319	12/2004	Kim	N/A	N/A
6884475	12/2004	Basceri	N/A	N/A
D505590	12/2004	Greiner	N/A	N/A
6889211	12/2004	Yoshiura et al.	N/A	N/A
6889864	12/2004	Lindfors et al.	N/A	N/A
6890596	12/2004	Sarigiannis et al.	N/A	N/A
6895158	12/2004	Alyward et al.	N/A	N/A
6899507	12/2004	Yamagishi et al.	N/A	N/A
6902395	12/2004	Oosterlaken et al.	N/A	N/A
6902647	12/2004	Hasper	N/A	N/A
6909839	12/2004	Wang et al.	N/A	N/A
6911092	12/2004	Sneh	N/A	N/A
6913152	12/2004	Zuk	N/A	N/A
6913796	12/2004	Albano et al.	N/A	N/A
6916398	12/2004	Chen et al.	N/A	N/A
6916559	12/2004	Murakawa et al.	N/A	N/A
6917755	12/2004	Nguyen et al.	N/A	N/A
6924078	12/2004	Lee et al.	N/A	N/A
6928890	12/2004	Gehner et al.	N/A	N/A
6929699	12/2004	Whitesell	N/A	N/A
6929700	12/2004	Tan et al.	N/A	N/A
6930041	12/2004	Agarwal	N/A	N/A
6930059	12/2004	Conley, Jr. et al.	N/A	N/A
6935269	12/2004	Lee et al.	N/A	N/A
6939579	12/2004	Bondestam et al.	N/A	N/A
6939817	12/2004	Sandhu et al.	N/A	N/A
6942753	12/2004	Choi et al.	N/A	N/A
6949204	12/2004	Lenz et al.	N/A	N/A
6951587	12/2004	Narushima	N/A	N/A
6952656	12/2004	Cordova et al.	N/A	N/A
6953609	12/2004	Carollo	N/A	N/A
6955741	12/2004	Yamagishi	N/A	N/A
6955836	12/2004	Kumagai et al.	N/A	N/A
6955928	12/2004	Brennan	N/A	N/A
6963052	12/2004	Kuibira et al.	N/A	N/A

6972055	12/2004	Sferlazzo	N/A	N/A
6972478	12/2004	Waite et al.	N/A	N/A
6974781	12/2004	Timmermans et al.	N/A	N/A
6975921	12/2004	Verhaar	N/A	N/A
6976822	12/2004	Woodruff	N/A	N/A
RE38937	12/2005	Nakamura	N/A	N/A
6981832	12/2005	Zinger et al.	N/A	N/A
6982046	12/2005	Srivastava et al.	N/A	N/A
6982103	12/2005	Basceri et al.	N/A	N/A
6984591	12/2005	Buchanan et al.	N/A	N/A
6984595	12/2005	Yamazaki	N/A	N/A
6985788	12/2005	Haanstra et al.	N/A	N/A
6986914	12/2005	Elers et al.	N/A	N/A
6987155	12/2005	Roh et al.	N/A	N/A
6990430	12/2005	Hosek	N/A	N/A
7005227	12/2005	Yueh et al.	N/A	N/A
7005391	12/2005	Min	N/A	N/A
7008879	12/2005	Lee et al.	N/A	N/A
7010580	12/2005	Fu et al.	N/A	N/A
7017514	12/2005	Shepherd et al.	N/A	N/A
7018941	12/2005	Cui et al.	N/A	N/A
7021330	12/2005	Maula et al.	N/A	N/A
7021881	12/2005	Yamagishi	N/A	N/A
7036453	12/2005	Ishikawa et al.	N/A	N/A
7041609	12/2005	Vaartstra	N/A	N/A
7045430	12/2005	Ahn et al.	N/A	N/A
7049226	12/2005	Chung et al.	N/A	N/A
7049247	12/2005	Gates et al.	N/A	N/A
7052584	12/2005	Basceri	N/A	N/A
7053009	12/2005	Conley, Jr. et al.	N/A	N/A
7055263	12/2005	Wu et al.	N/A	N/A
7055875	12/2005	Bonora	N/A	N/A
7062161	12/2005	Kusuda et al.	N/A	N/A
D524600	12/2005	Austin et al.	N/A	N/A
D525127	12/2005	Cogley et al.	N/A	N/A
7070178	12/2005	Van Der Toorn et al.	N/A	N/A
7071051	12/2005	Jeon et al.	N/A	N/A
7073834	12/2005	Matsumoto et al.	N/A	N/A
7074690	12/2005	Gauri et al.	N/A	N/A
7077614	12/2005	Hasper et al.	N/A	N/A
7080545	12/2005	Dimeo et al.	N/A	N/A
7084060	12/2005	Furukawa	N/A	N/A
7084079	12/2005	Conti et al.	N/A	N/A
7085623	12/2005	Siegers	N/A	N/A
7086347	12/2005	Howald et al.	N/A	N/A
7088003	12/2005	Gates et al.	N/A	N/A
7090394	12/2005	Hashikura et al.	N/A	N/A
7092287	12/2005	Beulens et al.	N/A	N/A
7098149	12/2005	Lukas	N/A	N/A
7098150	12/2005	Misra et al.	N/A	N/A

7100459	12/2005	Gehner et al.	N/A	N/A
7101763	12/2005	Anderson et al.	N/A	N/A
7108753	12/2005	Wood	N/A	N/A
7109098	12/2005	Ramaswamy et al.	N/A	N/A
7109114	12/2005	Chen et al.	N/A	N/A
7111232	12/2005	Bascom	N/A	N/A
7115305	12/2005	Bronikowski et al.	N/A	N/A
7115838	12/2005	Kurara et al.	N/A	N/A
7122085	12/2005	Shero et al.	N/A	N/A
7122222	12/2005	Xiao et al.	N/A	N/A
7122844	12/2005	Nakamura et al.	N/A	N/A
7129165	12/2005	Basol et al.	N/A	N/A
7132360	12/2005	Schaeffer et al.	N/A	N/A
7135421	12/2005	Ahn et al.	N/A	N/A
7140558	12/2005	McCracken et al.	N/A	N/A
7141499	12/2005	Raaijmakers	N/A	N/A
7143897	12/2005	Guzman et al.	N/A	N/A
7144806	12/2005	Fair et al.	N/A	N/A
7144809	12/2005	Elers et al.	N/A	N/A
7147766	12/2005	Uzoh et al.	N/A	N/A
7153542	12/2005	Nguyen et al.	N/A	N/A
D535673	12/2006	Conway et al.	N/A	N/A
7156380	12/2006	Soininen	N/A	N/A
7163393	12/2006	Adachi et al.	N/A	N/A
7163721	12/2006	Zhang et al.	N/A	N/A
7163900	12/2006	Weber	N/A	N/A
7168852	12/2006	Linnarsson	N/A	N/A
7172497	12/2006	Basol et al.	N/A	N/A
7173216	12/2006	Ptak	N/A	N/A
7183229	12/2006	Yamanaka	N/A	N/A
7186648	12/2006	Rozbicki	N/A	N/A
7192824	12/2006	Ahn et al.	N/A	N/A
7192892	12/2006	Ahn et al.	N/A	N/A
7195479	12/2006	Beatty et al.	N/A	N/A
7195693	12/2006	Cowans	N/A	N/A
D541125	12/2006	Gaudron	N/A	N/A
7198447	12/2006	Morimitsu et al.	N/A	N/A
7199513	12/2006	Huber et al.	N/A	N/A
7201943	12/2006	Park et al.	N/A	N/A
7202148	12/2006	Chen et al.	N/A	N/A
7202512	12/2006	Chen et al.	N/A	N/A
7204886	12/2006	Chen et al.	N/A	N/A
7204887	12/2006	Kawamura et al.	N/A	N/A
7205246	12/2006	MacNeil et al.	N/A	N/A
7205247	12/2006	Lee et al.	N/A	N/A
7207763	12/2006	Lee	N/A	N/A
7208198	12/2006	Basceri et al.	N/A	N/A
7208389	12/2006	Tipton et al.	N/A	N/A
7208413	12/2006	Byun et al.	N/A	N/A
7210925	12/2006	Adachi	N/A	N/A

7211524	12/2006	Ryu et al.	N/A	N/A
7211525	12/2006	Shanker	N/A	N/A
7214630	12/2006	Varadarajan et al.	N/A	N/A
7217617	12/2006	Basceri	N/A	N/A
7223014	12/2006	Lojen	N/A	N/A
7229502	12/2006	Wang et al.	N/A	N/A
7234476	12/2006	Arai	N/A	N/A
7235137	12/2006	Kitayama et al.	N/A	N/A
7235482	12/2006	Wu	N/A	N/A
7235501	12/2006	Ahn et al.	N/A	N/A
7238596	12/2006	Kouvetakis et al.	N/A	N/A
7238616	12/2006	Agarwal	N/A	N/A
7238653	12/2006	Lee et al.	N/A	N/A
D549815	12/2006	Murphy	N/A	N/A
7256375	12/2006	Oosterlaken	N/A	N/A
7265061	12/2006	Cho et al.	N/A	N/A
7274867	12/2006	Peukert	N/A	N/A
D553104	12/2006	Oohashi et al.	N/A	N/A
7279256	12/2006	Son	N/A	N/A
7290813	12/2006	Bonora	N/A	N/A
7294581	12/2006	Haverkort et al.	N/A	N/A
7296460	12/2006	Dimeo et al.	N/A	N/A
7297641	12/2006	Todd et al.	N/A	N/A
7298009	12/2006	Yan et al.	N/A	N/A
7301623	12/2006	Madsen et al.	N/A	N/A
D556704	12/2006	Nakamura et al.	N/A	N/A
D557226	12/2006	Uchino et al.	N/A	N/A
D558021	12/2006	Lawrence	N/A	N/A
7307028	12/2006	Goto et al.	N/A	N/A
7307178	12/2006	Kiyomori et al.	N/A	N/A
7311977	12/2006	Yokota et al.	N/A	N/A
7312148	12/2006	Ramaswamy et al.	N/A	N/A
7312162	12/2006	Ramaswamy et al.	N/A	N/A
7312494	12/2006	Ahn et al.	N/A	N/A
D559993	12/2007	Nagakubo et al.	N/A	N/A
D559994	12/2007	Nagakubo et al.	N/A	N/A
7320544	12/2007	Hsieh	N/A	N/A
7323401	12/2007	Ramaswamy et al.	N/A	N/A
D562357	12/2007	Hardy	N/A	N/A
7326656	12/2007	Brask et al.	N/A	N/A
7326657	12/2007	Xia et al.	N/A	N/A
7327948	12/2007	Shrinivasan	N/A	N/A
7329947	12/2007	Adachi et al.	N/A	N/A
7335611	12/2007	Ramaswamy et al.	N/A	N/A
7351057	12/2007	Berenbak et al.	N/A	N/A
7354482	12/2007	Konishi et al.	N/A	N/A
7354847	12/2007	Chan et al.	N/A	N/A
7354873	12/2007	Fukazawa et al.	N/A	N/A
7356762	12/2007	van Driel	N/A	N/A
7357138	12/2007	Ji et al.	N/A	N/A

7361447	12/2007	Jung	N/A	N/A
7375035	12/2007	Heden et al.	N/A	N/A
7376520	12/2007	Wong	N/A	N/A
7378618	12/2007	Sorabji et al.	N/A	N/A
7379785	12/2007	Higashi et al.	N/A	N/A
D571383	12/2007	Ota et al.	N/A	N/A
D571831	12/2007	Ota et al.	N/A	N/A
7381644	12/2007	Soubramonium et al.	N/A	N/A
7387685	12/2007	Choi et al.	N/A	N/A
7393207	12/2007	Imai	N/A	N/A
7393418	12/2007	Yokogawa	N/A	N/A
7393736	12/2007	Ahn et al.	N/A	N/A
7393765	12/2007	Hanawa et al.	N/A	N/A
7396491	12/2007	Marking et al.	N/A	N/A
7399388	12/2007	Moghadam et al.	N/A	N/A
7399570	12/2007	Lee et al.	N/A	N/A
7402534	12/2007	Mahajani	N/A	N/A
7405166	12/2007	Liang et al.	N/A	N/A
7405454	12/2007	Ahn et al.	N/A	N/A
D575713	12/2007	Ratcliffe	N/A	N/A
7408225	12/2007	Shinriki et al.	N/A	N/A
7410290	12/2007	Tanaka	N/A	N/A
7410666	12/2007	Elers	N/A	N/A
7411352	12/2007	Madocks	N/A	N/A
7414281	12/2007	Fastow	N/A	N/A
D576001	12/2007	Brunderman	N/A	N/A
7422635	12/2007	Zheng et al.	N/A	N/A
7422636	12/2007	Ishizaka	N/A	N/A
7422653	12/2007	Blahnik et al.	N/A	N/A
7422775	12/2007	Ramaswamy et al.	N/A	N/A
7425224	12/2007	Nguyen	N/A	N/A
7427571	12/2007	Lindeboom et al.	N/A	N/A
7429532	12/2007	Ramaswamy et al.	N/A	N/A
7431966	12/2007	Derderian et al.	N/A	N/A
7432476	12/2007	Morita et al.	N/A	N/A
7437060	12/2007	Wang et al.	N/A	N/A
7442275	12/2007	Cowans	N/A	N/A
7456429	12/2007	Levy	N/A	N/A
D583395	12/2007	Ueda	N/A	N/A
7467632	12/2007	Lee et al.	N/A	N/A
7473655	12/2008	Wang et al.	N/A	N/A
7475588	12/2008	Dimeo et al.	N/A	N/A
7476291	12/2008	Wang et al.	N/A	N/A
7479198	12/2008	Guffrey	N/A	N/A
7482247	12/2008	Papasouliotis	N/A	N/A
7482283	12/2008	Yamasaki et al.	N/A	N/A
D585968	12/2008	Elkins et al.	N/A	N/A
7489389	12/2008	Shibazaki et al.	N/A	N/A
7494882	12/2008	Vitale	N/A	N/A
7497614	12/2008	Gaff	N/A	N/A

7498242	12/2008	Kumar et al.	N/A	N/A
7501292	12/2008	Matsushita et al.	N/A	N/A
7501355	12/2008	Bhatia et al.	N/A	N/A
7503980	12/2008	Kida et al.	N/A	N/A
7504344	12/2008	Matsuki et al.	N/A	N/A
D590933	12/2008	Vansell	N/A	N/A
7514058	12/2008	Hitzman et al.	N/A	N/A
7514375	12/2008	Shanker et al.	N/A	N/A
D593585	12/2008	Ota et al.	N/A	N/A
D593969	12/2008	Li	N/A	N/A
7541297	12/2008	Mallick et al.	N/A	N/A
7544398	12/2008	Chang et al.	N/A	N/A
7547363	12/2008	Tomiyasu et al.	N/A	N/A
7547633	12/2008	Ranish et al.	N/A	N/A
7550396	12/2008	Frohberg et al.	N/A	N/A
D596476	12/2008	Welch	N/A	N/A
7561982	12/2008	Rund et al.	N/A	N/A
7563715	12/2008	Haukka et al.	N/A	N/A
7566891	12/2008	Rocha-Alvarez et al.	N/A	N/A
7569193	12/2008	Ferron et al.	N/A	N/A
7575968	12/2008	Sadaka et al.	N/A	N/A
7579285	12/2008	Zimmerman et al.	N/A	N/A
7579785	12/2008	Shinmen et al.	N/A	N/A
D600223	12/2008	Aggarwal	N/A	N/A
7582555	12/2008	Lang	N/A	N/A
7582575	12/2008	Fukazawa et al.	N/A	N/A
7589003	12/2008	Kouvetakis et al.	N/A	N/A
7589028	12/2008	Cho et al.	N/A	N/A
7589029	12/2008	Derderian et al.	N/A	N/A
7591601	12/2008	Matsuoka et al.	N/A	N/A
7591907	12/2008	Chen et al.	N/A	N/A
D602575	12/2008	Breda	N/A	N/A
7598513	12/2008	Kouvetakis et al.	N/A	N/A
7601223	12/2008	Lindfors et al.	N/A	N/A
7601225	12/2008	Tuominen et al.	N/A	N/A
7601652	12/2008	Singh et al.	N/A	N/A
7611640	12/2008	Howald et al.	N/A	N/A
7611751	12/2008	Elers	N/A	N/A
7611980	12/2008	Wells et al.	N/A	N/A
7618226	12/2008	Takizawa	N/A	N/A
7621672	12/2008	Ripley	N/A	N/A
7622369	12/2008	Lee et al.	N/A	N/A
7622378	12/2008	Liu et al.	N/A	N/A
7623940	12/2008	Huskamp et al.	N/A	N/A
D606952	12/2008	Lee et al.	N/A	N/A
7625820	12/2008	Papasouliotis	N/A	N/A
7629277	12/2008	Ghatnagar	N/A	N/A
7632549	12/2008	Goundar	N/A	N/A
7638951	12/2008	DeVincentis et al.	N/A	N/A
7640142	12/2008	Tachikawa et al.	N/A	N/A

7645341	12/2009	Kennedy et al.	N/A	N/A
7645484	12/2009	Ishizaka	N/A	N/A
7648895	12/2009	Kurokawa et al.	N/A	N/A
7648927	12/2009	Singh et al.	N/A	N/A
7651269	12/2009	Comendant	N/A	N/A
7651583	12/2009	Kent et al.	N/A	N/A
7651955	12/2009	Ranish et al.	N/A	N/A
7651959	12/2009	Fukazawa et al.	N/A	N/A
7651961	12/2009	Clark	N/A	N/A
D609652	12/2009	Nagasaka et al.	N/A	N/A
D609655	12/2009	Sugimoto	N/A	N/A
7661299	12/2009	Kusunoki	N/A	N/A
7662689	12/2009	Boyanov et al.	N/A	N/A
7670432	12/2009	Li	N/A	N/A
7674726	12/2009	Hasper et al.	N/A	N/A
7678197	12/2009	Maki	N/A	N/A
7678715	12/2009	Mungekar et al.	N/A	N/A
7682454	12/2009	Sneh	N/A	N/A
7682657	12/2009	Sherman	N/A	N/A
D613829	12/2009	Griffin et al.	N/A	N/A
D614153	12/2009	Fondurulia et al.	N/A	N/A
D614258	12/2009	Kojima	N/A	N/A
D614267	12/2009	Breda	N/A	N/A
D614268	12/2009	Breda	N/A	N/A
D614593	12/2009	Lee et al.	N/A	N/A
7690881	12/2009	Yamagishi	N/A	N/A
7691205	12/2009	Ikedo	N/A	N/A
7692171	12/2009	Kaszuba et al.	N/A	N/A
7695808	12/2009	Tuma	N/A	N/A
D616390	12/2009	Sato	N/A	N/A
D616394	12/2009	Sato	N/A	N/A
7712435	12/2009	Yoshizaki et al.	N/A	N/A
7713874	12/2009	Milligan	N/A	N/A
7716993	12/2009	Ozawa et al.	N/A	N/A
7718930	12/2009	Kawasaki et al.	N/A	N/A
7720560	12/2009	Menser et al.	N/A	N/A
7723648	12/2009	Tsukamoto et al.	N/A	N/A
7725012	12/2009	Aggarwal et al.	N/A	N/A
7727864	12/2009	Elers	N/A	N/A
7727880	12/2009	Chattopadhyay et al.	N/A	N/A
7732343	12/2009	Niroomand et al.	N/A	N/A
7736437	12/2009	Cadwell et al.	N/A	N/A
7736528	12/2009	Okita et al.	N/A	N/A
7736600	12/2009	Clark et al.	N/A	N/A
7737035	12/2009	Lind et al.	N/A	N/A
7740437	12/2009	de Ridder et al.	N/A	N/A
7740705	12/2009	Li	N/A	N/A
7745346	12/2009	Hausmann et al.	N/A	N/A
7748760	12/2009	Kushida	N/A	N/A
7749563	12/2009	Zheng et al.	N/A	N/A

7753584	12/2009	Gambino et al.	N/A	N/A
7754621	12/2009	Putjkonen	N/A	N/A
7758698	12/2009	Bang et al.	N/A	N/A
7763869	12/2009	Matsushita et al.	N/A	N/A
7767262	12/2009	Clark	N/A	N/A
7771796	12/2009	Kohno et al.	N/A	N/A
7780440	12/2009	Shibagaki et al.	N/A	N/A
7780789	12/2009	Wu et al.	N/A	N/A
7781352	12/2009	Fukazawa et al.	N/A	N/A
7789559	12/2009	Waser et al.	N/A	N/A
7789965	12/2009	Matsushita et al.	N/A	N/A
7790633	12/2009	Tarafdar et al.	N/A	N/A
7794546	12/2009	Li	N/A	N/A
7795160	12/2009	Wang et al.	N/A	N/A
7798096	12/2009	Mahajani et al.	N/A	N/A
7799300	12/2009	Lindfors et al.	N/A	N/A
7799706	12/2009	Yeom et al.	N/A	N/A
7803722	12/2009	Liang	N/A	N/A
D625977	12/2009	Watson et al.	N/A	N/A
7806587	12/2009	Kobayashi	N/A	N/A
7807566	12/2009	Tsuji et al.	N/A	N/A
7807578	12/2009	Bencher et al.	N/A	N/A
7816278	12/2009	Reed et al.	N/A	N/A
7824492	12/2009	Tois et al.	N/A	N/A
7825040	12/2009	Fukazawa et al.	N/A	N/A
7829460	12/2009	Streck et al.	N/A	N/A
7833348	12/2009	Wada et al.	N/A	N/A
7833353	12/2009	Furukawahara et al.	N/A	N/A
7838084	12/2009	Derderian et al.	N/A	N/A
7842518	12/2009	Miyajima	N/A	N/A
7842622	12/2009	Lee et al.	N/A	N/A
D629874	12/2009	Hermans	N/A	N/A
7850449	12/2009	Yang et al.	N/A	N/A
7851019	12/2009	Tuominen et al.	N/A	N/A
7851232	12/2009	van Schravendijk et al.	N/A	N/A
7858519	12/2009	Liu et al.	N/A	N/A
7858533	12/2009	Liu et al.	N/A	N/A
7858898	12/2009	Bailey et al.	N/A	N/A
7865070	12/2010	Nakamura	N/A	N/A
7871198	12/2010	Rempe et al.	N/A	N/A
7874726	12/2010	Jacobs et al.	N/A	N/A
7884918	12/2010	Hattori	N/A	N/A
7888233	12/2010	Gauri	N/A	N/A
7894474	12/2010	Bell	N/A	N/A
D633452	12/2010	Namiki et al.	N/A	N/A
D634329	12/2010	Wastrom	N/A	N/A
D634719	12/2010	Yasuda et al.	N/A	N/A
7897215	12/2010	Fair et al.	N/A	N/A
7897217	12/2010	Faguet	N/A	N/A
7902009	12/2010	Simonelli et al.	N/A	N/A

7902582	12/2010	Forbes et al.	N/A	N/A
7906174	12/2010	Wu et al.	N/A	N/A
7910288	12/2010	Abatchev et al.	N/A	N/A
7910452	12/2010	Roh et al.	N/A	N/A
7910494	12/2010	Dip et al.	N/A	N/A
7915139	12/2010	Lang	N/A	N/A
7915667	12/2010	Knoefler et al.	N/A	N/A
7919142	12/2010	Yeom et al.	N/A	N/A
7919416	12/2010	Lee et al.	N/A	N/A
7923382	12/2010	Huotari et al.	N/A	N/A
7925378	12/2010	Gilchrist et al.	N/A	N/A
7935940	12/2010	Smargiassi	N/A	N/A
7939447	12/2010	Bauer et al.	N/A	N/A
7942969	12/2010	Riker et al.	N/A	N/A
7946762	12/2010	Yednak	N/A	N/A
7951262	12/2010	Koshiishi et al.	N/A	N/A
7955516	12/2010	Chandrachood	N/A	N/A
7955646	12/2010	Cruse et al.	N/A	N/A
7955650	12/2010	Tsuji	N/A	N/A
7957708	12/2010	Karschnia et al.	N/A	N/A
7963736	12/2010	Takizawa et al.	N/A	N/A
7967913	12/2010	Hua et al.	N/A	N/A
7972980	12/2010	Lee et al.	N/A	N/A
7977256	12/2010	Liu et al.	N/A	N/A
7981751	12/2010	Zhu et al.	N/A	N/A
D643055	12/2010	Takahashi	N/A	N/A
7989365	12/2010	Park et al.	N/A	N/A
7989736	12/2010	Park et al.	N/A	N/A
7992318	12/2010	Kawaji	N/A	N/A
7993457	12/2010	Krotov et al.	N/A	N/A
7994070	12/2010	Dip et al.	N/A	N/A
7994721	12/2010	Espiau et al.	N/A	N/A
7997795	12/2010	Schwagerman et al.	N/A	N/A
7998875	12/2010	DeYoung	N/A	N/A
8003174	12/2010	Fukazawa	N/A	N/A
8003919	12/2010	Goto et al.	N/A	N/A
8004198	12/2010	Bakre et al.	N/A	N/A
8020315	12/2010	Nishimura	N/A	N/A
8030129	12/2010	Jeong	N/A	N/A
8033771	12/2010	Gage et al.	N/A	N/A
8038835	12/2010	Hayashi et al.	N/A	N/A
8041197	12/2010	Kasai et al.	N/A	N/A
8041450	12/2010	Takizawa et al.	N/A	N/A
8043972	12/2010	Liu et al.	N/A	N/A
8046193	12/2010	Yetter et al.	N/A	N/A
8047711	12/2010	Ploechinger	N/A	N/A
8048783	12/2010	Chung et al.	N/A	N/A
8051799	12/2010	Itagaki et al.	N/A	N/A
8052794	12/2010	Sumakeris et al.	N/A	N/A
8055378	12/2010	Numakura	N/A	N/A

8060252	12/2010	Gage et al.	N/A	N/A
RE43023	12/2010	Nakashima et al.	N/A	N/A
D649986	12/2010	Fujikata et al.	N/A	N/A
D651291	12/2010	Liebson et al.	N/A	N/A
8071451	12/2010	Berry	N/A	N/A
8071452	12/2010	Raisanen	N/A	N/A
8072578	12/2010	Yasuda et al.	N/A	N/A
8076230	12/2010	Wei	N/A	N/A
8076237	12/2010	Uzoh	N/A	N/A
8076250	12/2010	Rajagopalan	N/A	N/A
8076251	12/2010	Akae et al.	N/A	N/A
8078310	12/2010	Nishimoto et al.	N/A	N/A
8082946	12/2010	Laverdiere et al.	N/A	N/A
8083853	12/2010	Choi et al.	N/A	N/A
8084104	12/2010	Shinriki et al.	N/A	N/A
8084372	12/2010	You et al.	N/A	N/A
D652896	12/2011	Gether	N/A	N/A
8092604	12/2011	Tomiyasu et al.	N/A	N/A
8092606	12/2011	Park et al.	N/A	N/A
8100583	12/2011	Aggarwal	N/A	N/A
D653734	12/2011	Sisk	N/A	N/A
D654882	12/2011	Honma et al.	N/A	N/A
D654884	12/2011	Honma	N/A	N/A
D655055	12/2011	Toll	N/A	N/A
8110099	12/2011	Hersey et al.	N/A	N/A
8114734	12/2011	Yang et al.	N/A	N/A
8119466	12/2011	Avouris	N/A	N/A
8119527	12/2011	Chadrashekar et al.	N/A	N/A
D655260	12/2011	Honma et al.	N/A	N/A
D655261	12/2011	Honma et al.	N/A	N/A
D655599	12/2011	Durham	N/A	N/A
8128333	12/2011	Aburatani	N/A	N/A
8129290	12/2011	Balseanu et al.	N/A	N/A
8137462	12/2011	Fondurulia et al.	N/A	N/A
8137465	12/2011	Shrinivasan et al.	N/A	N/A
8138104	12/2011	Balseanu et al.	N/A	N/A
8138676	12/2011	Mills	N/A	N/A
8142862	12/2011	Lee et al.	N/A	N/A
8143174	12/2011	Xia et al.	N/A	N/A
8147242	12/2011	Shibagaki et al.	N/A	N/A
8152922	12/2011	Schmidt et al.	N/A	N/A
8158512	12/2011	Ji et al.	N/A	N/A
8172947	12/2011	Shibata et al.	N/A	N/A
8173554	12/2011	Lee et al.	N/A	N/A
8174400	12/2011	Park et al.	N/A	N/A
8178436	12/2011	King et al.	N/A	N/A
8187679	12/2011	Dickey et al.	N/A	N/A
8187951	12/2011	Wang	N/A	N/A
8192901	12/2011	Kageyama	N/A	N/A
8196234	12/2011	Glunk	N/A	N/A

8197915	12/2011	Oka et al.	N/A	N/A
8198168	12/2011	Tanioku	N/A	N/A
8202575	12/2011	Monsma et al.	N/A	N/A
8206506	12/2011	Kadkhodayan et al.	N/A	N/A
8216380	12/2011	White et al.	N/A	N/A
8227032	12/2011	Dussarrat et al.	N/A	N/A
8231799	12/2011	Bera et al.	N/A	N/A
D665055	12/2011	Yanagisawa et al.	N/A	N/A
8241991	12/2011	Hsieh et al.	N/A	N/A
8241992	12/2011	Clevenger et al.	N/A	N/A
8242028	12/2011	van Schravendijk et al.	N/A	N/A
8242031	12/2011	Mallick et al.	N/A	N/A
8246900	12/2011	Kasai et al.	N/A	N/A
8252114	12/2011	Vukovic	N/A	N/A
8252659	12/2011	Huyghabaert et al.	N/A	N/A
8252691	12/2011	Beynet et al.	N/A	N/A
8253204	12/2011	Lee et al.	N/A	N/A
8267633	12/2011	Obikane	N/A	N/A
8272516	12/2011	Salvador	N/A	N/A
8278176	12/2011	Bauer et al.	N/A	N/A
8278224	12/2011	Mui et al.	N/A	N/A
8282769	12/2011	Iizuka	N/A	N/A
8282847	12/2011	Romano	N/A	N/A
8282992	12/2011	Myo et al.	N/A	N/A
8287648	12/2011	Reed et al.	N/A	N/A
8291857	12/2011	Lam et al.	N/A	N/A
8293016	12/2011	Bahng et al.	N/A	N/A
8293642	12/2011	Kim	N/A	N/A
8298336	12/2011	Wang et al.	N/A	N/A
8298951	12/2011	Nakano	N/A	N/A
8307472	12/2011	Saxon et al.	N/A	N/A
8309173	12/2011	Tuominen et al.	N/A	N/A
8318327	12/2011	O'Donnell	N/A	N/A
8318584	12/2011	Li et al.	N/A	N/A
8323413	12/2011	Son	N/A	N/A
8324699	12/2011	Ichijo et al.	N/A	N/A
8328939	12/2011	Choi et al.	N/A	N/A
8329599	12/2011	Fukazawa et al.	N/A	N/A
8333839	12/2011	Oh	N/A	N/A
8334219	12/2011	Lee et al.	N/A	N/A
8338809	12/2011	Yang et al.	N/A	N/A
8349083	12/2012	Takasuka et al.	N/A	N/A
D676943	12/2012	Kluss et al.	N/A	N/A
8367528	12/2012	Bauer et al.	N/A	N/A
8372204	12/2012	Nakamura	N/A	N/A
8378464	12/2012	Kato et al.	N/A	N/A
8382370	12/2012	Aggarwal et al.	N/A	N/A
8382939	12/2012	Kutney et al.	N/A	N/A
8393091	12/2012	Kawamoto	N/A	N/A
8394466	12/2012	Hong et al.	N/A	N/A

8398773	12/2012	Jdira et al.	N/A	N/A
8402918	12/2012	Kadkhodayan et al.	N/A	N/A
8404044	12/2012	Arai	N/A	N/A
8404499	12/2012	Moffatt	N/A	N/A
8415258	12/2012	Akae	N/A	N/A
8415259	12/2012	Lee et al.	N/A	N/A
8415587	12/2012	Millman et al.	N/A	N/A
8419959	12/2012	Bettencourt et al.	N/A	N/A
8425682	12/2012	Wang et al.	N/A	N/A
8430620	12/2012	Blank et al.	N/A	N/A
8435894	12/2012	Chandrashekar et al.	N/A	N/A
8440259	12/2012	Chiang et al.	N/A	N/A
8443484	12/2012	Ozaki et al.	N/A	N/A
8444120	12/2012	Gregg et al.	N/A	N/A
8445075	12/2012	Xu et al.	N/A	N/A
8450191	12/2012	Wang	N/A	N/A
8454749	12/2012	Li	N/A	N/A
8465811	12/2012	Ueda	N/A	N/A
8465903	12/2012	Weidman et al.	N/A	N/A
8466411	12/2012	Arai	N/A	N/A
8470187	12/2012	Ha	N/A	N/A
8470718	12/2012	Lee	N/A	N/A
8484846	12/2012	Dhindsa	N/A	N/A
8492170	12/2012	Xie et al.	N/A	N/A
8496377	12/2012	Harr et al.	N/A	N/A
8496756	12/2012	Cruse et al.	N/A	N/A
8497213	12/2012	Yasui et al.	N/A	N/A
8501599	12/2012	Ueno et al.	N/A	N/A
8506162	12/2012	Schick et al.	N/A	N/A
8506713	12/2012	Takagi	N/A	N/A
8507720	12/2012	Shay	N/A	N/A
8524612	12/2012	Li et al.	N/A	N/A
8529701	12/2012	Morita	N/A	N/A
8535767	12/2012	Kimura	N/A	N/A
8536068	12/2012	Weidman et al.	N/A	N/A
D691974	12/2012	Osada et al.	N/A	N/A
8551892	12/2012	Nakano	N/A	N/A
8557712	12/2012	Antonelli et al.	N/A	N/A
8562272	12/2012	Lenz	N/A	N/A
8563443	12/2012	Fukazawa	N/A	N/A
8569184	12/2012	Oka	N/A	N/A
D693200	12/2012	Saunders	N/A	N/A
D693782	12/2012	Mori et al.	N/A	N/A
8573152	12/2012	de la Llera et al.	N/A	N/A
8573154	12/2012	Yorozuya	N/A	N/A
8586484	12/2012	Matsuyama et al.	N/A	N/A
8591659	12/2012	Fang et al.	N/A	N/A
8592005	12/2012	Ueda	N/A	N/A
D694790	12/2012	Matsumoto et al.	N/A	N/A
D695240	12/2012	Iida et al.	N/A	N/A

8608885	12/2012	Goto et al.	N/A	N/A
8614047	12/2012	Ayothi et al.	N/A	N/A
8616765	12/2012	Darabnia et al.	N/A	N/A
8617411	12/2012	Singh	N/A	N/A
D697038	12/2013	Matsumoto et al.	N/A	N/A
8623770	12/2013	Gao et al.	N/A	N/A
8633115	12/2013	Chang et al.	N/A	N/A
8637384	12/2013	Ando et al.	N/A	N/A
D698904	12/2013	Milligan et al.	N/A	N/A
8642488	12/2013	Liu et al.	N/A	N/A
8647439	12/2013	Sanchez et al.	N/A	N/A
8647722	12/2013	Kobayashi et al.	N/A	N/A
8647993	12/2013	Lavoie et al.	N/A	N/A
8651788	12/2013	Budde	N/A	N/A
8664127	12/2013	Bhatia et al.	N/A	N/A
8664627	12/2013	Ishikawa et al.	N/A	N/A
8667654	12/2013	Gros-Jean	N/A	N/A
8668957	12/2013	Dussarrat et al.	N/A	N/A
8669185	12/2013	Onizawa	N/A	N/A
8679958	12/2013	Takamure et al.	N/A	N/A
D702188	12/2013	Jacobs	N/A	N/A
8683943	12/2013	Onodera et al.	N/A	N/A
8697198	12/2013	Lee	N/A	N/A
8703002	12/2013	Matsudo et al.	N/A	N/A
8709162	12/2013	Leung et al.	N/A	N/A
8710580	12/2013	Sakuma et al.	N/A	N/A
8711338	12/2013	Liu et al.	N/A	N/A
D705745	12/2013	Kurs et al.	N/A	N/A
D705762	12/2013	Yu	N/A	N/A
8720965	12/2013	Hino et al.	N/A	N/A
8721791	12/2013	Choi et al.	N/A	N/A
8722510	12/2013	Watanabe et al.	N/A	N/A
8722546	12/2013	Fukazawa et al.	N/A	N/A
8726837	12/2013	Patalay et al.	N/A	N/A
8728832	12/2013	Raisanen et al.	N/A	N/A
8728956	12/2013	Lavoie et al.	N/A	N/A
8741062	12/2013	Lindfors et al.	N/A	N/A
8741065	12/2013	Odagiri et al.	N/A	N/A
8742668	12/2013	Nakano et al.	N/A	N/A
8758512	12/2013	Lee	N/A	N/A
8759223	12/2013	Sapre et al.	N/A	N/A
D709536	12/2013	Yoshimura et al.	N/A	N/A
D709537	12/2013	Kuwabara et al.	N/A	N/A
8764085	12/2013	Urabe	N/A	N/A
8771791	12/2013	Lee et al.	N/A	N/A
8771807	12/2013	Xiao et al.	N/A	N/A
8779502	12/2013	Sakuma et al.	N/A	N/A
8784676	12/2013	Guha et al.	N/A	N/A
8784950	12/2013	Fukazawa et al.	N/A	N/A
8784951	12/2013	Fukazawa et al.	N/A	N/A

8785215	12/2013	Kobayashi et al.	N/A	N/A
8785311	12/2013	Miyoshi	N/A	N/A
8790743	12/2013	Omari	N/A	N/A
8790749	12/2013	Omori et al.	N/A	N/A
8802201	12/2013	Raisanen et al.	N/A	N/A
8809170	12/2013	Bauer	N/A	N/A
D712358	12/2013	Allen et al.	N/A	N/A
D712359	12/2013	Allen et al.	N/A	N/A
8820809	12/2013	Ando et al.	N/A	N/A
8821640	12/2013	Cleary et al.	N/A	N/A
8821985	12/2013	Shao et al.	N/A	N/A
8823672	12/2013	Kim	N/A	N/A
8828886	12/2013	Samukawa et al.	N/A	N/A
8841182	12/2013	Chen et al.	N/A	N/A
8845806	12/2013	Aida et al.	N/A	N/A
8846502	12/2013	Haukka et al.	N/A	N/A
D715410	12/2013	Lohmann	N/A	N/A
8859368	12/2013	Deniz	N/A	N/A
8860955	12/2013	Rodnick et al.	N/A	N/A
8864202	12/2013	Schrameyer	N/A	N/A
8864375	12/2013	Abe et al.	N/A	N/A
D716742	12/2013	Jang et al.	N/A	N/A
8876974	12/2013	Wan	N/A	N/A
8877300	12/2013	Lee	N/A	N/A
8877655	12/2013	Shero et al.	N/A	N/A
8882923	12/2013	Saido et al.	N/A	N/A
8883270	12/2013	Shero et al.	N/A	N/A
8895108	12/2013	Lee	N/A	N/A
8895395	12/2013	Kerber et al.	N/A	N/A
8900935	12/2013	Guo et al.	N/A	N/A
8900999	12/2013	Wu et al.	N/A	N/A
8901016	12/2013	Jeongseok et al.	N/A	N/A
8901537	12/2013	Murthy et al.	N/A	N/A
8911553	12/2013	Baluja et al.	N/A	N/A
8911826	12/2013	Adachi et al.	N/A	N/A
8912101	12/2013	Tsuji et al.	N/A	N/A
D720838	12/2014	Yamagishi et al.	N/A	N/A
8927906	12/2014	Tadokoro et al.	N/A	N/A
8932802	12/2014	Wu et al.	N/A	N/A
8933375	12/2014	Dunn et al.	N/A	N/A
8937800	12/2014	Lubomirsky et al.	N/A	N/A
8940646	12/2014	Chandrasekharan	N/A	N/A
D723153	12/2014	Borkholder	N/A	N/A
8945305	12/2014	Marsh	N/A	N/A
8945306	12/2014	Tsuda	N/A	N/A
8945339	12/2014	Kakimoto	N/A	N/A
8946830	12/2014	Jung et al.	N/A	N/A
8956971	12/2014	Huakka	N/A	N/A
8956983	12/2014	Swaminathan	N/A	N/A
D723330	12/2014	York	N/A	N/A

D724553	12/2014	Choi	N/A	N/A
D724701	12/2014	Yamagishi et al.	N/A	N/A
D725168	12/2014	Yamagishi	N/A	N/A
8967608	12/2014	Mitsumori et al.	N/A	N/A
8968989	12/2014	Ouattara et al.	N/A	N/A
8969934	12/2014	Cheng et al.	N/A	N/A
8974868	12/2014	Ishikawa et al.	N/A	N/A
8980006	12/2014	Huh et al.	N/A	N/A
8986456	12/2014	Fondurulia et al.	N/A	N/A
8991214	12/2014	Hoshino et al.	N/A	N/A
8991887	12/2014	Shin et al.	N/A	N/A
8993054	12/2014	Jung et al.	N/A	N/A
8993072	12/2014	Xiao et al.	N/A	N/A
8993457	12/2014	Ramkumar et al.	N/A	N/A
D726365	12/2014	Weigensberg	N/A	N/A
D726884	12/2014	Yamagishi et al.	N/A	N/A
8999102	12/2014	Miyoshi et al.	N/A	N/A
9004744	12/2014	Kemp	N/A	N/A
9005539	12/2014	Halpin et al.	N/A	N/A
9017481	12/2014	Pettinger et al.	N/A	N/A
9017933	12/2014	Liu et al.	N/A	N/A
9018093	12/2014	Tsuji et al.	N/A	N/A
9018111	12/2014	Milligan et al.	N/A	N/A
9018567	12/2014	de Ridder et al.	N/A	N/A
9021985	12/2014	Alokozai et al.	N/A	N/A
9023737	12/2014	Beynet et al.	N/A	N/A
9023738	12/2014	Kato et al.	N/A	N/A
9029244	12/2014	Won et al.	N/A	N/A
9029253	12/2014	Milligan et al.	N/A	N/A
9029272	12/2014	Nakano	N/A	N/A
D732145	12/2014	Yamagishi	N/A	N/A
D732644	12/2014	Yamagishi et al.	N/A	N/A
D733257	12/2014	Schoenherr et al.	N/A	N/A
D733261	12/2014	Yamagishi et al.	N/A	N/A
D733262	12/2014	Kang	N/A	N/A
9057388	12/2014	Comeau et al.	N/A	N/A
9064815	12/2014	Zhang et al.	N/A	N/A
D733843	12/2014	Yamagishi	N/A	N/A
D734377	12/2014	Hirakida	N/A	N/A
9076635	12/2014	Gross et al.	N/A	N/A
9076726	12/2014	Kauerauf et al.	N/A	N/A
D735836	12/2014	Yamagishi et al.	N/A	N/A
D736348	12/2014	Tan	N/A	N/A
9095869	12/2014	Kilpi et al.	N/A	N/A
9096931	12/2014	Yednak et al.	N/A	N/A
9099423	12/2014	Weeks et al.	N/A	N/A
9099505	12/2014	Kusakabe et al.	N/A	N/A
9111972	12/2014	Takeshita et al.	N/A	N/A
9117657	12/2014	Nakano et al.	N/A	N/A
9117866	12/2014	Marquardt et al.	N/A	N/A

D739222	12/2014	Chadbourne	N/A	N/A
9123510	12/2014	Nakano et al.	N/A	N/A
9123577	12/2014	Fujimoto et al.	N/A	N/A
9127358	12/2014	Inoue et al.	N/A	N/A
9127362	12/2014	Scheible et al.	N/A	N/A
9129897	12/2014	Pore et al.	N/A	N/A
9136108	12/2014	Matsushita et al.	N/A	N/A
9136180	12/2014	Machkaoutsan	N/A	N/A
9142393	12/2014	Okabe et al.	N/A	N/A
9142437	12/2014	Fosnight et al.	N/A	N/A
9142764	12/2014	Wang	N/A	N/A
9153441	12/2014	Takamure et al.	N/A	N/A
9163310	12/2014	Lee	N/A	N/A
9166012	12/2014	Sim et al.	N/A	N/A
9169975	12/2014	Sarin et al.	N/A	N/A
9171714	12/2014	Mori	N/A	N/A
9171715	12/2014	Matero	N/A	N/A
9171716	12/2014	Fukuda	N/A	N/A
D742202	12/2014	Cyphers et al.	N/A	N/A
D743357	12/2014	Vyne	N/A	N/A
D743513	12/2014	Yamagishi	N/A	N/A
9174178	12/2014	Berger et al.	N/A	N/A
9175394	12/2014	Yudovsky et al.	N/A	N/A
9177784	12/2014	Raisanen et al.	N/A	N/A
9184047	12/2014	Liu et al.	N/A	N/A
9184054	12/2014	Huang et al.	N/A	N/A
9190263	12/2014	Ishikawa et al.	N/A	N/A
9190264	12/2014	Yuasa et al.	N/A	N/A
9196483	12/2014	Lee et al.	N/A	N/A
D745641	12/2014	Blum	N/A	N/A
9202727	12/2014	Dunn et al.	N/A	N/A
9214333	12/2014	Sims et al.	N/A	N/A
9214340	12/2014	Kurita et al.	N/A	N/A
9219006	12/2014	Chatterjee	N/A	N/A
9223203	12/2014	Farm et al.	N/A	N/A
9228259	12/2015	Haukka et al.	N/A	N/A
9240412	12/2015	Xie et al.	N/A	N/A
9245742	12/2015	Haukka	N/A	N/A
9252024	12/2015	Lam et al.	N/A	N/A
9252238	12/2015	Trevino et al.	N/A	N/A
9257274	12/2015	Kang et al.	N/A	N/A
9263298	12/2015	Matsumoto et al.	N/A	N/A
9267204	12/2015	Honma	N/A	N/A
9267850	12/2015	Aggarwal	N/A	N/A
D751176	12/2015	Schoenherr et al.	N/A	N/A
9275834	12/2015	Park et al.	N/A	N/A
9281223	12/2015	Hara	N/A	N/A
9281277	12/2015	Baek et al.	N/A	N/A
9284642	12/2015	Nakano	N/A	N/A
9287273	12/2015	Ragnarsson et al.	N/A	N/A

9297705	12/2015	Aggarwal	N/A	N/A
9299557	12/2015	Tolle et al.	N/A	N/A
9299595	12/2015	Dunn et al.	N/A	N/A
D753269	12/2015	Yamagishi et al.	N/A	N/A
D753629	12/2015	Plattard	N/A	N/A
9305836	12/2015	Gates et al.	N/A	N/A
9309598	12/2015	Wang et al.	N/A	N/A
9309978	12/2015	Hatch et al.	N/A	N/A
9310684	12/2015	Meyers et al.	N/A	N/A
9312155	12/2015	Mori	N/A	N/A
9315897	12/2015	Byun	N/A	N/A
9324811	12/2015	Weeks	N/A	N/A
9324846	12/2015	Camillo	N/A	N/A
D756929	12/2015	Harck et al.	N/A	N/A
9331200	12/2015	Wang et al.	N/A	N/A
9337031	12/2015	Kim et al.	N/A	N/A
9337054	12/2015	Hunks et al.	N/A	N/A
9337057	12/2015	Park et al.	N/A	N/A
9341296	12/2015	Yednak	N/A	N/A
9343297	12/2015	Fukazawa et al.	N/A	N/A
9343308	12/2015	Isii	N/A	N/A
9343343	12/2015	Mori	N/A	N/A
9343350	12/2015	Arai	N/A	N/A
9349620	12/2015	Kamata et al.	N/A	N/A
9353440	12/2015	Ge et al.	N/A	N/A
9353441	12/2015	Chung	N/A	N/A
9355876	12/2015	Reuter et al.	N/A	N/A
9355882	12/2015	Wu et al.	N/A	N/A
D759137	12/2015	Hassan	N/A	N/A
D759193	12/2015	Gutierrez et al.	N/A	N/A
9362107	12/2015	Thadani et al.	N/A	N/A
9362137	12/2015	Kang et al.	N/A	N/A
9362180	12/2015	Lee et al.	N/A	N/A
9365924	12/2015	Nonaka	N/A	N/A
9368352	12/2015	Takamure et al.	N/A	N/A
9370863	12/2015	Tsuji et al.	N/A	N/A
9378969	12/2015	Hsu et al.	N/A	N/A
D761325	12/2015	Abed	N/A	N/A
9384987	12/2015	Jung et al.	N/A	N/A
9390909	12/2015	Pasquale et al.	N/A	N/A
9394608	12/2015	Shero et al.	N/A	N/A
9396934	12/2015	Tolle	N/A	N/A
9396956	12/2015	Fukazawa	N/A	N/A
9399228	12/2015	Breiling et al.	N/A	N/A
D764196	12/2015	Handler et al.	N/A	N/A
9404587	12/2015	Shugrue	N/A	N/A
9412564	12/2015	Milligan	N/A	N/A
9412581	12/2015	Thadani et al.	N/A	N/A
9412582	12/2015	Sasaki et al.	N/A	N/A
9418885	12/2015	Sung et al.	N/A	N/A

9425078	12/2015	Tang et al.	N/A	N/A
9428833	12/2015	Duvall et al.	N/A	N/A
9443725	12/2015	Liu et al.	N/A	N/A
9447498	12/2015	Shiba et al.	N/A	N/A
9449793	12/2015	Shaji et al.	N/A	N/A
9449795	12/2015	Sabri et al.	N/A	N/A
9449843	12/2015	Korolik et al.	N/A	N/A
9449987	12/2015	Miyata et al.	N/A	N/A
9455138	12/2015	Fukazawa	N/A	N/A
9455177	12/2015	Park et al.	N/A	N/A
9460954	12/2015	De Jong et al.	N/A	N/A
9464352	12/2015	Nakano et al.	N/A	N/A
9472410	12/2015	Sadjadi et al.	N/A	N/A
9472432	12/2015	Blank	N/A	N/A
9474163	12/2015	Tolle et al.	N/A	N/A
9478414	12/2015	Kobayashi et al.	N/A	N/A
9478415	12/2015	Kimura	N/A	N/A
D770993	12/2015	Yoshida et al.	N/A	N/A
9484191	12/2015	Winkler	N/A	N/A
9496225	12/2015	Adusumilli et al.	N/A	N/A
9514927	12/2015	Tolle et al.	N/A	N/A
9514932	12/2015	Mallick et al.	N/A	N/A
9520289	12/2015	Park et al.	N/A	N/A
9523148	12/2015	Pore et al.	N/A	N/A
D777546	12/2016	Ishii et al.	N/A	N/A
9543180	12/2016	Kamiya	N/A	N/A
9556516	12/2016	Takamure	N/A	N/A
9558931	12/2016	Tang	N/A	N/A
9564312	12/2016	Henri et al.	N/A	N/A
9564314	12/2016	Takamure et al.	N/A	N/A
9570302	12/2016	Chang et al.	N/A	N/A
9574268	12/2016	Dunn et al.	N/A	N/A
9576952	12/2016	Joshi et al.	N/A	N/A
9583333	12/2016	Chatterjee	N/A	N/A
9583345	12/2016	Chen et al.	N/A	N/A
D782419	12/2016	Willette	N/A	N/A
9589770	12/2016	Winkler	N/A	N/A
9605342	12/2016	Alokozai et al.	N/A	N/A
9605343	12/2016	Winkler	N/A	N/A
9605736	12/2016	Foshage et al.	N/A	N/A
9607837	12/2016	Namba	N/A	N/A
D783351	12/2016	Fujino et al.	N/A	N/A
D784276	12/2016	Tiner et al.	N/A	N/A
9613801	12/2016	Carcasi et al.	N/A	N/A
9618846	12/2016	Shamma et al.	N/A	N/A
9627221	12/2016	Zaitsu et al.	N/A	N/A
D785766	12/2016	Sato	N/A	N/A
D787458	12/2016	Kim et al.	N/A	N/A
9640416	12/2016	Arai	N/A	N/A
9640448	12/2016	Ikegawa et al.	N/A	N/A

9640542	12/2016	Lee et al.	N/A	N/A
9644266	12/2016	Nasu et al.	N/A	N/A
9647114	12/2016	Margetis	N/A	N/A
9653267	12/2016	Carducci et al.	N/A	N/A
9657845	12/2016	Shugrue	N/A	N/A
9659799	12/2016	Lawson	N/A	N/A
9663857	12/2016	Nakano et al.	N/A	N/A
9666528	12/2016	Bergendahl et al.	N/A	N/A
D789888	12/2016	Jang et al.	N/A	N/A
D790041	12/2016	Jang et al.	N/A	N/A
9680268	12/2016	Finona	N/A	N/A
9684234	12/2016	Darling et al.	N/A	N/A
9685320	12/2016	Kang et al.	N/A	N/A
9691668	12/2016	Chang et al.	N/A	N/A
9691771	12/2016	Lansalot-Matras	N/A	N/A
9698031	12/2016	Kobayashi et al.	N/A	N/A
9708707	12/2016	Ditizio et al.	N/A	N/A
9708708	12/2016	Isobe et al.	N/A	N/A
9711345	12/2016	Shiba et al.	N/A	N/A
D793352	12/2016	Hill	N/A	N/A
D793526	12/2016	Behdjat	N/A	N/A
D793572	12/2016	Kozuka et al.	N/A	N/A
D793976	12/2016	Fukushima et al.	N/A	N/A
D794753	12/2016	Miller	N/A	N/A
D795208	12/2016	Sasaki et al.	N/A	N/A
9735024	12/2016	Zaitsu	N/A	N/A
9741546	12/2016	Carducci et al.	N/A	N/A
9741559	12/2016	Shimura et al.	N/A	N/A
9745658	12/2016	Kang et al.	N/A	N/A
9748104	12/2016	Sasaki et al.	N/A	N/A
9748145	12/2016	Kannan et al.	N/A	N/A
D796458	12/2016	Jang et al.	N/A	N/A
D796670	12/2016	Dolk et al.	N/A	N/A
D797067	12/2016	Zhang et al.	N/A	N/A
D798248	12/2016	Hanson et al.	N/A	N/A
9754779	12/2016	Ishikawa	N/A	N/A
9754818	12/2016	Shiu et al.	N/A	N/A
9759489	12/2016	Kaneko	N/A	N/A
9765432	12/2016	Ge et al.	N/A	N/A
9773818	12/2016	Kimura	N/A	N/A
D800782	12/2016	Bever et al.	N/A	N/A
9778561	12/2016	Marks et al.	N/A	N/A
9780225	12/2016	Venkatasubramanian et al.	N/A	N/A
9786491	12/2016	Suzuki et al.	N/A	N/A
9786570	12/2016	Kang et al.	N/A	N/A
9790595	12/2016	Jung et al.	N/A	N/A
9793115	12/2016	Tolle	N/A	N/A
9793135	12/2016	Zaitsu et al.	N/A	N/A
9793148	12/2016	Yamagishi et al.	N/A	N/A

9798308	12/2016	Mimura	N/A	N/A
9799736	12/2016	Ebrish et al.	N/A	N/A
9803926	12/2016	Kikuchi et al.	N/A	N/A
D801942	12/2016	Riker et al.	N/A	N/A
D802472	12/2016	Sasaki et al.	N/A	N/A
D802546	12/2016	Jang et al.	N/A	N/A
D803802	12/2016	Sasaki et al.	N/A	N/A
9808246	12/2016	Shelton et al.	N/A	N/A
9812319	12/2016	Fukazawa et al.	N/A	N/A
9812320	12/2016	Pore et al.	N/A	N/A
9812372	12/2016	Choi et al.	N/A	N/A
9820289	12/2016	Pawar et al.	N/A	N/A
9824881	12/2016	Niskanen et al.	N/A	N/A
9824884	12/2016	Sims et al.	N/A	N/A
9824893	12/2016	Smith et al.	N/A	N/A
9837355	12/2016	Briggs et al.	N/A	N/A
9842835	12/2016	Cheng et al.	N/A	N/A
9847221	12/2016	McLaughlin et al.	N/A	N/A
9847247	12/2016	Huang et al.	N/A	N/A
9850573	12/2016	Sun	N/A	N/A
D807494	12/2017	Kim et al.	N/A	N/A
D808254	12/2017	Deleu	N/A	N/A
9859151	12/2017	Zhu	N/A	N/A
9865455	12/2017	Sims et al.	N/A	N/A
9865456	12/2017	Pandey et al.	N/A	N/A
9865815	12/2017	Hausmann	N/A	N/A
9868131	12/2017	Kilpi et al.	N/A	N/A
9870964	12/2017	Yoshino et al.	N/A	N/A
9875891	12/2017	Henri et al.	N/A	N/A
9875893	12/2017	Takamure et al.	N/A	N/A
9881788	12/2017	Kim et al.	N/A	N/A
D810705	12/2017	Krishnan et al.	N/A	N/A
9887082	12/2017	Pore et al.	N/A	N/A
9890456	12/2017	Tolle et al.	N/A	N/A
9891521	12/2017	Kang et al.	N/A	N/A
9892908	12/2017	Pettinger et al.	N/A	N/A
9892913	12/2017	Margetis et al.	N/A	N/A
9895715	12/2017	Haukka et al.	N/A	N/A
9899291	12/2017	Kato	N/A	N/A
9899405	12/2017	Kim	N/A	N/A
9905420	12/2017	Margetis et al.	N/A	N/A
9905492	12/2017	Tang et al.	N/A	N/A
9909214	12/2017	Suemori	N/A	N/A
9909492	12/2017	Jeswine	N/A	N/A
9911595	12/2017	Smith et al.	N/A	N/A
9911676	12/2017	Tang	N/A	N/A
9916980	12/2017	Knaepen	N/A	N/A
9920451	12/2017	Sivaramakrishnan et al.	N/A	N/A
9922824	12/2017	Okada	N/A	N/A

9929005	12/2017	Shimamoto et al.	N/A	N/A
9929011	12/2017	Hawryluk et al.	N/A	N/A
9929055	12/2017	Dube et al.	N/A	N/A
9951421	12/2017	Lind	N/A	N/A
9960033	12/2017	Nozawa	N/A	N/A
9960072	12/2017	Coomer	N/A	N/A
9966299	12/2017	Tang et al.	N/A	N/A
9970112	12/2017	Koshi et al.	N/A	N/A
9984869	12/2017	Blanquart	N/A	N/A
D819580	12/2017	Krishnan et al.	N/A	N/A
9987747	12/2017	Hwang et al.	N/A	N/A
9991138	12/2017	Lin et al.	N/A	N/A
9996004	12/2017	Smith et al.	N/A	N/A
9997357	12/2017	Arghavani et al.	N/A	N/A
9997373	12/2017	Hudson	N/A	N/A
10014212	12/2017	Chen et al.	N/A	N/A
10017856	12/2017	Arnepalli et al.	N/A	N/A
10018920	12/2017	Chang et al.	N/A	N/A
10023960	12/2017	Alokozai	N/A	N/A
10032628	12/2017	Xie et al.	N/A	N/A
10032792	12/2017	Kim et al.	N/A	N/A
D825505	12/2017	Hanson et al.	N/A	N/A
D825614	12/2017	Bever et al.	N/A	N/A
10043661	12/2017	Kato et al.	N/A	N/A
10047435	12/2017	Haukka et al.	N/A	N/A
10053774	12/2017	Tolle et al.	N/A	N/A
10060473	12/2017	Davey et al.	N/A	N/A
D827592	12/2017	Ichino et al.	N/A	N/A
D829306	12/2017	Ikedo et al.	N/A	N/A
10083836	12/2017	Milligan	N/A	N/A
D830981	12/2017	Jeong et al.	N/A	N/A
10087522	12/2017	Raisanen et al.	N/A	N/A
10087525	12/2017	Schmotzer et al.	N/A	N/A
10090316	12/2017	Ootsuka	N/A	N/A
10103040	12/2017	Oosterlaken et al.	N/A	N/A
10106892	12/2017	Siddiqui et al.	N/A	N/A
RE47145	12/2017	Hashimoto	N/A	N/A
D834686	12/2017	Yamada et al.	N/A	N/A
10121671	12/2017	Fu et al.	N/A	N/A
10134617	12/2017	Gurary et al.	N/A	N/A
10134757	12/2017	Chun et al.	N/A	N/A
RE47170	12/2017	Beynet et al.	N/A	N/A
10147600	12/2017	Takamure et al.	N/A	N/A
10167557	12/2018	Hawkins et al.	N/A	N/A
10177024	12/2018	Gomm et al.	N/A	N/A
10177025	12/2018	Pore	N/A	N/A
10179947	12/2018	Fukazawa	N/A	N/A
10186420	12/2018	Fukazawa	N/A	N/A
10190213	12/2018	Zhu et al.	N/A	N/A
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10193429	12/2018	Smith et al.	N/A	N/A
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10211308	12/2018	Zhu et al.	N/A	N/A
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10236177	12/2018	Kohen et al.	N/A	N/A
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10249524	12/2018	den Hartog Besselink et al.	N/A	N/A
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10262859	12/2018	Margetis et al.	N/A	N/A
10269558	12/2018	Blanquart et al.	N/A	N/A
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10283353	12/2018	Kobayashi et al.	N/A	N/A
10287684	12/2018	Yanai et al.	N/A	N/A
10290508	12/2018	Kubota et al.	N/A	N/A
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10312055	12/2018	Suzuki	N/A	N/A
10312129	12/2018	Coomer	N/A	N/A
10319588	12/2018	Mattinen et al.	N/A	N/A
10322384	12/2018	Stumpf et al.	N/A	N/A
10332747	12/2018	Watanabe et al.	N/A	N/A
10332963	12/2018	Xie	N/A	N/A
D855089	12/2018	Hopkins	N/A	N/A
10340125	12/2018	Winkler	N/A	N/A
10340135	12/2018	Blanquart	N/A	N/A
10343920	12/2018	Haukka	N/A	N/A
10347547	12/2018	Varadarajan et al.	N/A	N/A
10354873	12/2018	Ko et al.	N/A	N/A
10361201	12/2018	Xie et al.	N/A	N/A
10361366	12/2018	Hakamata et al.	N/A	N/A
10367080	12/2018	Tang et al.	N/A	N/A
10388513	12/2018	Blanquart	N/A	N/A
10395917	12/2018	Niskanen et al.	N/A	N/A
10395919	12/2018	Masaru et al.	N/A	N/A
10395963	12/2018	Cooke	N/A	N/A
D859136	12/2018	Tenander et al.	N/A	N/A
10400335	12/2018	Ge et al.	N/A	N/A
10410943	12/2018	Jiang et al.	N/A	N/A
10424476	12/2018	Suzuki et al.	N/A	N/A
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D864134	12/2018	Watarai et al.	N/A	N/A
10428419	12/2018	Huotari et al.	N/A	N/A

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10468244	12/2018	Li et al.	N/A	N/A
10468251	12/2018	Ishikawa et al.	N/A	N/A
10483154	12/2018	Smith et al.	N/A	N/A
10510529	12/2018	Suzuki et al.	N/A	N/A
10510871	12/2018	More et al.	N/A	N/A
10529554	12/2019	Ishikawa et al.	N/A	N/A
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10590531	12/2019	Shirako et al.	N/A	N/A
10590535	12/2019	Huggare	N/A	N/A
10600637	12/2019	Suzuki et al.	N/A	N/A
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D881338	12/2019	Chen	N/A	N/A
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10622236	12/2019	Kuo et al.	N/A	N/A
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10662525	12/2019	Jang et al.	N/A	N/A
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10714335	12/2019	Kim et al.	N/A	N/A
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10734497	12/2019	Zhu et al.	N/A	N/A
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10770336	12/2019	Hill et al.	N/A	N/A
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2001/0000141	12/2000	Zhou et al.	N/A	N/A
2001/0001953	12/2000	Griffiths et al.	N/A	N/A
2001/0002581	12/2000	Nishikawa et al.	N/A	N/A
2001/0003015	12/2000	Chang et al.	N/A	N/A
2001/0003191	12/2000	Kovacs et al.	N/A	N/A
2001/0003271	12/2000	Otsuki	N/A	N/A
2001/0004880	12/2000	Cho et al.	N/A	N/A
2001/0006070	12/2000	Shang	N/A	N/A
2001/0007244	12/2000	Matsuse	N/A	N/A
2001/0007246	12/2000	Ueda et al.	N/A	N/A
2001/0007645	12/2000	Honma	N/A	N/A
2001/0014267	12/2000	Yamaga et al.	N/A	N/A
2001/0014514	12/2000	Geusic	N/A	N/A
2001/0015343	12/2000	Sprey et al.	N/A	N/A
2001/0016273	12/2000	Narasimhan et al.	N/A	N/A
2001/0017103	12/2000	Takeshita et al.	N/A	N/A
2001/0018267	12/2000	Shinriki et al.	N/A	N/A

2001/0019347	12/2000	Hauck	N/A	N/A
2001/0019777	12/2000	Tanaka et al.	N/A	N/A
2001/0019900	12/2000	Hasegawa	N/A	N/A
2001/0020715	12/2000	Yamasaki	N/A	N/A
2001/0021591	12/2000	Srinivasan et al.	N/A	N/A
2001/0022215	12/2000	Donohoe	N/A	N/A
2001/0024387	12/2000	Raaijmakers et al.	N/A	N/A
2001/0027026	12/2000	Dhindsa et al.	N/A	N/A
2001/0027585	12/2000	Lee	N/A	N/A
2001/0028924	12/2000	Sherman	N/A	N/A
2001/0031535	12/2000	Agnello et al.	N/A	N/A
2001/0031541	12/2000	Madan et al.	N/A	N/A
2001/0034097	12/2000	Lim et al.	N/A	N/A
2001/0038783	12/2000	Nakashima et al.	N/A	N/A
2001/0039922	12/2000	Nakahara	N/A	N/A
2001/0039966	12/2000	Walpole et al.	N/A	N/A
2001/0040511	12/2000	Bushner et al.	N/A	N/A
2001/0041250	12/2000	Werkhoven et al.	N/A	N/A
2001/0042511	12/2000	Liu et al.	N/A	N/A
2001/0042514	12/2000	Mizuno et al.	N/A	N/A
2001/0042594	12/2000	Shamouilian et al.	N/A	N/A
2001/0046765	12/2000	Cappellani et al.	N/A	N/A
2001/0047761	12/2000	Wijck et al.	N/A	N/A
2001/0048981	12/2000	Suzuki	N/A	N/A
2001/0049080	12/2000	Asano	N/A	N/A
2001/0049202	12/2000	Maeda et al.	N/A	N/A
2001/0052556	12/2000	Ting et al.	N/A	N/A
2001/0054381	12/2000	Umotoy et al.	N/A	N/A
2001/0054388	12/2000	Qian	N/A	N/A
2002/0000195	12/2001	Bang et al.	N/A	N/A
2002/0000202	12/2001	Yuda et al.	N/A	N/A
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2002/0001974	12/2001	Chan	N/A	N/A
2002/0001976	12/2001	Danek	N/A	N/A
2002/0005400	12/2001	Gat et al.	N/A	N/A
2002/0005943	12/2001	Voinalovich	N/A	N/A
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2002/0009119	12/2001	Matthew et al.	N/A	N/A
2002/0009560	12/2001	Ozono	N/A	N/A
2002/0009861	12/2001	Narwankar et al.	N/A	N/A
2002/0011210	12/2001	Satoh et al.	N/A	N/A
2002/0011211	12/2001	Halpin	N/A	N/A
2002/0011310	12/2001	Kamarehi et al.	N/A	N/A
2002/0013792	12/2001	Imielinski et al.	N/A	N/A
2002/0014204	12/2001	Pyo	N/A	N/A
2002/0014483	12/2001	Suzuki et al.	N/A	N/A
2002/0015853	12/2001	Wataya et al.	N/A	N/A
2002/0016829	12/2001	Defosse	N/A	N/A
2002/0020429	12/2001	Selbrede et al.	N/A	N/A
2002/0022347	12/2001	Park et al.	N/A	N/A

2002/0023677	12/2001	Zheng	N/A	N/A
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2002/0027945	12/2001	Hirano et al.	N/A	N/A
2002/0030047	12/2001	Shao et al.	N/A	N/A
2002/0031644	12/2001	Malofsky et al.	N/A	N/A
2002/0033183	12/2001	Sun et al.	N/A	N/A
2002/0036065	12/2001	Yamagishi et al.	N/A	N/A
2002/0041931	12/2001	Suntola et al.	N/A	N/A
2002/0043337	12/2001	Goodman et al.	N/A	N/A
2002/0045336	12/2001	Locati et al.	N/A	N/A
2002/0047705	12/2001	Tada et al.	N/A	N/A
2002/0048634	12/2001	Basceri	N/A	N/A
2002/0050648	12/2001	Kishida et al.	N/A	N/A
2002/0052119	12/2001	Cleemput	N/A	N/A
2002/0061716	12/2001	Korovin et al.	N/A	N/A
2002/0062633	12/2001	Denker et al.	N/A	N/A
2002/0064592	12/2001	Datta et al.	N/A	N/A
2002/0064598	12/2001	Wang et al.	N/A	N/A
2002/0066532	12/2001	Shih et al.	N/A	N/A
2002/0068458	12/2001	Chiang et al.	N/A	N/A
2002/0069222	12/2001	McNeely	N/A	N/A
2002/0073922	12/2001	Frankel et al.	N/A	N/A
2002/0073923	12/2001	Saito et al.	N/A	N/A
2002/0076490	12/2001	Chiang et al.	N/A	N/A
2002/0076507	12/2001	Chiang et al.	N/A	N/A
2002/0076944	12/2001	Wang et al.	N/A	N/A
2002/0078893	12/2001	Van Os et al.	N/A	N/A
2002/0079056	12/2001	Kudo et al.	N/A	N/A
2002/0079714	12/2001	Soucy et al.	N/A	N/A
2002/0081826	12/2001	Rotondaro et al.	N/A	N/A
2002/0086501	12/2001	O'Donnell et al.	N/A	N/A
2002/0088542	12/2001	Nishikawa et al.	N/A	N/A
2002/0090735	12/2001	Kishkovich et al.	N/A	N/A
2002/0094378	12/2001	O'Donnell et al.	N/A	N/A
2002/0094388	12/2001	Fonash et al.	N/A	N/A
2002/0096211	12/2001	Zheng	N/A	N/A
2002/0098627	12/2001	Pomarede et al.	N/A	N/A
2002/0099470	12/2001	Zinger et al.	N/A	N/A
2002/0100418	12/2001	Sandhu et al.	N/A	N/A
2002/0104481	12/2001	Chiang et al.	N/A	N/A
2002/0104751	12/2001	Drewery et al.	N/A	N/A
2002/0106909	12/2001	Kato et al.	N/A	N/A
2002/0108570	12/2001	Lindfors	N/A	N/A
2002/0108670	12/2001	Baker et al.	N/A	N/A
2002/0108714	12/2001	Doering et al.	N/A	N/A
2002/0109115	12/2001	Cederstav et al.	N/A	N/A
2002/0110695	12/2001	Yang et al.	N/A	N/A
2002/0110991	12/2001	Li	N/A	N/A
2002/0112114	12/2001	Blair et al.	N/A	N/A
2002/0114886	12/2001	Chou et al.	N/A	N/A

2002/0115252	12/2001	Haukka et al.	N/A	N/A
2002/0117113	12/2001	Tsuruno et al.	N/A	N/A
2002/0117262	12/2001	Pang et al.	N/A	N/A
2002/0122885	12/2001	Ahn	N/A	N/A
2002/0123200	12/2001	Yamamoto et al.	N/A	N/A
2002/0123230	12/2001	Hubacek	N/A	N/A
2002/0123237	12/2001	Nguyen et al.	N/A	N/A
2002/0124883	12/2001	Zheng	N/A	N/A
2002/0124906	12/2001	Suzuki et al.	N/A	N/A
2002/0127350	12/2001	Ishikawa et al.	N/A	N/A
2002/0127956	12/2001	Ashjaee et al.	N/A	N/A
2002/0129768	12/2001	Carpenter et al.	N/A	N/A
2002/0132408	12/2001	Ma et al.	N/A	N/A
2002/0134511	12/2001	Ushioda et al.	N/A	N/A
2002/0136214	12/2001	Do et al.	N/A	N/A
2002/0136909	12/2001	Yang	N/A	N/A
2002/0139775	12/2001	Chang	N/A	N/A
2002/0146512	12/2001	Rossman	N/A	N/A
2002/0151327	12/2001	Levitt	N/A	N/A
2002/0152244	12/2001	Dean et al.	N/A	N/A
2002/0155219	12/2001	Wang et al.	N/A	N/A
2002/0155708	12/2001	Lo et al.	N/A	N/A
2002/0157611	12/2001	Bondestam et al.	N/A	N/A
2002/0160112	12/2001	Sakai et al.	N/A	N/A
2002/0164420	12/2001	Derderian et al.	N/A	N/A
2002/0164423	12/2001	Chiang et al.	N/A	N/A
2002/0168870	12/2001	Matsuki	N/A	N/A
2002/0170676	12/2001	Mitrovic et al.	N/A	N/A
2002/0172768	12/2001	Endo et al.	N/A	N/A
2002/0174106	12/2001	Martin	N/A	N/A
2002/0179011	12/2001	Jonnalagadda et al.	N/A	N/A
2002/0181612	12/2001	Warble et al.	N/A	N/A
2002/0184111	12/2001	Swanson	N/A	N/A
2002/0187650	12/2001	Blalock et al.	N/A	N/A
2002/0187656	12/2001	Tan et al.	N/A	N/A
2002/0188376	12/2001	Derderian et al.	N/A	N/A
2002/0192370	12/2001	Metzner et al.	N/A	N/A
2002/0197402	12/2001	Chiang et al.	N/A	N/A
2002/0197849	12/2001	Mandal	N/A	N/A
2003/0000647	12/2002	Yudovsky et al.	N/A	N/A
2003/0002562	12/2002	Yerlikaya et al.	N/A	N/A
2003/0003607	12/2002	Kagoshima	N/A	N/A
2003/0003635	12/2002	Paranjpe et al.	N/A	N/A
2003/0003696	12/2002	Gelatos et al.	N/A	N/A
2003/0003719	12/2002	Lim et al.	N/A	N/A
2003/0008528	12/2002	Xia et al.	N/A	N/A
2003/0008602	12/2002	Ashjaee et al.	N/A	N/A
2003/0010355	12/2002	Nowak et al.	N/A	N/A
2003/0010451	12/2002	Tzu	N/A	N/A
2003/0010452	12/2002	Park et al.	N/A	N/A

2003/0012632	12/2002	Saeki	N/A	N/A
2003/0013314	12/2002	Ying et al.	N/A	N/A
2003/0015141	12/2002	Takagi	N/A	N/A
2003/0015294	12/2002	Wang	N/A	N/A
2003/0015596	12/2002	Evans	N/A	N/A
2003/0017265	12/2002	Basceri et al.	N/A	N/A
2003/0017266	12/2002	Basceri et al.	N/A	N/A
2003/0017268	12/2002	Hu	N/A	N/A
2003/0019428	12/2002	Ku et al.	N/A	N/A
2003/0019580	12/2002	Strang	N/A	N/A
2003/0022468	12/2002	Shioya et al.	N/A	N/A
2003/0022523	12/2002	Irino et al.	N/A	N/A
2003/0023338	12/2002	Chin et al.	N/A	N/A
2003/0024901	12/2002	Ishikawa	N/A	N/A
2003/0025146	12/2002	Narwankar et al.	N/A	N/A
2003/0026904	12/2002	Yadav et al.	N/A	N/A
2003/0027431	12/2002	Sneh et al.	N/A	N/A
2003/0029303	12/2002	Hasegawa et al.	N/A	N/A
2003/0029381	12/2002	Nishibayashi	N/A	N/A
2003/0029475	12/2002	Hua et al.	N/A	N/A
2003/0029563	12/2002	Kaushal et al.	N/A	N/A
2003/0032297	12/2002	Lindstrom et al.	N/A	N/A
2003/0035002	12/2002	Moles	N/A	N/A
2003/0035705	12/2002	Johnson	N/A	N/A
2003/0036272	12/2002	Shamouilian et al.	N/A	N/A
2003/0037800	12/2002	Bailey et al.	N/A	N/A
2003/0040120	12/2002	Allen et al.	N/A	N/A
2003/0040158	12/2002	Saitoh	N/A	N/A
2003/0040196	12/2002	Lim et al.	N/A	N/A
2003/0040841	12/2002	Nasr et al.	N/A	N/A
2003/0041971	12/2002	Kido et al.	N/A	N/A
2003/0042419	12/2002	Katsumata et al.	N/A	N/A
2003/0045961	12/2002	Nakao	N/A	N/A
2003/0049372	12/2002	Cook et al.	N/A	N/A
2003/0049375	12/2002	Nguyen et al.	N/A	N/A
2003/0049499	12/2002	Murakawa et al.	N/A	N/A
2003/0049571	12/2002	Hallock et al.	N/A	N/A
2003/0049580	12/2002	Goodman	N/A	N/A
2003/0049937	12/2002	Suzuki	N/A	N/A
2003/0054670	12/2002	Wang et al.	N/A	N/A
2003/0056726	12/2002	Holst et al.	N/A	N/A
2003/0057848	12/2002	Yuasa et al.	N/A	N/A
2003/0059535	12/2002	Luo et al.	N/A	N/A
2003/0059980	12/2002	Chen et al.	N/A	N/A
2003/0062359	12/2002	Ho et al.	N/A	N/A
2003/0065413	12/2002	Liteplo et al.	N/A	N/A
2003/0066482	12/2002	Pokharna et al.	N/A	N/A
2003/0066541	12/2002	Sun et al.	N/A	N/A
2003/0066826	12/2002	Lee et al.	N/A	N/A
2003/0070617	12/2002	Kim et al.	N/A	N/A

2003/0071015	12/2002	Chinn et al.	N/A	N/A
2003/0072882	12/2002	Niinisto et al.	N/A	N/A
2003/0075107	12/2002	Miyano et al.	N/A	N/A
2003/0075925	12/2002	Lindfors et al.	N/A	N/A
2003/0077857	12/2002	Xia et al.	N/A	N/A
2003/0077883	12/2002	Ohtake	N/A	N/A
2003/0082296	12/2002	Elers et al.	N/A	N/A
2003/0082307	12/2002	Chung et al.	N/A	N/A
2003/0085663	12/2002	Horsky	N/A	N/A
2003/0091938	12/2002	Fairbairn et al.	N/A	N/A
2003/0094133	12/2002	Yoshidome et al.	N/A	N/A
2003/0101938	12/2002	Ronsse et al.	N/A	N/A
2003/0109107	12/2002	Hsieh et al.	N/A	N/A
2003/0109951	12/2002	Hsiung et al.	N/A	N/A
2003/0111012	12/2002	Takeshima	N/A	N/A
2003/0111013	12/2002	Oosterlaken et al.	N/A	N/A
2003/0111963	12/2002	Tolmachev et al.	N/A	N/A
2003/0113995	12/2002	Xia et al.	N/A	N/A
2003/0116087	12/2002	Nguyen	N/A	N/A
2003/0121608	12/2002	Chen	N/A	N/A
2003/0124792	12/2002	Jeon et al.	N/A	N/A
2003/0124818	12/2002	Luo et al.	N/A	N/A
2003/0124820	12/2002	Johnsgard et al.	N/A	N/A
2003/0124842	12/2002	Hytros et al.	N/A	N/A
2003/0127049	12/2002	Han et al.	N/A	N/A
2003/0132319	12/2002	Hytros et al.	N/A	N/A
2003/0133854	12/2002	Tabata et al.	N/A	N/A
2003/0134038	12/2002	Paranjpe	N/A	N/A
2003/0140851	12/2002	Janakiraman et al.	N/A	N/A
2003/0141527	12/2002	Joo et al.	N/A	N/A
2003/0141820	12/2002	White et al.	N/A	N/A
2003/0143328	12/2002	Chen	N/A	N/A
2003/0143846	12/2002	Sekiya et al.	N/A	N/A
2003/0145789	12/2002	Bauch et al.	N/A	N/A
2003/0149506	12/2002	Haanstra et al.	N/A	N/A
2003/0150386	12/2002	Shimada	N/A	N/A
2003/0153177	12/2002	Tepman et al.	N/A	N/A
2003/0153186	12/2002	Bar-Gadda	N/A	N/A
2003/0157345	12/2002	Beldi et al.	N/A	N/A
2003/0157432	12/2002	Rottsegge	N/A	N/A
2003/0157436	12/2002	Manger et al.	N/A	N/A
2003/0159653	12/2002	Dando et al.	N/A	N/A
2003/0159656	12/2002	Tan	N/A	N/A
2003/0162412	12/2002	Chung	N/A	N/A
2003/0168001	12/2002	Sneh	N/A	N/A
2003/0168008	12/2002	Ohmi et al.	N/A	N/A
2003/0168012	12/2002	Tamura et al.	N/A	N/A
2003/0168174	12/2002	Foree	N/A	N/A
2003/0168699	12/2002	Honda	N/A	N/A
2003/0168750	12/2002	Basceri et al.	N/A	N/A

2003/0168948	12/2002	Yamagishi et al.	N/A	N/A
2003/0170153	12/2002	Bar-Gadda	N/A	N/A
2003/0170583	12/2002	Nakashima	N/A	N/A
2003/0170945	12/2002	Igeta et al.	N/A	N/A
2003/0173030	12/2002	Ishii et al.	N/A	N/A
2003/0173490	12/2002	Lappen	N/A	N/A
2003/0176074	12/2002	Paterson et al.	N/A	N/A
2003/0178145	12/2002	Anderson et al.	N/A	N/A
2003/0180458	12/2002	Sneh	N/A	N/A
2003/0181065	12/2002	O'Donnell	N/A	N/A
2003/0183156	12/2002	Dando	N/A	N/A
2003/0183856	12/2002	Wieczorek et al.	N/A	N/A
2003/0188682	12/2002	Tois et al.	N/A	N/A
2003/0188685	12/2002	Wang	N/A	N/A
2003/0190804	12/2002	Glenn et al.	N/A	N/A
2003/0192875	12/2002	Bieker et al.	N/A	N/A
2003/0198587	12/2002	Kaloyeros	N/A	N/A
2003/0200926	12/2002	Dando et al.	N/A	N/A
2003/0201541	12/2002	Kim	N/A	N/A
2003/0205096	12/2002	Gehner et al.	N/A	N/A
2003/0205202	12/2002	Funaki et al.	N/A	N/A
2003/0205237	12/2002	Sakuma	N/A	N/A
2003/0205327	12/2002	Howald et al.	N/A	N/A
2003/0207032	12/2002	Ahn et al.	N/A	N/A
2003/0209323	12/2002	Yokogaki	N/A	N/A
2003/0209326	12/2002	Lee et al.	N/A	N/A
2003/0209746	12/2002	Horii	N/A	N/A
2003/0210901	12/2002	Donald et al.	N/A	N/A
2003/0211735	12/2002	Rossmann	N/A	N/A
2003/0213435	12/2002	Okuda et al.	N/A	N/A
2003/0213560	12/2002	Wang et al.	N/A	N/A
2003/0213562	12/2002	Gondhalekar et al.	N/A	N/A
2003/0215963	12/2002	AmRhein et al.	N/A	N/A
2003/0217915	12/2002	Ouellet	N/A	N/A
2003/0219972	12/2002	Green	N/A	N/A
2003/0221780	12/2002	Lei et al.	N/A	N/A
2003/0226840	12/2002	Dalton	N/A	N/A
2003/0228772	12/2002	Cowans	N/A	N/A
2003/0230986	12/2002	Horsky et al.	N/A	N/A
2003/0231698	12/2002	Yamaguchi	N/A	N/A
2003/0232138	12/2002	Tuominen et al.	N/A	N/A
2003/0232491	12/2002	Yamaguchi	N/A	N/A
2003/0232497	12/2002	Xi et al.	N/A	N/A
2003/0232511	12/2002	Metzner et al.	N/A	N/A
2003/0234371	12/2002	Ziegler	N/A	N/A
2004/0002224	12/2003	Chono et al.	N/A	N/A
2004/0005147	12/2003	Wang et al.	N/A	N/A
2004/0005753	12/2003	Kostamo et al.	N/A	N/A
2004/0009307	12/2003	Koh et al.	N/A	N/A
2004/0009679	12/2003	Yeo et al.	N/A	N/A

2004/0010772	12/2003	McKenna et al.	N/A	N/A
2004/0011504	12/2003	Ku et al.	N/A	N/A
2004/0013577	12/2003	Ganguli et al.	N/A	N/A
2004/0013818	12/2003	Moon et al.	N/A	N/A
2004/0015300	12/2003	Ganguli et al.	N/A	N/A
2004/0016637	12/2003	Yang	N/A	N/A
2004/0018304	12/2003	Chung et al.	N/A	N/A
2004/0018307	12/2003	Park et al.	N/A	N/A
2004/0018694	12/2003	Lee et al.	N/A	N/A
2004/0018723	12/2003	Byun et al.	N/A	N/A
2004/0018750	12/2003	Sophie et al.	N/A	N/A
2004/0023125	12/2003	Nozawa et al.	N/A	N/A
2004/0023516	12/2003	Londergan et al.	N/A	N/A
2004/0025786	12/2003	Kontani et al.	N/A	N/A
2004/0025787	12/2003	Selbrede et al.	N/A	N/A
2004/0026372	12/2003	Takenaka et al.	N/A	N/A
2004/0029052	12/2003	Park et al.	N/A	N/A
2004/0031564	12/2003	Gottscho et al.	N/A	N/A
2004/0035358	12/2003	Basceri et al.	N/A	N/A
2004/0036129	12/2003	Forbes et al.	N/A	N/A
2004/0037339	12/2003	Watson et al.	N/A	N/A
2004/0037675	12/2003	Zinger et al.	N/A	N/A
2004/0038525	12/2003	Meng et al.	N/A	N/A
2004/0043149	12/2003	Gordon et al.	N/A	N/A
2004/0043544	12/2003	Asai et al.	N/A	N/A
2004/0048439	12/2003	Soman	N/A	N/A
2004/0048452	12/2003	Sugawara et al.	N/A	N/A
2004/0048492	12/2003	Ishikawa et al.	N/A	N/A
2004/0050325	12/2003	Samoilov	N/A	N/A
2004/0050496	12/2003	Iwai et al.	N/A	N/A
2004/0052972	12/2003	Schmitt	N/A	N/A
2004/0056017	12/2003	Renken	N/A	N/A
2004/0058517	12/2003	Nallan et al.	N/A	N/A
2004/0062081	12/2003	Drewes	N/A	N/A
2004/0063289	12/2003	Ohta	N/A	N/A
2004/0065255	12/2003	Yang et al.	N/A	N/A
2004/0069226	12/2003	Yoshida et al.	N/A	N/A
2004/0071897	12/2003	Verplancken et al.	N/A	N/A
2004/0077182	12/2003	Lim et al.	N/A	N/A
2004/0079286	12/2003	Lindfors	N/A	N/A
2004/0079960	12/2003	Shakuda	N/A	N/A
2004/0080697	12/2003	Song	N/A	N/A
2004/0082171	12/2003	Shin et al.	N/A	N/A
2004/0083961	12/2003	Basceri	N/A	N/A
2004/0083962	12/2003	Bang et al.	N/A	N/A
2004/0083964	12/2003	Ingle et al.	N/A	N/A
2004/0083975	12/2003	Tong et al.	N/A	N/A
2004/0087141	12/2003	Ramanathan et al.	N/A	N/A
2004/0087168	12/2003	Granneman et al.	N/A	N/A
2004/0089078	12/2003	Gehner et al.	N/A	N/A

2004/0089236	12/2003	Yokogawa et al.	N/A	N/A
2004/0092073	12/2003	Cabral et al.	N/A	N/A
2004/0092120	12/2003	Wicker	N/A	N/A
2004/0093963	12/2003	Gehner et al.	N/A	N/A
2004/0094206	12/2003	Ishida	N/A	N/A
2004/0094402	12/2003	Gopalraja	N/A	N/A
2004/0095074	12/2003	Ishii et al.	N/A	N/A
2004/0099213	12/2003	Adomaitis et al.	N/A	N/A
2004/0099635	12/2003	Nishikawa	N/A	N/A
2004/0101622	12/2003	Park et al.	N/A	N/A
2004/0103914	12/2003	Cheng et al.	N/A	N/A
2004/0104439	12/2003	Haukka et al.	N/A	N/A
2004/0105738	12/2003	Ahn et al.	N/A	N/A
2004/0106249	12/2003	Huotari	N/A	N/A
2004/0112288	12/2003	Whitesell	N/A	N/A
2004/0115936	12/2003	DePetrillo et al.	N/A	N/A
2004/0118342	12/2003	Cheng et al.	N/A	N/A
2004/0121620	12/2003	Pomarede et al.	N/A	N/A
2004/0124131	12/2003	Aitchison	N/A	N/A
2004/0124549	12/2003	Curran	N/A	N/A
2004/0126213	12/2003	Pelzmann et al.	N/A	N/A
2004/0126929	12/2003	Tang et al.	N/A	N/A
2004/0126990	12/2003	Ohta	N/A	N/A
2004/0127069	12/2003	Yamazaki et al.	N/A	N/A
2004/0129211	12/2003	Blonigan et al.	N/A	N/A
2004/0129671	12/2003	Ji et al.	N/A	N/A
2004/0134429	12/2003	Yamanaka	N/A	N/A
2004/0137756	12/2003	Li et al.	N/A	N/A
2004/0142577	12/2003	Sugawara et al.	N/A	N/A
2004/0144311	12/2003	Chen	N/A	N/A
2004/0144323	12/2003	Kai	N/A	N/A
2004/0144980	12/2003	Ahn et al.	N/A	N/A
2004/0146644	12/2003	Xia et al.	N/A	N/A
2004/0151844	12/2003	Zhang et al.	N/A	N/A
2004/0151845	12/2003	Nguyen et al.	N/A	N/A
2004/0152287	12/2003	Sherrill et al.	N/A	N/A
2004/0154746	12/2003	Park	N/A	N/A
2004/0159343	12/2003	Shimbara et al.	N/A	N/A
2004/0163590	12/2003	Tran et al.	N/A	N/A
2004/0168627	12/2003	Conley et al.	N/A	N/A
2004/0168742	12/2003	Kim et al.	N/A	N/A
2004/0168769	12/2003	Matsuoka et al.	N/A	N/A
2004/0169032	12/2003	Murayama et al.	N/A	N/A
2004/0185177	12/2003	Basceri et al.	N/A	N/A
2004/0187304	12/2003	Chen et al.	N/A	N/A
2004/0187777	12/2003	Okamoto et al.	N/A	N/A
2004/0187784	12/2003	Sferlazzo	N/A	N/A
2004/0187790	12/2003	Bader	N/A	N/A
2004/0187928	12/2003	Ambrosina	N/A	N/A
2004/0198069	12/2003	Metzner et al.	N/A	N/A

2004/0200499	12/2003	Harvey et al.	N/A	N/A
2004/0202786	12/2003	Wongsenakhum et al.	N/A	N/A
2004/0203251	12/2003	Kawaguchi et al.	N/A	N/A
2004/0206305	12/2003	Choi et al.	N/A	N/A
2004/0208228	12/2003	Hashikura et al.	N/A	N/A
2004/0209477	12/2003	Buxbaum et al.	N/A	N/A
2004/0211357	12/2003	Gadgil	N/A	N/A
2004/0212947	12/2003	Nguyen	N/A	N/A
2004/0213921	12/2003	Leu	N/A	N/A
2004/0214399	12/2003	Ahn et al.	N/A	N/A
2004/0214445	12/2003	Shimizu et al.	N/A	N/A
2004/0217217	12/2003	Han et al.	N/A	N/A
2004/0219793	12/2003	Hishiya et al.	N/A	N/A
2004/0220699	12/2003	Heden et al.	N/A	N/A
2004/0221807	12/2003	Vergheze et al.	N/A	N/A
2004/0221808	12/2003	Kawano	N/A	N/A
2004/0223893	12/2003	Tabata et al.	N/A	N/A
2004/0224478	12/2003	Chudzik et al.	N/A	N/A
2004/0226507	12/2003	Carpenter et al.	N/A	N/A
2004/0226515	12/2003	Yendler et al.	N/A	N/A
2004/0228968	12/2003	Basceri	N/A	N/A
2004/0231600	12/2003	Lee	N/A	N/A
2004/0231799	12/2003	Lee et al.	N/A	N/A
2004/0238523	12/2003	Kuibira et al.	N/A	N/A
2004/0241322	12/2003	Basceri et al.	N/A	N/A
2004/0241341	12/2003	Lin	N/A	N/A
2004/0241998	12/2003	Hanson	N/A	N/A
2004/0245091	12/2003	Karim et al.	N/A	N/A
2004/0247779	12/2003	Selvamanickam et al.	N/A	N/A
2004/0250600	12/2003	Bevers et al.	N/A	N/A
2004/0253790	12/2003	Ootsuka	N/A	N/A
2004/0253867	12/2003	Matsumoto	N/A	N/A
2004/0261492	12/2003	Zarkar et al.	N/A	N/A
2004/0261706	12/2003	Lindfors et al.	N/A	N/A
2004/0261712	12/2003	Hayashi et al.	N/A	N/A
2004/0261946	12/2003	Endoh et al.	N/A	N/A
2004/0266011	12/2003	Lee et al.	N/A	N/A
2005/0000428	12/2004	Shero et al.	N/A	N/A
2005/0001062	12/2004	McCracken et al.	N/A	N/A
2005/0003089	12/2004	Won et al.	N/A	N/A
2005/0003600	12/2004	Kasai et al.	N/A	N/A
2005/0003662	12/2004	Jurisch et al.	N/A	N/A
2005/0006682	12/2004	Bae et al.	N/A	N/A
2005/0008799	12/2004	Tomiyasu et al.	N/A	N/A
2005/0009325	12/2004	Chung et al.	N/A	N/A
2005/0016452	12/2004	Ryu et al.	N/A	N/A
2005/0016470	12/2004	Kang et al.	N/A	N/A
2005/0016956	12/2004	Liu et al.	N/A	N/A
2005/0017272	12/2004	Yamashita et al.	N/A	N/A
2005/0019026	12/2004	Wang et al.	N/A	N/A

2005/0019494	12/2004	Moghadam et al.	N/A	N/A
2005/0019960	12/2004	Lee et al.	N/A	N/A
2005/0020071	12/2004	Sonobe et al.	N/A	N/A
2005/0023231	12/2004	Huang et al.	N/A	N/A
2005/0023624	12/2004	Ahn et al.	N/A	N/A
2005/0026402	12/2004	Jurgensen	N/A	N/A
2005/0033075	12/2004	Chi et al.	N/A	N/A
2005/0034664	12/2004	Koh et al.	N/A	N/A
2005/0034674	12/2004	Ono	N/A	N/A
2005/0037154	12/2004	Koh et al.	N/A	N/A
2005/0037578	12/2004	Chen et al.	N/A	N/A
2005/0037610	12/2004	Cha	N/A	N/A
2005/0037619	12/2004	Granneman et al.	N/A	N/A
2005/0040144	12/2004	Sellers	N/A	N/A
2005/0042778	12/2004	Peukert	N/A	N/A
2005/0046825	12/2004	Powell et al.	N/A	N/A
2005/0048797	12/2004	Fukazawa	N/A	N/A
2005/0051093	12/2004	Makino et al.	N/A	N/A
2005/0051100	12/2004	Chiang et al.	N/A	N/A
2005/0051854	12/2004	Cabral et al.	N/A	N/A
2005/0054175	12/2004	Bauer	N/A	N/A
2005/0054198	12/2004	Um	N/A	N/A
2005/0054228	12/2004	March	N/A	N/A
2005/0056218	12/2004	Sun et al.	N/A	N/A
2005/0056780	12/2004	Miller et al.	N/A	N/A
2005/0059261	12/2004	Basceri et al.	N/A	N/A
2005/0059262	12/2004	Yin et al.	N/A	N/A
2005/0059264	12/2004	Cheung	N/A	N/A
2005/0061964	12/2004	Nagano et al.	N/A	N/A
2005/0062773	12/2004	Fouet	N/A	N/A
2005/0063451	12/2004	Abe et al.	N/A	N/A
2005/0064207	12/2004	Senzaki et al.	N/A	N/A
2005/0064719	12/2004	Liu	N/A	N/A
2005/0066893	12/2004	Soininen	N/A	N/A
2005/0069651	12/2004	Miyoshi	N/A	N/A
2005/0070123	12/2004	Hirano	N/A	N/A
2005/0070128	12/2004	Xia et al.	N/A	N/A
2005/0070729	12/2004	Kiyomori et al.	N/A	N/A
2005/0072357	12/2004	Shero et al.	N/A	N/A
2005/0074576	12/2004	Chaiken et al.	N/A	N/A
2005/0074983	12/2004	Shinriki et al.	N/A	N/A
2005/0079124	12/2004	Sanderson	N/A	N/A
2005/0079691	12/2004	Kim et al.	N/A	N/A
2005/0081786	12/2004	Kubista et al.	N/A	N/A
2005/0085090	12/2004	Mui et al.	N/A	N/A
2005/0090123	12/2004	Nishimura et al.	N/A	N/A
2005/0092247	12/2004	Schmidt	N/A	N/A
2005/0092249	12/2004	Kilpela et al.	N/A	N/A
2005/0092439	12/2004	Keeton et al.	N/A	N/A
2005/0092733	12/2004	Ito et al.	N/A	N/A

2005/0095770	12/2004	Kumagai et al.	N/A	N/A
2005/0095779	12/2004	Park et al.	N/A	N/A
2005/0095859	12/2004	Chen et al.	N/A	N/A
2005/0098107	12/2004	Du Bois et al.	N/A	N/A
2005/0100669	12/2004	Kools et al.	N/A	N/A
2005/0101154	12/2004	Huang	N/A	N/A
2005/0101843	12/2004	Quinn et al.	N/A	N/A
2005/0104112	12/2004	Haukka et al.	N/A	N/A
2005/0106762	12/2004	Chakrapani et al.	N/A	N/A
2005/0106893	12/2004	Wilk	N/A	N/A
2005/0107627	12/2004	Dussarrat et al.	N/A	N/A
2005/0109461	12/2004	Sun	N/A	N/A
2005/0110069	12/2004	Kil et al.	N/A	N/A
2005/0112282	12/2004	Gordon et al.	N/A	N/A
2005/0115946	12/2004	Shim et al.	N/A	N/A
2005/0118804	12/2004	Byun et al.	N/A	N/A
2005/0118837	12/2004	Todd	N/A	N/A
2005/0120805	12/2004	Lane	N/A	N/A
2005/0120962	12/2004	Ushioda et al.	N/A	N/A
2005/0121145	12/2004	Du Bois et al.	N/A	N/A
2005/0123690	12/2004	Derderian et al.	N/A	N/A
2005/0130427	12/2004	Seok-Jun	N/A	N/A
2005/0132957	12/2004	El-Raghy	N/A	N/A
2005/0133160	12/2004	Kennedy et al.	N/A	N/A
2005/0133161	12/2004	Carpenter et al.	N/A	N/A
2005/0133166	12/2004	Satitpunwaycha et al.	N/A	N/A
2005/0136188	12/2004	Chang	N/A	N/A
2005/0136657	12/2004	Yokoi et al.	N/A	N/A
2005/0139160	12/2004	Lei et al.	N/A	N/A
2005/0141591	12/2004	Sakano	N/A	N/A
2005/0142361	12/2004	Nakanishi	N/A	N/A
2005/0145338	12/2004	Park et al.	N/A	N/A
2005/0148162	12/2004	Chen et al.	N/A	N/A
2005/0150601	12/2004	Srivastava	N/A	N/A
2005/0151184	12/2004	Lee et al.	N/A	N/A
2005/0153571	12/2004	Senzaki	N/A	N/A
2005/0153573	12/2004	Okudaira et al.	N/A	N/A
2005/0160987	12/2004	Kasai et al.	N/A	N/A
2005/0161434	12/2004	Sugawara et al.	N/A	N/A
2005/0164469	12/2004	Haupt	N/A	N/A
2005/0170306	12/2004	Oosterlaken et al.	N/A	N/A
2005/0172895	12/2004	Kijima et al.	N/A	N/A
2005/0173003	12/2004	Laverdiere et al.	N/A	N/A
2005/0175789	12/2004	Helms	N/A	N/A
2005/0178333	12/2004	Loke et al.	N/A	N/A
2005/0181535	12/2004	Yun et al.	N/A	N/A
2005/0181555	12/2004	Haukka et al.	N/A	N/A
2005/0183827	12/2004	White et al.	N/A	N/A
2005/0186688	12/2004	Basceri	N/A	N/A
2005/0187647	12/2004	Wang et al.	N/A	N/A

2005/0191828	12/2004	Al-Bayati et al.	N/A	N/A
2005/0193948	12/2004	Oohirabaru et al.	N/A	N/A
2005/0193952	12/2004	Goodman et al.	N/A	N/A
2005/0199013	12/2004	Vandroux et al.	N/A	N/A
2005/0199342	12/2004	Shajii et al.	N/A	N/A
2005/0201908	12/2004	Nakamura et al.	N/A	N/A
2005/0208217	12/2004	Shinriki et al.	N/A	N/A
2005/0208219	12/2004	Basceri	N/A	N/A
2005/0208718	12/2004	Lim et al.	N/A	N/A
2005/0208778	12/2004	Li	N/A	N/A
2005/0211167	12/2004	Gunji	N/A	N/A
2005/0211384	12/2004	Hayashi	N/A	N/A
2005/0212119	12/2004	Shero	N/A	N/A
2005/0214457	12/2004	Schmitt et al.	N/A	N/A
2005/0214458	12/2004	Meiere	N/A	N/A
2005/0218462	12/2004	Ahn et al.	N/A	N/A
2005/0221021	12/2004	Strang	N/A	N/A
2005/0221618	12/2004	AmRhein et al.	N/A	N/A
2005/0223982	12/2004	Park et al.	N/A	N/A
2005/0223994	12/2004	Blomiley et al.	N/A	N/A
2005/0227502	12/2004	Schmitt et al.	N/A	N/A
2005/0229848	12/2004	Shinriki	N/A	N/A
2005/0229849	12/2004	Silvetti et al.	N/A	N/A
2005/0229972	12/2004	Hoshi et al.	N/A	N/A
2005/0233477	12/2004	Yamazaki et al.	N/A	N/A
2005/0238807	12/2004	Lin et al.	N/A	N/A
2005/0241176	12/2004	Shero et al.	N/A	N/A
2005/0241763	12/2004	Huang et al.	N/A	N/A
2005/0241765	12/2004	Dhindsa et al.	N/A	N/A
2005/0245058	12/2004	Lee et al.	N/A	N/A
2005/0249876	12/2004	Kawahara et al.	N/A	N/A
2005/0250340	12/2004	Chen et al.	N/A	N/A
2005/0251990	12/2004	Choi	N/A	N/A
2005/0252447	12/2004	Zhao et al.	N/A	N/A
2005/0252449	12/2004	Nguyen et al.	N/A	N/A
2005/0252455	12/2004	Moriya et al.	N/A	N/A
2005/0253061	12/2004	Cameron et al.	N/A	N/A
2005/0255257	12/2004	Choi et al.	N/A	N/A
2005/0255327	12/2004	Chaney et al.	N/A	N/A
2005/0258280	12/2004	Goto et al.	N/A	N/A
2005/0260347	12/2004	Narwankar et al.	N/A	N/A
2005/0260837	12/2004	Walther et al.	N/A	N/A
2005/0260850	12/2004	Loke	N/A	N/A
2005/0263072	12/2004	Balasubramanian et al.	N/A	N/A
2005/0263075	12/2004	Wang et al.	N/A	N/A
2005/0263719	12/2004	Ohdaira et al.	N/A	N/A
2005/0263932	12/2004	Heugel	N/A	N/A
2005/0268856	12/2004	Miller et al.	N/A	N/A
2005/0271812	12/2004	Myo et al.	N/A	N/A
2005/0271813	12/2004	Kher et al.	N/A	N/A

2005/0272247	12/2004	Ikeda et al.	N/A	N/A
2005/0274323	12/2004	Seidel et al.	N/A	N/A
2005/0276928	12/2004	Okumura et al.	N/A	N/A
2005/0277271	12/2004	Beintner	N/A	N/A
2005/0282101	12/2004	Adachi	N/A	N/A
2005/0282350	12/2004	Chou et al.	N/A	N/A
2005/0284573	12/2004	Egley et al.	N/A	N/A
2005/0284991	12/2004	Saez	N/A	N/A
2005/0285097	12/2004	Shang et al.	N/A	N/A
2005/0285208	12/2004	Ren et al.	N/A	N/A
2005/0287725	12/2004	Kitagawa	N/A	N/A
2005/0287771	12/2004	Seamons et al.	N/A	N/A
2006/0000411	12/2005	Seo	N/A	N/A
2006/0006538	12/2005	Allman et al.	N/A	N/A
2006/0008997	12/2005	Jang et al.	N/A	N/A
2006/0009044	12/2005	Igeta et al.	N/A	N/A
2006/0013674	12/2005	Elliott et al.	N/A	N/A
2006/0013946	12/2005	Park et al.	N/A	N/A
2006/0014384	12/2005	Lee et al.	N/A	N/A
2006/0014397	12/2005	Seamons et al.	N/A	N/A
2006/0016783	12/2005	Wu et al.	N/A	N/A
2006/0019033	12/2005	Muthukrishnan et al.	N/A	N/A
2006/0019495	12/2005	Marcadal et al.	N/A	N/A
2006/0019502	12/2005	Park et al.	N/A	N/A
2006/0021572	12/2005	Wolden	N/A	N/A
2006/0021573	12/2005	Monsma et al.	N/A	N/A
2006/0021703	12/2005	Umotoy et al.	N/A	N/A
2006/0024439	12/2005	Tuominen et al.	N/A	N/A
2006/0026314	12/2005	Franchuk et al.	N/A	N/A
2006/0032443	12/2005	Hasebe et al.	N/A	N/A
2006/0040054	12/2005	Pearlstein et al.	N/A	N/A
2006/0040508	12/2005	Ji	N/A	N/A
2006/0046518	12/2005	Hill et al.	N/A	N/A
2006/0048710	12/2005	Horiguchi et al.	N/A	N/A
2006/0051505	12/2005	Kortshagen et al.	N/A	N/A
2006/0051520	12/2005	Behle et al.	N/A	N/A
2006/0051925	12/2005	Ahn et al.	N/A	N/A
2006/0057799	12/2005	Horiguchi et al.	N/A	N/A
2006/0057828	12/2005	Omura	N/A	N/A
2006/0057858	12/2005	Chung et al.	N/A	N/A
2006/0060930	12/2005	Metz et al.	N/A	N/A
2006/0062910	12/2005	Meiere	N/A	N/A
2006/0063346	12/2005	Lee et al.	N/A	N/A
2006/0068097	12/2005	Yamasaki et al.	N/A	N/A
2006/0068104	12/2005	Ishizaka	N/A	N/A
2006/0068121	12/2005	Lee et al.	N/A	N/A
2006/0068125	12/2005	Radhakrishnan	N/A	N/A
2006/0081558	12/2005	Collins et al.	N/A	N/A
2006/0087638	12/2005	Hirayanagi	N/A	N/A
2006/0090702	12/2005	Koo et al.	N/A	N/A

2006/0093756	12/2005	Rajagopalan et al.	N/A	N/A
2006/0094236	12/2005	Elkins et al.	N/A	N/A
2006/0096540	12/2005	Choi	N/A	N/A
2006/0097220	12/2005	Kim et al.	N/A	N/A
2006/0097305	12/2005	Lee	N/A	N/A
2006/0099782	12/2005	Ritenour	N/A	N/A
2006/0102968	12/2005	Bojarczuk et al.	N/A	N/A
2006/0105566	12/2005	Waldfried et al.	N/A	N/A
2006/0107898	12/2005	Blomberg	N/A	N/A
2006/0108221	12/2005	Goodwin et al.	N/A	N/A
2006/0108331	12/2005	Nozawa et al.	N/A	N/A
2006/0110930	12/2005	Senzaki	N/A	N/A
2006/0110934	12/2005	Fukuchi	N/A	N/A
2006/0113038	12/2005	Gondhalekar et al.	N/A	N/A
2006/0113675	12/2005	Chang et al.	N/A	N/A
2006/0113806	12/2005	Tsuji et al.	N/A	N/A
2006/0115589	12/2005	Vukovic	N/A	N/A
2006/0118240	12/2005	Holber et al.	N/A	N/A
2006/0118241	12/2005	Ohmi et al.	N/A	N/A
2006/0125099	12/2005	Gordon et al.	N/A	N/A
2006/0127067	12/2005	Wintenberger et al.	N/A	N/A
2006/0128142	12/2005	Whelan et al.	N/A	N/A
2006/0128168	12/2005	Ahn et al.	N/A	N/A
2006/0130751	12/2005	Volfovski et al.	N/A	N/A
2006/0130767	12/2005	Herchen	N/A	N/A
2006/0133955	12/2005	Peters	N/A	N/A
2006/0137608	12/2005	Choi et al.	N/A	N/A
2006/0137609	12/2005	Puchacz et al.	N/A	N/A
2006/0141155	12/2005	Gordon et al.	N/A	N/A
2006/0141758	12/2005	Naumann et al.	N/A	N/A
2006/0144820	12/2005	Sawin et al.	N/A	N/A
2006/0147626	12/2005	Blomberg	N/A	N/A
2006/0148151	12/2005	Murthy et al.	N/A	N/A
2006/0148180	12/2005	Ahn et al.	N/A	N/A
2006/0151117	12/2005	Kasanami et al.	N/A	N/A
2006/0154424	12/2005	Yang et al.	N/A	N/A
2006/0156979	12/2005	Thakur et al.	N/A	N/A
2006/0156981	12/2005	Fondurulia	N/A	N/A
2006/0162658	12/2005	Weidman	N/A	N/A
2006/0162661	12/2005	Jung et al.	N/A	N/A
2006/0162863	12/2005	Kim et al.	N/A	N/A
2006/0163612	12/2005	Kouvetakis et al.	N/A	N/A
2006/0163683	12/2005	Roth et al.	N/A	N/A
2006/0165892	12/2005	Weidman	N/A	N/A
2006/0166428	12/2005	Kamioka	N/A	N/A
2006/0172531	12/2005	Lin et al.	N/A	N/A
2006/0175669	12/2005	Kim et al.	N/A	N/A
2006/0176928	12/2005	Nakamura et al.	N/A	N/A
2006/0177855	12/2005	Utermohlen	N/A	N/A
2006/0182885	12/2005	Lei et al.	N/A	N/A

2006/0185589	12/2005	Zehavi et al.	N/A	N/A
2006/0188360	12/2005	Bonora et al.	N/A	N/A
2006/0191555	12/2005	Yoshida et al.	N/A	N/A
2006/0193979	12/2005	Meiere et al.	N/A	N/A
2006/0193980	12/2005	Hasegawa	N/A	N/A
2006/0196418	12/2005	Lindfors et al.	N/A	N/A
2006/0196420	12/2005	Ushakov et al.	N/A	N/A
2006/0196421	12/2005	Ronsse et al.	N/A	N/A
2006/0199357	12/2005	Wan et al.	N/A	N/A
2006/0205194	12/2005	Bauer	N/A	N/A
2006/0205223	12/2005	Smayling	N/A	N/A
2006/0205231	12/2005	Chou et al.	N/A	N/A
2006/0208215	12/2005	Metzner et al.	N/A	N/A
2006/0211224	12/2005	Matsuda	N/A	N/A
2006/0211243	12/2005	Ishizaka et al.	N/A	N/A
2006/0211259	12/2005	Maes	N/A	N/A
2006/0213437	12/2005	Ishizaka et al.	N/A	N/A
2006/0213439	12/2005	Ishizaka	N/A	N/A
2006/0213441	12/2005	Kobrin et al.	N/A	N/A
2006/0216942	12/2005	Kim et al.	N/A	N/A
2006/0219169	12/2005	Chen et al.	N/A	N/A
2006/0219361	12/2005	Wang et al.	N/A	N/A
2006/0219363	12/2005	Matsumoto et al.	N/A	N/A
2006/0223301	12/2005	Vanhaelemeersch et al.	N/A	N/A
2006/0223337	12/2005	Ahn et al.	N/A	N/A
2006/0226117	12/2005	Bertram et al.	N/A	N/A
2006/0228496	12/2005	Choi	N/A	N/A
2006/0228863	12/2005	Zhang et al.	N/A	N/A
2006/0228888	12/2005	Lee et al.	N/A	N/A
2006/0228898	12/2005	Wajda et al.	N/A	N/A
2006/0236934	12/2005	Choi et al.	N/A	N/A
2006/0240187	12/2005	Weidman	N/A	N/A
2006/0240574	12/2005	Yoshie	N/A	N/A
2006/0240662	12/2005	Conley et al.	N/A	N/A
2006/0247404	12/2005	Todd	N/A	N/A
2006/0249175	12/2005	Nowak et al.	N/A	N/A
2006/0249253	12/2005	Dando	N/A	N/A
2006/0251827	12/2005	Nowak	N/A	N/A
2006/0252228	12/2005	Jeng	N/A	N/A
2006/0252244	12/2005	Vaartstra et al.	N/A	N/A
2006/0252351	12/2005	Kundracik	N/A	N/A
2006/0254514	12/2005	Kang et al.	N/A	N/A
2006/0257563	12/2005	Doh et al.	N/A	N/A
2006/0257584	12/2005	Derderian et al.	N/A	N/A
2006/0258078	12/2005	Lee et al.	N/A	N/A
2006/0258173	12/2005	Xiao et al.	N/A	N/A
2006/0260545	12/2005	Ramaswamy et al.	N/A	N/A
2006/0263522	12/2005	Byun	N/A	N/A
2006/0263540	12/2005	Ramaswamy et al.	N/A	N/A
2006/0264060	12/2005	Ramaswamy et al.	N/A	N/A

2006/0264066	12/2005	Bartholomew	N/A	N/A
2006/0266289	12/2005	Verghese et al.	N/A	N/A
2006/0269690	12/2005	Watanabe et al.	N/A	N/A
2006/0269692	12/2005	Balseanu	N/A	N/A
2006/0275710	12/2005	Yamazaki et al.	N/A	N/A
2006/0275933	12/2005	Du Bois et al.	N/A	N/A
2006/0278524	12/2005	Stowell	N/A	N/A
2006/0283629	12/2005	Kikuchi et al.	N/A	N/A
2006/0286774	12/2005	Singh et al.	N/A	N/A
2006/0286775	12/2005	Singh et al.	N/A	N/A
2006/0286817	12/2005	Kato et al.	N/A	N/A
2006/0286818	12/2005	Wang et al.	N/A	N/A
2006/0286819	12/2005	Seutter	N/A	N/A
2006/0291982	12/2005	Tanaka	N/A	N/A
2006/0292310	12/2005	Le et al.	N/A	N/A
2006/0292845	12/2005	Chiang et al.	N/A	N/A
2007/0004204	12/2006	Fukazawa et al.	N/A	N/A
2007/0006806	12/2006	Imai	N/A	N/A
2007/0010072	12/2006	Bailey et al.	N/A	N/A
2007/0012402	12/2006	Sneh	N/A	N/A
2007/0014919	12/2006	Hamalainen et al.	N/A	N/A
2007/0020160	12/2006	Berkman et al.	N/A	N/A
2007/0020167	12/2006	Han et al.	N/A	N/A
2007/0020830	12/2006	Speranza	N/A	N/A
2007/0020953	12/2006	Tsai et al.	N/A	N/A
2007/0022954	12/2006	Iizuka et al.	N/A	N/A
2007/0026148	12/2006	Arai et al.	N/A	N/A
2007/0026162	12/2006	Wei et al.	N/A	N/A
2007/0026540	12/2006	Nooten et al.	N/A	N/A
2007/0026654	12/2006	Huotari et al.	N/A	N/A
2007/0028842	12/2006	Inagawa et al.	N/A	N/A
2007/0031598	12/2006	Okuyama et al.	N/A	N/A
2007/0031599	12/2006	Gschwandtner et al.	N/A	N/A
2007/0032004	12/2006	Ramaswamy et al.	N/A	N/A
2007/0032045	12/2006	Kasahara et al.	N/A	N/A
2007/0032054	12/2006	Ramaswamy et al.	N/A	N/A
2007/0032082	12/2006	Ramaswamy et al.	N/A	N/A
2007/0032095	12/2006	Ramaswamy et al.	N/A	N/A
2007/0034477	12/2006	Inui	N/A	N/A
2007/0037343	12/2006	Colombo et al.	N/A	N/A
2007/0037412	12/2006	Dip et al.	N/A	N/A
2007/0042117	12/2006	Kupurao et al.	N/A	N/A
2007/0042581	12/2006	Sano et al.	N/A	N/A
2007/0044716	12/2006	Tetsuka et al.	N/A	N/A
2007/0045244	12/2006	Lee et al.	N/A	N/A
2007/0047384	12/2006	McLaughlin et al.	N/A	N/A
2007/0048953	12/2006	Gealy et al.	N/A	N/A
2007/0049053	12/2006	Mahajani	N/A	N/A
2007/0051299	12/2006	Ong et al.	N/A	N/A
2007/0051312	12/2006	Sneh	N/A	N/A

2007/0051471	12/2006	Kawaguchi et al.	N/A	N/A
2007/0054049	12/2006	Lindfors et al.	N/A	N/A
2007/0054499	12/2006	Jang	N/A	N/A
2007/0056843	12/2006	Ye et al.	N/A	N/A
2007/0056850	12/2006	Ye et al.	N/A	N/A
2007/0059948	12/2006	Metzner et al.	N/A	N/A
2007/0062439	12/2006	Wada et al.	N/A	N/A
2007/0062453	12/2006	Ishikawa	N/A	N/A
2007/0062646	12/2006	Ogawa et al.	N/A	N/A
2007/0065578	12/2006	McDougall	N/A	N/A
2007/0065597	12/2006	Kaido et al.	N/A	N/A
2007/0066010	12/2006	Ando	N/A	N/A
2007/0066038	12/2006	Sadjadi et al.	N/A	N/A
2007/0066079	12/2006	Kolster et al.	N/A	N/A
2007/0066084	12/2006	Wajda et al.	N/A	N/A
2007/0074665	12/2006	Chacin et al.	N/A	N/A
2007/0077355	12/2006	Chacin et al.	N/A	N/A
2007/0082132	12/2006	Shinriki	N/A	N/A
2007/0082500	12/2006	Norman et al.	N/A	N/A
2007/0082508	12/2006	Chiang et al.	N/A	N/A
2007/0084405	12/2006	Kim	N/A	N/A
2007/0087296	12/2006	Kim et al.	N/A	N/A
2007/0087515	12/2006	Yieh et al.	N/A	N/A
2007/0087579	12/2006	Kitayama et al.	N/A	N/A
2007/0089670	12/2006	Ikedo	N/A	N/A
2007/0092696	12/2006	Tsukatani et al.	N/A	N/A
2007/0095283	12/2006	Galewski	N/A	N/A
2007/0095286	12/2006	Baek et al.	N/A	N/A
2007/0096194	12/2006	Streck et al.	N/A	N/A
2007/0098527	12/2006	Hall et al.	N/A	N/A
2007/0107845	12/2006	Ishizawa et al.	N/A	N/A
2007/0111030	12/2006	Nakano et al.	N/A	N/A
2007/0111470	12/2006	Smythe	N/A	N/A
2007/0111545	12/2006	Lee et al.	N/A	N/A
2007/0113788	12/2006	Nozawa et al.	N/A	N/A
2007/0116872	12/2006	Li et al.	N/A	N/A
2007/0116873	12/2006	Li et al.	N/A	N/A
2007/0116887	12/2006	Faguet	N/A	N/A
2007/0116888	12/2006	Faguet	N/A	N/A
2007/0117414	12/2006	Moffatt et al.	N/A	N/A
2007/0119370	12/2006	Ma et al.	N/A	N/A
2007/0120275	12/2006	Liu	N/A	N/A
2007/0123037	12/2006	Lee et al.	N/A	N/A
2007/0123060	12/2006	Rahtu	N/A	N/A
2007/0123189	12/2006	Saito et al.	N/A	N/A
2007/0125762	12/2006	Cui et al.	N/A	N/A
2007/0128538	12/2006	Fairbairn et al.	N/A	N/A
2007/0128570	12/2006	Goto et al.	N/A	N/A
2007/0128858	12/2006	Haukka et al.	N/A	N/A
2007/0128876	12/2006	Fukiage	N/A	N/A

2007/0128888	12/2006	Goto et al.	N/A	N/A
2007/0129621	12/2006	Kellogg et al.	N/A	N/A
2007/0131168	12/2006	Gomi et al.	N/A	N/A
2007/0134821	12/2006	Thakur et al.	N/A	N/A
2007/0134919	12/2006	Gunji et al.	N/A	N/A
2007/0134942	12/2006	Ahn et al.	N/A	N/A
2007/0137794	12/2006	Qiu et al.	N/A	N/A
2007/0144442	12/2006	Migita	N/A	N/A
2007/0146621	12/2006	Yeom	N/A	N/A
2007/0148347	12/2006	Hatanpaa et al.	N/A	N/A
2007/0148350	12/2006	Rahtu	N/A	N/A
2007/0148990	12/2006	Deboer et al.	N/A	N/A
2007/0153625	12/2006	Lundgren et al.	N/A	N/A
2007/0155138	12/2006	Tomasini et al.	N/A	N/A
2007/0157466	12/2006	Kida et al.	N/A	N/A
2007/0157683	12/2006	Li	N/A	N/A
2007/0158026	12/2006	Amikura	N/A	N/A
2007/0163440	12/2006	Kim et al.	N/A	N/A
2007/0163490	12/2006	Habel et al.	N/A	N/A
2007/0163625	12/2006	Lee	N/A	N/A
2007/0163996	12/2006	Horiguchi	N/A	N/A
2007/0166457	12/2006	Yamoto et al.	N/A	N/A
2007/0166459	12/2006	Chang et al.	N/A	N/A
2007/0166966	12/2006	Todd et al.	N/A	N/A
2007/0166999	12/2006	Vaarstra	N/A	N/A
2007/0170372	12/2006	Horsky	N/A	N/A
2007/0173071	12/2006	Afzali-Ardakani et al.	N/A	N/A
2007/0175393	12/2006	Nishimura et al.	N/A	N/A
2007/0175397	12/2006	Tomiyasu et al.	N/A	N/A
2007/0178235	12/2006	Yamada et al.	N/A	N/A
2007/0181066	12/2006	Cadwell et al.	N/A	N/A
2007/0184179	12/2006	Waghray et al.	N/A	N/A
2007/0186849	12/2006	Furuya	N/A	N/A
2007/0186952	12/2006	Honda et al.	N/A	N/A
2007/0187362	12/2006	Nakagawa et al.	N/A	N/A
2007/0187363	12/2006	Oka et al.	N/A	N/A
2007/0190266	12/2006	Fu et al.	N/A	N/A
2007/0190362	12/2006	Weidman	N/A	N/A
2007/0190744	12/2006	Hiraiwa et al.	N/A	N/A
2007/0190782	12/2006	Park	N/A	N/A
2007/0199510	12/2006	Weiner et al.	N/A	N/A
2007/0202678	12/2006	Plombon et al.	N/A	N/A
2007/0205788	12/2006	Natsuhara et al.	N/A	N/A
2007/0207275	12/2006	Nowak et al.	N/A	N/A
2007/0209588	12/2006	Li et al.	N/A	N/A
2007/0209590	12/2006	Li	N/A	N/A
2007/0210890	12/2006	Hsu et al.	N/A	N/A
2007/0212484	12/2006	Li	N/A	N/A
2007/0212510	12/2006	Hieslmair	257/E31.13	H01L 31/1804

2007/0212811	12/2006	Hanawa et al.	N/A	N/A
2007/0212827	12/2006	Girotra et al.	N/A	N/A
2007/0215048	12/2006	Suzuki et al.	N/A	N/A
2007/0215278	12/2006	Furuse et al.	N/A	N/A
2007/0215580	12/2006	Koshiishi et al.	N/A	N/A
2007/0218200	12/2006	Suzuki et al.	N/A	N/A
2007/0218705	12/2006	Matsuki et al.	N/A	N/A
2007/0222131	12/2006	Fukumoto et al.	N/A	N/A
2007/0224777	12/2006	Hamelin	N/A	N/A
2007/0224833	12/2006	Morisada et al.	N/A	N/A
2007/0227665	12/2006	Matsumoto et al.	N/A	N/A
2007/0231488	12/2006	Von Kaenel	N/A	N/A
2007/0232031	12/2006	Singh et al.	N/A	N/A
2007/0232071	12/2006	Balseanu et al.	N/A	N/A
2007/0232501	12/2006	Tonomura	N/A	N/A
2007/0234955	12/2006	Suzuki et al.	N/A	N/A
2007/0237697	12/2006	Clark	N/A	N/A
2007/0237698	12/2006	Clark	N/A	N/A
2007/0237699	12/2006	Clark	N/A	N/A
2007/0238305	12/2006	Delgadino et al.	N/A	N/A
2007/0241688	12/2006	DeVancentis et al.	N/A	N/A
2007/0243317	12/2006	Du Bois et al.	N/A	N/A
2007/0247075	12/2006	Kim et al.	N/A	N/A
2007/0248767	12/2006	Okura	N/A	N/A
2007/0248832	12/2006	Maeda et al.	N/A	N/A
2007/0249131	12/2006	Allen et al.	N/A	N/A
2007/0251444	12/2006	Gros-Jean et al.	N/A	N/A
2007/0251456	12/2006	Herchen et al.	N/A	N/A
2007/0252233	12/2006	Yamazaki et al.	N/A	N/A
2007/0252244	12/2006	Srividya et al.	N/A	N/A
2007/0252532	12/2006	DeVancentis et al.	N/A	N/A
2007/0254414	12/2006	Miyanami	N/A	N/A
2007/0258506	12/2006	Schwagerman et al.	N/A	N/A
2007/0258855	12/2006	Turcot et al.	N/A	N/A
2007/0259778	12/2006	Spencer et al.	N/A	N/A
2007/0261868	12/2006	Gross	N/A	N/A
2007/0264427	12/2006	Shinriki et al.	N/A	N/A
2007/0264793	12/2006	Oh et al.	N/A	N/A
2007/0264807	12/2006	Leone et al.	N/A	N/A
2007/0266932	12/2006	Hiramatsu	N/A	N/A
2007/0266945	12/2006	Shuto et al.	N/A	N/A
2007/0269983	12/2006	Sneh	N/A	N/A
2007/0275166	12/2006	Thridandam et al.	N/A	N/A
2007/0277735	12/2006	Mokhesi et al.	N/A	N/A
2007/0281082	12/2006	Mokhesi et al.	N/A	N/A
2007/0281105	12/2006	Mokhesi et al.	N/A	N/A
2007/0281106	12/2006	Lubomirsky et al.	N/A	N/A
2007/0281496	12/2006	Ingle et al.	N/A	N/A
2007/0286957	12/2006	Suzuki et al.	N/A	N/A
2007/0289534	12/2006	Lubomirsky et al.	N/A	N/A

2007/0292974	12/2006	Mizuno et al.	N/A	N/A
2007/0295602	12/2006	Tiller et al.	N/A	N/A
2007/0298362	12/2006	Rocha-Alvarez et al.	N/A	N/A
2007/0298565	12/2006	Nieh et al.	N/A	N/A
2008/0003425	12/2007	Spencer et al.	N/A	N/A
2008/0003824	12/2007	Padhi et al.	N/A	N/A
2008/0003838	12/2007	Haukka et al.	N/A	N/A
2008/0006208	12/2007	Ueno et al.	N/A	N/A
2008/0018004	12/2007	Steidl	N/A	N/A
2008/0020591	12/2007	Balseanu et al.	N/A	N/A
2008/0020593	12/2007	Wang et al.	N/A	N/A
2008/0023436	12/2007	Gros-Jean et al.	N/A	N/A
2008/0026162	12/2007	Dickey et al.	N/A	N/A
2008/0026574	12/2007	Brcka	N/A	N/A
2008/0026597	12/2007	Munro et al.	N/A	N/A
2008/0029790	12/2007	Ahn et al.	N/A	N/A
2008/0031708	12/2007	Bonora et al.	N/A	N/A
2008/0032514	12/2007	Sano et al.	N/A	N/A
2008/0035055	12/2007	Dip et al.	N/A	N/A
2008/0035306	12/2007	White et al.	N/A	N/A
2008/0035607	12/2007	O'Hara et al.	N/A	N/A
2008/0036354	12/2007	Letz et al.	N/A	N/A
2008/0038485	12/2007	Fukazawa et al.	N/A	N/A
2008/0038934	12/2007	Vrtis et al.	N/A	N/A
2008/0042165	12/2007	Sugizaki	N/A	N/A
2008/0042192	12/2007	Park et al.	N/A	N/A
2008/0043803	12/2007	Bandoh	N/A	N/A
2008/0044932	12/2007	Samoilov et al.	N/A	N/A
2008/0044938	12/2007	England et al.	N/A	N/A
2008/0050536	12/2007	Aing et al.	N/A	N/A
2008/0050538	12/2007	Hirata	N/A	N/A
2008/0054332	12/2007	Kim et al.	N/A	N/A
2008/0054813	12/2007	Espiau et al.	N/A	N/A
2008/0056860	12/2007	Natume	N/A	N/A
2008/0057659	12/2007	Forbes et al.	N/A	N/A
2008/0061667	12/2007	Gaertner et al.	N/A	N/A
2008/0063798	12/2007	Kher et al.	N/A	N/A
2008/0066778	12/2007	Matsushita et al.	N/A	N/A
2008/0067146	12/2007	Onishi et al.	N/A	N/A
2008/0069951	12/2007	Chacin et al.	N/A	N/A
2008/0069955	12/2007	Hong et al.	N/A	N/A
2008/0072821	12/2007	Dalton et al.	N/A	N/A
2008/0075562	12/2007	Maria et al.	N/A	N/A
2008/0075838	12/2007	Inoue et al.	N/A	N/A
2008/0075881	12/2007	Won et al.	N/A	N/A
2008/0076070	12/2007	Koh et al.	N/A	N/A
2008/0076266	12/2007	Fukazawa et al.	N/A	N/A
2008/0076281	12/2007	Ciancanelli et al.	N/A	N/A
2008/0081104	12/2007	Hasebe et al.	N/A	N/A
2008/0081113	12/2007	Clark	N/A	N/A

2008/0081121	12/2007	Morita et al.	N/A	N/A
2008/0083710	12/2007	Chen et al.	N/A	N/A
2008/0083948	12/2007	Lin et al.	N/A	N/A
2008/0085226	12/2007	Fondurulia et al.	N/A	N/A
2008/0085610	12/2007	Wang et al.	N/A	N/A
2008/0087218	12/2007	Shimada et al.	N/A	N/A
2008/0087642	12/2007	Sawin et al.	N/A	N/A
2008/0087890	12/2007	Ahn et al.	N/A	N/A
2008/0092815	12/2007	Chen et al.	N/A	N/A
2008/0092821	12/2007	Otsuka et al.	N/A	N/A
2008/0099147	12/2007	Myo et al.	N/A	N/A
2008/0102203	12/2007	Wu	N/A	N/A
2008/0102205	12/2007	Barry et al.	N/A	N/A
2008/0102208	12/2007	Wu et al.	N/A	N/A
2008/0102603	12/2007	Kobayashi et al.	N/A	N/A
2008/0102630	12/2007	Saito	N/A	N/A
2008/0105276	12/2007	Yeh et al.	N/A	N/A
2008/0110401	12/2007	Fujikawa et al.	N/A	N/A
2008/0110568	12/2007	Son et al.	N/A	N/A
2008/0113094	12/2007	Casper	N/A	N/A
2008/0113096	12/2007	Mahajani	N/A	N/A
2008/0113097	12/2007	Mahajani et al.	N/A	N/A
2008/0118334	12/2007	Bonora	N/A	N/A
2008/0121177	12/2007	Bang et al.	N/A	N/A
2008/0121626	12/2007	Thomas et al.	N/A	N/A
2008/0121962	12/2007	Forbes et al.	N/A	N/A
2008/0124197	12/2007	van der Meulen et al.	N/A	N/A
2008/0124908	12/2007	Forbes et al.	N/A	N/A
2008/0124945	12/2007	Miya et al.	N/A	N/A
2008/0124946	12/2007	Xiao et al.	N/A	N/A
2008/0128726	12/2007	Sakata et al.	N/A	N/A
2008/0129209	12/2007	Deakins et al.	N/A	N/A
2008/0132046	12/2007	Walther	N/A	N/A
2008/0133154	12/2007	Krauss et al.	N/A	N/A
2008/0134887	12/2007	Sherer	N/A	N/A
2008/0135516	12/2007	Yokogawa et al.	N/A	N/A
2008/0135936	12/2007	Nakajima	N/A	N/A
2008/0142046	12/2007	Johnson et al.	N/A	N/A
2008/0142483	12/2007	Hua	N/A	N/A
2008/0146033	12/2007	Park	N/A	N/A
2008/0149031	12/2007	Chu et al.	N/A	N/A
2008/0149593	12/2007	Bai et al.	N/A	N/A
2008/0152463	12/2007	Chidambaram et al.	N/A	N/A
2008/0153308	12/2007	Ogawa et al.	N/A	N/A
2008/0153311	12/2007	Padhi et al.	N/A	N/A
2008/0156769	12/2007	Weiner et al.	N/A	N/A
2008/0157157	12/2007	Tomomura	N/A	N/A
2008/0157212	12/2007	Lavoie et al.	N/A	N/A
2008/0157365	12/2007	Ott et al.	N/A	N/A
2008/0173237	12/2007	Collins	N/A	N/A

2008/0173238	12/2007	Nakashima et al.	N/A	N/A
2008/0173240	12/2007	Furukawahara	N/A	N/A
2008/0173326	12/2007	Gu et al.	N/A	N/A
2008/0176335	12/2007	Alberti et al.	N/A	N/A
2008/0176375	12/2007	Erben et al.	N/A	N/A
2008/0176412	12/2007	Komeda	N/A	N/A
2008/0178608	12/2007	Tandou et al.	N/A	N/A
2008/0178805	12/2007	Paterson et al.	N/A	N/A
2008/0179104	12/2007	Zhang	N/A	N/A
2008/0179291	12/2007	Collins et al.	N/A	N/A
2008/0179715	12/2007	Coppa	N/A	N/A
2008/0182075	12/2007	Chopra	N/A	N/A
2008/0182390	12/2007	Lemmi et al.	N/A	N/A
2008/0182411	12/2007	Elers	N/A	N/A
2008/0191193	12/2007	Li et al.	N/A	N/A
2008/0193643	12/2007	Dip	N/A	N/A
2008/0194088	12/2007	Srinivasan et al.	N/A	N/A
2008/0194105	12/2007	Dominguez et al.	N/A	N/A
2008/0194113	12/2007	Kim et al.	N/A	N/A
2008/0194169	12/2007	Sterling et al.	N/A	N/A
2008/0199977	12/2007	Weigel et al.	N/A	N/A
2008/0202416	12/2007	Provencher	N/A	N/A
2008/0202689	12/2007	Kim	N/A	N/A
2008/0203487	12/2007	Hohage et al.	N/A	N/A
2008/0205483	12/2007	Rempe et al.	N/A	N/A
2008/0210162	12/2007	Yonebayashi	N/A	N/A
2008/0210278	12/2007	Orii et al.	N/A	N/A
2008/0211423	12/2007	Shinmen et al.	N/A	N/A
2008/0211526	12/2007	Shinma	N/A	N/A
2008/0213696	12/2007	Meeus et al.	N/A	N/A
2008/0214003	12/2007	Xia et al.	N/A	N/A
2008/0216077	12/2007	Emani et al.	N/A	N/A
2008/0216742	12/2007	Takebayashi	N/A	N/A
2008/0216958	12/2007	Goto et al.	N/A	N/A
2008/0220619	12/2007	Matsushita et al.	N/A	N/A
2008/0223130	12/2007	Snell et al.	N/A	N/A
2008/0223725	12/2007	Han et al.	N/A	N/A
2008/0224240	12/2007	Ahn et al.	N/A	N/A
2008/0228306	12/2007	Yetter et al.	N/A	N/A
2008/0229811	12/2007	Zhao et al.	N/A	N/A
2008/0230352	12/2007	Hirata	N/A	N/A
2008/0230371	12/2007	McHugh	N/A	N/A
2008/0233288	12/2007	Clark	N/A	N/A
2008/0237572	12/2007	Chui et al.	N/A	N/A
2008/0237604	12/2007	Alshareef et al.	N/A	N/A
2008/0241052	12/2007	Hooper et al.	N/A	N/A
2008/0241384	12/2007	Jeong	N/A	N/A
2008/0241387	12/2007	Keto	N/A	N/A
2008/0242116	12/2007	Clark	N/A	N/A
2008/0246101	12/2007	Li et al.	N/A	N/A

2008/0248310	12/2007	Kim et al.	N/A	N/A
2008/0248597	12/2007	Qin et al.	N/A	N/A
2008/0257102	12/2007	Packer	N/A	N/A
2008/0257494	12/2007	Hayashi et al.	N/A	N/A
2008/0260345	12/2007	Mertesdorf et al.	N/A	N/A
2008/0260963	12/2007	Yoon et al.	N/A	N/A
2008/0261405	12/2007	Yang et al.	N/A	N/A
2008/0261413	12/2007	Mahajani	N/A	N/A
2008/0264337	12/2007	Sano et al.	N/A	N/A
2008/0264443	12/2007	Shrinivasan et al.	N/A	N/A
2008/0267598	12/2007	Nakamura	N/A	N/A
2008/0268171	12/2007	Ma et al.	N/A	N/A
2008/0268635	12/2007	Yu et al.	N/A	N/A
2008/0272424	12/2007	Kim et al.	N/A	N/A
2008/0274369	12/2007	Lee et al.	N/A	N/A
2008/0276864	12/2007	Koelmel et al.	N/A	N/A
2008/0277647	12/2007	Kouvetakis et al.	N/A	N/A
2008/0277656	12/2007	Park et al.	N/A	N/A
2008/0277715	12/2007	Ohmi et al.	N/A	N/A
2008/0282970	12/2007	Heys et al.	N/A	N/A
2008/0283962	12/2007	Dyer	N/A	N/A
2008/0286463	12/2007	Tiner et al.	N/A	N/A
2008/0289574	12/2007	Jacobs et al.	N/A	N/A
2008/0289650	12/2007	Arena	N/A	N/A
2008/0289767	12/2007	Tandou et al.	N/A	N/A
2008/0291964	12/2007	Shrimpling	N/A	N/A
2008/0293198	12/2007	Kojima et al.	N/A	N/A
2008/0295872	12/2007	Riker et al.	N/A	N/A
2008/0298945	12/2007	Cox	N/A	N/A
2008/0299326	12/2007	Fukazawa	N/A	N/A
2008/0299758	12/2007	Harada et al.	N/A	N/A
2008/0302303	12/2007	Choi et al.	N/A	N/A
2008/0303744	12/2007	Hirayama et al.	N/A	N/A
2008/0305014	12/2007	Honda	N/A	N/A
2008/0305246	12/2007	Choi et al.	N/A	N/A
2008/0305443	12/2007	Nakamura	N/A	N/A
2008/0314319	12/2007	Hamano et al.	N/A	N/A
2008/0314892	12/2007	Graham	N/A	N/A
2008/0315292	12/2007	Ji et al.	N/A	N/A
2008/0317972	12/2007	Hendriks	N/A	N/A
2009/0000550	12/2008	Tran et al.	N/A	N/A
2009/0000551	12/2008	Choi et al.	N/A	N/A
2009/0000769	12/2008	Lin et al.	N/A	N/A
2009/0004875	12/2008	Shen et al.	N/A	N/A
2009/0011145	12/2008	Yun	N/A	N/A
2009/0011150	12/2008	Jeon et al.	N/A	N/A
2009/0011608	12/2008	Nabatame	N/A	N/A
2009/0014879	12/2008	Park et al.	N/A	N/A
2009/0017227	12/2008	Fu et al.	N/A	N/A
2009/0017631	12/2008	Bencher	N/A	N/A

2009/0017733	12/2008	Takahashi et al.	N/A	N/A
2009/0020072	12/2008	Mizunaga et al.	N/A	N/A
2009/0023229	12/2008	Matsushita	N/A	N/A
2009/0029503	12/2008	Arai	N/A	N/A
2009/0029528	12/2008	Sanchez et al.	N/A	N/A
2009/0029564	12/2008	Yamashita et al.	N/A	N/A
2009/0031954	12/2008	Nishikido et al.	N/A	N/A
2009/0033907	12/2008	Watson	N/A	N/A
2009/0035463	12/2008	Dip	N/A	N/A
2009/0035584	12/2008	Tran et al.	N/A	N/A
2009/0035927	12/2008	Olsen et al.	N/A	N/A
2009/0035946	12/2008	Pierreux et al.	N/A	N/A
2009/0035947	12/2008	Horii	N/A	N/A
2009/0036292	12/2008	Sun et al.	N/A	N/A
2009/0039475	12/2008	Shioya	N/A	N/A
2009/0041952	12/2008	Yoon et al.	N/A	N/A
2009/0041984	12/2008	Mayers et al.	N/A	N/A
2009/0042344	12/2008	Ye et al.	N/A	N/A
2009/0042408	12/2008	Maeda	N/A	N/A
2009/0045749	12/2008	Ganachev et al.	N/A	N/A
2009/0045829	12/2008	Awazu	N/A	N/A
2009/0047426	12/2008	Park et al.	N/A	N/A
2009/0047433	12/2008	Kim et al.	N/A	N/A
2009/0047447	12/2008	Sawin et al.	N/A	N/A
2009/0050621	12/2008	Awazu	N/A	N/A
2009/0052498	12/2008	Halpin et al.	N/A	N/A
2009/0053023	12/2008	Wakabayashi	N/A	N/A
2009/0053900	12/2008	Nozawa et al.	N/A	N/A
2009/0053906	12/2008	Miya et al.	N/A	N/A
2009/0056112	12/2008	Kobayashi	N/A	N/A
2009/0056629	12/2008	Katz et al.	N/A	N/A
2009/0057269	12/2008	Katz et al.	N/A	N/A
2009/0060480	12/2008	Herchen	N/A	N/A
2009/0061083	12/2008	Chiang et al.	N/A	N/A
2009/0061644	12/2008	Chiang et al.	N/A	N/A
2009/0061647	12/2008	Mallick et al.	N/A	N/A
2009/0075029	12/2008	Thomas	428/173	H01L 21/02447
2009/0075490	12/2008	Dussarrat	N/A	N/A
2009/0075491	12/2008	Liu et al.	N/A	N/A
2009/0080136	12/2008	Nagayama et al.	N/A	N/A
2009/0081879	12/2008	Sukekawa et al.	N/A	N/A
2009/0084317	12/2008	Wu	N/A	N/A
2009/0085156	12/2008	Dewey et al.	N/A	N/A
2009/0087585	12/2008	Lee et al.	N/A	N/A
2009/0087964	12/2008	Maeda et al.	N/A	N/A
2009/0087967	12/2008	Todd	N/A	N/A
2009/0090382	12/2008	Morisada	N/A	N/A
2009/0093080	12/2008	Choi et al.	N/A	N/A
2009/0093094	12/2008	Ye et al.	N/A	N/A

2009/0093100	12/2008	Xia et al.	N/A	N/A
2009/0095221	12/2008	Tam et al.	N/A	N/A
2009/0104351	12/2008	Kakegawa	N/A	N/A
2009/0104594	12/2008	Webb	N/A	N/A
2009/0104789	12/2008	Mallick et al.	N/A	N/A
2009/0107404	12/2008	Ogliari et al.	N/A	N/A
2009/0108308	12/2008	Yang et al.	N/A	N/A
2009/0111246	12/2008	Bauer	118/722	H01L 21/02535
2009/0112458	12/2008	Nakai	N/A	N/A
2009/0115064	12/2008	Sandhu et al.	N/A	N/A
2009/0116936	12/2008	Marubayashi et al.	N/A	N/A
2009/0117717	12/2008	Tomasini et al.	N/A	N/A
2009/0117723	12/2008	Kim et al.	N/A	N/A
2009/0117746	12/2008	Masuda	N/A	N/A
2009/0120580	12/2008	Kagoshima et al.	N/A	N/A
2009/0122293	12/2008	Shibazaki	N/A	N/A
2009/0122458	12/2008	Lischer et al.	N/A	N/A
2009/0124131	12/2008	Breunsbach et al.	N/A	N/A
2009/0130331	12/2008	Asai	N/A	N/A
2009/0130859	12/2008	Itatani et al.	N/A	N/A
2009/0136665	12/2008	Choi et al.	N/A	N/A
2009/0136668	12/2008	Gregg et al.	N/A	N/A
2009/0136683	12/2008	Fukasawa et al.	N/A	N/A
2009/0137055	12/2008	Bognar	N/A	N/A
2009/0139657	12/2008	Lee et al.	N/A	N/A
2009/0142905	12/2008	Yamazaki	N/A	N/A
2009/0142933	12/2008	Yajima et al.	N/A	N/A
2009/0142935	12/2008	Fukazawa et al.	N/A	N/A
2009/0146322	12/2008	Weling et al.	N/A	N/A
2009/0147819	12/2008	Goodman et al.	N/A	N/A
2009/0155488	12/2008	Nakano et al.	N/A	N/A
2009/0156015	12/2008	Park et al.	N/A	N/A
2009/0156017	12/2008	Fukazawa et al.	N/A	N/A
2009/0159000	12/2008	Aggarwal et al.	N/A	N/A
2009/0159002	12/2008	Bera et al.	N/A	N/A
2009/0159424	12/2008	Liu et al.	N/A	N/A
2009/0162647	12/2008	Sun et al.	N/A	N/A
2009/0162996	12/2008	Ramaswarthy et al.	N/A	N/A
2009/0163038	12/2008	Miyoshi	N/A	N/A
2009/0165715	12/2008	Oh	N/A	N/A
2009/0165721	12/2008	Pitney et al.	N/A	N/A
2009/0165722	12/2008	Ha	N/A	N/A
2009/0166616	12/2008	Uchiyama	N/A	N/A
2009/0176018	12/2008	Zou et al.	N/A	N/A
2009/0179365	12/2008	Lerner et al.	N/A	N/A
2009/0183520	12/2008	Yukimoto	N/A	N/A
2009/0186479	12/2008	Okabe et al.	N/A	N/A
2009/0186571	12/2008	Haro	N/A	N/A
2009/0194233	12/2008	Tamura et al.	N/A	N/A

2009/0197015	12/2008	Kudela et al.	N/A	N/A
2009/0197411	12/2008	Dussarrat et al.	N/A	N/A
2009/0200494	12/2008	Hatem	N/A	N/A
2009/0200547	12/2008	Griffin et al.	N/A	N/A
2009/0204403	12/2008	Hollander et al.	N/A	N/A
2009/0206056	12/2008	Xu	N/A	N/A
2009/0209081	12/2008	Matero	N/A	N/A
2009/0211523	12/2008	Kuppurao et al.	N/A	N/A
2009/0211525	12/2008	Sarigiannis et al.	N/A	N/A
2009/0214825	12/2008	Sun et al.	N/A	N/A
2009/0217871	12/2008	Kim et al.	N/A	N/A
2009/0221149	12/2008	Hammond et al.	N/A	N/A
2009/0221151	12/2008	Honda et al.	N/A	N/A
2009/0223441	12/2008	Arena et al.	N/A	N/A
2009/0227094	12/2008	Bateman	N/A	N/A
2009/0230211	12/2008	Kobayashi et al.	N/A	N/A
2009/0232985	12/2008	Dussarrat et al.	N/A	N/A
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2009/0236276	12/2008	Kurth et al.	N/A	N/A
2009/0236315	12/2008	Willwerth et al.	N/A	N/A
2009/0239386	12/2008	Suzaki et al.	N/A	N/A
2009/0242130	12/2008	Tian et al.	N/A	N/A
2009/0242957	12/2008	Ma et al.	N/A	N/A
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2009/0246399	12/2008	Goundar	N/A	N/A
2009/0246971	12/2008	Reid et al.	N/A	N/A
2009/0250004	12/2008	Yamada et al.	N/A	N/A
2009/0250955	12/2008	Aoki	N/A	N/A
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2009/0256127	12/2008	Feist et al.	N/A	N/A
2009/0261331	12/2008	Yang et al.	N/A	N/A
2009/0266296	12/2008	Tachibana et al.	N/A	N/A
2009/0267135	12/2008	Tanaka et al.	N/A	N/A
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2009/0269506	12/2008	Okura et al.	N/A	N/A
2009/0269507	12/2008	Yu et al.	N/A	N/A
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2009/0275205	12/2008	Kiehlbauch et al.	N/A	N/A
2009/0275210	12/2008	Shanker et al.	N/A	N/A
2009/0277510	12/2008	Shikata	N/A	N/A
2009/0277874	12/2008	Rui et al.	N/A	N/A
2009/0280248	12/2008	Goodman et al.	N/A	N/A
2009/0283041	12/2008	Tomiyasu et al.	N/A	N/A
2009/0283217	12/2008	Lubomirsky et al.	N/A	N/A
2009/0284156	12/2008	Banna et al.	N/A	N/A
2009/0286400	12/2008	Heo et al.	N/A	N/A
2009/0286402	12/2008	Xia et al.	N/A	N/A
2009/0286405	12/2008	Okesaku et al.	N/A	N/A
2009/0289300	12/2008	Sasaki et al.	N/A	N/A
2009/0291208	12/2008	Gordon et al.	N/A	N/A

2009/0291566	12/2008	Ueno et al.	N/A	N/A
2009/0297696	12/2008	Pore et al.	N/A	N/A
2009/0297710	12/2008	Lindfors	N/A	N/A
2009/0297731	12/2008	Goundar	N/A	N/A
2009/0298257	12/2008	Lee et al.	N/A	N/A
2009/0302002	12/2008	Collins et al.	N/A	N/A
2009/0302434	12/2008	Pallem et al.	N/A	N/A
2009/0304558	12/2008	Patton	N/A	N/A
2009/0308315	12/2008	de Ridder	N/A	N/A
2009/0308425	12/2008	Yednak	N/A	N/A
2009/0311857	12/2008	Todd et al.	N/A	N/A
2009/0314208	12/2008	Zhou et al.	N/A	N/A
2009/0314309	12/2008	Sankarakrishnan et al.	N/A	N/A
2009/0315093	12/2008	Li et al.	N/A	N/A
2009/0317214	12/2008	Hsiao et al.	N/A	N/A
2009/0320754	12/2008	Oya	N/A	N/A
2009/0320881	12/2008	Aitchison	N/A	N/A
2009/0324971	12/2008	De Vries et al.	N/A	N/A
2009/0324989	12/2008	Witz et al.	N/A	N/A
2009/0325391	12/2008	De Vusser et al.	N/A	N/A
2009/0325469	12/2008	Koo et al.	N/A	N/A
2010/0000608	12/2009	Goto et al.	N/A	N/A
2010/0001409	12/2009	Humbert et al.	N/A	N/A
2010/0003406	12/2009	Lam et al.	N/A	N/A
2010/0006031	12/2009	Choi et al.	N/A	N/A
2010/0006923	12/2009	Fujitsuka	N/A	N/A
2010/0008656	12/2009	Sorabji et al.	N/A	N/A
2010/0012036	12/2009	Silva et al.	N/A	N/A
2010/0012153	12/2009	Shigemoto et al.	N/A	N/A
2010/0014479	12/2009	Kim	N/A	N/A
2010/0015813	12/2009	McGinnis et al.	N/A	N/A
2010/0018460	12/2009	Singh et al.	N/A	N/A
2010/0018913	12/2009	Blum	N/A	N/A
2010/0022097	12/2009	Yamoto et al.	N/A	N/A
2010/0024727	12/2009	Kim et al.	N/A	N/A
2010/0024872	12/2009	Kishimoto	N/A	N/A
2010/0025766	12/2009	Nuttinck et al.	N/A	N/A
2010/0025796	12/2009	Dabiran	N/A	N/A
2010/0031884	12/2009	Aggarwal et al.	N/A	N/A
2010/0032587	12/2009	Hosch et al.	N/A	N/A
2010/0032842	12/2009	Herdt et al.	N/A	N/A
2010/0034719	12/2009	Dussarrat et al.	N/A	N/A
2010/0038687	12/2009	Klaus et al.	N/A	N/A
2010/0040441	12/2009	Obikane	N/A	N/A
2010/0041179	12/2009	Lee	N/A	N/A
2010/0041243	12/2009	Cheng et al.	N/A	N/A
2010/0046321	12/2009	McLaughlin et al.	N/A	N/A
2010/0050943	12/2009	Kato et al.	N/A	N/A
2010/0051584	12/2009	Okita et al.	N/A	N/A
2010/0051597	12/2009	Morita et al.	N/A	N/A

2010/0055312	12/2009	Kato et al.	N/A	N/A
2010/0055316	12/2009	Honma	N/A	N/A
2010/0055318	12/2009	Volf et al.	N/A	N/A
2010/0055442	12/2009	Kellock	N/A	N/A
2010/0055898	12/2009	Chang et al.	N/A	N/A
2010/0058984	12/2009	Marubayashi	N/A	N/A
2010/0065758	12/2009	Liu et al.	N/A	N/A
2010/0068009	12/2009	Kimura	N/A	N/A
2010/0068383	12/2009	Kato et al.	N/A	N/A
2010/0068414	12/2009	Takahashi et al.	N/A	N/A
2010/0068891	12/2009	Hatanaka et al.	N/A	N/A
2010/0075037	12/2009	Marsh et al.	N/A	N/A
2010/0075488	12/2009	Collins et al.	N/A	N/A
2010/0075507	12/2009	Chang et al.	N/A	N/A
2010/0078601	12/2009	Pallem et al.	N/A	N/A
2010/0078746	12/2009	Jung	N/A	N/A
2010/0081094	12/2009	Hasebe et al.	N/A	N/A
2010/0081293	12/2009	Mallick et al.	N/A	N/A
2010/0086703	12/2009	Mangum et al.	N/A	N/A
2010/0087069	12/2009	Miya et al.	N/A	N/A
2010/0089320	12/2009	Kim	N/A	N/A
2010/0089870	12/2009	Hiroshima et al.	N/A	N/A
2010/0090149	12/2009	Thompson et al.	N/A	N/A
2010/0092679	12/2009	Lee et al.	N/A	N/A
2010/0092696	12/2009	Shinriki	N/A	N/A
2010/0093187	12/2009	Lee et al.	N/A	N/A
2010/0094430	12/2009	Krumdieck	N/A	N/A
2010/0098862	12/2009	Xu et al.	N/A	N/A
2010/0101728	12/2009	Iwasaki	N/A	N/A
2010/0102417	12/2009	Ganguli et al.	N/A	N/A
2010/0105936	12/2009	Tada et al.	N/A	N/A
2010/0111648	12/2009	Tamura et al.	N/A	N/A
2010/0112496	12/2009	Nakajima et al.	N/A	N/A
2010/0116207	12/2009	Givens	N/A	N/A
2010/0116208	12/2009	Sangam	N/A	N/A
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2010/0119727	12/2009	Takagi	N/A	N/A
2010/0119844	12/2009	Sun et al.	N/A	N/A
2010/0120261	12/2009	Kim et al.	N/A	N/A
2010/0121100	12/2009	Shay	N/A	N/A
2010/0124610	12/2009	Aikawa et al.	N/A	N/A
2010/0124618	12/2009	Kobayashi et al.	N/A	N/A
2010/0124621	12/2009	Kobayashi et al.	N/A	N/A
2010/0126415	12/2009	Ishino et al.	N/A	N/A
2010/0126539	12/2009	Lee et al.	N/A	N/A
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2010/0126666	12/2009	Tandou et al.	N/A	N/A
2010/0129548	12/2009	Sneh	N/A	N/A
2010/0129670	12/2009	Sun et al.	N/A	N/A

2010/0129990	12/2009	Nishizawa et al.	N/A	N/A
2010/0130015	12/2009	Nakajima et al.	N/A	N/A
2010/0130017	12/2009	Luo et al.	N/A	N/A
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2010/0133255	12/2009	Bahng et al.	N/A	N/A
2010/0134023	12/2009	Mills	N/A	N/A
2010/0136216	12/2009	Tsuei et al.	N/A	N/A
2010/0140221	12/2009	Kikuchi et al.	N/A	N/A
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2010/0143609	12/2009	Fukazawa et al.	N/A	N/A
2010/0144150	12/2009	Sills et al.	N/A	N/A
2010/0144162	12/2009	Lee et al.	N/A	N/A
2010/0144968	12/2009	Lee et al.	N/A	N/A
2010/0145547	12/2009	Darabnia et al.	N/A	N/A
2010/0147396	12/2009	Yamagishi et al.	N/A	N/A
2010/0151206	12/2009	Wu et al.	N/A	N/A
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2010/0159707	12/2009	Huang et al.	N/A	N/A
2010/0162752	12/2009	Tabata et al.	N/A	N/A
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2010/0163187	12/2009	Yokogawa et al.	N/A	N/A
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2010/0166630	12/2009	Gu et al.	N/A	N/A
2010/0168404	12/2009	Girolami et al.	N/A	N/A
2010/0170441	12/2009	Won et al.	N/A	N/A
2010/0170868	12/2009	Lin et al.	N/A	N/A
2010/0173432	12/2009	White et al.	N/A	N/A
2010/0178137	12/2009	Chintalapati et al.	N/A	N/A
2010/0178423	12/2009	Shimizu et al.	N/A	N/A
2010/0180819	12/2009	Hatanaka et al.	N/A	N/A
2010/0183825	12/2009	Becker et al.	N/A	N/A
2010/0184302	12/2009	Lee et al.	N/A	N/A
2010/0186669	12/2009	Shin et al.	N/A	N/A
2010/0189923	12/2009	Goundar et al.	N/A	N/A
2010/0193501	12/2009	Zucker et al.	N/A	N/A
2010/0193955	12/2009	Milligan et al.	N/A	N/A
2010/0195392	12/2009	Freeman	N/A	N/A
2010/0195690	12/2009	Moench et al.	N/A	N/A
2010/0197107	12/2009	Matsuzaki	N/A	N/A
2010/0202860	12/2009	Reed	N/A	N/A
2010/0203242	12/2009	Borden	N/A	N/A
2010/0206767	12/2009	Odashima et al.	N/A	N/A
2010/0209598	12/2009	Xu et al.	N/A	N/A
2010/0210108	12/2009	Ishizaka et al.	N/A	N/A
2010/0219757	12/2009	Benzerrouk et al.	N/A	N/A
2010/0221452	12/2009	Kang	N/A	N/A
2010/0224130	12/2009	Smith et al.	N/A	N/A
2010/0227458	12/2009	Chung et al.	N/A	N/A
2010/0229795	12/2009	Tanabe	N/A	N/A

2010/0229965	12/2009	Kashima et al.	N/A	N/A
2010/0230051	12/2009	Iizuka	N/A	N/A
2010/0230863	12/2009	Moench et al.	N/A	N/A
2010/0233885	12/2009	Kushibiki et al.	N/A	N/A
2010/0233886	12/2009	Yang et al.	N/A	N/A
2010/0236691	12/2009	Yamazaki	N/A	N/A
2010/0243166	12/2009	Hayashi et al.	N/A	N/A
2010/0244688	12/2009	Braun et al.	N/A	N/A
2010/0246630	12/2009	Kaszynski et al.	N/A	N/A
2010/0248465	12/2009	Yi et al.	N/A	N/A
2010/0252434	12/2009	Roy	N/A	N/A
2010/0255196	12/2009	Geisler et al.	N/A	N/A
2010/0255198	12/2009	Cleary et al.	N/A	N/A
2010/0255218	12/2009	Oka et al.	N/A	N/A
2010/0255625	12/2009	De Vries	N/A	N/A
2010/0255658	12/2009	Aggarwal	N/A	N/A
2010/0258809	12/2009	Muller	N/A	N/A
2010/0259152	12/2009	Yasuda et al.	N/A	N/A
2010/0266765	12/2009	White et al.	N/A	N/A
2010/0267224	12/2009	Choi et al.	N/A	N/A
2010/0267248	12/2009	Ma et al.	N/A	N/A
2010/0270675	12/2009	Harada	N/A	N/A
2010/0275846	12/2009	Kitagawa	N/A	N/A
2010/0279008	12/2009	Takagi	N/A	N/A
2010/0279512	12/2009	Udea et al.	N/A	N/A
2010/0282163	12/2009	Aggarwal et al.	N/A	N/A
2010/0282170	12/2009	Nishizawa	N/A	N/A
2010/0282645	12/2009	Wang	N/A	N/A
2010/0285237	12/2009	Ditizio et al.	N/A	N/A
2010/0285319	12/2009	Kwak et al.	N/A	N/A
2010/0294199	12/2009	Tran et al.	N/A	N/A
2010/0297391	12/2009	Kley	N/A	N/A
2010/0301752	12/2009	Bakre et al.	N/A	N/A
2010/0304047	12/2009	Yang et al.	N/A	N/A
2010/0307415	12/2009	Shero et al.	N/A	N/A
2010/0317177	12/2009	Huang et al.	N/A	N/A
2010/0317198	12/2009	Antonelli	N/A	N/A
2010/0322604	12/2009	Fondurulia et al.	N/A	N/A
2010/0322822	12/2009	Fritchie et al.	N/A	N/A
2010/0326358	12/2009	Choi	N/A	N/A
2011/0000619	12/2010	Suh	N/A	N/A
2011/0003087	12/2010	Soininen et al.	N/A	N/A
2011/0003450	12/2010	Lee et al.	N/A	N/A
2011/0005684	12/2010	Hayami et al.	N/A	N/A
2011/0006402	12/2010	Zhou	N/A	N/A
2011/0006406	12/2010	Urbanowicz et al.	N/A	N/A
2011/0008950	12/2010	Xu	N/A	N/A
2011/0014359	12/2010	Hashim	N/A	N/A
2011/0014795	12/2010	Lee	N/A	N/A
2011/0017139	12/2010	Chiang et al.	N/A	N/A

2011/0021033	12/2010	Ikeuchi et al.	N/A	N/A
2011/0027725	12/2010	Tsutsumi et al.	N/A	N/A
2011/0027999	12/2010	Sparks et al.	N/A	N/A
2011/0031562	12/2010	Lin et al.	N/A	N/A
2011/0034039	12/2010	Liang et al.	N/A	N/A
2011/0039419	12/2010	Date et al.	N/A	N/A
2011/0041764	12/2010	Webb et al.	N/A	N/A
2011/0042200	12/2010	Wilby	N/A	N/A
2011/0045610	12/2010	van Schravendijk et al.	N/A	N/A
2011/0045676	12/2010	Park et al.	N/A	N/A
2011/0046314	12/2010	Klipp et al.	N/A	N/A
2011/0048642	12/2010	Mihara et al.	N/A	N/A
2011/0048769	12/2010	Fujiwara	N/A	N/A
2011/0049100	12/2010	Han et al.	N/A	N/A
2011/0052833	12/2010	Hanawa et al.	N/A	N/A
2011/0053383	12/2010	Shero et al.	N/A	N/A
2011/0056513	12/2010	Hombach et al.	N/A	N/A
2011/0056626	12/2010	Brown et al.	N/A	N/A
2011/0057248	12/2010	Ma et al.	N/A	N/A
2011/0061810	12/2010	Ganguly et al.	N/A	N/A
2011/0065289	12/2010	Asai	N/A	N/A
2011/0067522	12/2010	Lai	N/A	N/A
2011/0070380	12/2010	Shero et al.	N/A	N/A
2011/0070740	12/2010	Bettencourt et al.	N/A	N/A
2011/0076401	12/2010	Chao et al.	N/A	N/A
2011/0081519	12/2010	Dillingh	N/A	N/A
2011/0083496	12/2010	Lin et al.	N/A	N/A
2011/0086516	12/2010	Lee et al.	N/A	N/A
2011/0089166	12/2010	Hunter et al.	N/A	N/A
2011/0089419	12/2010	Yamazaki et al.	N/A	N/A
2011/0089469	12/2010	Merckling	N/A	N/A
2011/0091650	12/2010	Noguchi et al.	N/A	N/A
2011/0092077	12/2010	Xu et al.	N/A	N/A
2011/0097901	12/2010	Banna et al.	N/A	N/A
2011/0098841	12/2010	Tsuda	N/A	N/A
2011/0100489	12/2010	Orito et al.	N/A	N/A
2011/0104395	12/2010	Kumagai et al.	N/A	N/A
2011/0107512	12/2010	Gilbert	N/A	N/A
2011/0108194	12/2010	Yoshioka et al.	N/A	N/A
2011/0108424	12/2010	Puget et al.	N/A	N/A
2011/0108741	12/2010	Ingram	N/A	N/A
2011/0108929	12/2010	Meng	N/A	N/A
2011/0114261	12/2010	Matsumoto et al.	N/A	N/A
2011/0114601	12/2010	Lubomirsky et al.	N/A	N/A
2011/0115378	12/2010	Lubomirsky et al.	N/A	N/A
2011/0117490	12/2010	Bae et al.	N/A	N/A
2011/0117492	12/2010	Yamada et al.	N/A	N/A
2011/0117728	12/2010	Su et al.	N/A	N/A
2011/0117732	12/2010	Bauer et al.	N/A	N/A
2011/0117737	12/2010	Agarwala et al.	N/A	N/A

2011/0117749	12/2010	Sheu	N/A	N/A
2011/0121503	12/2010	Burrows et al.	N/A	N/A
2011/0121736	12/2010	Hirayama et al.	N/A	N/A
2011/0124196	12/2010	Lee	N/A	N/A
2011/0127702	12/2010	Gautam et al.	N/A	N/A
2011/0132542	12/2010	Ilzuka	N/A	N/A
2011/0135842	12/2010	Faguet et al.	N/A	N/A
2011/0139272	12/2010	Matsumoto et al.	N/A	N/A
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2011/0155264	12/2010	Minami et al.	N/A	N/A
2011/0155322	12/2010	Himori et al.	N/A	N/A
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2011/0159202	12/2010	Matsushita	N/A	N/A
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2011/0171380	12/2010	Higashi et al.	N/A	N/A
2011/0171775	12/2010	Yamamoto et al.	N/A	N/A
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2011/0177648	12/2010	Tanner et al.	N/A	N/A
2011/0180233	12/2010	Bera et al.	N/A	N/A
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2011/0186984	12/2010	Saito et al.	N/A	N/A
2011/0192820	12/2010	Yeom et al.	N/A	N/A
2011/0195574	12/2010	Blasco et al.	N/A	N/A
2011/0198034	12/2010	Sun et al.	N/A	N/A
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2011/0207332	12/2010	Liu et al.	N/A	N/A
2011/0210468	12/2010	Shannon et al.	N/A	N/A
2011/0212625	12/2010	Toyoda et al.	N/A	N/A
2011/0217838	12/2010	Hsieh et al.	N/A	N/A
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2011/0223334	12/2010	Yudovsky et al.	N/A	N/A
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2011/0232678	12/2010	Shih et al.	N/A	N/A
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2011/0236594	12/2010	Haverkamp et al.	N/A	N/A
2011/0236600	12/2010	Fox et al.	N/A	N/A
2011/0237040	12/2010	Ng et al.	N/A	N/A
2011/0237082	12/2010	Nakajima et al.	N/A	N/A
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2011/0239940	12/2010	Benvenuti et al.	N/A	N/A
2011/0244673	12/2010	Cho et al.	N/A	N/A
2011/0253044	12/2010	Tam et al.	N/A	N/A
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2011/0256692	12/2010	Tam et al.	N/A	N/A
2011/0256724	12/2010	Chandrasekharan et al.	N/A	N/A
2011/0256726	12/2010	Lavoie et al.	N/A	N/A
2011/0256727	12/2010	Beynet et al.	N/A	N/A
2011/0256734	12/2010	Hausmann et al.	N/A	N/A
2011/0259519	12/2010	Kenworthy et al.	N/A	N/A
2011/0262642	12/2010	Xiao et al.	N/A	N/A
2011/0263107	12/2010	Chung et al.	N/A	N/A
2011/0263115	12/2010	Ganguli et al.	N/A	N/A
2011/0264250	12/2010	Nishimura et al.	N/A	N/A
2011/0265549	12/2010	Cruse et al.	N/A	N/A
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2011/0265951	12/2010	Xu et al.	N/A	N/A
2011/0266611	12/2010	Kim et al.	N/A	N/A
2011/0269314	12/2010	Lee et al.	N/A	N/A
2011/0275018	12/2010	Matteo et al.	N/A	N/A
2011/0275166	12/2010	Shero et al.	N/A	N/A
2011/0277690	12/2010	Rozenzon et al.	N/A	N/A
2011/0281417	12/2010	Gordon et al.	N/A	N/A
2011/0283933	12/2010	Makarov et al.	N/A	N/A
2011/0286819	12/2010	Shibata et al.	N/A	N/A
2011/0291243	12/2010	Seamons	N/A	N/A
2011/0294075	12/2010	Chen et al.	N/A	N/A
2011/0294288	12/2010	Lee et al.	N/A	N/A
2011/0297088	12/2010	Song et al.	N/A	N/A
2011/0298062	12/2010	Ganguli et al.	N/A	N/A
2011/0300720	12/2010	Fu	N/A	N/A
2011/0305102	12/2010	Berger et al.	N/A	N/A
2011/0305856	12/2010	Bonn	N/A	N/A
2011/0308453	12/2010	Su et al.	N/A	N/A
2011/0308460	12/2010	Hong et al.	N/A	N/A
2011/0312191	12/2010	Ohkura et al.	N/A	N/A
2011/0318142	12/2010	Gage et al.	N/A	N/A
2011/0318888	12/2010	Komatsu et al.	N/A	N/A
2011/0318935	12/2010	Oyabu et al.	N/A	N/A
2012/0001172	12/2011	Shang et al.	N/A	N/A
2012/0003500	12/2011	Yoshida et al.	N/A	N/A
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2012/0012556	12/2011	Matsumoto et al.	N/A	N/A
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2012/0024223	12/2011	Torres et al.	N/A	N/A
2012/0024227	12/2011	Takasuka et al.	N/A	N/A
2012/0024231	12/2011	Horino	N/A	N/A
2012/0024478	12/2011	Huang et al.	N/A	N/A
2012/0024479	12/2011	Palagashvili et al.	N/A	N/A
2012/0027547	12/2011	Jager et al.	N/A	N/A
2012/0028454	12/2011	Swaminathan et al.	N/A	N/A
2012/0028469	12/2011	Onizawa et al.	N/A	N/A
2012/0031333	12/2011	Kurita et al.	N/A	N/A
2012/0031500	12/2011	Hirose et al.	N/A	N/A
2012/0032311	12/2011	Gates	N/A	N/A
2012/0033695	12/2011	Hayashi et al.	N/A	N/A
2012/0034793	12/2011	Narushima et al.	N/A	N/A
2012/0036732	12/2011	Varadarajan	N/A	N/A
2012/0040097	12/2011	Volf et al.	N/A	N/A
2012/0040528	12/2011	Kim et al.	N/A	N/A
2012/0043556	12/2011	Dube et al.	N/A	N/A
2012/0043617	12/2011	Nakagawa et al.	N/A	N/A
2012/0046421	12/2011	Darling et al.	N/A	N/A
2012/0052681	12/2011	Marsh	N/A	N/A
2012/0055401	12/2011	Tozawa	N/A	N/A
2012/0058270	12/2011	Winter et al.	N/A	N/A
2012/0058630	12/2011	Quinn	N/A	N/A
2012/0064690	12/2011	Hirota et al.	N/A	N/A
2012/0064726	12/2011	Nozawa et al.	N/A	N/A
2012/0064764	12/2011	Islam	N/A	N/A
2012/0068242	12/2011	Shin et al.	N/A	N/A
2012/0070136	12/2011	Koelmel et al.	N/A	N/A
2012/0070997	12/2011	Larson	N/A	N/A
2012/0073400	12/2011	Wang	N/A	N/A
2012/0074533	12/2011	Aoyama	N/A	N/A
2012/0077349	12/2011	Li et al.	N/A	N/A
2012/0077350	12/2011	Miya et al.	N/A	N/A
2012/0080756	12/2011	Suzuki	N/A	N/A
2012/0083134	12/2011	Wu et al.	N/A	N/A
2012/0088031	12/2011	Neel	N/A	N/A
2012/0088369	12/2011	Weidman et al.	N/A	N/A
2012/0090704	12/2011	Laverdiere et al.	N/A	N/A
2012/0091522	12/2011	Ozaki et al.	N/A	N/A
2012/0094010	12/2011	Sugiura et al.	N/A	N/A
2012/0094468	12/2011	Bhatia et al.	N/A	N/A
2012/0098107	12/2011	Raisanen et al.	N/A	N/A
2012/0100464	12/2011	Kageyama	N/A	N/A
2012/0103264	12/2011	Choi et al.	N/A	N/A
2012/0103522	12/2011	Hohenwarter	N/A	N/A
2012/0103939	12/2011	Wu et al.	N/A	N/A
2012/0104514	12/2011	Park et al.	N/A	N/A
2012/0107607	12/2011	Takaki et al.	N/A	N/A
2012/0108039	12/2011	Zajaji	N/A	N/A
2012/0108048	12/2011	Lim et al.	N/A	N/A

2012/0111271	12/2011	Begarney et al.	N/A	N/A
2012/0114877	12/2011	Lee	N/A	N/A
2012/0115250	12/2011	Ariga et al.	N/A	N/A
2012/0115257	12/2011	Matsuyam et al.	N/A	N/A
2012/0119337	12/2011	Sasaki et al.	N/A	N/A
2012/0121823	12/2011	Chhabra	N/A	N/A
2012/0122275	12/2011	Koo et al.	N/A	N/A
2012/0122302	12/2011	Weisman et al.	N/A	N/A
2012/0122319	12/2011	Shimizu	N/A	N/A
2012/0125258	12/2011	Lee	N/A	N/A
2012/0126300	12/2011	Park et al.	N/A	N/A
2012/0128897	12/2011	Xiao et al.	N/A	N/A
2012/0135145	12/2011	Je et al.	N/A	N/A
2012/0139009	12/2011	Ning et al.	N/A	N/A
2012/0145078	12/2011	Huang et al.	N/A	N/A
2012/0146113	12/2011	Suzuki et al.	N/A	N/A
2012/0149207	12/2011	Graff	N/A	N/A
2012/0149213	12/2011	Nittala	N/A	N/A
2012/0156108	12/2011	Fondurulia et al.	N/A	N/A
2012/0156890	12/2011	Yim et al.	N/A	N/A
2012/0160172	12/2011	Wamura et al.	N/A	N/A
2012/0160809	12/2011	Ishibashi et al.	N/A	N/A
2012/0161126	12/2011	Yamazaki	N/A	N/A
2012/0161405	12/2011	Mohn	N/A	N/A
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2012/0164837	12/2011	Tan et al.	N/A	N/A
2012/0164842	12/2011	Watanabe	N/A	N/A
2012/0164846	12/2011	Ha et al.	N/A	N/A
2012/0170170	12/2011	Gros-Jean	N/A	N/A
2012/0171391	12/2011	Won	N/A	N/A
2012/0171874	12/2011	Thridandam et al.	N/A	N/A
2012/0175518	12/2011	Godet et al.	N/A	N/A
2012/0175751	12/2011	Gatineau et al.	N/A	N/A
2012/0177845	12/2011	Odedra et al.	N/A	N/A
2012/0180719	12/2011	Inoue et al.	N/A	N/A
2012/0180954	12/2011	Yang et al.	N/A	N/A
2012/0183689	12/2011	Suzuki et al.	N/A	N/A
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2012/0214318	12/2011	Fukazawa et al.	N/A	N/A
2012/0216743	12/2011	Itoh et al.	N/A	N/A
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2012/0241089	12/2011	Dielmann et al.	N/A	N/A
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2012/0247386	12/2011	Sanchez et al.	N/A	N/A
2012/0252229	12/2011	Timans et al.	N/A	N/A
2012/0258257	12/2011	Nguyen et al.	N/A	N/A
2012/0263875	12/2011	Brenninger et al.	N/A	N/A
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2012/0269967	12/2011	Yudovsky et al.	N/A	N/A
2012/0270339	12/2011	Xie et al.	N/A	N/A
2012/0270384	12/2011	Sanchez et al.	N/A	N/A
2012/0270393	12/2011	Pore et al.	N/A	N/A
2012/0270407	12/2011	Werner et al.	N/A	N/A
2012/0273052	12/2011	Ye et al.	N/A	N/A
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2012/0280275	12/2011	Hata	257/E29.081	H01L 21/0262
2012/0288625	12/2011	Furuya et al.	N/A	N/A
2012/0289053	12/2011	Holland et al.	N/A	N/A
2012/0289056	12/2011	Bergman et al.	N/A	N/A
2012/0289057	12/2011	DeDontney	N/A	N/A
2012/0295427	12/2011	Bauer	438/494	H01L 29/66636
2012/0295449	12/2011	Fukazawa	N/A	N/A
2012/0302055	12/2011	Pore et al.	N/A	N/A
2012/0303313	12/2011	Moroi et al.	N/A	N/A
2012/0304935	12/2011	Oosterlaken et al.	N/A	N/A

2012/0305026	12/2011	Nomura et al.	N/A	N/A
2012/0305196	12/2011	Mori et al.	N/A	N/A
2012/0305987	12/2011	Hirler et al.	N/A	N/A
2012/0307588	12/2011	Hanada et al.	N/A	N/A
2012/0309181	12/2011	Machkaoutsan et al.	N/A	N/A
2012/0310440	12/2011	Darabnia et al.	N/A	N/A
2012/0315113	12/2011	Hiroki	N/A	N/A
2012/0315404	12/2011	Li et al.	N/A	N/A
2012/0318334	12/2011	Bedell et al.	N/A	N/A
2012/0318457	12/2011	Nguyen et al.	N/A	N/A
2012/0318773	12/2011	Wu et al.	N/A	N/A
2012/0319279	12/2011	Isobayashi	N/A	N/A
2012/0320491	12/2011	Doh et al.	N/A	N/A
2012/0321786	12/2011	Satitpunwaycha et al.	N/A	N/A
2012/0322252	12/2011	Son et al.	N/A	N/A
2012/0325148	12/2011	Yamagishi et al.	N/A	N/A
2012/0328780	12/2011	Yamagishi et al.	N/A	N/A
2012/0329208	12/2011	Pore et al.	N/A	N/A
2013/0002121	12/2012	Ma	N/A	N/A
2013/0005122	12/2012	Schwarzenbach et al.	N/A	N/A
2013/0005147	12/2012	Angyal et al.	N/A	N/A
2013/0008607	12/2012	Matsumoto et al.	N/A	N/A
2013/0011630	12/2012	Sullivan et al.	N/A	N/A
2013/0011983	12/2012	Tsai	N/A	N/A
2013/0011984	12/2012	Wang et al.	N/A	N/A
2013/0012003	12/2012	Haukka et al.	N/A	N/A
2013/0012034	12/2012	Ahn et al.	N/A	N/A
2013/0014697	12/2012	Kanayama	N/A	N/A
2013/0014896	12/2012	Shoji et al.	N/A	N/A
2013/0017503	12/2012	De Ridder et al.	N/A	N/A
2013/0019944	12/2012	Hekmatshoar-Tabai et al.	N/A	N/A
2013/0019945	12/2012	Hekmatshoar-Tabai et al.	N/A	N/A
2013/0019960	12/2012	Choi et al.	N/A	N/A
2013/0020246	12/2012	Hoots et al.	N/A	N/A
2013/0023120	12/2012	Yaehashi et al.	N/A	N/A
2013/0023124	12/2012	Nemani et al.	N/A	N/A
2013/0023129	12/2012	Reed	N/A	N/A
2013/0025538	12/2012	Collins et al.	N/A	N/A
2013/0025786	12/2012	Davidkovich et al.	N/A	N/A
2013/0026451	12/2012	Bangsaruntip et al.	N/A	N/A
2013/0032085	12/2012	Hanawa et al.	N/A	N/A
2013/0037532	12/2012	Volfovski et al.	N/A	N/A
2013/0037858	12/2012	Hong et al.	N/A	N/A
2013/0037886	12/2012	Tsai et al.	N/A	N/A
2013/0040447	12/2012	Swaminathan et al.	N/A	N/A
2013/0040481	12/2012	Vallely et al.	N/A	N/A
2013/0042811	12/2012	Shanker et al.	N/A	N/A
2013/0048606	12/2012	Mao et al.	N/A	N/A

2013/0052585	12/2012	Ayothi et al.	N/A	N/A
2013/0052836	12/2012	Hirose et al.	N/A	N/A
2013/0059078	12/2012	Gatineau et al.	N/A	N/A
2013/0059415	12/2012	Kato et al.	N/A	N/A
2013/0061755	12/2012	Frederick	N/A	N/A
2013/0062753	12/2012	Nguyen et al.	N/A	N/A
2013/0062839	12/2012	Tschinderle et al.	N/A	N/A
2013/0064973	12/2012	Chen et al.	N/A	N/A
2013/0065189	12/2012	Yoshii et al.	N/A	N/A
2013/0068391	12/2012	Mazzocco et al.	N/A	N/A
2013/0068727	12/2012	Okita	N/A	N/A
2013/0068970	12/2012	Matsushita	N/A	N/A
2013/0069052	12/2012	Sandhu	N/A	N/A
2013/0070456	12/2012	Jang et al.	N/A	N/A
2013/0075746	12/2012	Mallikarjunaswamy et al.	N/A	N/A
2013/0075788	12/2012	Tomabechi	N/A	N/A
2013/0078376	12/2012	Higashino et al.	N/A	N/A
2013/0078392	12/2012	Xiao et al.	N/A	N/A
2013/0081702	12/2012	Mohammed et al.	N/A	N/A
2013/0082274	12/2012	Yang	N/A	N/A
2013/0084156	12/2012	Shimamoto	N/A	N/A
2013/0084408	12/2012	Nakao et al.	N/A	N/A
2013/0084711	12/2012	Liang et al.	N/A	N/A
2013/0084714	12/2012	Oka et al.	N/A	N/A
2013/0085618	12/2012	Ding	N/A	N/A
2013/0089667	12/2012	Lai et al.	N/A	N/A
2013/0089716	12/2012	Krishnamurthy et al.	N/A	N/A
2013/0089988	12/2012	Wang et al.	N/A	N/A
2013/0092085	12/2012	Lee	N/A	N/A
2013/0093048	12/2012	Chang et al.	N/A	N/A
2013/0093321	12/2012	Yoshikawa et al.	N/A	N/A
2013/0095664	12/2012	Matero et al.	N/A	N/A
2013/0095973	12/2012	Kroneberger et al.	N/A	N/A
2013/0099318	12/2012	Adam et al.	N/A	N/A
2013/0104988	12/2012	Yednak et al.	N/A	N/A
2013/0104992	12/2012	Yednak et al.	N/A	N/A
2013/0107415	12/2012	Banna et al.	N/A	N/A
2013/0109172	12/2012	Collins et al.	N/A	N/A
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2013/0119018	12/2012	Kanarik et al.	N/A	N/A
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2013/0134148	12/2012	Tachikawa	N/A	N/A
2013/0137279	12/2012	Yamamoto et al.	N/A	N/A
2013/0143401	12/2012	Yu et al.	N/A	N/A
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2013/0145587	12/2012	Adhiprakash	N/A	N/A
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2013/0147050	12/2012	Bonner, III et al.	N/A	N/A
2013/0149874	12/2012	Hirose et al.	N/A	N/A
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2013/0157409	12/2012	Vaidya	N/A	N/A
2013/0157521	12/2012	Aldrich et al.	N/A	N/A
2013/0160709	12/2012	White et al.	N/A	N/A
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2013/0164458	12/2012	Soininen et al.	N/A	N/A
2013/0168353	12/2012	Okita et al.	N/A	N/A
2013/0168354	12/2012	Kanarik	N/A	N/A
2013/0171818	12/2012	Kim et al.	N/A	N/A
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2013/0196507	12/2012	Ma et al.	N/A	N/A
2013/0200518	12/2012	Ahmed et al.	N/A	N/A
2013/0202387	12/2012	Hiroki	N/A	N/A
2013/0203258	12/2012	Chen et al.	N/A	N/A
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2013/0220550	12/2012	Koo et al.	N/A	N/A
2013/0224964	12/2012	Fukazawa	N/A	N/A
2013/0228225	12/2012	Leeser	N/A	N/A
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2013/0280891	12/2012	Kim et al.	N/A	N/A
2013/0285155	12/2012	Glass	N/A	N/A
2013/0287526	12/2012	Bluck et al.	N/A	N/A
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2013/0302520	12/2012	Wang et al.	N/A	N/A
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2014/0015186	12/2013	Wessel et al.	N/A	N/A
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2015/0024567	12/2014	Tsai et al.	N/A	N/A
2015/0024609	12/2014	Milligan et al.	N/A	N/A
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2015/0072509	12/2014	Chi et al.	N/A	N/A
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2015/0086316	12/2014	Greenberg	N/A	N/A
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2015/0087139	12/2014	O'Neill et al.	N/A	N/A
2015/0087154	12/2014	Guha et al.	N/A	N/A
2015/0091057	12/2014	Xie et al.	N/A	N/A
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2015/0110147	12/2014	Mizuno	N/A	N/A
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2015/0144060	12/2014	Park	N/A	N/A
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2015/0147483	12/2014	Fukazawa	N/A	N/A
2015/0147488	12/2014	Choi et al.	N/A	N/A
2015/0147875	12/2014	Takamure et al.	N/A	N/A
2015/0147877	12/2014	Jung	N/A	N/A
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2015/0155177	12/2014	Zhang et al.	N/A	N/A
2015/0155370	12/2014	Tsai et al.	N/A	N/A
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2015/0167161	12/2014	Canizares et al.	N/A	N/A
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2015/0167165	12/2014	Lindfors	N/A	N/A
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2015/0170945	12/2014	Segawa et al.	N/A	N/A
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2016/0085003	12/2015	Jaiswal	N/A	N/A
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2017/0092535	12/2016	Kimihiko et al.	N/A	N/A
2017/0092847	12/2016	Kim et al.	N/A	N/A
2017/0100742	12/2016	Pore et al.	N/A	N/A
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2017/0103907	12/2016	Chu et al.	N/A	N/A
2017/0104061	12/2016	Peng et al.	N/A	N/A
2017/0107621	12/2016	Suemori	N/A	N/A
2017/0110313	12/2016	Tang et al.	N/A	N/A
2017/0110601	12/2016	Blomberg et al.	N/A	N/A
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2017/0114464	12/2016	Iriuda et al.	N/A	N/A
2017/0114465	12/2016	Kalutarage et al.	N/A	N/A
2017/0117141	12/2016	Zhu et al.	N/A	N/A
2017/0117202	12/2016	Tang et al.	N/A	N/A
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2017/0117222	12/2016	Kim et al.	N/A	N/A
2017/0121845	12/2016	Grutzmacher et al.	N/A	N/A
2017/0130332	12/2016	Stumpf	N/A	N/A
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2017/0148918	12/2016	Ye et al.	N/A	N/A
2017/0152968	12/2016	Raj et al.	N/A	N/A
2017/0154757	12/2016	Winkler et al.	N/A	N/A
2017/0154770	12/2016	Margetis et al.	N/A	N/A
2017/0154895	12/2016	Huo	N/A	N/A
2017/0159177	12/2016	Monsma et al.	N/A	N/A
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2017/0173696	12/2016	Sheinman	N/A	N/A
2017/0175290	12/2016	Chen et al.	N/A	N/A
2017/0178899	12/2016	Kabansky et al.	N/A	N/A
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2017/0186621	12/2016	Zaitsu	N/A	N/A
2017/0186754	12/2016	Blomberg et al.	N/A	N/A
2017/0191159	12/2016	Polyak et al.	N/A	N/A
2017/0191164	12/2016	Alokozai et al.	N/A	N/A
2017/0191685	12/2016	Ronne et al.	N/A	N/A
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2017/0200622	12/2016	Shiokawa et al.	N/A	N/A
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2017/0213960	12/2016	de Araujo et al.	N/A	N/A
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2017/0218515	12/2016	Shin et al.	N/A	N/A
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2017/0256429	12/2016	Lawson et al.	N/A	N/A
2017/0260649	12/2016	Coomer	N/A	N/A
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2018/0171475	12/2017	Maes et al.	N/A	N/A
2018/0171477	12/2017	Kim et al.	N/A	N/A
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2018/0254211	12/2017	Kosakai et al.	N/A	N/A
2018/0258532	12/2017	Kato et al.	N/A	N/A
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2019/0027605	12/2018	Tolle	N/A	H01L 21/02529
2019/0032209	12/2018	Huggare	N/A	N/A
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2019/0035810	12/2018	Chun et al.	N/A	N/A
2019/0040529	12/2018	Verbaas et al.	N/A	N/A
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2019/0204029	12/2018	Tanabe	N/A	N/A
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2019/0217277	12/2018	Jeon et al.	N/A	N/A
2019/0221433	12/2018	Raisanen et al.	N/A	N/A
2019/0229008	12/2018	Rokkam et al.	N/A	N/A
2019/0233446	12/2018	MacDonald et al.	N/A	N/A
2019/0233940	12/2018	Guo et al.	N/A	N/A
2019/0237325	12/2018	Wang et al.	N/A	N/A
2019/0237327	12/2018	Kohen et al.	N/A	N/A
2019/0244803	12/2018	Suzuki	N/A	N/A
2019/0249300	12/2018	Hatanpaa et al.	N/A	N/A
2019/0249303	12/2018	Kuroda et al.	N/A	N/A
2019/0252195	12/2018	Haukka	N/A	N/A
2019/0252196	12/2018	Vayrynen et al.	N/A	N/A
2019/0259611	12/2018	Nakano et al.	N/A	N/A
2019/0259612	12/2018	Nozawa et al.	N/A	N/A
2019/0264324	12/2018	Shugrue et al.	N/A	N/A
2019/0271078	12/2018	Raisanen et al.	N/A	N/A

2019/0272981	12/2018	Xu et al.	N/A	N/A
2019/0272993	12/2018	Mattinen et al.	N/A	N/A
2019/0273133	12/2018	Agrawal et al.	N/A	N/A
2019/0276934	12/2018	Verghese et al.	N/A	N/A
2019/0287769	12/2018	Blomberg et al.	N/A	N/A
2019/0295837	12/2018	Pore et al.	N/A	N/A
2019/0301014	12/2018	Pierreux et al.	N/A	N/A
2019/0304776	12/2018	Choi	N/A	N/A
2019/0304780	12/2018	Kohen et al.	N/A	N/A
2019/0304790	12/2018	Mousa et al.	N/A	N/A
2019/0304821	12/2018	Pierreux et al.	N/A	N/A
2019/0311894	12/2018	Girard et al.	N/A	N/A
2019/0311897	12/2018	Kang et al.	N/A	N/A
2019/0311940	12/2018	Choi et al.	N/A	N/A
2019/0318910	12/2018	Mori	N/A	N/A
2019/0318923	12/2018	Blanquart et al.	N/A	N/A
2019/0319100	12/2018	Chen et al.	N/A	N/A
2019/0322812	12/2018	Wojtecki et al.	N/A	N/A
2019/0330740	12/2018	Klaver	N/A	N/A
2019/0333753	12/2018	Ueda et al.	N/A	N/A
2019/0333761	12/2018	Tois et al.	N/A	N/A
2019/0338418	12/2018	Goradia et al.	N/A	N/A
2019/0346300	12/2018	Kim et al.	N/A	N/A
2019/0348261	12/2018	Lin et al.	N/A	N/A
2019/0348273	12/2018	Tang et al.	N/A	N/A
2019/0348515	12/2018	Li et al.	N/A	N/A
2019/0362970	12/2018	Wang et al.	N/A	N/A
2019/0362989	12/2018	Reuter et al.	N/A	N/A
2019/0363006	12/2018	Min	N/A	N/A
2019/0363015	12/2018	Cheng et al.	N/A	N/A
2019/0368040	12/2018	Kachel et al.	N/A	N/A
2019/0368041	12/2018	Sreeram et al.	N/A	N/A
2019/0371594	12/2018	Niskanen et al.	N/A	N/A
2019/0371640	12/2018	Raisanen et al.	N/A	N/A
2019/0375638	12/2018	Haukka	N/A	N/A
2019/0376180	12/2018	Niskanen	N/A	N/A
2019/0378711	12/2018	Suzuki et al.	N/A	N/A
2019/0378916	12/2018	Tang et al.	N/A	N/A
2019/0385907	12/2018	Gottheim et al.	N/A	N/A
2019/0390338	12/2018	Raisanen et al.	N/A	N/A
2019/0390343	12/2018	Min et al.	N/A	N/A
2019/0393304	12/2018	Guillorn et al.	N/A	N/A
2019/0393308	12/2018	Lo et al.	N/A	N/A
2020/0002811	12/2019	Sreeram et al.	N/A	N/A
2020/0002812	12/2019	Lee et al.	N/A	N/A
2020/0012081	12/2019	Komai	N/A	N/A
2020/0013612	12/2019	Blanquart et al.	N/A	N/A
2020/0013613	12/2019	Blanquart	N/A	N/A
2020/0013626	12/2019	Longrie et al.	N/A	N/A
2020/0013629	12/2019	de Roest et al.	N/A	N/A

2020/0018421	12/2019	Shugrue	N/A	N/A
2020/0035489	12/2019	Huang et al.	N/A	N/A
2020/0040458	12/2019	Ma et al.	N/A	N/A
2020/0048768	12/2019	Wiegers et al.	N/A	N/A
2020/0052056	12/2019	Park et al.	N/A	N/A
2020/0052089	12/2019	Yu et al.	N/A	N/A
2020/0056282	12/2019	Raisanen et al.	N/A	N/A
2020/0056286	12/2019	Shero et al.	N/A	N/A
2020/0058469	12/2019	Ranjan et al.	N/A	N/A
2020/0063262	12/2019	Katou et al.	N/A	N/A
2020/0064737	12/2019	de Roest	N/A	N/A
2020/0066512	12/2019	Tois et al.	N/A	N/A
2020/0066552	12/2019	Susa	N/A	N/A
2020/0080200	12/2019	Um et al.	N/A	N/A
2020/0083054	12/2019	Väyrynen et al.	N/A	N/A
2020/0083375	12/2019	Tolle et al.	N/A	N/A
2020/0083469	12/2019	Lhullier	N/A	N/A
2020/0102649	12/2019	Reed	N/A	N/A
2020/0102653	12/2019	Muralidhar et al.	N/A	N/A
2020/0105579	12/2019	Zhu	N/A	N/A
2020/0105594	12/2019	Cheng et al.	N/A	N/A
2020/0105895	12/2019	Tang et al.	N/A	N/A
2020/0109472	12/2019	Um et al.	N/A	N/A
2020/0111669	12/2019	Zaitsu et al.	N/A	N/A
2020/0111690	12/2019	Oosterlaken	N/A	N/A
2020/0118811	12/2019	Utsuno et al.	N/A	N/A
2020/0118815	12/2019	Fukazawa et al.	N/A	N/A
2020/0118817	12/2019	Blomberg et al.	N/A	N/A
2020/0119038	12/2019	Hopkins et al.	N/A	N/A
2020/0126771	12/2019	Roh et al.	N/A	N/A
2020/0126840	12/2019	Roh et al.	N/A	N/A
2020/0127122	12/2019	Lee	N/A	H01L 29/0847
2020/0131634	12/2019	Gao et al.	N/A	N/A
2020/0135512	12/2019	Oosterlaken et al.	N/A	N/A
2020/0135915	12/2019	Savant et al.	N/A	N/A
2020/0140995	12/2019	Blanquart et al.	N/A	N/A
2020/0141003	12/2019	Lee et al.	N/A	N/A
2020/0144058	12/2019	Kohen	N/A	N/A
2020/0152473	12/2019	Tapily et al.	N/A	N/A
2020/0152750	12/2019	Morrow et al.	N/A	N/A
2020/0161129	12/2019	Mattinen et al.	N/A	N/A
2020/0161438	12/2019	Tang et al.	N/A	N/A
2020/0168485	12/2019	Fluit	N/A	N/A
2020/0173020	12/2019	Huggare	N/A	N/A
2020/0176244	12/2019	Lee et al.	N/A	N/A
2020/0176246	12/2019	Huotari et al.	N/A	N/A
2020/0181770	12/2019	Longrie et al.	N/A	N/A
2020/0185249	12/2019	Rice et al.	N/A	N/A
2020/0185257	12/2019	Nishiwaki	N/A	N/A

2020/0194253	12/2019	Banerjee et al.	N/A	N/A
2020/0194268	12/2019	Sharma	N/A	N/A
2020/0203157	12/2019	Su	N/A	N/A
2020/0224309	12/2019	Sreeram et al.	N/A	N/A
2020/0224311	12/2019	Niskanen et al.	N/A	N/A
2020/0227243	12/2019	Kim et al.	N/A	N/A
2020/0227250	12/2019	Pore et al.	N/A	N/A
2020/0227325	12/2019	Xie et al.	N/A	N/A
2020/0232096	12/2019	Hatanpää et al.	N/A	N/A
2020/0234989	12/2019	Jeong	N/A	N/A
2020/0251328	12/2019	Zaitsu et al.	N/A	N/A
2020/0258766	12/2019	Garssen et al.	N/A	N/A
2020/0266057	12/2019	Lee et al.	N/A	N/A
2020/0266097	12/2019	Liu et al.	N/A	N/A
2020/0266098	12/2019	Pore et al.	N/A	N/A
2020/0266208	12/2019	Kim et al.	N/A	N/A
2020/0270749	12/2019	Li et al.	N/A	N/A
2020/0270752	12/2019	Pierreux et al.	N/A	N/A
2020/0273728	12/2019	Benjaminson et al.	N/A	N/A
2020/0273729	12/2019	Verbaas et al.	N/A	N/A
2020/0279721	12/2019	White et al.	N/A	N/A
2020/0279758	12/2019	Sharma	N/A	N/A
2020/0283893	12/2019	Byun et al.	N/A	N/A
2020/0283894	12/2019	Lehn et al.	N/A	N/A
2020/0284467	12/2019	Lindeboom et al.	N/A	N/A
2020/0286725	12/2019	Sharma et al.	N/A	N/A
2020/0286726	12/2019	Shero et al.	N/A	N/A
2020/0294789	12/2019	Woodruff et al.	N/A	N/A
2020/0303180	12/2019	Kim et al.	N/A	N/A
2020/0303196	12/2019	Chen et al.	N/A	N/A
2020/0309455	12/2019	Fluit	N/A	N/A
2020/0312652	12/2019	Kang et al.	N/A	N/A
2020/0312681	12/2019	Tanaka et al.	N/A	N/A
2020/0318237	12/2019	Fukazawa	N/A	N/A
2020/0321209	12/2019	Ishikawa et al.	N/A	N/A
2020/0332416	12/2019	Fluit	N/A	N/A
2020/0340113	12/2019	Hatanpää et al.	N/A	N/A
2020/0340138	12/2019	Tolle et al.	N/A	N/A
2020/0343134	12/2019	Kovalgin et al.	N/A	N/A
2020/0343358	12/2019	Zhu et al.	N/A	N/A
2020/0350193	12/2019	Garssen	N/A	N/A
2020/0354836	12/2019	Yednak et al.	N/A	N/A
2020/0355296	12/2019	Shugrue et al.	N/A	N/A
2020/0357631	12/2019	Ueda	N/A	N/A
2020/0365391	12/2019	Blanquart et al.	N/A	N/A
2020/0365433	12/2019	de Ridder et al.	N/A	N/A
2020/0365434	12/2019	de Ridder	N/A	N/A
2020/0365444	12/2019	Hill et al.	N/A	N/A
2020/0373152	12/2019	Blanquart	N/A	N/A
2020/0385861	12/2019	Deye et al.	N/A	N/A

2020/0385867	12/2019	Kim et al.	N/A	N/A
2020/0385868	12/2019	Kim et al.	N/A	N/A
2020/0395199	12/2019	Miyama	N/A	N/A
2020/0395209	12/2019	Yoshimoto et al.	N/A	N/A
2020/0395444	12/2019	Lo et al.	N/A	N/A
2020/0402846	12/2019	Collins et al.	N/A	N/A
2021/0002762	12/2020	Jun et al.	N/A	N/A
2021/0005449	12/2020	Blanquart et al.	N/A	N/A
2021/0005450	12/2020	Mattinen et al.	N/A	N/A
2021/0005723	12/2020	Tang et al.	N/A	N/A
2021/0013010	12/2020	Yoshikawa et al.	N/A	N/A
2021/0013034	12/2020	Wu et al.	N/A	N/A
2021/0013037	12/2020	Sun et al.	N/A	N/A
2021/0013042	12/2020	Väyrynen et al.	N/A	N/A
2021/0013085	12/2020	Roh et al.	N/A	N/A
2021/0017648	12/2020	Kubota et al.	N/A	N/A
2021/0020429	12/2020	Khazaka et al.	N/A	N/A
2021/0020431	12/2020	Blanquart et al.	N/A	N/A
2021/0020432	12/2020	Blanquart	N/A	N/A
2021/0020467	12/2020	Sekiguchi	N/A	N/A
2021/0020470	12/2020	Lee	N/A	N/A
2021/0028021	12/2020	Mousa et al.	N/A	N/A
2021/0032754	12/2020	White et al.	N/A	N/A
2021/0033977	12/2020	Raaijmakers et al.	N/A	N/A
2021/0035785	12/2020	Jeong et al.	N/A	N/A
2021/0035786	12/2020	Jeong et al.	N/A	N/A
2021/0035802	12/2020	Tolle et al.	N/A	N/A
2021/0035824	12/2020	de Ridder	N/A	N/A
2021/0035839	12/2020	de Ridder	N/A	N/A
2021/0035840	12/2020	de Ridder et al.	N/A	N/A
2021/0035841	12/2020	Fluit	N/A	N/A
2021/0035842	12/2020	de Ridder	N/A	N/A
2021/0035854	12/2020	Yoo et al.	N/A	N/A
2021/0040613	12/2020	White et al.	N/A	N/A
2021/0040615	12/2020	White et al.	N/A	N/A
2021/0041284	12/2020	Yednak, III	N/A	N/A
2021/0043444	12/2020	Blanquart	N/A	N/A
2021/0050231	12/2020	Oosterlaken	N/A	N/A
2021/0054500	12/2020	Zope et al.	N/A	N/A
2021/0054504	12/2020	Wang et al.	N/A	N/A
2021/0057214	12/2020	Kengoyama et al.	N/A	N/A
2021/0057223	12/2020	Stevens et al.	N/A	N/A
2021/0057275	12/2020	Pierreux et al.	N/A	N/A
2021/0057570	12/2020	Lin et al.	N/A	N/A
2021/0066075	12/2020	Zhang et al.	N/A	N/A
2021/0066079	12/2020	Lima et al.	N/A	N/A
2021/0066080	12/2020	Mattinen et al.	N/A	N/A
2021/0066083	12/2020	Haukka	N/A	N/A
2021/0066084	12/2020	Raisanen et al.	N/A	N/A
2021/0070783	12/2020	Odedra et al.	N/A	N/A

2021/0071296	12/2020	Watarai et al.	N/A	N/A
2021/0071298	12/2020	Maes et al.	N/A	N/A
2021/0074527	12/2020	Lee et al.	N/A	N/A
2021/0082692	12/2020	Kikuchi	N/A	N/A
2021/0090878	12/2020	Kang et al.	N/A	N/A
2021/0095372	12/2020	Minjauw et al.	N/A	N/A
2021/0102289	12/2020	Tsuji et al.	N/A	N/A
2021/0102290	12/2020	Acosta et al.	N/A	N/A
2021/0102292	12/2020	Lin et al.	N/A	N/A
2021/0104384	12/2020	Parkhe	N/A	N/A
2021/0104399	12/2020	Kuroda et al.	N/A	N/A
2021/0108328	12/2020	Yanagisawa	N/A	N/A
2021/0111025	12/2020	Zyulkov et al.	N/A	N/A
2021/0111053	12/2020	De Ridder	N/A	N/A
2021/0118667	12/2020	Fukazawa et al.	N/A	N/A
2021/0118679	12/2020	Lima et al.	N/A	N/A
2021/0118687	12/2020	Wang et al.	N/A	N/A
2021/0119032	12/2020	Chang	N/A	H01L 29/66795
2021/0125827	12/2020	Khazaka et al.	N/A	N/A
2021/0125832	12/2020	Bhatnagar	N/A	N/A
2021/0134588	12/2020	Kohen et al.	N/A	N/A
2021/0134959	12/2020	Lima et al.	N/A	N/A
2021/0140043	12/2020	Thombare et al.	N/A	N/A
2021/0143003	12/2020	Fukuda et al.	N/A	N/A
2021/0151315	12/2020	Pierreux et al.	N/A	N/A
2021/0151348	12/2020	Utsuno et al.	N/A	N/A
2021/0151352	12/2020	Zope et al.	N/A	N/A
2021/0156024	12/2020	Roh et al.	N/A	N/A
2021/0156030	12/2020	Shugrue	N/A	N/A
2021/0159077	12/2020	Longrie et al.	N/A	N/A
2021/0166910	12/2020	Kim et al.	N/A	N/A
2021/0166924	12/2020	Moon et al.	N/A	N/A
2021/0166925	12/2020	Moon et al.	N/A	N/A
2021/0166940	12/2020	Nozawa	N/A	N/A
2021/0172064	12/2020	Moon	N/A	N/A
2021/0175052	12/2020	Takahashi et al.	N/A	N/A
2021/0175069	12/2020	Hikavy	N/A	H01L 29/7848
2021/0180184	12/2020	Verni et al.	N/A	N/A
2021/0180188	12/2020	Kim et al.	N/A	N/A
2021/0180189	12/2020	Shugrue et al.	N/A	N/A
2021/0193458	12/2020	Salmi et al.	N/A	N/A
2021/0205758	12/2020	Kimtee et al.	N/A	N/A
2021/0207269	12/2020	Huang et al.	N/A	N/A
2021/0207270	12/2020	de Ridder et al.	N/A	N/A
2021/0210373	12/2020	Singu et al.	N/A	N/A
2021/0214842	12/2020	Yoon et al.	N/A	N/A
2021/0225615	12/2020	Yoshida	N/A	N/A
2021/0225622	12/2020	Shoji	N/A	N/A

2021/0225642	12/2020	Utsuno et al.	N/A	N/A
2021/0225643	12/2020	Kuroda et al.	N/A	N/A
2021/0230744	12/2020	Kimtee et al.	N/A	N/A
2021/0230746	12/2020	Shiba	N/A	N/A
2021/0233772	12/2020	Zhu et al.	N/A	N/A
2021/0238736	12/2020	Butail et al.	N/A	N/A
2021/0238742	12/2020	Susa et al.	N/A	N/A
2021/0239614	12/2020	Muralidhar et al.	N/A	N/A
2021/0242011	12/2020	Shero et al.	N/A	N/A
2021/0246556	12/2020	Mori	N/A	N/A
2021/0247693	12/2020	Maes et al.	N/A	N/A
2021/0249303	12/2020	Blanquart	N/A	N/A
2021/0254216	12/2020	Mori et al.	N/A	N/A
2021/0254238	12/2020	Khazaka et al.	N/A	N/A
2021/0257213	12/2020	Kikuchi et al.	N/A	N/A
2021/0257509	12/2020	Nishiwaki	N/A	N/A
2021/0265134	12/2020	Singh et al.	N/A	N/A
2021/0265158	12/2020	Kaneko	N/A	N/A
2021/0268554	12/2020	Mori	N/A	N/A
2021/0269914	12/2020	Väyrynen et al.	N/A	N/A
2021/0272821	12/2020	Oosterlaken	N/A	N/A
2021/0273109	12/2020	Yamazaki et al.	N/A	N/A
2021/0280448	12/2020	Ganguli et al.	N/A	N/A
2021/0285102	12/2020	Yoon et al.	N/A	N/A
2021/0287878	12/2020	Um et al.	N/A	N/A
2021/0287912	12/2020	Shiba et al.	N/A	N/A
2021/0287928	12/2020	Kim et al.	N/A	N/A
2021/0288476	12/2020	Wei	N/A	N/A
2021/0292902	12/2020	Kajbafvala et al.	N/A	N/A
2021/0296130	12/2020	Longrie et al.	N/A	N/A
2021/0296144	12/2020	Lin et al.	N/A	N/A
2021/0310125	12/2020	Ma et al.	N/A	N/A
2021/0313150	12/2020	Kang et al.	N/A	N/A
2021/0313167	12/2020	Pore et al.	N/A	N/A
2021/0313170	12/2020	Suzuki	N/A	N/A
2021/0313178	12/2020	Nakano	N/A	N/A
2021/0313182	12/2020	Zhu et al.	N/A	N/A
2021/0317576	12/2020	Väyrynen et al.	N/A	N/A
2021/0319982	12/2020	Kim et al.	N/A	N/A
2021/0320003	12/2020	Sugiura et al.	N/A	N/A
2021/0320010	12/2020	Wang et al.	N/A	N/A
2021/0320020	12/2020	Oosterlaken et al.	N/A	N/A
2021/0324510	12/2020	Kuwano et al.	N/A	N/A
2021/0324518	12/2020	de Ridder	N/A	N/A
2021/0327704	12/2020	Kajbafvala et al.	N/A	N/A
2021/0327714	12/2020	Lee et al.	N/A	N/A
2021/0327715	12/2020	Xie et al.	N/A	N/A
2021/0328036	12/2020	Li et al.	N/A	N/A
2021/0391172	12/2020	Barbosa Lima	N/A	H01L 21/0262

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
520629	12/2018	AT	N/A
1186873	12/1997	CN	N/A
2588350	12/2002	CN	N/A
1563483	12/2004	CN	N/A
1655362	12/2004	CN	N/A
1664987	12/2004	CN	N/A
1825535	12/2005	CN	N/A
101047143	12/2006	CN	N/A
101142012	12/2007	CN	N/A
101609858	12/2008	CN	N/A
101681873	12/2009	CN	N/A
101308794	12/2009	CN	N/A
102094183	12/2010	CN	N/A
102383106	12/2011	CN	N/A
103014846	12/2012	CN	N/A
102539019	12/2012	CN	N/A
103515222	12/2013	CN	N/A
102373440	12/2013	CN	N/A
203721699	12/2013	CN	N/A
104244620	12/2013	CN	N/A
104307264	12/2014	CN	N/A
104498895	12/2014	CN	N/A
104517892	12/2014	CN	N/A
204629865	12/2014	CN	N/A
105253917	12/2015	CN	N/A
205448240	12/2015	CN	N/A
104342637	12/2016	CN	N/A
206145834	12/2016	CN	N/A
104233226	12/2016	CN	N/A
106895521	12/2016	CN	N/A
104201108	12/2016	CN	N/A
104630735	12/2016	CN	N/A
107675144	12/2017	CN	N/A
108389798	12/2017	CN	N/A
106011785	12/2017	CN	N/A
108910843	12/2017	CN	N/A
109000352	12/2017	CN	N/A
8902307	12/1988	DE	N/A
3836696	12/1988	DE	N/A
3626724	12/1993	DE	N/A
10133013	12/2002	DE	N/A
102008052750	12/2008	DE	N/A
0058571	12/1981	EP	N/A
0499004	12/1991	EP	N/A
0634785	12/1994	EP	N/A
0550058	12/1997	EP	N/A

0887632	12/1997	EP	N/A
678909	12/1998	EP	N/A
1061567	12/1999	EP	N/A
1889817	12/2007	EP	N/A
2036600	12/2008	EP	N/A
2426233	12/2011	EP	N/A
686869	12/1929	FR	N/A
1408266	12/1964	FR	N/A
2233614	12/1974	FR	N/A
2517790	12/1982	FR	N/A
2610007	12/1989	FR	N/A
2708624	12/1994	FR	N/A
400010	12/1932	GB	N/A
752-277	12/1955	GB	N/A
1186889	12/1965	GB	N/A
1514921	12/1977	GB	N/A
2051875	12/1980	GB	N/A
58-19462	12/1982	JP	N/A
S5979545	12/1983	JP	N/A
S59127847	12/1983	JP	N/A
59-211779	12/1983	JP	N/A
S60110133	12/1984	JP	N/A
61038863	12/1985	JP	N/A
S62237236	12/1986	JP	N/A
S63136532	12/1987	JP	N/A
H0165766	12/1988	JP	N/A
H01185176	12/1988	JP	N/A
H01-296613	12/1988	JP	N/A
H01-307229	12/1988	JP	N/A
H01313954	12/1988	JP	N/A
H02-93071	12/1989	JP	N/A
H02-185038	12/1989	JP	N/A
H02-217469	12/1989	JP	N/A
H03-044472	12/1990	JP	N/A
H03-155625	12/1990	JP	N/A
H03211753	12/1990	JP	N/A
H03-248427	12/1990	JP	N/A
H04-29313	12/1991	JP	N/A
H04-115531	12/1991	JP	N/A
H05-6880	12/1992	JP	N/A
H05-23079	12/1992	JP	N/A
H05-118928	12/1992	JP	N/A
H05-171446	12/1992	JP	N/A
H05-291142	12/1992	JP	N/A
H06-053210	12/1993	JP	N/A
H06-84888	12/1993	JP	N/A
H06-140399	12/1993	JP	N/A
H06-16433	12/1993	JP	N/A
6204231	12/1993	JP	N/A
H06-319177	12/1993	JP	N/A

H06-338497	12/1993	JP	N/A
H07-29836	12/1994	JP	N/A
H07-297271	12/1994	JP	N/A
H07-066267	12/1994	JP	N/A
H07-74162	12/1994	JP	N/A
H07-109576	12/1994	JP	N/A
H07-034936	12/1994	JP	N/A
H07-225214	12/1994	JP	N/A
7-272694	12/1994	JP	N/A
H07-283149	12/1994	JP	N/A
H07-209093	12/1994	JP	N/A
H08-181135	12/1995	JP	N/A
H08-335558	12/1995	JP	N/A
H09-064149	12/1996	JP	N/A
9-89676	12/1996	JP	N/A
H09-148322	12/1996	JP	N/A
H10-41096	12/1997	JP	N/A
H10-50635	12/1997	JP	N/A
H10-50800	12/1997	JP	N/A
H10-064696	12/1997	JP	N/A
H10-153494	12/1997	JP	N/A
H10-227703	12/1997	JP	N/A
H10-0261620	12/1997	JP	N/A
H11-097163	12/1998	JP	N/A
H11-118615	12/1998	JP	N/A
H11-183264	12/1998	JP	N/A
H11-183265	12/1998	JP	N/A
H11-195688	12/1998	JP	N/A
H11-287715	12/1998	JP	N/A
2000068355	12/1999	JP	N/A
2000182949	12/1999	JP	N/A
2000269163	12/1999	JP	N/A
2001015698	12/2000	JP	N/A
2001023872	12/2000	JP	N/A
2001207265	12/2000	JP	N/A
2001207268	12/2000	JP	N/A
2001210602	12/2000	JP	N/A
2001220677	12/2000	JP	N/A
2001257199	12/2000	JP	N/A
2001287180	12/2000	JP	N/A
2002118066	12/2001	JP	N/A
2002164342	12/2001	JP	N/A
2002170781	12/2001	JP	N/A
2002237375	12/2001	JP	N/A
2003035574	12/2002	JP	N/A
2003053688	12/2002	JP	N/A
2003133300	12/2002	JP	N/A
2003153706	12/2002	JP	N/A
2003303814	12/2002	JP	N/A
2004014952	12/2003	JP	N/A

2004023043	12/2003	JP	N/A
2004088077	12/2003	JP	N/A
2004091848	12/2003	JP	N/A
2004113270	12/2003	JP	N/A
2004128019	12/2003	JP	N/A
2004134553	12/2003	JP	N/A
2004163293	12/2003	JP	N/A
2004244298	12/2003	JP	N/A
2004244661	12/2003	JP	N/A
2004294638	12/2003	JP	N/A
3589954	12/2003	JP	N/A
2004310019	12/2003	JP	N/A
2005033221	12/2004	JP	N/A
2005079254	12/2004	JP	N/A
2005507030	12/2004	JP	N/A
2005172489	12/2004	JP	N/A
3725100	12/2004	JP	N/A
2006028572	12/2005	JP	N/A
2006049352	12/2005	JP	N/A
2006059931	12/2005	JP	N/A
2006090762	12/2005	JP	N/A
2006124831	12/2005	JP	N/A
2006124832	12/2005	JP	N/A
2006153706	12/2005	JP	N/A
2006186271	12/2005	JP	N/A
2006188729	12/2005	JP	N/A
2006278058	12/2005	JP	N/A
2006319261	12/2005	JP	N/A
2007027777	12/2006	JP	N/A
2007287902	12/2006	JP	N/A
3140111	12/2007	JP	N/A
2008060304	12/2007	JP	N/A
2008066159	12/2007	JP	N/A
2008085129	12/2007	JP	N/A
2008089320	12/2007	JP	N/A
2008172083	12/2007	JP	N/A
2008198629	12/2007	JP	N/A
2008202107	12/2007	JP	N/A
2009016815	12/2008	JP	N/A
2009088421	12/2008	JP	N/A
2009099938	12/2008	JP	N/A
2009194248	12/2008	JP	N/A
2009239082	12/2008	JP	N/A
2009251216	12/2008	JP	N/A
2009252851	12/2008	JP	N/A
2010067940	12/2009	JP	N/A
2010097834	12/2009	JP	N/A
2010205967	12/2009	JP	N/A
2010251444	12/2009	JP	N/A
2010255218	12/2009	JP	N/A

2011049592	12/2010	JP	N/A
2011162830	12/2010	JP	N/A
2011181681	12/2010	JP	N/A
2009010412	12/2011	JP	N/A
2012146939	12/2011	JP	N/A
2012164736	12/2011	JP	N/A
2012195513	12/2011	JP	N/A
2013026479	12/2012	JP	N/A
2013196822	12/2012	JP	N/A
2013235912	12/2012	JP	N/A
D1422467	12/2013	JP	N/A
2014222693	12/2013	JP	N/A
2015021175	12/2014	JP	N/A
2015115461	12/2014	JP	N/A
2016098406	12/2015	JP	N/A
2010123843	12/2015	JP	N/A
2016174158	12/2015	JP	N/A
2017183242	12/2016	JP	N/A
2017220011	12/2016	JP	N/A
6519897	12/2018	JP	N/A
1998-0026850	12/1997	KR	N/A
20000000946	12/1999	KR	N/A
10-0253664	12/1999	KR	N/A
10-2000-0031098	12/1999	KR	N/A
10-2000-0045257	12/1999	KR	N/A
10-0273261	12/1999	KR	N/A
10-0295043	12/2000	KR	N/A
10-2002-0027695	12/2001	KR	N/A
10-2002-0064028	12/2001	KR	N/A
2002-0086763	12/2001	KR	N/A
10-0377095	12/2002	KR	N/A
2003-0092305	12/2002	KR	N/A
10-2005-0054122	12/2004	KR	N/A
10-0547248	12/2005	KR	N/A
10-0593960	12/2005	KR	N/A
10-2006-0129566	12/2005	KR	N/A
10-0688484	12/2006	KR	N/A
20070041701	12/2006	KR	N/A
10-2007-0079797	12/2006	KR	N/A
10-2007-0084683	12/2006	KR	N/A
10-2007-0117817	12/2006	KR	N/A

2008-0058620	12/2007	KR	N/A
10-2009-0039936	12/2008	KR	N/A
10-2009-0055443	12/2008	KR	N/A
10-2009-0056475	12/2008	KR	N/A
10-2009-0086790	12/2008	KR	N/A
10-2009-0105977	12/2008	KR	N/A
10-0936694	12/2009	KR	N/A
10-2010-0015073	12/2009	KR	N/A
10-2010-0020834	12/2009	KR	N/A
10-2010-0032812	12/2009	KR	N/A
10-2010-0077442	12/2009	KR	N/A
10-2010-0079920	12/2009	KR	N/A
10-2010-0122701	12/2009	KR	N/A
10-2010-0132779	12/2009	KR	N/A
10-2011-0058534	12/2010	KR	N/A
10-1114219	12/2011	KR	N/A
10-2012-0111060	12/2011	KR	N/A
10-2013-0007806	12/2012	KR	N/A
10-2013-0067600	12/2012	KR	N/A
10-2013-0129149	12/2012	KR	N/A
10-1347962	12/2013	KR	N/A
10-1491726	12/2014	KR	N/A
10-1535573	12/2014	KR	N/A
101758892	12/2016	KR	N/A
10-2019-0032077	12/2018	KR	N/A
8600255	12/1986	NL	N/A
8701549	12/1988	NL	N/A
1786406	12/1992	RU	N/A
494614	12/1975	SU	N/A
1408319	12/1987	SU	N/A
538327	12/2002	TW	N/A
540093	12/2002	TW	N/A

M292692	12/2005	TW	N/A
200731357	12/2006	TW	N/A
200903625	12/2008	TW	N/A
201213596	12/2011	TW	N/A
201234453	12/2011	TW	N/A
201247690	12/2011	TW	N/A
M446412	12/2012	TW	N/A
201330086	12/2012	TW	N/A
D162593	12/2013	TW	N/A
D164568	12/2013	TW	N/A
201531587	12/2014	TW	N/A
M512254	12/2014	TW	N/A
I514509	12/2014	TW	N/A
201613231	12/2015	TW	N/A
M553518	12/2016	TW	N/A
D196097	12/2018	TW	N/A
1987001508	12/1986	WO	N/A
1990004045	12/1989	WO	N/A
1991004522	12/1990	WO	N/A
1991006975	12/1990	WO	N/A
1992000477	12/1991	WO	N/A
1994027315	12/1993	WO	N/A
1996017107	12/1995	WO	N/A
1997003223	12/1996	WO	N/A
1998032893	12/1997	WO	N/A
1999023690	12/1998	WO	N/A
DM/048579	12/1998	WO	N/A
2004008491	12/2001	WO	N/A
2004007800	12/2003	WO	N/A
2004008827	12/2003	WO	N/A
2004010467	12/2003	WO	N/A
2004106584	12/2003	WO	N/A
2005112082	12/2004	WO	N/A
2006035281	12/2005	WO	N/A
2006054854	12/2005	WO	N/A
2006056091	12/2005	WO	N/A
2006078666	12/2005	WO	N/A
2006080782	12/2005	WO	N/A
2006097525	12/2005	WO	N/A
2006101857	12/2005	WO	N/A
2006114781	12/2005	WO	N/A
2007024720	12/2006	WO	N/A
2007027165	12/2006	WO	N/A
2007076195	12/2006	WO	N/A
2007088940	12/2006	WO	N/A
2007117718	12/2006	WO	N/A
2007131051	12/2006	WO	N/A
2007140376	12/2006	WO	N/A
2008045972	12/2007	WO	N/A
2008091900	12/2007	WO	N/A

2008121463	12/2007	WO	N/A
2008147731	12/2007	WO	N/A
2009028619	12/2008	WO	N/A
2009029532	12/2008	WO	N/A
2009039251	12/2008	WO	N/A
2009099776	12/2008	WO	N/A
2009154889	12/2008	WO	N/A
2009154896	12/2008	WO	N/A
2010039363	12/2009	WO	N/A
2010077533	12/2009	WO	N/A
2010100702	12/2009	WO	N/A
2010110558	12/2009	WO	N/A
2010118051	12/2009	WO	N/A
2010129428	12/2009	WO	N/A
2010129430	12/2009	WO	N/A
2010129431	12/2009	WO	N/A
2011019950	12/2010	WO	N/A
2011149640	12/2010	WO	N/A
2012056592	12/2011	WO	N/A
2012077590	12/2011	WO	N/A
2013043330	12/2012	WO	N/A
2013078065	12/2012	WO	N/A
2013078066	12/2012	WO	N/A
2013177269	12/2012	WO	N/A
2014107290	12/2013	WO	N/A
2015026230	12/2014	WO	N/A
2015107009	12/2014	WO	N/A
2015112728	12/2014	WO	N/A
2015127614	12/2014	WO	N/A
2016019795	12/2015	WO	N/A
2018109553	12/2015	WO	N/A
2018109554	12/2015	WO	N/A
2017108713	12/2016	WO	N/A
2017108714	12/2016	WO	N/A
2017125401	12/2016	WO	N/A
2017212546	12/2016	WO	N/A
2018003072	12/2017	WO	N/A
2018008088	12/2017	WO	N/A
2018013778	12/2017	WO	N/A
2018020316	12/2017	WO	N/A
2018020318	12/2017	WO	N/A
2018020320	12/2017	WO	N/A
2018020327	12/2017	WO	N/A
2018105349	12/2017	WO	N/A
2018109551	12/2017	WO	N/A
2018109552	12/2017	WO	N/A
2018178771	12/2017	WO	N/A
2019030565	12/2018	WO	N/A
2019103610	12/2018	WO	N/A
2019103613	12/2018	WO	N/A

2019142055	12/2018	WO	N/A
2019158960	12/2018	WO	N/A
2019214578	12/2018	WO	N/A
2019217749	12/2018	WO	N/A
2019229537	12/2018	WO	N/A
2020002995	12/2019	WO	N/A
2020003000	12/2019	WO	N/A
2020118100	12/2019	WO	N/A
2021072042	12/2020	WO	N/A

OTHER PUBLICATIONS

CNIPA; Office Action dated Jul. 21, 2021 in Application No. 201610131743.1. cited by applicant

CNIPA; Office Action dated Jul. 23, 2021 in Application No. 201710131319.1. cited by applicant

CNIPA; Notice of Allowance dated Jun. 3, 2021 in Application No. 201711057929.8. cited by applicant

CNIPA; Office Action dated Sep. 13, 2021 in Application No. 201780076230.8. cited by applicant

CNIPA; Office Action dated Sep. 13, 2021 in Application No. 201780076321.1. cited by applicant

CNIPA; Office Action dated Apr. 30, 2021 in Application No. 201810018936.5. cited by applicant

CNIPA; Office Action dated Apr. 16, 2021 in Application No. 201810116717.0. cited by applicant

CNIPA; Notice of Allowance dated Sep. 15, 2021 in Application No. 201810116717.0. cited by applicant

CNIPA; Office Action dated May 26, 2021 in Application No. 201810215131.X. cited by applicant

CNIPA; Office Action dated Jun. 16, 2021 in Application No. 201810344382.8. cited by applicant

CNIPA; Notice of Allowance dated Oct. 11, 2021 in Application No. 201810344382.8. cited by applicant

CNIPA; Office Action dated Jul. 5, 2021 in Application No. 201810530514.6. cited by applicant

CNIPA; Office Action dated Jun. 30, 2021 in Application No. 201810755771.X. cited by applicant

CNIPA; Office Action dated Sep. 23, 2021 in Application No. 201880048547.5. cited by applicant

CNIPA; Office Action dated May 19, 2021 in Application No. 201880072606.2. cited by applicant

CNIPA; Office Action dated May 31, 2021 in Application No. 201910857144.1. cited by applicant

CNIPA; Office Action dated Jun. 2, 2021 in Application No. 201910920881.1. cited by applicant

CNIPA; Office Action dated Jul. 5, 2021 in Application No. 201911042495.3. cited by applicant

CNIPA; Office Action dated Apr. 27, 2021 in Application No. 201911250100.9. cited by applicant

CNIPA; Office Action dated Jul. 7, 2021 in Application No. 201980011788.7. cited by applicant

CNIPA; Notice of Allowance dated Jun. 16, 2021 in Application No. 202130111248.6. cited by applicant

EPO; Extended European Search Report dated Aug. 11, 2021 in Application No. 21169226.4. cited by applicant

JPO; Notice of Allowance dated Aug. 11, 2021 in Application No. 2018-024655. cited by applicant

JPO; Office Action dated Aug. 2, 2021 in Application No. 2019-504674. cited by applicant

JPO; Office Action dated Sep. 10, 2021 in Application No. 2019-531445. cited by applicant

JPO; Notice of Allowance dated Jun. 30, 2021 in Application No. 2021-004208. cited by applicant

KIPO; Office Action dated Jun. 28, 2021 in Application No. 10-2014-0027217. cited by applicant

KIPO; Notice of Allowance dated May 27, 2021 in Application No. 10-2014-0105478. cited by applicant

KIPO; Notice of Allowance dated Aug. 17, 2021 in Application No. 10-2014-0122903. cited by applicant

KIPO; Notice of Allowance dated Sep. 8, 2021 in Application No. 10-2014-0136089. cited by applicant

KIPO; Notice of Allowance dated Jul. 6, 2021 in Application No. 10-2014-0145220. cited by applicant

applicant

KIPO; Notice of Allowance dated Sep. 23, 2021 in Application No. 2014-0156196. cited by applicant

KIPO; Notice of Allowance dated Jul. 21, 2021 in Application No. 10-2015-0025314. cited by applicant

KIPO; Office Action dated May 19, 2021 in Application No. 10-2015-0036819. cited by applicant

KIPO; Office Action dated Jun. 23, 2021 in Application No. 10-2015-0037658. cited by applicant

KIPO; Office Action dated Sep. 6, 2021 in Application No. 10-2015-0046393. cited by applicant

KIPO; Office Action dated Jun. 10, 2021 in Application No. 2015-61391. cited by applicant

KIPO; Office Action dated Sep. 3, 2021 in Application No. 2017-37847. cited by applicant

KIPO; Office Action dated Aug. 2, 2021 in Application No. 2017-43865. cited by applicant

KIPO; Office Action dated Oct. 1, 2021 in Application No. 2017-43919. cited by applicant

KIPO; Office Action dated Sep. 28, 2021 in Application No. 2017-49172. cited by applicant

KIPO; Notice of Allowance dated Sep. 28, 2021 in Application No. 2017-54647. cited by applicant

KIPO; Office Action dated in Aug. 30, 2021 in Application No. 2017-81515. cited by applicant

KIPO; Office Action dated Sep. 10, 2021 in Application No. 2017-86083. cited by applicant

KIPO; Office Action dated Jun. 29, 2021 in Application No. 10-2017-0066979. cited by applicant

KIPO; Notice of Allowance dated Jun. 24, 2021 in Application No. 10-2019-0044213. cited by applicant

KIPO; Office Action dated Jun. 17, 2021 in Application No. 10-2021-0051860. cited by applicant

KIPO; Office Action dated Aug. 18, 2021 in Application No. 10-2021-0090283. cited by applicant

KIPO; Notice of Allowance dated Sep. 2, 2021 in Application No. 30-2020-0047043. cited by applicant

TIPO; Office Action dated Jun. 22, 2021 in Application No. 104108277. cited by applicant

TIPO; Notice of Allowance dated Sep. 22, 2021 in Application No. 104108277. cited by applicant

TIPO; Office Action dated Jun. 2, 2021 in Application No. 105129977. cited by applicant

TIPO; Notice of Allowance dated Sep. 9, 2021 in Application No. 105129977. cited by applicant

TIPO; Notice of Allowance dated Jun. 4, 2021 in Application No. 105131896. cited by applicant

TIPO; Notice of Allowance dated Jul. 30, 2021 in Application No. 105134275. cited by applicant

TIPO; Notice of Allowance dated Feb. 24, 2021 in Application No. 106113604. cited by applicant

TIPO; Notice of Allowance dated Sep. 3, 2021 in Application No. 106115126. cited by applicant

TIPO; Office Action dated Jun. 21, 2021 in Application No. 106120902. cited by applicant

TIPO; Notice of Allowance dated Jun. 30, 2021 in Application No. 106121797. cited by applicant

TIPO; Notice of Allowance dated Sep. 1, 2021 in Application No. 106122231. cited by applicant

TIPO; Notice of Allowance dated Jun. 3, 2021 in Application No. 106123203. cited by applicant

TIPO; Notice of Allowance dated Aug. 24, 2021 in Application No. 106124129. cited by applicant

TIPO; Notice of Allowance dated Jul. 28, 2021 in Application No. 106129491. cited by applicant

TIPO; Notice of Allowance dated Aug. 10, 2021 in Application No. 106129971. cited by applicant

TIPO; Office Action dated Jun. 15, 2021 in Application No. 106133152. cited by applicant

TIPO; Notice of Allowance dated Oct. 12, 2021 in Application No. 106135925. cited by applicant

TIPO; Office Action dated Jul. 13, 2021 in Application No. 106138996. cited by applicant

TIPO; Notice of Allowance dated Sep. 24, 2021 in Application No. 106143559. cited by applicant

TIPO; Office Action dated Aug. 5, 2021 in Application No. 106143566. cited by applicant

TIPO; Office Action dated May 26, 2021 in Application No. 106143568. cited by applicant

TIPO; Notice of Allowance dated Jul. 30, 2021 in Application No. 106143570. cited by applicant

TIPO; Office Action dated Jun. 7, 2021 in Application No. 107103230. cited by applicant

TIPO; Office Action dated Aug. 10, 2021 in Application No. 107103230. cited by applicant

TIPO; Office Action dated Sep. 8, 2021 in Application No. 107105788. cited by applicant

TIPO; Office Action dated Aug. 3, 2021 in Application No. 107105982. cited by applicant

TIPO; Office Action dated Aug. 16, 2021 in Application No. 107114888. cited by applicant

TIPO; Office Action dated Sep. 28, 2021 in Application No. 109112983. cited by applicant
TIPO; Office Action dated Sep. 13, 2021 in Application No. 109119438. cited by applicant
TIPO; Office Action dated Oct. 18, 2021 in Application No. 109300594. cited by applicant
TIPO; Notice of Allowance dated Jun. 9, 2021 in Application No. 109303437. cited by applicant
TIPO; Notice of Allowance dated Oct. 1, 2021 in Application No. 110118827. cited by applicant
TIPO; Notice of Allowance dated Jun. 30, 2021 in Application No. 110301018. cited by applicant
TIPO; Notice of Allowance dated Jul. 29, 2021 in Application No. 110301101. cited by applicant
TIPO; Notice of Allowance dated Jul. 27, 2021 in Application No. 110302670. cited by applicant
USPTO; Final Office Action dated Aug. 4, 2021 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Non-Final Office Action dated Jun. 10, 2021 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Final Office Action dated Oct. 1, 2021 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Non-Final Office Action dated Jul. 9, 2021 in U.S. Appl. No. 15/286,503. cited by applicant
USPTO; Notice of Allowance dated Jun. 29, 2021 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Non-Final Office Action dated Jun. 25, 2021 in U.S. Appl. No. 15/380,909. cited by applicant
USPTO; Final Office Action dated Jun. 21, 2021 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO; Advisory Action dated Aug. 30, 2021 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO; Non-Final Office Action dated Oct. 6, 2021 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO; Advisory Action dated Jul. 27, 2021 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Final Office Action dated Jun. 9, 2021 in U.S. Appl. No. 15/491,726. cited by applicant
USPTO; Advisory Action dated Aug. 24, 2021 in U.S. Appl. No. 15/491,726. cited by applicant
USPTO; Non-Final Office Action dated Sep. 28, 2021 in U.S. Appl. No. 15/491,726. cited by applicant
USPTO; Notice of Allowance dated Aug. 25, 2021 in U.S. Appl. No. 15/611,707. cited by applicant
USPTO; Non-Final Office Action dated Sep. 16, 2021 in U.S. Appl. No. 15/636,307. cited by applicant
USPTO; Advisory Action dated Jun. 11, 2021 in U.S. Appl. No. 15/690,017. cited by applicant
USPTO; Non-Final Office Action dated Jun. 17, 2021 in U.S. Appl. No. 15/690,017. cited by applicant
USPTO; Final Office Action dated Jul. 15, 2021 in U.S. Appl. No. 15/726,959. cited by applicant
USPTO; Advisory Action dated Sep. 21, 2021 in U.S. Appl. No. 15/726,959. cited by applicant
USPTO; Notice of Allowance dated Sep. 15, 2021 in U.S. Appl. No. 15/835,328. cited by applicant
USPTO; Advisory Action dated Jun. 11, 2021 in U.S. Appl. No. 15/909,705. cited by applicant
USPTO; Notice of Allowance dated Jun. 4, 2021 in U.S. Appl. No. 15/923,834. cited by applicant
USPTO; Final Office Action dated Jun. 28, 2021 in U.S. Appl. No. 15/925,532. cited by applicant
USPTO; Advisory Action dated Aug. 27, 2021 in U.S. Appl. No. 15/925,532. cited by applicant
USPTO; Non-Final Office Action dated Oct. 6, 2021 in U.S. Appl. No. 15/925,532. cited by applicant
USPTO; Non-Final Office Action dated Jun. 16, 2021 in U.S. Appl. No. 15/940,729. cited by applicant
USPTO; Notice of Allowance dated Sep. 28, 2021 in U.S. Appl. No. 15/940,729. cited by applicant
USPTO; Non-Final Office Action dated Aug. 2, 2021 in U.S. Appl. No. 15/962,980. cited by applicant
USPTO; Non-Final Office Action dated Oct. 4, 2021 in U.S. Appl. No. 15/967,146. cited by applicant
USPTO; Final Office Action dated Aug. 3, 2021 in U.S. Appl. No. 15/974,948. cited by applicant
USPTO; Advisory Action dated Oct. 21, 2021 in U.S. Appl. No. 15/974,948. cited by applicant

USPTO; Final Office Action dated Oct. 12, 2021 in U.S. Appl. No. 15/996,286. cited by applicant
USPTO; Non-Final Office Action dated Jul. 9, 2021 in U.S. Appl. No. 15/997,445. cited by applicant
USPTO; Final Office Action dated Aug. 5, 2021 in U.S. Appl. No. 16/000,109. cited by applicant
USPTO; Final Office Action dated Oct. 20, 2021 in U.S. Appl. No. 16/000,125. cited by applicant
USPTO; Final Office Action dated Sep. 3, 2021 in U.S. Appl. No. 16/004,041. cited by applicant
USPTO; Non-Final Office Action dated Oct. 22, 2021 in U.S. Appl. No. 16/042,791. cited by applicant
USPTO; Non-Final Office Action dated Sep. 28, 2021 in U.S. Appl. No. 16/105,745. cited by applicant
USPTO; Non-Final Office Action dated Sep. 29, 2021 in U.S. Appl. No. 16/105,761. cited by applicant
USPTO; Non-Final Office Action dated Jun. 25, 2021 in U.S. Appl. No. 16/105,802. cited by applicant
USPTO; Final Office Action dated Jul. 27, 2021 in U.S. Appl. No. 16/108,950. cited by applicant
USPTO; Advisory Action dated Oct. 1, 2021 in U.S. Appl. No. 16/108,950. cited by applicant
USPTO; Non-Final Office Action dated Jun. 18, 2021 in U.S. Appl. No. 16/117,530. cited by applicant
USPTO; Final Office Action dated Jul. 8, 2021 in U.S. Appl. No. 16/151,074. cited by applicant
USPTO; Notice of Allowance dated Sep. 17, 2021 in U.S. Appl. No. 16/151,074. cited by applicant
USPTO; Non-Final Office Action dated Sep. 16, 2021 in U.S. Appl. No. 16/152,260. cited by applicant
USPTO; Advisory Action dated Aug. 2, 2021 in U.S. Appl. No. 16/172,535. cited by applicant
USPTO; Non-Final Office Action dated Oct. 15, 2021 in U.S. Appl. No. 16/172,535. cited by applicant
USPTO; Final Office Action dated Jun. 14, 2021 in U.S. Appl. No. 16/202,941. cited by applicant
USPTO; Advisory Action dated Aug. 16, 2021 in U.S. Appl. No. 16/202,941. cited by applicant
USPTO; Non-Final Office Action dated Sep. 20, 2021 in U.S. Appl. No. 16/202,941. cited by applicant
USPTO; Final Office Action dated Jul. 26, 2021 in U.S. Appl. No. 16/206,589. cited by applicant
USPTO; Notice of Allowance dated Sep. 13, 2021 in U.S. Appl. No. 16/206,589. cited by applicant
USPTO; Non-Final Office Action dated Aug. 25, 2021 in U.S. Appl. No. 16/210,922. cited by applicant
USPTO; Final Office Action dated Sep. 1, 2021 in U.S. Appl. No. 16/240,392. cited by applicant
USPTO; Notice of Allowance dated Oct. 15, 2021 in U.S. Appl. No. 16/240,392. cited by applicant
USPTO; Final Office Action dated Sep. 7, 2021 in U.S. Appl. No. 16/251,534. cited by applicant
USPTO; Final Office Action dated Aug. 18, 2021 in U.S. Appl. No. 16/252,567. cited by applicant
USPTO; Final Office Action dated Jul. 21, 2021 in U.S. Appl. No. 16/252,569. cited by applicant
USPTO; Advisory Action dated Jul. 15, 2021 in U.S. Appl. No. 16/468,258. cited by applicant
USPTO; Non-Final Office Action dated Oct. 21, 2021 in U.S. Appl. No. 16/468,258. cited by applicant
USPTO; Advisory Action dated Jul. 8, 2021 in U.S. Appl. No. 16/546,543. cited by applicant
USPTO; Ex Parte Quayle Action dated Aug. 6, 2021 in U.S. Appl. No. 16/563,473. cited by applicant
USPTO; Notice of Allowance dated Sep. 30, 2021 in U.S. Appl. No. 16/563,473. cited by applicant
USPTO; Non-Final Office Action dated Oct. 5, 2021 in U.S. Appl. No. 16/588,807. cited by applicant
USPTO; Notice of Allowance dated Oct. 18, 2021 in U.S. Appl. No. 16/601,593. cited by applicant
USPTO; Non-Final Office Action dated Jun. 11, 2021 in U.S. Appl. No. 16/671,847. cited by applicant

USPTO; Non-Final Office Action dated Jun. 24, 2021 in U.S. Appl. No. 16/673,860. cited by applicant

USPTO; Non-Final Office Action dated Sep. 28, 2021 in U.S. Appl. No. 16/692,859. cited by applicant

USPTO; Notice of Allowance dated Jul. 20, 2021 in U.S. Appl. No. 16/704,835. cited by applicant

USPTO; Non-Final Office Action dated Aug. 31, 2021 in U.S. Appl. No. 16/712,707. cited by applicant

USPTO; Non-Final Office Action dated Sep. 21, 2021 in U.S. Appl. No. 16/736,336. cited by applicant

USPTO; Final Office Action dated Sep. 15, 2021 in U.S. Appl. No. 16/789,138. cited by applicant

USPTO; Non-Final Office Action dated Jun. 24, 2021 in U.S. Appl. No. 16/792,058. cited by applicant

USPTO; Notice of Allowance dated Oct. 19, 2021 in U.S. Appl. No. 16/792,058. cited by applicant

USPTO; Non-Final Office Action dated Oct. 6, 2021 in U.S. Appl. No. 16/792,544. cited by applicant

USPTO; Notice of Allowance dated Sep. 27, 2021 in U.S. Appl. No. 16/792,571. cited by applicant

USPTO; Non-Final Office Action dated Aug. 17, 2021 in U.S. Appl. No. 16/797,346. cited by applicant

USPTO; Notice of Allowance dated Jul. 21, 2021 in U.S. Appl. No. 16/816,078. cited by applicant

USPTO; Non-Final Office Action dated Jun. 23, 2021 in U.S. Appl. No. 16/827,012. cited by applicant

USPTO; Non-Final Office Action dated Feb. 23, 2021 in U.S. Appl. No. 16/827,506. cited by applicant

USPTO; Notice of Allowance dated May 27, 2021 in U.S. Appl. No. 16/827,506. cited by applicant

USPTO; Final Office Action dated Oct. 6, 2021 in U.S. Appl. No. 16/828,753. cited by applicant

USPTO; Non-Final Office Action dated Jul. 28, 2021 in U.S. Appl. No. 16/835,283. cited by applicant

USPTO; Non-Final Office Action dated Oct. 4, 2021 in U.S. Appl. No. 16/840,960. cited by applicant

USPTO; Non-Final Office Action dated Aug. 3, 2021 in U.S. Appl. No. 16/849,793. cited by applicant

USPTO; Non-Final Office Action dated Jun. 15, 2021 in U.S. Appl. No. 16/861,144. cited by applicant

USPTO; Final Office Action dated Sep. 17, 2021 in U.S. Appl. No. 16/861,144. cited by applicant

USPTO; Non-Final Office Action dated Sep. 1, 2021 in U.S. Appl. No. 16/867,385. cited by applicant

USPTO; Final Office Action dated Oct. 7, 2021 in U.S. Appl. No. 16/872,045. cited by applicant

USPTO; Final Office Action dated Oct. 14, 2021 in U.S. Appl. No. 16/878,443. cited by applicant

USPTO; Non-Final Office Action dated Oct. 4, 2021 in U.S. Appl. No. 16/886,405. cited by applicant

USPTO; Non-Final Office Action dated Jul. 28, 2021 in U.S. Appl. No. 16/888,423. cited by applicant

USPTO; Non-Final Office Action dated Oct. 13, 2021 in U.S. Appl. No. 16/893,206. cited by applicant

USPTO; Notice of Allowance dated Jun. 17, 2021 in U.S. Appl. No. 16/924,595. cited by applicant

USPTO; Non-Final Office Action dated Jul. 27, 2021 in U.S. Appl. No. 16/930,193. cited by applicant

USPTO; Non-Final Office Action dated Jun. 25, 2021 in U.S. Appl. No. 16/930,305. cited by applicant

USPTO; Non-Final Office Action dated Sep. 14, 2021 in U.S. Appl. No. 16/932,707. cited by applicant

applicant
USPTO; Non-Final Office Action dated Oct. 4, 2021 in U.S. Appl. No. 16/935,275. cited by applicant
USPTO; Notice of Allowance dated Sep. 22, 2021 in U.S. Appl. No. 16/935,280. cited by applicant
USPTO; Non-Final Office Action dated Aug. 23, 2021 in U.S. Appl. No. 16/938,868. cited by applicant
USPTO; Non-Final Office Action dated Oct. 12, 2021 in U.S. Appl. No. 16/944,763. cited by applicant
USPTO; Non-Final Office Action dated Aug. 19, 2021 in U.S. Appl. No. 16/992,806. cited by applicant
USPTO; Non-Final Office Action dated Jul. 7, 2021 in U.S. Appl. No. 16/999,065. cited by applicant
USPTO; Final Office Action dated Oct. 21, 2021 in U.S. Appl. No. 16/999,065. cited by applicant
USPTO; Final Office Action dated Jul. 23, 2021 in U.S. Appl. No. 17/009,093. cited by applicant
USPTO; Advisory Action dated Oct. 5, 2021 in U.S. Appl. No. 17/009,093. cited by applicant
USPTO; Non-Final Office Action dated Sep. 28, 2021 in U.S. Appl. No. 17/023,129. cited by applicant
USPTO; Notice of Allowance dated Oct. 14, 2021 in U.S. Appl. No. 17/024,092. cited by applicant
USPTO; Notice of Allowance dated Oct. 4, 2021 in U.S. Appl. No. 17/028,066. cited by applicant
USPTO; Non-Final Office Action dated Sep. 10, 2021 in U.S. Appl. No. 17/038,514. cited by applicant
USPTO; Non-Final Office Action dated Oct. 22, 2021 in U.S. Appl. No. 17/073,544. cited by applicant
USPTO; Non-Final Office Action dated Sep. 30, 2021 in U.S. Appl. No. 17/093,224. cited by applicant
USPTO; Non-Final Office Action dated Sep. 22, 2021 in U.S. Appl. No. 17/126,275. cited by applicant
USPTO; Non-Final Office Action dated Sep. 23, 2021 in U.S. Appl. No. 17/254,111. cited by applicant
USPTO; Non-Final Office Action dated Sep. 27, 2021 in U.S. Appl. No. 17/254,366. cited by applicant
USPTO; Notice of Allowance dated Jul. 1, 2021 in U.S. Appl. No. 29/679,620. cited by applicant
USPTO; Notice of Allowance dated Jul. 9, 2021 in U.S. Appl. No. 29/692,490. cited by applicant
USPTO; Non-Final Office Action dated Jul. 14, 2021 in U.S. Appl. No. 29/695,044. cited by applicant
USPTO; Notice of Allowance dated Jun. 8, 2021 in U.S. Appl. No. 29/696,472. cited by applicant
USPTO; Notice of Allowance dated Sep. 21, 2021 in U.S. Appl. No. 29/702,865. cited by applicant
Alen et al. "Atomic layer deposition of molybdenum nitride thin films for cu metallizations" J of The Electrochemical Society, 152(5) G361-G366 (2005). cited by applicant
Best et al. "Complex Halides of the Transition Metals. 24.1 Reactions of Dimeric Molybdenum (II) Halide Complexes Containing Strong Metal-Metal Bonds with Bidentate Tertiary Phosphines and Arsines" Inorganic Chemistry, vol. 17, No. 1, pp. 99-104 (1978). cited by applicant
Blakeney et al. "Atomic Layer Deposition of Aluminum Metal Films Using a Thermally Stable Aluminum Hydride Reducing Agent" Chem. Mater., 30, pp. 1844-1848 (2018). cited by applicant
Buitrago et al. "SnOx high-efficiency EUV interference lithography gratings towards the ultimate resolution in photolithography" Abstract, 1 page (2016). cited by applicant
Buitrago et al. "State-of-the-art EUV materials and processes for the 7 nm node and beyond" Proc of SPIE, vol. 10143, 8 pages (2017). cited by applicant
Cheng et al. "Improved High-Temperature Leakage in High-Density MIM Capacitors by Using a TiLaO Dielectric and an Ir Electrode" IEEE Electron Device Letters, vol. 28, No. 12, 3 pages

(2007). cited by applicant

Firestop Support Plate, Type B, 6 In Apr. 4, 2012, Amazon. Com, May 10, 2021, <https://www.amazon.com/AmeriVent-Firestop-Support-Plate-Type/dp/B007R7W951/> (2012). cited by applicant

Gertsch et al. "SF₄ as the Fluorination Reactant for Al₂O₃ and VO₂ Thermal Atomic Layer Etching" *Chem. Mater.*, 31, pp. 3624-3635 (2019). cited by applicant

Han et al. "Synthesis and characterization of novel zinc precursors for ZnO thin film deposition by atomic layer deposition" *Dalton Transactions* 49.14 (2020): 4306-4314. cited by applicant

Hayashi et al. "2,2-Difluoro-1,3-dimethylimidazolidine (DFI). A new fluorinating agent" *Chem. Commun.* pp. 1618-1619 (2002). cited by applicant

Imai et al. "Energetic stability and magnetic moment of tri-, tetra-, and octa-ferromagnetic element nitrides predicted by first-principle calculations" *J of Alloys and Compounds*, vol. 611, 19 pages (2019). cited by applicant

Jacoby "Chemical deposition methods to the rescue" *Cen. Acs. Org.*, pp. 29-32 (2018). cited by applicant

Jiang et al. "Sensitizer for EUV Chemically Amplified Resist: Metal versus Halogen" *J of Photopolymer Science and Technology*, vol. 32, No. 1, pp. 21-25 (2019). cited by applicant

Kang et al. "Optical Performance of Extreme Ultraviolet Lithography Mask with an Indium Tin Oxide Absorber" *J of Nanoscience and Nanotechnology*, vol. 12, pp. 3330-3333 (2012). cited by applicant

Kim et al. "Atomic layer deposition of transition metals for silicide contact formation: Growth characteristics and silicidation" *Microelectronic Engineering*, 106, pp. 69-75 (2013). cited by applicant

Lim et al. "Atomic layer deposition of transition metals" *Nature Materials*, vol. 2 pp. 749-754 (2003). cited by applicant

Maina et al. "Atomic layer deposition of transition metal films and nanostructures for electronic and catalytic applications" *Critical Reviews in Solid State and Materials Sciences*, Abstract, 2 pages (2020). cited by applicant

Miikkulainen et al. "Crystallinity of inorganic films grown by atomic layer deposition: Overview and general trends" *Journal of Applied Physics*, 112, 102 pages (2013). cited by applicant

Niskanen et al. "Radical-enhanced atomic layer deposition of metallic copper thin films" *Journal of the Electrochemical Society* 152(1) pp. G25-G28 (2004). cited by applicant

O'Hanlon "A User's Guide to Vacuum Technology" Third Edition, Chapter 19, pp. 359-378 (2003). cited by applicant

Ovanesyan et al. "Atomic Layer Deposition of SiC_xN_y Using Si₂Cl₆ and CH₃NH₂ Plasma" *Chem. Mater.* 2017, 29, pp. 6269-6278 (2017). cited by applicant

Park et al. "Superfilling CVD of copper using a catalytic surfactant" *Proceedings of the IEEE 2001 International Interconnect Technology Conference*, 3 pages (2001). cited by applicant

Petrov et al. "1,1,2,2-Tetrafluoroethyl-N,N-dimethylamine: a new selective fluorinating agent" *J of Fluorine Chemistry*, 109, pp. 25-31 (2001). cited by applicant

Popov et al. "Atomic Layer Deposition of PbI₂ Thin Films" *Chem. Mater.* 31, pp. 1101-1109 (2019). cited by applicant

Popovici et al. "High-performance (EOT<0.4nm, J_g~10⁻⁷ A/cm²) ALD-deposited Ru/SrTiO₃ stack for next generations DRAM pillar capacitor" *2018 IEEE International Electron Devices Meeting (IEDM)*, 4 pages (2018). cited by applicant

Puurunen "Surface Chemistry of Atomic Layer Deposition: A Case Study for the Trimethylaluminum/Water Process" *Journal of Applied Physics*, 97, 55 pages (2005). cited by applicant

Rahemi et al. "Variation in electron work function with temperature and its effect on the Young's modulus of metals" *Scripta Materialia*, 99, pp. 41-44 (2015). cited by applicant

SciFinder Search Results on hydrazido-based precursor for boron nitride films, search conducted Nov. 5, 2020, 5 pages (2020). cited by applicant

Shiba et al. "Stable yttrium oxyfluoride used in plasma process chamber" J. Vac. Sci. Technol. A, 35(2), 6 pages (2017). cited by applicant

Shigemoto et al. "Thermal cleaning of silicon nitride with fluorine and additive mixture" 1 page (2007). cited by applicant

Takaoka et al. "F-Propene-Dialkylamine Reaction Products as Fluorinating Agents" Bulletin of the Chemical Society of Japan, vol. 52 (11), pp. 3377-3380 (1979). cited by applicant

Tsoutsou et al. "Atomic layer deposition of $\text{La}_x\text{Zr}_{1-x}\text{O}_2$ -d ($x=0.25$) high-k dielectrics for advanced gate stacks" Applied Physics Letters, 94, 3 pages (2009). cited by applicant

U.S. Appl. No. 60/545,181, filed Feb. 13, 2004 in the names of Matthew G. Goodman et al., and entitled "Forced Flow Susceptor with Exit Holes and Veins for Improved Process" pp. 1-15 (2004). cited by applicant

U.S. Appl. No. 60/591,258, filed Jul. 26, 2004 in the names of Jeroen Stoutjesdijk et al., and entitled "Susceptor Support for Eliminating Backside Nodules" pp. 1-71 (2004). cited by applicant

U.S. Appl. No. 62/504,470, filed May 10, 2017 in the names of Jennifer Y. Sun et al., and entitled "Metal-Oxy-Fluoride Films for Chamber Components" pp. 1-82 (2017). cited by applicant

Van Asselt et al. "New Palladium Complexes of Cis-Fixed Bidentate Nitrogen Ligands as Catalysts for Carbon-Carbon Bond Formation" Organometallics, 11, pp. 1999-2001 (1992). cited by applicant

Van Asselt et al. "On the Mechanism of Formation of Homocoupled Products in the Carbon-Carbon Cross-Coupling Reaction Catalyzed by Palladium Complexes Containing Rigid Bidentate Nitrogen Ligands" Organometallics, 13, pp. 1972-1980 (1994). cited by applicant

Vayrynen et al. "Atomic Layer Deposition of Nickel Nitride Thin Films Using NiCl_2 (TMPDA) and Tert-Butylhydrazine as Precursors" Phy. Status Solidi A, 216, 9 pages (2019). cited by applicant

Vesters et al. "Sensitizers in EUV Chemically Amplified Resist: Mechanism of sensitivity improvement" Proc. SPIE 10583 Extreme Ultraviolet (EUV) Lithography IX, 1058307, 11 pages (2018). cited by applicant

Wilklow-Marnell et al. "First-row transitional-metal oxalate resists for EUV" J. Micro/Nanolith. MEMS MOEMS 17(4) Oct.-Dec. 2018, 9 pages (2018). cited by applicant

Zhang et al. "Mechanical Stability of Air-gap Interconnects" Proc. Future Fab International, pp. 81-87 (2008). cited by applicant

Zientara et al. Journal of the European Ceramic Society, 27, Abstract, 1 page (2007). cited by applicant

CNIPA; Office Action dated Mar. 30, 2021 in Application No. 201610131743.1. cited by applicant

CNIPA; Office Action dated Mar. 1, 2021 in Application No. 201710131319.1. cited by applicant

CNIPA; Notice of Allowance dated Mar. 30, 2021 in Application No. 201710762817.6. cited by applicant

CNIPA; Office Action dated Jan. 28, 2021 in Application No. 201711057557.9. cited by applicant

CNIPA; Office Action dated Jan. 26, 2021 in Application No. 201711057929.8. cited by applicant

CNIPA; Office Action dated Apr. 19, 2021 in Application No. 201711328250.8. cited by applicant

CNIPA; Notice of Allowance dated Apr. 7, 2021 in Application No. 201780044761.9. cited by applicant

CNIPA; Office Action dated Jan. 29, 2021 in Application No. 201780076230.8. cited by applicant

CNIPA; Office Action dated Jan. 27, 2021 in Application No. 201780076321.1. cited by applicant

CNIPA; Office Action dated Mar. 24, 2021 in Application No. 201880048547.5. cited by applicant

CNIPA; Office Action dated Dec. 22, 2020 in Application No. 201910378791.4. cited by applicant

CNIPA; Notice of Allowance dated Apr. 7, 2021 in Application No. 202030579755.8. cited by applicant

EPO; Extended European Search Report dated Apr. 6, 2021 in Application No. 21150514.4. cited by applicant

JPO; Notice of Allowance dated Apr. 6, 2021 in Application No. 2017-139817. cited by applicant

JPO; Office Action dated Mar. 31, 2021 in Application No. 2018-024655. cited by applicant

JPO; Notice of Allowance dated Mar. 17, 2021 in Application No. 2020-010953. cited by applicant

KIPO; Office Action dated Mar. 23, 2021 in Application No. 10-2014-0011765. cited by applicant

KIPO; Office Action dated Apr. 27, 2021 in Application No. 10-2014-0027217. cited by applicant

KIPO; Notice of Allowance dated Apr. 15, 2021 in Application No. 10-2014-0103853. cited by applicant

KIPO; Notice of Allowance dated Mar. 25, 2021 in Application No. 10-2014-0128626. cited by applicant

KIPO; Office Action dated Apr. 20, 2021 in Application No. 10-2014-0136089. cited by applicant

KIPO; Office Action dated Mar. 19, 2021 in Application No. 10-2014-0156196. cited by applicant

KIPO; Office Action dated Mar. 29, 2021 in Application No. 10-2014-0165685. cited by applicant

KIPO; Office Action dated Apr. 5, 2021 in Application No. 10-2015-0031720. cited by applicant

KIPO; Office Action dated Apr. 19, 2021 in Application No. 10-2015-0035094. cited by applicant

KIPO; Office Action dated Mar. 8, 2021 in Application No. 10-2017-0054647. cited by applicant

KIPO; Office Action dated Mar. 10, 2021 in Application No. 10-2017-0055703. cited by applicant

KIPO; Notice of Allowance dated May 24, 2021 in Application No. 10-2020-0101096. cited by applicant

KIPO; Notice of Allowance dated Apr. 1, 2021 in Application No. 30-2020-0030139 (M001). cited by applicant

KIPO; Notice of Allowance dated Apr. 1, 2021 in Application No. 30-2020-0030139 (M002). cited by applicant

TIPO; Notice of Allowance dated May 13, 2021 in Application No. 105122394. cited by applicant

TIPO; Notice of Allowance dated Mar. 5, 2021 in Application No. 105131284. cited by applicant

TIPO; Office Action dated Feb. 25, 2021 in Application No. 105134275. cited by applicant

TIPO; Notice of Allowance dated Mar. 4, 2021 in Application No. 106100823. cited by applicant

TIPO; Notice of Allowance dated May 6, 2021 in Application No. 106108522. cited by applicant

TIPO; Notice of Allowance dated Mar. 4, 2021 in Application No. 106111693. cited by applicant

TIPO; Office Action dated Feb. 25, 2021 in Application No. 106121797. cited by applicant

TIPO; Office Action dated Apr. 26, 2021 in Application No. 106122231. cited by applicant

TIPO; Office Action dated Apr. 22, 2021 in Application No. 106124126. cited by applicant

TIPO; Office Action dated Apr. 22, 2021 in Application No. 106124128. cited by applicant

TIPO; Notice of Allowance dated Mar. 25, 2021 in Application No. 106124130. cited by applicant

TIPO; Office Action dated Mar. 4, 2021 in Application No. 106127948. cited by applicant

TIPO; Office Action dated Mar. 15, 2021 in Application No. 106129971. cited by applicant

TIPO; Office Action dated Apr. 7, 2021 in Application No. 106135925. cited by applicant

TIPO; Office Action dated Apr. 7, 2021 in Application No. 106136905. cited by applicant

TIPO; Office Action dated Mar. 29, 2021 in Application No. 106143559. cited by applicant

TIPO; Office Action dated Mar. 31, 2021 in Application No. 106143570. cited by applicant

TIPO; Office Action dated Jan. 15, 2021 in Application No. 108142842. cited by applicant

TIPO; Office Action dated Jan. 25, 21 in Application No. 108143562. cited by applicant

TIPO; Notice of Allowance dated May 18, 2021 in Application No. 109300595. cited by applicant

TIPO; Notice of Allowance dated Mar. 30, 2021 in Application No. 109305460. cited by applicant

USPTO; Notice of Allowance dated Feb. 10, 2021 in U.S. Appl. No. 14/219,839. cited by applicant

USPTO; Advisory Action dated Apr. 13, 2021 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Non-Final Office Action dated May 12, 2021 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Final Office Action dated Feb. 24, 2021 in U.S. Appl. No. 15/262,990. cited by applicant

USPTO; Advisory Action dated Apr. 28, 2021 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Final Office Action dated Mar. 25, 2021 in U.S. Appl. No. 15/286,503. cited by applicant
USPTO; Advisory Action dated Apr. 30, 2021 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Final Office Action dated Jan. 7, 2021 in U.S. Appl. No. 15/380,909. cited by applicant
USPTO; Non-Final Office Action dated Feb. 9, 2021 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Final Office Action dated May 21, 2021 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Final Office Action dated Jun. 2, 2021 in U.S. Appl. No. 15/611,707. cited by applicant
USPTO; Advisory Action dated Mar. 25, 2021 in U.S. Appl. No. 15/636,307. cited by applicant
USPTO; Non-Final Office Action dated Apr. 21, 2021 in U.S. Appl. No. 15/636,307. cited by applicant
USPTO; Final Office Action dated Mar. 10, 2021 in U.S. Appl. No. 15/690,017. cited by applicant
USPTO; Notice of Allowance dated Apr. 16, 2021 in U.S. Appl. No. 15/691,241. cited by applicant
USPTO; Non-Final Office Action dated Mar. 18, 2021 in U.S. Appl. No. 15/835,328. cited by applicant
USPTO; Notice of Allowance dated Mar. 19, 2021 in U.S. Appl. No. 15/890,037. cited by applicant
USPTO; Final Office Action dated Apr. 19, 2021 in U.S. Appl. No. 15/909,705. cited by applicant
USPTO; Non-Final Office Action dated May 20, 2021 in U.S. Appl. No. 15/917,224. cited by applicant
USPTO; Non-Final Office Action dated Feb. 18, 2021 in U.S. Appl. No. 15/923,834. cited by applicant
USPTO; Advisory Action dated Apr. 28, 2021 in U.S. Appl. No. 15/940,729. cited by applicant
USPTO; Notice of Allowance dated Apr. 7, 2021 in U.S. Appl. No. 15/940,759. cited by applicant
USPTO; Advisory Action dated Feb. 22, 2021 in U.S. Appl. No. 15/962,980. cited by applicant
USPTO; Final Office Action dated Apr. 13, 2021 in U.S. Appl. No. 15/967,146. cited by applicant
USPTO; Advisory Action dated Jun. 2, 2021 in U.S. Appl. No. 15/967,146. cited by applicant
USPTO; Non-Final Office Action dated Feb. 19, 2021 in U.S. Appl. No. 15/974,948. cited by applicant
USPTO; Non-Final Office Action dated May 25, 2021 in U.S. Appl. No. 15/985,539. cited by applicant
USPTO; Non-Final Office Action dated Apr. 5, 2021 in U.S. Appl. No. 15/996,286. cited by applicant
USPTO; Non-Final Office Action dated Feb. 4, 2021 in U.S. Appl. No. 16/000,109. cited by applicant
USPTO; Non-Final Office Action dated May 19, 2021 in U.S. Appl. No. 16/000,125. cited by applicant
USPTO; Notice of Allowance dated Feb. 5, 2021 in U.S. Appl. No. 16/000,156. cited by applicant
USPTO; Non-Final Office Action dated Mar. 19, 2021 in U.S. Appl. No. 16/004,041. cited by applicant
USPTO; Final Office Action dated Mar. 23, 2021 in U.S. Appl. No. 16/039,817. cited by applicant
USPTO; Advisory Action dated May 28, 2021 in U.S. Appl. No. 16/039,817. cited by applicant
USPTO; Final Office Action dated Mar. 8, 2021 in U.S. Appl. No. 16/042,791. cited by applicant
USPTO; Advisory Action dated May 14, 2021 in U.S. Appl. No. 16/042,791. cited by applicant
USPTO; Notice of Allowance dated Mar. 10, 2021 in U.S. Appl. No. 16/055,532. cited by applicant
USPTO; Final Office Action dated Mar. 23, 2021 in U.S. Appl. No. 16/105,745. cited by applicant
USPTO; Final Office Action dated Mar. 24, 2021 in U.S. Appl. No. 16/105,761. cited by applicant
USPTO; Advisory Action dated May 20, 2021 in U.S. Appl. No. 16/105,761. cited by applicant
USPTO; Final Office Action dated Mar. 18, 2021 in U.S. Appl. No. 16/105,802. cited by applicant
USPTO; Advisory Action dated May 27, 2021 in U.S. Appl. No. 16/105,802. cited by applicant
USPTO; Advisory Action dated Feb. 17, 2021 in U.S. Appl. No. 16/108,950. cited by applicant

USPTO; Non-Final Office Action dated Apr. 13, 2021 in U.S. Appl. No. 16/108,950. cited by applicant

USPTO; Notice of Allowance dated Mar. 23, 2021 in U.S. Appl. No. 16/116,708. cited by applicant

USPTO; Advisory Action dated Mar. 10, 2021 in U.S. Appl. No. 16/117,530. cited by applicant

USPTO; Non-Final Office Action dated Mar. 23, 2021 in U.S. Appl. No. 16/151,074. cited by applicant

USPTO; Final Office Action dated May 6, 2021 in U.S. Appl. No. 16/152,260. cited by applicant

USPTO; Notice of Allowance dated Feb. 24, 2021 in U.S. Appl. No. 16/167,164. cited by applicant

USPTO; Non-Final Office Action dated Feb. 19, 2021 in U.S. Appl. No. 16/172,535. cited by applicant

USPTO; Final Office Action dated May 27, 2021 in U.S. Appl. No. 16/172,535. cited by applicant

USPTO; Advisory Action dated Apr. 5, 2021 in U.S. Appl. No. 16/176,517. cited by applicant

USPTO; Notice of Allowance dated May 13, 2021 in U.S. Appl. No. 16/176,517. cited by applicant

USPTO; Notice of Allowance dated Feb. 19, 2021 in U.S. Appl. No. 16/183,258. cited by applicant

USPTO; Notice of Allowance dated Feb. 10, 2021 in U.S. Appl. No. 16/205,899. cited by applicant

USPTO; Non-Final Office Action dated Mar. 29, 2021 in U.S. Appl. No. 16/206,589. cited by applicant

USPTO; Advisory Action dated Feb. 25, 2021 in U.S. Appl. No. 16/210,922. cited by applicant

USPTO; Final Office Action dated Mar. 29, 2021 in U.S. Appl. No. 16/219,555. cited by applicant

USPTO; Non-Final Office Action dated May 2, 2021 in U.S. Appl. No. 16/240,392. cited by applicant

USPTO; Advisory Action dated Jan. 26, 2021 in U.S. Appl. No. 16/251,534. cited by applicant

USPTO; Non-Final Office Action dated Feb. 23, 2021 in U.S. Appl. No. 16/251,534. cited by applicant

USPTO; Non-Final Office Action dated Mar. 30, 2021 in U.S. Appl. No. 16/252,567. cited by applicant

USPTO; Non-Final Office Action dated Mar. 4, 2021 in U.S. Appl. No. 16/252,569. cited by applicant

USPTO; Non-Final Office Action dated Apr. 20, 2021 in U.S. Appl. No. 16/397,045. cited by applicant

USPTO; Notice of Allowance dated Mar. 10, 2021 in U.S. Appl. No. 16/400,814. cited by applicant

USPTO; Non-Final Office Action dated May 7, 2021 in U.S. Appl. No. 16/423,824. cited by applicant

USPTO; Notice of Allowance dated Apr. 28, 2021 in U.S. Appl. No. 16/453,249. cited by applicant

USPTO; Final Office Action dated May 20, 2021 in U.S. Appl. No. 16/468,258. cited by applicant

USPTO; Notice of Allowance dated Apr. 26, 2021 in U.S. Appl. No. 16/517,122. cited by applicant

USPTO; Final Office Action dated May 12, 2021 in U.S. Appl. No. 16/546,543. cited by applicant

USPTO; Non-Final Office Action dated Feb. 1, 2021 in U.S. Appl. No. 16/563,473. cited by applicant

USPTO; Non-Final Office Action dated May 10, 2021 in U.S. Appl. No. 16/601,593. cited by applicant

USPTO; Non-Final Office Action dated Feb. 24, 2021 in U.S. Appl. No. 16/637,134. cited by applicant

USPTO; Notice of Allowance dated Jun. 2, 2021 in U.S. Appl. No. 16/637,134. cited by applicant

USPTO; Notice of Allowance dated Apr. 30, 2021 in U.S. Appl. No. 16/685,787. cited by applicant

USPTO; Non-Final Office Action dated Apr. 5, 2021 in U.S. Appl. No. 16/704,835. cited by applicant

USPTO; Final Office Action dated Feb. 22, 2021 in U.S. Appl. No. 16/713,311. cited by applicant

USPTO; Notice of Allowance dated May 20, 2021 in U.S. Appl. No. 16/752,514. cited by applicant

USPTO; Notice of Allowance dated May 24, 2021 in U.S. Appl. No. 16/765,125. cited by applicant

USPTO; Non-Final Office Action dated Apr. 15, 2021 in U.S. Appl. No. 16/789,138. cited by applicant

USPTO; Notice of Allowance dated May 12, 2021 in U.S. Appl. No. 16/800,114. cited by applicant

USPTO; Non-Final Office Action dated Apr. 5, 2021 in U.S. Appl. No. 16/816,078. cited by applicant

USPTO; Non-Final Office Action dated May 19, 2021 in U.S. Appl. No. 16/828,753. cited by applicant

USPTO; Non-Final Office Action dated May 4, 2021 in U.S. Appl. No. 16/872,045. cited by applicant

USPTO; Non-Final Office Action dated May 3, 2021 in U.S. Appl. No. 16/878,443. cited by applicant

USPTO; Non-Final Office Action dated Dec. 31, 2020 in U.S. Appl. No. 16/924,595. cited by applicant

USPTO; Non-Final Office Action dated May 25, 2021 in U.S. Appl. No. 16/935,280. cited by applicant

USPTO; Non-Final Office Action dated Apr. 14, 2021 in U.S. Appl. No. 17/009,093. cited by applicant

USPTO; Ex Parte Quayle Action dated Apr. 13, 2021 in U.S. Appl. No. 29/679,620. cited by applicant

USPTO; Notice of Allowance dated May 19, 2021 in U.S. Appl. No. 29/702,881. cited by applicant

Aubin et al. "Very low temperature (450° C.) selective epitaxial growth of heavily in situ boron-doped SiGe layers" *Semiconductor Science and Technology*, 30, 10 pages (2015). cited by applicant

Barnscheidt et al. "Highly boron-doped germanium layers on Si(001) grown by carbon-mediated epitaxy" *Semiconductor Science and Technology*, 33, 9 pages (2018). cited by applicant

Belyansky et al. "Low Temperature Borophosphosilicate Glass (BPSG) Process for High Aspect Ratio Gap Fill" www.electrochem.org/dl/ma/201/pdfs/0705.pdf, downloaded May 15, 2021, 1 page. cited by applicant

Cheremisin et al. "UV-laser modification and selective ion-beam etching of amorphous vanadium pentoxide thin films" *Phys. Status Solidi A, Applications and materials science*, 206 (7), pp. 1484-1487 (2009). cited by applicant

Dingemans et al. "Plasma-Assisted ALD for the Conformal Deposition of SiO₂: Process, Material and Electronic Properties" *J of the Electrochemical Society*, 159(3), H277-H285 (2012). cited by applicant

G02-1152 "Atomic Layer Deposition of Al₂O₃ with Alcohol Oxidants for Impeding Substrate Oxidation" Abstract. Oct. 16, 2019 (2019). cited by applicant

Imamura et al. "Cyclic C₄F₈ and O₂ plasma etching of TiO₂ for high-aspect-ratio three-dimensional devices" *Template for JJAP Regular Papers*, Jan. 2014, pp. 29 (2014). cited by applicant

Kim et al. "A process for topographically selective deposition on 3D nanostructures by ion implantation" *ACS Nano*, 10, 4, 4451-4458 (2016). cited by applicant

Lee et al. "Ultraviolet light enhancement of Ta₂O₅ dry etch rates" *J. of Vacuum Science & Technology B: Microelectronics and Nanometer Structures Processing, Measurement, and Phenomena*, 18, pp. 293-295 (2000). cited by applicant

Musschoot et al. "Atomic layer deposition of titanium nitride from TDMAT precursor" *Microelectronic Engineering*, 86, pp. 72-77 (2009). cited by applicant

Noircler et al. "Transmission electron microscopy characterization of low temperature boron doped silicon epitaxial films" *CrystEngComm*, 22(33), pp. 5464-5472 (2020). cited by applicant

Oyama et al. "Topotactic synthesis of vanadium nitride solid foams" *Journal of Materials Research* vol. 8. No. 6, pp. 1450-1454 (1993). cited by applicant

Rimoldi et al. "Atomic Layer Deposition of Rhenium-Aluminum Oxide Thin Films and ReO_x Incorporation in a Metal-Organic Framework" *Applied Materials & Interfaces*, 9, pp. 35067-35074 (2017). cited by applicant

Standard Motor Products LX249 Ignition Pick Up, Nov. 11, 2005, Amazon.com, May 10, 2021. URL: <https://www.amazon.com/Standard-Motor-Products-LX249-Ignition/dp/B000C7ZTS4/> (2005). cited by applicant

Tao et al. "Improved performance of GeON as charge storage layer in flash memory by optimal annealing" *Microelectronics Reliability*, vol. 52, pp. 2597-2601 (2012). cited by applicant

Wirths et al. "Low temperature RPCVD epitaxial growth of Si₂H₆ and Ge₂H₆" *Solid-State Electronics*, 88, pp. 2-9 (2013). cited by applicant

Yanguas-Gil et al. "Modulation of the Growth Per Cycle in Atomic Layer Deposition Using Reversible Surface Functionalization" *Chemistry of Materials*, 25, pp. 4849-4860 (2013). cited by applicant

CNIPA; Office Action dated Jan. 10, 2013 in Application No. 201080015699.9. cited by applicant

CNIPA; Office Action dated Aug. 1, 2013 in Application No. 201080015699.9. cited by applicant

CNIPA; Office Action dated Jan. 21, 2014 in Application No. 201080015699.9. cited by applicant

CNIPA; Office Action dated Jul. 25, 2014 in Application No. 201080015699.9. cited by applicant

CNIPA; Office Action dated Jan. 12, 2015 in Application No. 201080015699.9. cited by applicant

CNIPA; Notice of Allowance dated May 8, 2015 in Application No. 201080015699.9. cited by applicant

CNIPA; Office Action dated Dec. 10, 2013 in Application No. 201080020267.7. cited by applicant

CNIPA; Notice of Allowance dated Aug. 22, 2014 in Application No. 201080020267.7. cited by applicant

CNIPA; Office Action dated Jan. 21, 2013 in Application No. 201080020268.1. cited by applicant

CNIPA; Office Action dated Sep. 26, 2013 in Application No. 201080020268.1. cited by applicant

CNIPA; Office Action dated Apr. 3, 2014 in Application No. 201080020268.1. cited by applicant

CNIPA; Office Action dated Sep. 23, 2014 in Application No. 201080020268.1. cited by applicant

CNIPA; Office Action dated Apr. 7, 2015 in Application No. 201080020268.1. cited by applicant

CNIPA; Notice of Allowance dated Oct. 16, 2015 in Application No. 201080020268.1. cited by applicant

CNIPA; Office Action dated May 24, 2013 in Application No. 201080036764.6. cited by applicant

CNIPA; Office Action dated Jan. 2, 2014 in Application No. 201080036764.6. cited by applicant

CNIPA; Office Action dated Jul. 1, 2014 in Application No. 201080036764.6. cited by applicant

CNIPA; Notice of Allowance dated Oct. 24, 2014 in Application No. 201080036764.6. cited by applicant

CNIPA; Office Action dated Feb. 8, 2014 in Application No. 201110155056.0. cited by applicant

CNIPA; Office Action dated Sep. 16, 2014 in Application No. 201110155056.0. cited by applicant

CNIPA; Office Action dated Feb. 9, 2015 in Application No. 201110155056.0. cited by applicant

CNIPA; Notice of Allowance dated Aug. 26, 2015 in Application No. 201110155056.0. cited by applicant

CNIPA; Office Action dated Dec. 4, 2015 in Application No. 201210201995.9. cited by applicant

CNIPA; Office Action dated Jul. 14, 2016 in Application No. 201210201995.9. cited by applicant

CNIPA; Office Action dated Jan. 20, 2017 in Application No. 201210201995.9. cited by applicant

CNIPA; Notice of Allowance dated Apr. 13, 2017 in Application No. 201210201995.9. cited by applicant

CNIPA; Office Action dated Dec. 24, 2015 in Application No. 201280057466.4. cited by applicant

CNIPA; Notice of Allowance dated Jun. 16, 2016 in Application No. 201280057466.4. cited by applicant

CNIPA; Office Action dated Dec. 4, 2015 in Application No. 201280057542.1. cited by applicant

CNIPA; Office Action dated May 16, 2016 in Application No. 201280057542.1. cited by applicant

CNIPA; Office Action dated Sep. 9, 2016 in Application No. 201280057542.1. cited by applicant
CNIPA; Notice of Allowance dated Jan. 3, 2017 in Application No. 201280057542.1. cited by applicant

CNIPA; Office Action dated Dec. 5, 2016 in Application No. 201310412808.6. cited by applicant
CNIPA; Notice of Allowance dated Jul. 20, 2017 in Application No. 201310412808.6. cited by applicant

CNIPA; Office Action dated Feb. 5, 2018 in Application No. 201410331047.6. cited by applicant
CNIPA; Office Action dated Dec. 14, 2018 in Application No. 201410331047.6. cited by applicant
CNIPA; Notice of Allowance dated Jun. 14, 2019 in Application No. 201410331047.6. cited by applicant

CNIPA; Office Action dated Oct. 19, 2018 in Application No. 201510765170.3. cited by applicant
CNIPA; Office Action dated Jun. 28, 2019 in Application No. 201510765170.3. cited by applicant
CNIPA; Notice of Allowance dated Sep. 30, 2019 in Application No. 201510765170.3. cited by applicant

CNIPA; Office Action dated Oct. 31, 2018 in Application No. 201510765406.3. cited by applicant
CNIPA; Office Action dated Jun. 28, 2019 in Application No. 201510765406.3. cited by applicant
CNIPA; Notice of Allowance dated Dec. 27, 2019 in Application No. 201510765406.3. cited by applicant

CNIPA; Office Action dated Jan. 21, 2020 in Application No. 201610028064.1. cited by applicant
CNIPA; Notice of Allowance dated May 22, 2020 in Application No. 201610028064.1. cited by applicant

CNIPA; Office Action dated Mar. 26, 2020 in Application No. 201610131743.1. cited by applicant
CNIPA; Office Action dated Oct. 16, 2020 in Application No. 201610131743.1. cited by applicant
CNIPA; Office Action dated Mar. 14, 2019 in Application No. 201610141027.1. cited by applicant
CNIPA; Notice of Allowance dated Sep. 3, 2019 in Application No. 201610141027.1. cited by applicant

CNIPA; Office Action dated Jul. 23, 2019 in Application No. 201610897958.4. cited by applicant
CNIPA; Office Action dated Feb. 28, 2020 in Application No. 201610897958.4. cited by applicant
CNIPA; Office Action dated Jun. 16, 2020 in Application No. 201610897958.4. cited by applicant
CNIPA; Office Action dated Nov. 4, 2019 in Application No. 201610898822.5. cited by applicant
CNIPA; Office Action dated Jul. 3, 2020 in Application No. 201610898822.5. cited by applicant
CNIPA; Notice of Allowance dated Oct. 9, 2020 in Application No. 201610898822.5. cited by applicant

CNIPA; Office Action dated May 18, 2020 in Application No. 201610982040.X. cited by applicant
CNIPA; Notice of Allowance dated Dec. 21, 2020 in Application No. 201610982040.X. cited by applicant

CNIPA; Office Action dated May 29, 2020 in Application No. 201710131319.1. cited by applicant
CNIPA; Office Action dated Jun. 1, 2020 in Application No. 201710173292.2. cited by applicant
CNIPA; Notice of Allowance dated Nov. 4, 2020 in Application No. 201710173292.2. cited by applicant

CNIPA; Office Action dated Dec. 20, 2018 in Application No. 201710738549.4. cited by applicant
CNIPA; Office Action dated Nov. 20, 2020 in Application No. 201710762817.6. cited by applicant
CNIPA; Office Action dated Jun. 20, 2019 in Application No. 201711120632.1. cited by applicant
CNIPA; Notice of Allowance dated May 25, 2017 in Application No. 201730010308.9. cited by applicant

CNIPA; Office Action dated Aug. 31, 2020 in Application No. 201780044761.9. cited by applicant
CNIPA; Office Action dated Nov. 11, 2019 in Application No. 201810379112.0. cited by applicant
CNIPA; Office Action dated Apr. 27, 2020 in Application No. 201810379112.0. cited by applicant
CNIPA; Office Action dated Apr. 27, 2020 in Application No. 201810836604.8. cited by applicant
CNIPA; Office Action dated Jul. 30, 2020 in Application No. 201810836604.8. cited by applicant

CNIPA; Office Action dated Apr. 22, 2020 in Application No. 201810994464.7. cited by applicant
CNIPA; Notice of Allowance dated Oct. 24, 2018 in Application No. 201830060972.9. cited by applicant
CNIPA; Notice of Allowance dated Nov. 1, 2018 in Application No. 201830397219.9. cited by applicant
CNIPA; Notice of Allowance dated May 14, 2020 in Application No. 201930615780.4. cited by applicant
CNIPA; Notice of Allowance dated Apr. 27, 2020 in Application No. 201930660355.7. cited by applicant
CNIPA; Notice of Allowance dated Dec. 28, 2020 in Application No. 201930704935.1. cited by applicant
CNIPA; Notice of Allowance dated Jun. 2, 2020 in Application No. 202030048819.1. cited by applicant
CNIPA; Notice of Allowance dated May 15, 2020 in Application No. 202030048854.3. cited by applicant
CNIPA; Notice of Allowance dated May 15, 2020 in Application No. 202030053560.X. cited by applicant
CNIPA; Notice of Allowance dated Aug. 7, 2020 in Application No. 202030053609.1. cited by applicant
CNIPA; Notice of Allowance dated May 20, 2020 in Application No. 202030053634.X. cited by applicant
CNIPA; Notice of Allowance dated Dec. 10, 2020 in Application No. 202030352604.9. cited by applicant
CNIPA; Office Action dated Jan. 25, 2021 in Application No. 202030579755.8. cited by applicant
EPO; Office Action dated Jul. 29, 2008 in Application No. 00902194.0. cited by applicant
EPO; Supplementary European Search Report and Opinion dated Nov. 9, 2012 in Application No. 08798519.8. cited by applicant
EPO; Office Action dated Jul. 18, 2016 in Application No. 08798519.8. cited by applicant
EPO; Extended European Search Report dated Dec. 9, 2016 in Application No. 09767208.3. cited by applicant
EPO; Office Action dated Aug. 10, 2018 in Application No. 09767208.3. cited by applicant
EPO; Notice of Allowance dated Aug. 1, 2019 in Application No. 09767208.3. cited by applicant
EPO; Office Action dated Jun. 27, 2003 in Application No. 97929595.3. cited by applicant
EPO; Office Action dated Feb. 18, 2004 in Application No. 97929595.3. cited by applicant
EPO; Notice of Allowance dated Apr. 29, 2004 in Application No. 97929595.3. cited by applicant
EPO; Supplementary European Search Report and Opinion dated Jan. 5, 2017 in Application No. 09836647.9. cited by applicant
EPO; Office Action dated Feb. 28, 2018 in Application No. 09836647.9. cited by applicant
EPO; Office Action dated Jan. 11, 2019 in Application No. 09836647.9. cited by applicant
EPO; Notice of Allowance dated Oct. 16, 2019 in Application No. 09836647.9. cited by applicant
EPO; Notice of Allowance dated Jan. 10, 2001 in Application No. 98911263.6. cited by applicant
EPO; Office Action dated Jun. 6, 2001 in Application No. 98911277.6. cited by applicant
EPO; Office Action dated Jun. 27, 2003 in Application No. 98967122.7. cited by applicant
EPO; Notice of Allowance dated Feb. 23, 2004 in Application No. 98967122.7. cited by applicant
EPO; Office Action dated Sep. 22, 2003 in Application No. 99946447.2. cited by applicant
EPO; Office Action dated Nov. 25, 2003 in Application No. 99946447.2. cited by applicant
EPO; Office Action dated Mar. 30, 2004 in Application No. 99946447.2. cited by applicant
EPO; Office Action dated Jul. 19, 2004 in Application No. 99946447.2. cited by applicant
EPO; Notice of Allowance dated Sep. 22, 2004 in Application No. 99946447.2. cited by applicant
EPO; Supplementary European Search Report dated May 27, 2002 in Application No. 99973123.

cited by applicant

EPO; Notice of Allowance dated Jun. 11, 2004 in Application No. 99973123. cited by applicant

EPO; Extended European Search Report dated Apr. 28, 2014 in Application No. 11162225.4. cited by applicant

EPO; Notice of Allowance dated Feb. 3, 2015 in Application No. 11162225.4. cited by applicant

EPO; Extended European Search Report dated Nov. 29, 2019 in Application No. 19188826.2. cited by applicant

EPO; Extended European Search Report dated Mar. 3, 2020 in Application No. 19205558.0. cited by applicant

IPOS; Notice of Allowance dated Aug. 14, 2017 in Application No. 10201401237. cited by applicant

JPO; Office Action dated Aug. 10, 2009 in Application No. 2003029767. cited by applicant

JPO; Office Action dated Apr. 13, 2010 in Application No. 2003029767. cited by applicant

JPO; Notice of Allowance dated Jun. 24, 2010 in Application No. 2003029767. cited by applicant

JPO; Office Action dated Oct. 30, 2008 in Application No. 2004558313. cited by applicant

JPO; Office Action dated Feb. 19, 2009 in Application No. 2004558313. cited by applicant

JPO; Notice of Allowance dated Jun. 30, 2009 in Application No. 2004558313. cited by applicant

JPO; Office Action dated Mar. 16, 2012 in Application No. 2009-532567. cited by applicant

JPO; Notice of Allowance dated Jul. 23, 2012 in Application No. 2009-532567. cited by applicant

JPO; Notice of Allowance dated Mar. 29, 2013 in Application No. 2010-509478. cited by applicant

JPO; Office Action dated Dec. 20, 2011 in Application No. 2010-522075. cited by applicant

JPO; Office Action dated Apr. 11, 2012 in Application No. 2010-522075. cited by applicant

JPO; Office Action dated May 15, 2013 in Application No. 2010058415. cited by applicant

JPO; Office Action dated Oct. 30, 2013 in Application No. 2010058415. cited by applicant

JPO; Office Action dated Aug. 7, 2014 in Application No. 2010058415. cited by applicant

JPO; Notice of Allowance dated Dec. 18, 2014 in Application No. 2010058415. cited by applicant

JPO; Office Action dated Aug. 22, 2013 in Application No. 2010-153754. cited by applicant

JPO; Office Action dated Oct. 30, 2013 in Application No. 2010-193285. cited by applicant

JPO; Office Action dated Aug. 26, 2015 in Application No. 2011-284831. cited by applicant

JPO; Notice of Allowance dated Mar. 3, 2016 in Application No. 2011-284831. cited by applicant

JPO; Office Action dated May 31, 2012 in Application No. 2011-514650. cited by applicant

JPO; Office Action dated Sep. 11, 2012 in Application No. 2011-514650. cited by applicant

JPO; Notice of Allowance dated Dec. 10, 2012 in Application No. 2011-514650. cited by applicant

JPO; Office Action dated Dec. 10, 2014 in Application No. 2011090067. cited by applicant

JPO; Notice of Allowance dated Apr. 28, 2015 in Application No. 2011090067. cited by applicant

JPO; Office Action dated Jul. 14, 2016 in Application No. 2012-153698. cited by applicant

JPO; Notice of Allowance dated Oct. 21, 2016 in Application No. 2012-153698. cited by applicant

JPO; Office Action dated Dec. 20, 2013 in Application No. 2012-504786. cited by applicant

JPO; Office Action dated Jan. 25, 2014 in Application No. 2012-504786. cited by applicant

JPO; Office Action dated Dec. 1, 2014 in Application No. 2012-504786. cited by applicant

JPO; Notice of Allowance dated Jun. 12, 2015 in Application No. 2012504786. cited by applicant

JPO; Office Action dated Mar. 11, 2013 in Application No. 2012-509857. cited by applicant

JPO; Notice of Allowance dated Jun. 29, 2013 in Application No. 2012-509857. cited by applicant

JPO; Office Action dated May 19, 2017 in Application No. 2013-160173. cited by applicant

JPO; Notice of Allowance dated Aug. 23, 2017 in Application No. 2013-160173. cited by applicant

JPO; Office Action dated Aug. 14, 2017 in Application No. 2013-178344. cited by applicant

JPO; Office Action dated Jan. 23, 2018 in Application No. 2013-178344. cited by applicant

JPO; Notice of Allowance dated Jul. 24, 2018 in Application No. 2013-178344. cited by applicant

JPO; Office Action dated Apr. 3, 2018 in Application No. 2014-120675. cited by applicant

JPO; Notice of Allowance dated Jun. 6, 2018 in Application No. 2014-120675. cited by applicant

JPO; Office Action dated Mar. 28, 2018 in Application No. 2014-188835. cited by applicant
JPO; Notice of Allowance dated Jun. 6, 2018 in Application No. 2014-188835. cited by applicant
JPO; Office Action dated Apr. 12, 2018 in Application No. 2014-205548. cited by applicant
JPO; Notice of Allowance dated Dec. 19, 2019 in Application No. 2014205548. cited by applicant
JPO; Notice of Allowance dated May 14, 2018 in Application No. 2014-216540. cited by applicant
JPO; Office Action dated Jul. 20, 2018 in Application No. 2015-034774. cited by applicant
JPO; Office Action dated Jun. 27, 2019 in Application No. 2015034774. cited by applicant
JPO; Notice of Allowance dated Nov. 20, 2019 in Application No. 2015034774. cited by applicant
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KIPO; Notice of Allowance dated Oct. 22, 2020 in Application No. 30-2019-0058566 (M001). cited by applicant
KIPO; Notice of Allowance dated Oct. 22, 2020 in Application No. 30-2019-0058566 (M002).

cited by applicant
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TIPO; Notice of Allowance dated Dec. 10, 2020 in Application No. 106108152. cited by applicant
TIPO; Office Action dated Oct. 7, 2020 in Application No. 106108522. cited by applicant
TIPO; Office Action dated Aug. 21, 2020 in Application No. 106111548. cited by applicant
TIPO; Notice of Allowance dated Jan. 25, 2021 in Application No. 106111548. cited by applicant
TIPO; Office Action dated Oct. 8, 2020 in Application No. 106111693. cited by applicant
TIPO; Office Action dated Sep. 30, 2020 in Application No. 106113604. cited by applicant
TIPO; Office Action dated Jan. 6, 2021 in Application No. 106115126. cited by applicant
TIPO; Office Action dated Nov. 6, 2017 in Application No. 106117181. cited by applicant
TIPO; Notice of Allowance dated Jun. 5, 2018 in Application No. 106117181. cited by applicant
TIPO; Office Action dated Sep. 28, 2018 in Application No. 106119537. cited by applicant
TIPO; Office Action dated Jul. 15, 2020 in Application No. 106120365. cited by applicant
TIPO; Office Action dated Nov. 12, 2020 in Application No. 106120365. cited by applicant
TIPO; Office Action dated Nov. 18, 2020 in Application No. 106120902. cited by applicant
TIPO; Office Action dated Dec. 21, 2020 in Application No. 106122231. cited by applicant
TIPO; Office Action dated Oct. 8, 2020 in Application No. 106123203. cited by applicant
TIPO; Office Action dated Dec. 29, 2020 in Application No. 106124126. cited by applicant
TIPO; Office Action dated Dec. 9, 2020 in Application No. 106124128. cited by applicant
TIPO; Office Action dated Jan. 25, 2021 in Application No. 106124129. cited by applicant
TIPO; Office Action dated Dec. 9, 2020 in Application No. 106124130. cited by applicant
TIPO; Office Action dated Dec. 26, 2018 in Application No. 106127690. cited by applicant
TIPO; Office Action dated Jan. 20, 2021 in Application No. 106129491. cited by applicant
TIPO; Office Action dated Feb. 8, 2021 in Application No. 106133152. cited by applicant
TIPO; Office Action dated Aug. 31, 2018 in Application No. 106138119. cited by applicant
TIPO; Office Action dated Jun. 25, 2018 in Application No. 106138800. cited by applicant
TIPO; Office Action dated Jan. 7, 2019 in Application No. 106138800. cited by applicant
TIPO; Notice of Allowance dated Sep. 20, 2019 in Application No. 106138800. cited by applicant
TIPO; Office Action dated Oct. 3, 2018 in Application No. 106142731. cited by applicant
TIPO; Office Action dated Sep. 28, 2018 in Application No. 107112951. cited by applicant
TIPO; Office Action dated Aug. 27, 2019 in Application No. 107116804. cited by applicant
TIPO; Notice of Allowance dated Jan. 16, 2020 in Application No. 107116804. cited by applicant
TIPO; Office Action dated Nov. 20, 2018 in Application No. 107118271. cited by applicant
TIPO; Office Action dated Jun. 4, 2019 in Application No. 107123992. cited by applicant
TIPO; Office Action dated May 28, 2019 in Application No. 107125586. cited by applicant
TIPO; Office Action dated Jan. 17, 2020 in Application No. 107127688. cited by applicant
TIPO; Notice of Allowance dated Aug. 29, 2018 in Application No. 107300633. cited by applicant
TIPO; Notice of Allowance dated Feb. 21, 2019 in Application No. 107303723. cited by applicant
TIPO; Office Action dated Jun. 28, 2019 in Application No. 108102948. cited by applicant
TIPO; Office Action dated Dec. 24, 2019 in Application No. 108105002. cited by applicant
TIPO; Office Action dated Feb. 3, 2020 in Application No. 108114221. cited by applicant
TIPO; Office Action dated Nov. 13, 2019 in Application No. 108115406. cited by applicant
TIPO; Office Action dated Feb. 21, 2020 in Application No. 108120947. cited by applicant
TIPO; Notice of Allowance dated Mar. 12, 2020 in Application No. 108129100. cited by applicant
TIPO; Office Action dated Jun. 24, 2020 in Application No. 108135004. cited by applicant
TIPO; Office Action dated Jun. 29, 2020 in Application No. 108137226. cited by applicant

TIPO; Office Action dated Jul. 9, 2020 in Application No. 108137227. cited by applicant
TIPO; Office Action dated Aug. 17, 2020 in Application No. 108139018. cited by applicant
TIPO; Notice of Allowance dated Oct. 19, 2020 in Application No. 108306935. cited by applicant
TIPO; Notice of Allowance dated Oct. 19, 2020 in Application No. 108306935D01. cited by applicant
TIPO; Notice of Allowance dated Jul. 15, 2020 in Application No. 108307027. cited by applicant
TIPO; Notice of Allowance dated Oct. 19, 2020 in Application No. 108307027D01. cited by applicant
TIPO; Notice of Allowance dated Sep. 15, 2020 in Application No. 108307301. cited by applicant
TIPO; Notice of Allowance dated Nov. 24, 2020 in Application No. 108307599. cited by applicant
TIPO; Notice of Allowance dated Jan. 11, 2021 in Application No. 108307599D01. cited by applicant
TIPO; Notice of Allowance dated Sep. 25, 2020 in Application No. 108307759. cited by applicant
TIPO; Notice of Allowance dated Jan. 11, 2021 in Application No. 109300591. cited by applicant
TIPO; Notice of Allowance dated Jan. 11, 2021 in Application No. 109300593. cited by applicant
TIPO; Office Action dated Jan. 11, 2021 in Application No. 109300594. cited by applicant
USPTO; Non-Final Office Action dated Mar. 28, 2001 in U.S. Appl. No. 09/452,844. cited by applicant
USPTO; Final Office Action dated Sep. 11, 2001 in U.S. Appl. No. 09/452,844. cited by applicant
USPTO; Advisory Action dated Jan. 24, 2002 in U.S. Appl. No. 09/452,844. cited by applicant
USPTO; Non-Final Office Action dated Apr. 9, 2002 in U.S. Appl. No. 09/452,844. cited by applicant
USPTO; Final Office Action dated Sep. 27, 2002 in U.S. Appl. No. 09/452,844. cited by applicant
USPTO; Notice of Allowance dated Jun. 17, 2003 in U.S. Appl. No. 09/452,844. cited by applicant
USPTO; Non-Final Office Action dated Oct. 31, 2003 in U.S. Appl. No. 09/452,844. cited by applicant
USPTO; Notice of Allowance dated Mar. 26, 2004 in U.S. Appl. No. 09/452,844. cited by applicant
USPTO; Non-Final Office Action dated Dec. 10, 2002 in U.S. Appl. No. 09/771,673. cited by applicant
USPTO; Final Office Action dated May 29, 2003 in U.S. Appl. No. 09/771,673. cited by applicant
USPTO; Final Office Action dated Feb. 6, 2004 in U.S. Appl. No. 09/771,673. cited by applicant
USPTO; Advisory Action dated Apr. 20, 2004 in U.S. Appl. No. 09/771,673. cited by applicant
USPTO; Notice of Allowance dated Jul. 29, 2004 in U.S. Appl. No. 09/771,673. cited by applicant
USPTO; Non-Final Office Action dated Jan. 3, 2002 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Final Office Action dated Jun. 21, 2002 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Advisory Action dated Aug. 27, 2002 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Non-Final Action dated Nov. 27, 2002 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Final Office Action dated May 27, 2003 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Advisory Action dated Oct. 22, 2003 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Examiner's Answer to Appeal Brief dated May 6, 2004 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Decision on Appeal dated Feb. 16, 2005 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Non-Final Action dated May 19, 2005 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Final Action dated Nov. 21, 2005 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Notice of Allowance dated Feb. 23, 2006 in U.S. Appl. No. 09/807,580. cited by applicant
USPTO; Notice of Allowance dated Jul. 26, 2005 in U.S. Appl. No. 10/033,058. cited by applicant
USPTO; Non-Final Office Action dated Aug. 25, 2005 in U.S. Appl. No. 10/191,635. cited by applicant
USPTO; Final Office Action dated Apr. 25, 2006 in U.S. Appl. No. 10/191,635. cited by applicant

USPTO; Non-Final Office Action dated Nov. 20, 2006 in U.S. Appl. No. 10/191,635. cited by applicant

USPTO; Notice of Allowance dated May 21, 2007 in U.S. Appl. No. 10/191,635. cited by applicant

USPTO; Notice of Allowance dated Feb. 20, 2008 in U.S. Appl. No. 10/191,635. cited by applicant

USPTO; Non-Final Office Action dated May 13, 2003 in U.S. Appl. No. 10/222,229. cited by applicant

USPTO; Non-Final Office Action dated Oct. 22, 2003 in U.S. Appl. No. 10/222,229. cited by applicant

USPTO; Final Office Action dated Mar. 22, 2004 in U.S. Appl. No. 10/222,229. cited by applicant

USPTO; Advisory Action dated Oct. 7, 2004 in U.S. Appl. No. 10/222,229. cited by applicant

USPTO; Non-Final Office Action dated Dec. 22, 2004 in U.S. Appl. No. 10/222,229. cited by applicant

USPTO; Final Office Action dated Jun. 20, 2005 in U.S. Appl. No. 10/222,229. cited by applicant

USPTO; Advisory Action dated Nov. 16, 2005 in U.S. Appl. No. 10/222,229. cited by applicant

USPTO; Notice of Allowance dated Mar. 8, 2006 in U.S. Appl. No. 10/222,229. cited by applicant

USPTO; Non-Final Office Action dated Feb. 9, 2004 in U.S. Appl. No. 10/603,515. cited by applicant

USPTO; Notice of Allowance dated Jun. 18, 2004 in U.S. Appl. No. 10/603,515. cited by applicant

USPTO; Non-Final Office Action dated Jan. 26, 2005 in U.S. Appl. No. 10/838,510. cited by applicant

USPTO; Notice of Allowance dated Jul. 12, 2005 in U.S. Appl. No. 10/838,510. cited by applicant

USPTO; Non-Final Office Action dated Oct. 17, 2006 in U.S. Appl. No. 10/932,633. cited by applicant

USPTO; Non-Final Office Action dated Apr. 19, 2007 in U.S. Appl. No. 10/932,633. cited by applicant

USPTO; Notice of Allowance dated Sep. 10, 2007 in U.S. Appl. No. 10/932,633. cited by applicant

USPTO; Office Action dated Feb. 15, 2011 in U.S. Appl. No. 12/118,596. cited by applicant

USPTO; Notice of Allowance dated Aug. 4, 2011 in U.S. Appl. No. 12/118,596. cited by applicant

USPTO; Non-Final Office Action dated Apr. 28, 2010 in U.S. Appl. No. 12/121,085. cited by applicant

USPTO; Notice of Allowance dated Jul. 26, 2010 in U.S. Appl. No. 12/121,085. cited by applicant

USPTO; Notice of Allowance dated Oct. 4, 2010 in U.S. Appl. No. 12/121,085. cited by applicant

USPTO; Non-Final Office Action dated Sep. 13, 2010 in U.S. Appl. No. 12/140,809. cited by applicant

USPTO; Final Office Action dated Dec. 28, 2010 in U.S. Appl. No. 12/140,809. cited by applicant

USPTO; Notice of Allowance dated Mar. 17, 2011 in U.S. Appl. No. 12/140,809. cited by applicant

USPTO; Non-Final Office Action dated Mar. 15, 2011 in U.S. Appl. No. 12/193,924. cited by applicant

USPTO; Foma; Office Action dated Sep. 30, 2011 in U.S. Appl. No. 12/193,924. cited by applicant

USPTO; Non-Final Office Action dated Oct. 24, 2012 in U.S. Appl. No. 12/193,924. cited by applicant

USPTO; Final Office Action dated Apr. 17, 2013 in U.S. Appl. No. 12/193,924. cited by applicant

USPTO; Advisory Action dated Jul. 9, 2013 in U.S. Appl. No. 12/193,924. cited by applicant

USPTO; Non-Final Office Action dated Jul. 28, 2011 in U.S. Appl. No. 12/330,096. cited by applicant

USPTO; Final Office Action dated Jan. 13, 2012 in U.S. Appl. No. 12/330,096. cited by applicant

USPTO; Notice of Allowance dated Mar. 6, 2012 in U.S. Appl. No. 12/330,096. cited by applicant

USPTO; Non-Final Office Action dated Mar. 20, 2012 in U.S. Appl. No. 12/330,096. cited by applicant

USPTO; Notice of Allowance dated Jun. 7, 2012 in U.S. Appl. No. 12/330,096. cited by applicant

USPTO; Non-Final Office Action dated Apr. 1, 2010 in U.S. Appl. No. 12/357,174. cited by applicant

USPTO; Final Office Action dated Sep. 1, 2010 in U.S. Appl. No. 12/357,174. cited by applicant

USPTO; Notice of Allowance dated Dec. 13, 2010 in U.S. Appl. No. 12/357,174. cited by applicant

USPTO; Non-Final Office Action dated Dec. 29, 2010 in U.S. Appl. No. 12/362,023. cited by applicant

USPTO; Non-Final Office Action dated Jul. 26, 2011 in U.S. Appl. No. 12/416,809. cited by applicant

USPTO; Final Office Action dated Dec. 6, 2011 in U.S. Appl. No. 12/416,809. cited by applicant

USPTO; Notice of Allowance dated Apr. 2, 2012 in U.S. Appl. No. 12/416,809. cited by applicant

USPTO; Advisory Action dated Feb. 3, 2012 in U.S. Appl. No. 12/416,809. cited by applicant

USPTO; Notice of Allowance dated Jun. 16, 2011 in U.S. Appl. No. 12/430,751. cited by applicant

USPTO; Notice of Allowance dated Jul. 27, 2011 in U.S. Appl. No. 12/430,751. cited by applicant

USPTO; Non-Final Office Action dated Aug. 3, 2011 in U.S. Appl. No. 12/436,300. cited by applicant

USPTO; Final Office Action dated Jan. 23, 2012 in U.S. Appl. No. 12/436,300. cited by applicant

USPTO; Advisory Action dated Mar. 6, 2012 in U.S. Appl. No. 12/436,300. cited by applicant

USPTO; Non-Final Office Action dated May 22, 2012 in U.S. Appl. No. 12/436,300. cited by applicant

USPTO; Notice of Allowance dated Nov. 28, 2012 in U.S. Appl. No. 12/436,300. cited by applicant

USPTO; Non-Final Office Action dated Apr. 11, 2012 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Final Office Action dated Sep. 26, 2012 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Non-Final Office Action dated May 31, 2013 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Final Office Action dated Oct. 17, 2013 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Non-Final Office Action dated Feb. 4, 2014 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Final Office Action dated Jun. 23, 2014 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Advisory Action dated Oct. 1, 2014 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Non-Final Office Action dated Feb. 3, 2015 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Final Office Action dated May 13, 2015 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Non-Final Office Action dated Oct. 14, 2015 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Final Office Action dated Dec. 31, 2015 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Notice of Allowance dated Feb. 3, 2016 in U.S. Appl. No. 12/436,306. cited by applicant

USPTO; Non-Final Office Action dated Aug. 3, 2011 in U.S. Appl. No. 12/436,315. cited by applicant

USPTO; Notice of Allowance dated Nov. 17, 2011 in U.S. Appl. No. 12/436,315. cited by applicant

USPTO; Notice of Allowance dated Oct. 1, 2010 in U.S. Appl. No. 12/467,017. cited by applicant

USPTO; Non-Final Office Action dated Mar. 18, 2010 in U.S. Appl. No. 12/489,252. cited by applicant

USPTO; Notice of Allowance dated Sep. 2, 2010 in U.S. Appl. No. 12/489,252. cited by applicant

USPTO; Non-Final Office Action dated Dec. 15, 2010 in U.S. Appl. No. 12/553,759. cited by applicant

USPTO; Final Office Action dated May 4, 2011 in U.S. Appl. No. 12/553,759. cited by applicant

USPTO; Advisory Action dated Jul. 13, 2011 in U.S. Appl. No. 12/553,759. cited by applicant

USPTO; Non-Final Office Action dated Sep. 6, 2011 in U.S. Appl. No. 12/553,759. cited by applicant

USPTO; Notice of Allowance dated Jan. 27, 2012 in U.S. Appl. No. 12/553,759. cited by applicant
USPTO; Non-Final Office Action dated Oct. 19, 2012 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Final Office Action dated May 8, 2013 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Advisory Action dated Jul. 23, 2013 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Non-Final Office Action dated Apr. 8, 2015 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Final Office Action dated Oct. 22, 2015 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Advisory Action dated Mar. 4, 2016 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Non-Final Office Action dated Jun. 30, 2016 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Final Office Action dated Feb. 10, 2017 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Advisory Action dated May 16, 2017 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Non-Final Office Action dated Nov. 29, 2017 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Final Office Action dated Aug. 10, 2018 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Notice of Allowance dated Apr. 4, 2019 in U.S. Appl. No. 12/618,355. cited by applicant
USPTO; Non-Final Office Action dated Feb. 16, 2012 in U.S. Appl. No. 12/618,419. cited by applicant
USPTO; Final Office Action dated Jun. 22, 2012 in U.S. Appl. No. 12/618,419. cited by applicant
USPTO; Non-Final Office Action dated Nov. 27, 2012 in U.S. Appl. No. 12/618,419. cited by applicant
USPTO; Advisory Action dated Aug. 9, 2012 in U.S. Appl. No. 12/618,419. cited by applicant
USPTO; Notice of Allowance dated Apr. 12, 2013 in U.S. Appl. No. 12/618,419. cited by applicant
USPTO; Non-Final Office Action dated Jun. 12, 2013 in U.S. Appl. No. 12/618,419. cited by applicant
USPTO; Notice of Allowance dated Oct. 9, 2013 in U.S. Appl. No. 12/618,419. cited by applicant
USPTO; Non-Final Office Action dated Dec. 6, 2011 in U.S. Appl. No. 12/718,731. cited by applicant
USPTO; Notice of Allowance dated Mar. 16, 2012 in U.S. Appl. No. 12/718,731. cited by applicant
USPTO; Office Action dated Feb. 26, 2013 in U.S. Appl. No. 12/754,223. cited by applicant
USPTO; Final Office Action dated Jun. 28, 2013 in U.S. Appl. No. 12/754,223. cited by applicant
USPTO; Office Action dated Feb. 25, 2014 in U.S. Appl. No. 12/754,223. cited by applicant
USPTO; Final Office Action dated Jul. 14, 2014 in U.S. Appl. No. 12/754,223. cited by applicant
USPTO; Non-Final Office Action dated Mar. 25, 2015 in U.S. Appl. No. 12/754,223. cited by applicant
USPTO; Final Office Action dated Aug. 12, 2015 in U.S. Appl. No. 12/754,223. cited by applicant
USPTO; Notice of Allowance dated May 23, 2016 in U.S. Appl. No. 12/754,223. cited by applicant
USPTO; Office Action dated Apr. 23, 2013 in U.S. Appl. No. 12/763,037. cited by applicant
USPTO; Final Office Action dated Oct. 21, 2013 in U.S. Appl. No. 12/763,037. cited by applicant
USPTO; Office Action dated Oct. 8, 2014 in U.S. Appl. No. 12/763,037. cited by applicant
USPTO; Notice of Allowance dated Jan. 27, 2015 in U.S. Appl. No. 12/763,037. cited by applicant
USPTO; Non-Final Office Action dated Jan. 24, 2011 in U.S. Appl. No. 12/778,808. cited by applicant
USPTO; Notice of Allowance dated May 9, 2011 in U.S. Appl. No. 12/778,808. cited by applicant
USPTO; Notice of Allowance dated Oct. 12, 2012 in U.S. Appl. No. 12/832,739. cited by applicant
USPTO; Non-Final Office Action dated Oct. 16, 2012 in U.S. Appl. No. 12/847,848. cited by applicant
USPTO; Final Office Action dated Apr. 22, 2013 in U.S. Appl. No. 12/847,848. cited by applicant
USPTO; Advisory Action dated Jul. 1, 2013 in U.S. Appl. No. 12/847,848. cited by applicant

USPTO; Notice of Allowance dated Jan. 16, 2014 in U.S. Appl. No. 12/847,848. cited by applicant
USPTO; Office Action dated Dec. 6, 2012 in U.S. Appl. No. 12/854,818. cited by applicant
USPTO; Final Office Action dated Mar. 13, 2013 in U.S. Appl. No. 12/854,818. cited by applicant
USPTO; Office Action dated Aug. 30, 2013 in U.S. Appl. No. 12/854,818. cited by applicant
USPTO; Final Office Action dated Mar. 26, 2014 in U.S. Appl. No. 12/854,818. cited by applicant
USPTO; Office Action dated Jun. 3, 2014 in U.S. Appl. No. 12/854,818. cited by applicant
USPTO; Non-Final Office Action dated Jul. 11, 2012 in U.S. Appl. No. 12/875,889. cited by applicant
USPTO; Notice of Allowance dated Jan. 4, 2013 in U.S. Appl. No. 12/875,889. cited by applicant
USPTO; Notice of Allowance dated Jan. 9, 2012 in U.S. Appl. No. 12/901,323. cited by applicant
USPTO; Non-Final Office Action dated Nov. 20, 2013 in U.S. Appl. No. 12/910,607. cited by applicant
USPTO; Final Office Action dated Apr. 28, 2014 in U.S. Appl. No. 12/910,607. cited by applicant
USPTO; Advisory Action dated Jul. 9, 2014 in U.S. Appl. No. 12/910,607. cited by applicant
USPTO; Notice of Allowance dated Aug. 15, 2014 in U.S. Appl. No. 12/910,607. cited by applicant
USPTO; Non-Final Office Action dated Oct. 24, 2012 in U.S. Appl. No. 12/940,906. cited by applicant
USPTO; Final Office Action dated Feb. 13, 2013 in U.S. Appl. No. 12/940,906. cited by applicant
USPTO; Notice of Allowance dated Apr. 23, 2013 in U.S. Appl. No. 12/940,906. cited by applicant
USPTO; Non-Final Office Action dated Dec. 7, 2012 in U.S. Appl. No. 12/953,870. cited by applicant
USPTO; Final Office Action dated Apr. 22, 2013 in U.S. Appl. No. 12/953,870. cited by applicant
USPTO; Advisory Action dated Jul. 8, 2013 in U.S. Appl. No. 12/953,870. cited by applicant
USPTO; Non-Final Office Action dated Aug. 28, 2013 in U.S. Appl. No. 12/953,870. cited by applicant
USPTO; Final Office Action dated Apr. 17, 2014 in U.S. Appl. No. 12/953,870. cited by applicant
USPTO; Non-Final Office Action dated Sep. 19, 2012 in U.S. Appl. No. 13/016,735. cited by applicant
USPTO; Final Office Action dated Feb. 11, 2013 in U.S. Appl. No. 13/016,735. cited by applicant
USPTO; Notice of Allowance dated Apr. 24, 2013 in U.S. Appl. No. 13/016,735. cited by applicant
USPTO; Non-Final Office Action dated Apr. 4, 2012 in U.S. Appl. No. 13/030,438. cited by applicant
USPTO; Final Office Action dated Aug. 22, 2012 in U.S. Appl. No. 13/030,438. cited by applicant
USPTO; Notice of Allowance dated Oct. 24, 2012 in U.S. Appl. No. 13/030,438. cited by applicant
USPTO; Non-Final Office Action dated Dec. 3, 2012 in U.S. Appl. No. 13/040,013. cited by applicant
USPTO; Notice of Allowance dated May 3, 2013 in U.S. Appl. No. 13/040,013. cited by applicant
USPTO; Non-Final Office Action dated Feb. 15, 2012 in U.S. Appl. No. 13/085,531. cited by applicant
USPTO; Notice of Allowance dated Jul. 12, 2012 in U.S. Appl. No. 13/085,531. cited by applicant
USPTO; Notice of Allowance dated Sep. 13, 2012 in U.S. Appl. No. 13/085,698. cited by applicant
USPTO; Non-Final Office Action dated Mar. 29, 2013 in U.S. Appl. No. 13/094,402. cited by applicant
USPTO; Final Office Action dated Jul. 17, 2013 in U.S. Appl. No. 13/094,402. cited by applicant
USPTO; Notice of Allowance dated Sep. 30, 2013 in U.S. Appl. No. 13/094,402. cited by applicant
USPTO; Office Action dated Oct. 7, 2013 in U.S. Appl. No. 13/102,980. cited by applicant
USPTO; Final Office Action dated Mar. 25, 2014 in U.S. Appl. No. 13/102,980. cited by applicant
USPTO; Advisory Action dated Jun. 12, 2014 in U.S. Appl. No. 13/102,980. cited by applicant
USPTO; Notice of Allowance dated Jul. 3, 2014 in U.S. Appl. No. 13/102,980. cited by applicant
USPTO; Notice of Allowance dated Sep. 17, 2014 in U.S. Appl. No. 13/102,980. cited by applicant

USPTO; Non-Final Office Action dated Jul. 17, 2014 in U.S. Appl. No. 13/154,271. cited by applicant

USPTO; Final Office Action dated Jan. 2, 2015 in U.S. Appl. No. 13/154,271. cited by applicant

USPTO; Non-Final Office Action dated May 27, 2015 in U.S. Appl. No. 13/154,271. cited by applicant

USPTO; Final Office Action dated Nov. 23, 2015 in U.S. Appl. No. 13/154,271. cited by applicant

USPTO; Notice of Allowance dated Feb. 10, 2016 in U.S. Appl. No. 13/154,271. cited by applicant

USPTO; Non-Final Office Action dated Jun. 27, 2016 in U.S. Appl. No. 13/166,367. cited by applicant

USPTO; Final Office Action dated Dec. 30, 2016 in U.S. Appl. No. 13/166,367. cited by applicant

USPTO; Advisory Action dated Apr. 21, 2017 in U.S. Appl. No. 13/166,367. cited by applicant

USPTO; Notice of Allowance dated Jun. 28, 2017 in U.S. Appl. No. 13/166,367. cited by applicant

USPTO; Non-Final Office Action dated Oct. 27, 2014 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Final Office Action dated May 26, 2015 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Non-Final Office Action dated Sep. 1, 2015 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Final Office Action dated Mar. 3, 2016 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Non-Final Office Action dated Jun. 9, 2016 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Final Office Action dated Dec. 9, 2016 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Advisory Action dated May 13, 2016 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Advisory Action dated Feb. 15, 2017 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Non-Final Office Action dated Apr. 26, 2017 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Final Office Action dated Nov. 2, 2017 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Advisory Action dated Feb. 8, 2018 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Non-Final Office Action dated Apr. 6, 2018 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Final Office Action dated Nov. 2, 2018 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Advisory Action dated Feb. 4, 2019 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Notice of Allowance dated Apr. 4, 2019 in U.S. Appl. No. 13/169,951. cited by applicant

USPTO; Non-Final Office Action dated Jun. 24, 2014 in U.S. Appl. No. 13/181,407. cited by applicant

USPTO; Final Office Action dated Sep. 24, 2014 in U.S. Appl. No. 13/181,407. cited by applicant

USPTO; Advisory Action dated Dec. 17, 2014 in U.S. Appl. No. 13/181,407. cited by applicant

USPTO; Non-Final Office Action dated Jan. 2, 2015 in U.S. Appl. No. 13/181,407. cited by applicant

USPTO; Final Office Action dated Apr. 8, 2015 in U.S. Appl. No. 13/181,407. cited by applicant

USPTO; Non-Final Office Action dated Jan. 23, 2013 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Final Office Action dated Jul. 29, 2013 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Advisory Action dated Nov. 7, 2013 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Non-Final Office Action dated Jul. 16, 2014 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Final Office Action dated Feb. 17, 2015 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Advisory Action dated May 18, 2015 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Non-Final Office Action dated Aug. 10, 2015 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Final Office Action dated Feb. 12, 2016 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Non-Final Office Action dated Dec. 15, 2016 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Final Office Action dated Jun. 15, 2017 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Advisory Action dated Oct. 4, 2017 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Non-Final Office Action dated Jul. 26, 2018 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Final Office Action dated Dec. 28, 2018 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Non-Final Office Action dated Aug. 19, 2019 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Notice of Allowance dated Jul. 6, 2020 in U.S. Appl. No. 13/184,351. cited by applicant

USPTO; Non-Final Office Action dated Sep. 17, 2014 in U.S. Appl. No. 13/187,300. cited by applicant

USPTO; Final Office Action dated Apr. 15, 2015 in U.S. Appl. No. 13/187,300. cited by applicant

USPTO; Non-Final Office Action dated Apr. 7, 2016 in U.S. Appl. No. 13/187,300. cited by applicant

USPTO; Final Office Action dated Sep. 23, 2016 in U.S. Appl. No. 13/187,300. cited by applicant

USPTO; Non-Final Office Action dated Jan. 30, 2017 in U.S. Appl. No. 13/187,300. cited by applicant

USPTO; Final Office Action dated Aug. 9, 2017 in U.S. Appl. No. 13/187,300. cited by applicant

USPTO; Examiner's Answer to Appeal Brief dated Apr. 20, 2018 in U.S. Appl. No. 13/187,300. cited by applicant

USPTO; Patent Board Decision dated Sep. 23, 2019 in U.S. Appl. No. 13/187,300. cited by applicant

USPTO; Non-Final Office Action dated Oct. 1, 2012 in U.S. Appl. No. 13/191,762. cited by applicant

USPTO; Final Office Action dated Apr. 10, 2013 in U.S. Appl. No. 13/191,762. cited by applicant

USPTO; Notice of Allowance dated Aug. 15, 2013 in U.S. Appl. No. 13/191,762. cited by applicant

USPTO; Non-Final Office Action dated Oct. 22, 2012 in U.S. Appl. No. 13/238,960. cited by applicant

USPTO; Final Office Action dated May 3, 2013 in U.S. Appl. No. 13/238,960. cited by applicant

USPTO; Non-Final Office Action dated Apr. 26, 2013 in U.S. Appl. No. 13/250,721. cited by applicant

USPTO; Notice of Allowance dated Sep. 11, 2013 in U.S. Appl. No. 13/250,721. cited by applicant

USPTO; Non-Final Office Action dated Jul. 2, 2014 in U.S. Appl. No. 13/283,408. cited by applicant

USPTO; Final Office Action dated Jan. 29, 2015 in U.S. Appl. No. 13/283,408. cited by applicant

USPTO; Non-Final Office Action dated Jun. 17, 2015 in U.S. Appl. No. 13/283,408. cited by applicant

USPTO; Final Office Action dated Dec. 18, 2015 in U.S. Appl. No. 13/283,408. cited by applicant

USPTO; Advisory Action dated Mar. 28, 2016 in U.S. Appl. No. 13/283,408. cited by applicant

USPTO; Notice of Allowance dated Mar. 28, 2016 in U.S. Appl. No. 13/283,408. cited by applicant

USPTO; Non-Final Office Action dated Jul. 30, 2014 in U.S. Appl. No. 13/284,642. cited by applicant

USPTO; Notice of Allowance dated Feb. 11, 2015 in U.S. Appl. No. 13/284,642. cited by applicant

USPTO; Non-Final Office Action dated Jan. 28, 2014 in U.S. Appl. No. 13/312,591. cited by applicant

USPTO; Final Office Action dated May 14, 2014 in U.S. Appl. No. 13/312,591. cited by applicant

USPTO; Advisory Action dated Aug. 26, 2014 in U.S. Appl. No. 13/312,591. cited by applicant

USPTO; Non-Final Office Action dated Nov. 26, 2014 in U.S. Appl. No. 13/312,591. cited by applicant

USPTO; Final Office Action dated Mar. 20, 2015 in U.S. Appl. No. 13/312,591. cited by applicant
USPTO; Notice of Allowance dated May 14, 2015 in U.S. Appl. No. 13/312,591. cited by applicant
USPTO; Notice of Allowance dated Jun. 11, 2015 in U.S. Appl. No. 13/312,591. cited by applicant
USPTO; Non-Final Office Action dated Apr. 9, 2014 in U.S. Appl. No. 13/333,420. cited by applicant
USPTO; Notice of Allowance dated Sep. 15, 2014 in U.S. Appl. No. 13/333,420. cited by applicant
USPTO; Office Action dated Feb. 11, 2013 in U.S. Appl. No. 13/339,609. cited by applicant
USPTO; Final Office Action dated May 17, 2013 in U.S. Appl. No. 13/339,609. cited by applicant
USPTO; Office Action dated Aug. 29, 2013 in U.S. Appl. No. 13/339,609. cited by applicant
USPTO; Final Office Action dated Dec. 18, 2013 in U.S. Appl. No. 13/339,609. cited by applicant
USPTO; Notice of Allowance dated Apr. 7, 2014 in U.S. Appl. No. 13/339,609. cited by applicant
USPTO; Non-Final Office Action dated Oct. 10, 2012 in U.S. Appl. No. 13/406,791. cited by applicant
USPTO; Final Office Action dated Jan. 31, 2013 in U.S. Appl. No. 13/406,791. cited by applicant
USPTO; Advisory Action dated Mar. 27, 2013 in U.S. Appl. No. 13/406,791. cited by applicant
USPTO; Non-Final Office Action dated Apr. 25, 2013 in U.S. Appl. No. 13/406,791. cited by applicant
USPTO; Final Office Action dated Aug. 23, 2013 in U.S. Appl. No. 13/406,791. cited by applicant
USPTO; Advisory Action dated Oct. 29, 2013 in U.S. Appl. No. 13/406,791. cited by applicant
USPTO; Non-Final Office Action dated Dec. 4, 2013 in U.S. Appl. No. 13/406,791. cited by applicant
USPTO; Final Office Action dated Apr. 21, 2014 in U.S. Appl. No. 13/406,791. cited by applicant
USPTO; Non-Final Office Action dated Jan. 14, 2013 in U.S. Appl. No. 13/410,970. cited by applicant
USPTO; Notice of Allowance dated Feb. 14, 2013 in U.S. Appl. No. 13/410,970. cited by applicant
USPTO; Non-Final Office Action dated Feb. 13, 2014 in U.S. Appl. No. 13/411,271. cited by applicant
USPTO; Non-Final Office Action dated Jul. 31, 2014 in U.S. Appl. No. 13/411,271. cited by applicant
USPTO; Advisory Action dated Apr. 22, 2015 in U.S. Appl. No. 13/411,271. cited by applicant
USPTO; Final Office Action dated Jan. 16, 2015 in U.S. Appl. No. 13/411,271. cited by applicant
USPTO; Notice of Allowance dated Oct. 6, 2015 in U.S. Appl. No. 13/411,271. cited by applicant
USPTO; Office Action dated Feb. 4, 2014 in U.S. Appl. No. 13/439,528. cited by applicant
USPTO; Final Office Action dated Jul. 8, 2014 in U.S. Appl. No. 13/439,528. cited by applicant
USPTO; Notice of Allowance dated Oct. 21, 2014 in U.S. Appl. No. 13/439,528. cited by applicant
USPTO; Non-Final Office Action dated Apr. 11, 2013 in U.S. Appl. No. 13/450,368. cited by applicant
USPTO; Notice of Allowance dated Jul. 17, 2013 in U.S. Appl. No. 13/450,368. cited by applicant
USPTO; Non-Final Office Action dated May 23, 2013 in U.S. Appl. No. 13/465,340. cited by applicant
USPTO; Final Office Action dated Oct. 30, 2013 in U.S. Appl. No. 13/465,340. cited by applicant
USPTO; Notice of Allowance dated Feb. 12, 2014 in U.S. Appl. No. 13/465,340. cited by applicant
USPTO; Non-Final Office Action dated Oct. 17, 2013 in U.S. Appl. No. 13/493,897. cited by applicant
USPTO; Notice of Allowance dated Mar. 20, 2014 in U.S. Appl. No. 13/493,897. cited by applicant
USPTO; Non-Final Office Action dated Dec. 20, 2013 in U.S. Appl. No. 13/535,214. cited by applicant
USPTO; Final Office Action dated Jun. 18, 2014 in U.S. Appl. No. 13/535,214. cited by applicant
USPTO; Notice of Allowance dated Oct. 23, 2014 in U.S. Appl. No. 13/535,214. cited by applicant
USPTO; Non-Final Office Action dated Sep. 11, 2013 in U.S. Appl. No. 13/550,419. cited by

applicant

USPTO; Final Office Action dated Jan. 27, 2014 in U.S. Appl. No. 13/550,419. cited by applicant

USPTO; Advisory Action dated Mar. 31, 2014 in U.S. Appl. No. 13/550,419. cited by applicant

USPTO; Notice of Allowance dated May 29, 2014 in U.S. Appl. No. 13/550,419. cited by applicant

USPTO; Non-Final Office Action dated Aug. 8, 2014 in U.S. Appl. No. 13/563,066. cited by applicant

USPTO; Final Office Action dated Feb. 12, 2015 in U.S. Appl. No. 13/563,066. cited by applicant

USPTO; Advisory Action dated Apr. 16, 2015 in U.S. Appl. No. 13/563,066. cited by applicant

USPTO; Notice of Allowance dated Jun. 12, 2015 in U.S. Appl. No. 13/563,066. cited by applicant

USPTO; Notice of Allowance dated Jul. 16, 2015 in U.S. Appl. No. 13/563,066. cited by applicant

USPTO; Non-Final Office Action dated May 28, 2013 in U.S. Appl. No. 13/563,274. cited by applicant

USPTO; Notice of Allowance dated Sep. 27, 2013 in U.S. Appl. No. 13/563,274. cited by applicant

USPTO; Non-Final Office Action dated Nov. 7, 2013 in U.S. Appl. No. 13/565,564. cited by applicant

USPTO; Final Office Action dated Feb. 28, 2014 in U.S. Appl. No. 13/565,564. cited by applicant

USPTO; Advisory Action dated May 5, 2014 in U.S. Appl. No. 13/565,564. cited by applicant

USPTO; Non-Final Office Action dated Jul. 2, 2014 in U.S. Appl. No. 13/565,564. cited by applicant

USPTO; Notice of Allowance dated Nov. 3, 2014 in U.S. Appl. No. 13/565,564. cited by applicant

USPTO; Notice of Allowance dated Sep. 13, 2013 in U.S. Appl. No. 13/566,069. cited by applicant

USPTO; Non-Final Office Action dated Aug. 30, 2013 in U.S. Appl. No. 13/570,067. cited by applicant

USPTO; Notice of Allowance dated Jan. 6, 2014 in U.S. Appl. No. 13/570,067. cited by applicant

USPTO; Non-Final Office Action dated Oct. 15, 2014 in U.S. Appl. No. 13/597,043. cited by applicant

USPTO; Final Office Action dated Mar. 13, 2015 in U.S. Appl. No. 13/597,043. cited by applicant

USPTO; Notice of Allowance dated Aug. 28, 2015 in U.S. Appl. No. 13/597,043. cited by applicant

USPTO; Non-Final Office Action dated Feb. 12, 2015 in U.S. Appl. No. 13/597,108. cited by applicant

USPTO; Final Office Action dated Jun. 1, 2015 in U.S. Appl. No. 13/597,108. cited by applicant

USPTO; Advisory Action dated Sep. 2, 2015 in U.S. Appl. No. 13/597,108. cited by applicant

USPTO; Non-Final Office Action dated Dec. 8, 2015 in U.S. Appl. No. 13/597,108. cited by applicant

USPTO; Final Office Action dated Jun. 2, 2016 in U.S. Appl. No. 13/597,108. cited by applicant

USPTO; Non-Final Office Action dated Sep. 15, 2016 in U.S. Appl. No. 13/597,108. cited by applicant

USPTO; Notice of Allowance dated Mar. 7, 2017 in U.S. Appl. No. 13/597,108. cited by applicant

USPTO; Notice of Allowance dated Mar. 27, 2014 in U.S. Appl. No. 13/604,498. cited by applicant

USPTO; Non-Final Office Action dated Nov. 15, 2013 in U.S. Appl. No. 13/612,538. cited by applicant

USPTO; Non-Final Office Action dated Jul. 10, 2014 in U.S. Appl. No. 13/612,538. cited by applicant

USPTO; Notice of Allowance dated Feb. 25, 2015 in U.S. Appl. No. 13/612,538. cited by applicant

USPTO; Non-Final Office Action dated Apr. 15, 2015 in U.S. Appl. No. 13/646,403. cited by applicant

USPTO; Final Office Action dated Oct. 15, 2015 in U.S. Appl. No. 13/646,403. cited by applicant

USPTO; Notice of Allowance dated Feb. 2, 2016 in U.S. Appl. No. 13/646,403. cited by applicant

USPTO; Non-Final Office Action dated May 15, 2014 in U.S. Appl. No. 13/646,471. cited by applicant

USPTO; Final Office Action dated Aug. 18, 2014 in U.S. Appl. No. 13/646,471. cited by applicant
USPTO; Advisory Action dated Nov. 14, 2014 in U.S. Appl. No. 13/646,471. cited by applicant
USPTO; Non-Final Office Action dated Dec. 16, 2014 in U.S. Appl. No. 13/646,471. cited by applicant
USPTO; Final Office Action dated Apr. 21, 2015 in U.S. Appl. No. 13/646,471. cited by applicant
USPTO; Non-Final Office Action dated Aug. 19, 2015 in U.S. Appl. No. 13/646,471. cited by applicant
USPTO; Final Office Action dated Jan. 22, 2016 in U.S. Appl. No. 13/646,471. cited by applicant
USPTO; Advisory Action dated Apr. 15, 2016 in U.S. Appl. No. 13/646,471. cited by applicant
USPTO; Non-Final Office Action dated Jun. 2, 2016 in U.S. Appl. No. 13/646,471. cited by applicant
USPTO; Final Office Action dated Oct. 20, 2016 in U.S. Appl. No. 13/646,471. cited by applicant
USPTO; Non-Final Office Action dated May 28, 2015 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Final Office Action dated Dec. 14, 2017 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Final Office Action dated Nov. 19, 2015 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Advisory Action dated Feb. 12, 2016 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Non-Final Office Action dated May 10, 2016 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Final Office Action dated Sep. 20, 2016 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Advisory Action dated Dec. 29, 2016 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Non-Final Office Action dated May 17, 2017 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Non-Final Office Action dated Dec. 14, 2017 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Advisory Action dated Apr. 19, 2018 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Non-Final Office Action dated Sep. 20, 2018 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Final Office Action dated Mar. 15, 2019 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Non-Final Office Action dated Sep. 18, 2019 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Notice of Allowance dated Feb. 10, 2020 in U.S. Appl. No. 13/651,144. cited by applicant
USPTO; Advisory Action dated Jun. 26, 2015 in U.S. Appl. No. 13/791,246. cited by applicant
USPTO; Non-Final Office Action dated Jun. 18, 2015 in U.S. Appl. No. 13/665,366. cited by applicant
USPTO; Final Office Action dated Mar. 1, 2016 in U.S. Appl. No. 13/665,366. cited by applicant
USPTO; Advisory Action dated May 13, 2016 in U.S. Appl. No. 13/665,366. cited by applicant
USPTO; Non-Final Office Action dated Jun. 17, 2016 in U.S. Appl. No. 13/665,366. cited by applicant
USPTO; Final Office Action dated May 3, 2017 in U.S. Appl. No. 13/665,366. cited by applicant
USPTO; Non-Final Office Action dated Apr. 3, 2015 in U.S. Appl. No. 13/677,133. cited by applicant
USPTO; Notice of Allowance dated Aug. 4, 2015 in U.S. Appl. No. 13/677,133. cited by applicant
USPTO; Notice of Allowance dated Aug. 24, 2015 in U.S. Appl. No. 13/677,133. cited by applicant
USPTO; Office Action dated Jun. 2, 2014 in U.S. Appl. No. 13/677,151. cited by applicant
USPTO; Final Office Action dated Nov. 14, 2014 in U.S. Appl. No. 13/677,151. cited by applicant
USPTO; Notice of Allowance dated Feb. 26, 2015 in U.S. Appl. No. 13/677,151. cited by applicant
USPTO; Notice of Allowance dated Mar. 17, 2015 in U.S. Appl. No. 13/677,151. cited by applicant
USPTO; Non-Final Office Action dated Aug. 20, 2013 in U.S. Appl. No. 13/679,502. cited by applicant

USPTO; Final Office Action dated Feb. 25, 2014 in U.S. Appl. No. 13/679,502. cited by applicant
USPTO; Notice of Allowance dated May 2, 2014 in U.S. Appl. No. 13/679,502. cited by applicant
USPTO; Non-Final Office Action dated Jul. 21, 2015 in U.S. Appl. No. 13/727,324. cited by applicant
USPTO; Final Office Action dated Jan. 22, 2016 in U.S. Appl. No. 13/727,324. cited by applicant
USPTO; Advisory Action dated Apr. 6, 2016 in U.S. Appl. No. 13/727,324. cited by applicant
USPTO; Non-Final Office Action dated May 25, 2016 in U.S. Appl. No. 13/727,324. cited by applicant
USPTO; Final Office Action dated Dec. 1, 2016 in U.S. Appl. No. 13/727,324. cited by applicant
USPTO; Notice of Allowance dated Mar. 1, 2017 in U.S. Appl. No. 13/727,324. cited by applicant
USPTO; Non-Final Office Action dated Oct. 24, 2013 in U.S. Appl. No. 13/749,878. cited by applicant
USPTO; Non-Final Office Action dated Jun. 18, 2014 in U.S. Appl. No. 13/749,878. cited by applicant
USPTO; Final Office Action dated Dec. 10, 2014 in U.S. Appl. No. 13/749,878. cited by applicant
USPTO; Notice of Allowance Mar. 13, 2015 dated in U.S. Appl. No. 13/749,878. cited by applicant
USPTO; Non-Final Office Action dated Sep. 16, 2013 in U.S. Appl. No. 13/760,160. cited by applicant
USPTO; Final Office Action dated Dec. 27, 2013 in U.S. Appl. No. 13/760,160. cited by applicant
USPTO; Non-Final Office Action dated Jun. 4, 2014 in U.S. Appl. No. 13/760,160. cited by applicant
USPTO; Final Office Action dated Sep. 25, 2014 in U.S. Appl. No. 13/760,160. cited by applicant
USPTO; Final Office Action dated Jan. 28, 2015 in U.S. Appl. No. 13/760,160. cited by applicant
USPTO; Final Office Action dated May 12, 2015 in U.S. Appl. No. 13/760,160. cited by applicant
USPTO; Notice of Allowance dated Oct. 21, 2015 in U.S. Appl. No. 13/760,160. cited by applicant
USPTO; Notice of Allowance dated Jan. 20, 2016 in U.S. Appl. No. 13/760,160. cited by applicant
USPTO; Office Action dated Apr. 23, 2014 in U.S. Appl. No. 13/784,362. cited by applicant
USPTO; Notice of Allowance dated Aug. 13, 2014 in U.S. Appl. No. 13/784,362. cited by applicant
USPTO; Non-Final Office Action dated Dec. 19, 2013 in U.S. Appl. No. 13/784,388. cited by applicant
USPTO; Notice of Allowance dated Jun. 4, 2014 in U.S. Appl. No. 13/784,388. cited by applicant
USPTO; Non-Final Office Action dated Sep. 19, 2014 in U.S. Appl. No. 13/791,246. cited by applicant
USPTO; Final Office Action dated Mar. 25, 2015 in U.S. Appl. No. 13/791,246. cited by applicant
USPTO; Non-Final Office Action dated Oct. 26, 2015 in U.S. Appl. No. 13/791,246. cited by applicant
USPTO; Final Office Action dated Apr. 20, 2016 in U.S. Appl. No. 13/791,246. cited by applicant
USPTO; Advisory Action dated Jul. 13, 2016 in U.S. Appl. No. 13/791,246. cited by applicant
USPTO; Non-Final Office Action dated Aug. 11, 2016 in U.S. Appl. No. 13/791,246. cited by applicant
USPTO; Notice of Allowance dated Oct. 19, 2016 in U.S. Appl. No. 13/791,246. cited by applicant
USPTO; Notice of Allowance dated Nov. 25, 2016 in U.S. Appl. No. 13/791,246. cited by applicant
USPTO; Non-Final Office Action dated Nov. 6, 2015 in U.S. Appl. No. 13/791,339. cited by applicant
USPTO; Final Office Action dated Apr. 12, 2016 in U.S. Appl. No. 13/791,339. cited by applicant
USPTO; Advisory Action dated Jul. 14, 2016 in U.S. Appl. No. 13/791,339. cited by applicant
USPTO; Notice of Allowance dated Aug. 24, 2016 in U.S. Appl. No. 13/791,339. cited by applicant
USPTO; Non-Final Office Action dated Mar. 21, 2014 in U.S. Appl. No. 13/799,708. cited by applicant
USPTO; Notice of Allowance dated Oct. 31, 2014 in U.S. Appl. No. 13/799,708. cited by applicant

USPTO; Non-Final Office Action dated Oct. 9, 2014 in U.S. Appl. No. 13/874,708. cited by applicant

USPTO; Notice of Allowance dated Mar. 10, 2015 in U.S. Appl. No. 13/874,708. cited by applicant

USPTO; Notice of Allowance dated Apr. 10, 2014 in U.S. Appl. No. 13/901,341. cited by applicant

USPTO; Notice of Allowance dated Jun. 6, 2014 in U.S. Appl. No. 13/901,341. cited by applicant

USPTO; Non-Final Office Action dated Jan. 2, 2015 in U.S. Appl. No. 13/901,372. cited by applicant

USPTO; Final Office Action dated Apr. 16, 2015 in U.S. Appl. No. 13/901,372. cited by applicant

USPTO; Notice of Allowance dated Aug. 5, 2015 in U.S. Appl. No. 13/901,372. cited by applicant

USPTO; Advisory Action dated Jun. 29, 2015 in U.S. Appl. No. 13/901,372. cited by applicant

USPTO; Non-Final Office Action dated Jul. 8, 2015 in U.S. Appl. No. 13/901,400. cited by applicant

USPTO; Final Office Action dated Jan. 14, 2016 in U.S. Appl. No. 13/901,400. cited by applicant

USPTO; Notice of Allowance dated Apr. 12, 2016 in U.S. Appl. No. 13/901,400. cited by applicant

USPTO; Non-Final Office Action dated Apr. 24, 2014 in U.S. Appl. No. 13/912,666. cited by applicant

USPTO; Final Office Action dated Sep. 25, 2014 in U.S. Appl. No. 13/912,666. cited by applicant

USPTO; Advisory Action dated Dec. 11, 2014 in U.S. Appl. No. 13/912,666. cited by applicant

USPTO; Non-Final Office Action dated Jan. 26, 2015 in U.S. Appl. No. 13/912,666. cited by applicant

USPTO; Notice of Allowance dated Jun. 25, 2015 in U.S. Appl. No. 13/912,666. cited by applicant

USPTO; Non-Final Office Action dated Dec. 16, 2014 in U.S. Appl. No. 13/915,732. cited by applicant

USPTO; Final Office Action dated Apr. 10, 2015 in U.S. Appl. No. 13/915,732. cited by applicant

USPTO; Notice of Allowance dated Jun. 19, 2015 in U.S. Appl. No. 13/915,732. cited by applicant

USPTO; Notice of Allowance dated Mar. 17, 2015 in U.S. Appl. No. 13/923,197. cited by applicant

USPTO; Non-Final Office Action dated Sep. 12, 2014 in U.S. Appl. No. 13/941,134. cited by applicant

USPTO; Notice of Allowance dated Jan. 20, 2015 in U.S. Appl. No. 13/941,134. cited by applicant

USPTO; Non-Final Office Action dated Jul. 30, 2015 in U.S. Appl. No. 13/941,216. cited by applicant

USPTO; Final Office Action dated Mar. 1, 2016 in U.S. Appl. No. 13/941,216. cited by applicant

USPTO; Non-Final Office Action dated Jun. 15, 2016 in U.S. Appl. No. 13/941,216. cited by applicant

USPTO; Notice of Allowance dated Sep. 13, 2016 in U.S. Appl. No. 13/941,216. cited by applicant

USPTO; Notice of Allowance dated Nov. 14, 2016 in U.S. Appl. No. 13/941,216. cited by applicant

USPTO; Non-Final Office Action dated Jan. 14, 2014 in U.S. Appl. No. 13/941,226. cited by applicant

USPTO; Non-Final Office Action dated Jul. 8, 2014 in U.S. Appl. No. 13/941,226. cited by applicant

USPTO; Non-Final Office Action dated Feb. 3, 2015 in U.S. Appl. No. 13/941,226. cited by applicant

USPTO; Final Office Action dated Feb. 12, 2016 in U.S. Appl. No. 13/941,226. cited by applicant

USPTO; Advisory Action dated Jul. 29, 2016 in U.S. Appl. No. 13/941,226. cited by applicant

USPTO; Non-Final Office Action dated Aug. 8, 2017 in U.S. Appl. No. 13/941,226. cited by applicant

USPTO; Notice of Allowance dated Aug. 13, 2018 in U.S. Appl. No. 13/941,226. cited by applicant

USPTO; Notice of Allowance dated Oct. 3, 2018 in U.S. Appl. No. 13/941,226. cited by applicant

USPTO; Non-Final Office Action dated Oct. 30, 2014 in U.S. Appl. No. 13/948,055. cited by applicant

USPTO; Notice of Allowance dated Feb. 27, 2015 in U.S. Appl. No. 13/948,055. cited by applicant
USPTO; Notice of Allowance dated Mar. 31, 2015 in U.S. Appl. No. 13/948,055. cited by applicant
USPTO; Non-Final Office Action dated Jun. 29, 2015 in U.S. Appl. No. 13/966,782. cited by applicant
USPTO; Final Office Action dated Jan. 4, 2016 in U.S. Appl. No. 13/966,782. cited by applicant
USPTO; Notice of Allowance dated Mar. 21, 2016 in U.S. Appl. No. 13/966,782. cited by applicant
USPTO; Notice of Allowance dated Oct. 7, 2015 in U.S. Appl. No. 13/973,777. cited by applicant
USPTO; Non-Final Office Action dated Feb. 20, 2015 in U.S. Appl. No. 14/018,231. cited by applicant
USPTO; Notice of Allowance dated Jul. 20, 2015 in U.S. Appl. No. 14/018,231. cited by applicant
USPTO; Non-Final Office Action dated Apr. 7, 2015 in U.S. Appl. No. 14/018,345. cited by applicant
USPTO; Final Office Action dated Sep. 14, 2015 in U.S. Appl. No. 14/018,345. cited by applicant
USPTO; Notice of Allowance dated Jan. 14, 2016 in U.S. Appl. No. 14/018,345. cited by applicant
USPTO; Notice of Allowance dated Mar. 17, 2016 in U.S. Appl. No. 14/018,345. cited by applicant
USPTO; Non-Final Office Action dated Mar. 26, 2015 in U.S. Appl. No. 14/031,982. cited by applicant
USPTO; Final Office Action dated Aug. 28, 2015 in U.S. Appl. No. 14/031,982. cited by applicant
USPTO; Notice of Allowance dated Nov. 17, 2015 in U.S. Appl. No. 14/031,982. cited by applicant
USPTO; Non-Final Office Action dated Jan. 2, 2015 in U.S. Appl. No. 14/040,196. cited by applicant
USPTO; Non-Final Office Action dated Apr. 28, 2015 in U.S. Appl. No. 14/040,196. cited by applicant
USPTO; Notice of Allowance dated Sep. 11, 2015 in U.S. Appl. No. 14/040,196. cited by applicant
USPTO; Non-Final Action dated Dec. 3, 2015 in U.S. Appl. No. 14/050,150. cited by applicant
USPTO; Final Office Action dated Jun. 15, 2016 in U.S. Appl. No. 14/050,150. cited by applicant
USPTO; Final Office Action dated Jul. 8, 2016 in U.S. Appl. No. 14/050,150. cited by applicant
USPTO; Notice of Allowance dated Oct. 20, 2016 in U.S. Appl. No. 14/050,150. cited by applicant
USPTO; Non-Final Office Action dated Dec. 15, 2014 in U.S. Appl. No. 14/065,114. cited by applicant
USPTO; Final Office Action dated Jun. 19, 2015 in U.S. Appl. No. 14/065,114. cited by applicant
USPTO; Advisory Action dated Aug. 24, 2015 in U.S. Appl. No. 14/065,114. cited by applicant
USPTO; Non-Final Office Action dated Oct. 7, 2015 in U.S. Appl. No. 14/065,114. cited by applicant
USPTO; Notice of Allowance dated Feb. 22, 2016 in U.S. Appl. No. 14/065,114. cited by applicant
USPTO; Non-Final Office Action dated Nov. 14, 2014 in U.S. Appl. No. 14/069,244. cited by applicant
USPTO; Notice of Allowance dated Mar. 25, 2015 in U.S. Appl. No. 14/069,244. cited by applicant
USPTO; Non-Final Office Action dated Mar. 19, 2015 in U.S. Appl. No. 14/079,302. cited by applicant
USPTO; Final Office Action dated Sep. 1, 2015 in U.S. Appl. No. 14/079,302. cited by applicant
USPTO; Non-Final Office Action dated Dec. 23, 2015 in U.S. Appl. No. 14/079,302. cited by applicant
USPTO; Non-Final Office Action dated Apr. 27, 2016 in U.S. Appl. No. 14/079,302. cited by applicant
USPTO; Final Office Action dated Aug. 22, 2016 in U.S. Appl. No. 14/079,302. cited by applicant
USPTO; Notice of Allowance dated Dec. 14, 2016 in U.S. Appl. No. 14/079,302. cited by applicant
USPTO; Advisory Action dated Nov. 10, 2015 in U.S. Appl. No. 14/079,302. cited by applicant
USPTO; Non-Final Office Action dated Sep. 9, 2015 in U.S. Appl. No. 14/090,750. cited by applicant

USPTO; Final Office Action dated Feb. 11, 2016 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Advisory Action dated May 5, 2016 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Non-Final Office Action dated Jun. 14, 2016 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Advisory Action dated Dec. 21, 2016 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Advisory Action dated Jan. 30, 2018 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Final Office Action dated Sep. 28, 2016 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Non-Final Office Action dated Jun. 23, 2017 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Final Office Action dated Nov. 17, 2017 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Non-Final Office Action dated Mar. 12, 2018 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Notice of Allowance dated Aug. 29, 2018 in U.S. Appl. No. 14/090,750. cited by applicant
USPTO; Non-Final Office Action dated Mar. 19, 2015 in U.S. Appl. No. 14/166,462. cited by applicant
USPTO; Notice of Allowance dated Sep. 3, 2015 in U.S. Appl. No. 14/166,462. cited by applicant
USPTO; Non-Final Office Action dated Nov. 17, 2015 in U.S. Appl. No. 14/172,220. cited by applicant
USPTO; Notice of Allowance dated Apr. 22, 2016 in U.S. Appl. No. 14/172,220. cited by applicant
USPTO; Office Action dated May 29, 2014 in U.S. Appl. No. 14/183,187. cited by applicant
USPTO; Final Office Action dated Nov. 7, 2014 in U.S. Appl. No. 14/183,187. cited by applicant
USPTO; Advisory Action dated Feb. 20, 2015 in U.S. Appl. No. 14/183,187. cited by applicant
USPTO; Non-Final Office Action dated Mar. 16, 2015 in U.S. Appl. No. 14/183,187. cited by applicant
USPTO; Final Office Action dated Jul. 10, 2015 in U.S. Appl. No. 14/183,187. cited by applicant
USPTO; Notice of Allowance dated Aug. 31, 2015 in U.S. Appl. No. 14/183,187. cited by applicant
USPTO; Non-Final Office Action dated Jan. 11, 2016 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Final Office Action dated Aug. 25, 2016 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Advisory Action dated Jan. 12, 2017 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Non-Final Office Action dated Mar. 23, 2017 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Final Office Action dated Oct. 5, 2017 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Advisory Action dated Jan. 3, 2018 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Non-Final Office Action dated Apr. 18, 2018 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Final Office Action dated Jan. 25, 2019 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Non-Final Office Action dated Aug. 8, 2019 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Notice of Allowance dated Feb. 10, 2020 in U.S. Appl. No. 14/188,760. cited by applicant
USPTO; Non-Final Office Action dated Oct. 8, 2015 in U.S. Appl. No. 14/218,374. cited by applicant
USPTO; Final Office Action dated Feb. 23, 2016 in U.S. Appl. No. 14/218,374. cited by applicant
USPTO; Advisory Action dated Apr. 29, 2016 in U.S. Appl. No. 14/218,374. cited by applicant
USPTO; Notice of Allowance dated Aug. 5, 2016 in U.S. Appl. No. 14/218,374. cited by applicant
USPTO; Non-Final Office Action dated Jul. 15, 2016 in U.S. Appl. No. 14/218,690. cited by applicant
USPTO; Final Office Action dated Nov. 14, 2016 in U.S. Appl. No. 14/218,690. cited by applicant
USPTO; Non-Final Office Action dated Apr. 6, 2017 in U.S. Appl. No. 14/218,690. cited by applicant

USPTO; Final Office Action dated Jul. 20, 2017 in U.S. Appl. No. 14/218,690. cited by applicant
USPTO; Non-Final Office Action dated Jan. 11, 2018 in U.S. Appl. No. 14/218,690. cited by applicant
USPTO; Final Office Action dated May 24, 2018 in U.S. Appl. No. 14/218,690. cited by applicant
USPTO; Notice of Allowance dated Sep. 24, 2018 in U.S. Appl. No. 14/218,690. cited by applicant
USPTO; Non-Final Office Action dated Sep. 22, 2015 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Final Office Action dated Mar. 25, 2016 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Non-Final Office Action dated Dec. 22, 2016 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Advisory Action dated Jun. 30, 2016 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Final Office Action dated Jul. 6, 2017 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Non-Final Office Action dated Mar. 27, 2018 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Final Office Action dated Nov. 1, 2018 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Advisory Action dated Jan. 22, 2019 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Non-Final Office Action dated Jul. 15, 2019 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Final Office Action dated Jan. 27, 2020 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Advisory Action dated Apr. 24, 2020 in U.S. Appl. No. 14/219,839. cited by applicant
USPTO; Advisory Action dated Apr. 24, 2020 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Non-Final Office Action dated Nov. 25, 2015 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Final Office action dated May 19, 2016 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Advisory Action dated Aug. 22, 2016 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Non-Final Office Action dated Dec. 23, 2016 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Final Office action dated Jul. 6, 2017 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Advisory Action dated Oct. 5, 2017 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Non-Final Office Action dated Apr. 6, 2018 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Final Office Action dated Nov. 2, 2018 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Advisory Action dated Jan. 22, 2019 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Non-Final Office Action dated Jun. 24, 2019 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Final Office Action dated Jan. 13, 2020 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Non-Final Office Action dated Dec. 1, 2020 in U.S. Appl. No. 14/219,879. cited by applicant
USPTO; Non-Final Office Action dated Sep. 18, 2015 in U.S. Appl. No. 14/244,689. cited by applicant
USPTO; Notice of Allowance dated Feb. 11, 2016 in U.S. Appl. No. 14/244,689. cited by applicant
USPTO; Non-Final Office Action dated Oct. 7, 2015 in U.S. Appl. No. 14/246,969. cited by applicant
USPTO; Final Office Action dated May 4, 2016 in U.S. Appl. No. 14/246,969. cited by applicant
USPTO; Advisory Action dated Aug. 2, 2016 in U.S. Appl. No. 14/246,969. cited by applicant
USPTO; Non-Final Office Action dated Aug. 12, 2016 in U.S. Appl. No. 14/246,969. cited by applicant
USPTO; Notice of Allowance dated Feb. 27, 2017 in U.S. Appl. No. 14/246,969. cited by applicant
USPTO; Non-Final Office Action dated Nov. 20, 2015 in U.S. Appl. No. 14/260,701. cited by applicant

USPTO; Notice of Allowance dated Jun. 2, 2016 in U.S. Appl. No. 14/260,701. cited by applicant
USPTO; Notice of Allowance dated Feb. 23, 2016 in U.S. Appl. No. 14/327,134. cited by applicant
USPTO; Non-Final Office Action dated Aug. 19, 2015 in U.S. Appl. No. 14/268,348. cited by applicant
USPTO; Non-Final Office Action dated Jan. 6, 2016 in U.S. Appl. No. 14/268,348. cited by applicant
USPTO; Final Office Action dated Apr. 29, 2016 in U.S. Appl. No. 14/268,348. cited by applicant
USPTO; Notice of Allowance dated Aug. 19, 2016 in U.S. Appl. No. 14/268,348. cited by applicant
USPTO; Notice of Allowance dated Aug. 30, 2016 in U.S. Appl. No. 14/268,348. cited by applicant
USPTO; Non-Final Office Action dated Oct. 20, 2015 in U.S. Appl. No. 14/281,477. cited by applicant
USPTO; Notice of Allowance dated Mar. 28, 2016 in U.S. Appl. No. 14/281,477. cited by applicant
USPTO; Non-Final Office Action dated Jan. 13, 2017 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Final Office Action dated Jul. 10, 2017 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Non-Final Office Action dated Nov. 29, 2017 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Final Office Action dated Mar. 28, 2018 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Non-Final Office Action Dated Jul. 27, 2018 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Final Office Action dated Feb. 7, 2019 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Non-Final Office Action dated Aug. 8, 2019 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Final Office Action dated Feb. 20, 2020 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Non-Final Office Action dated Apr. 28, 2020 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Notice of Allowance dated Sep. 3, 2020 in U.S. Appl. No. 14/444,744. cited by applicant
USPTO; Non-Final Office Action dated May 18, 2016 in U.S. Appl. No. 14/449,838. cited by applicant
USPTO; Notice of Allowance dated Nov. 28, 2016 in U.S. Appl. No. 14/449,838. cited by applicant
USPTO; Non-Final Office Action dated Feb. 12, 2015 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Final Office Action dated Jul. 14, 2015 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Non-Final Office Action dated Nov. 6, 2015 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Advisory Action dated Nov. 6, 2015 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Final Office Acton dated Jun. 17, 2016 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Advisory Action dated Sep. 21, 2016 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Non-Final Office Action dated Oct. 6, 2016 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Final Office Acton dated May 4, 2017 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Non-Final Office Action dated Oct. 19, 2017 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Final Office Action dated Jun. 14, 2018 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Non-Final Office Action dated Jan. 11, 2019 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Final Office Action dated Jun. 25, 2019 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Non-Final Office Action dated Oct. 31, 2019 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Final Office Action dated May 15, 2020 in U.S. Appl. No. 14/457,058. cited by applicant

USPTO; Notice of Allowance dated Aug. 6, 2020 in U.S. Appl. No. 14/457,058. cited by applicant
USPTO; Non-Final Office Action dated Sep. 16, 2016 in U.S. Appl. No. 14/465,252. cited by applicant
USPTO; Final Office Action dated Nov. 1, 2016 in U.S. Appl. No. 14/465,252. cited by applicant
USPTO; Non-Final Office Action dated Mar. 6, 2017 in U.S. Appl. No. 14/465,252. cited by applicant
USPTO; Final Office Action dated Jun. 9, 2017 in U.S. Appl. No. 14/465,252. cited by applicant
USPTO; Notice of Allowance dated Oct. 3, 2017 in U.S. Appl. No. 14/465,252. cited by applicant
USPTO; Non-Final Office Action dated Nov. 24, 2015 in U.S. Appl. No. 14/498,036. cited by applicant
USPTO; Final Office Action dated Apr. 5, 2016 in U.S. Appl. No. 14/498,036. cited by applicant
USPTO; Advisory Action dated Jun. 16, 2016 in U.S. Appl. No. 14/498,036. cited by applicant
USPTO; Notice of Allowance dated Aug. 17, 2016 in U.S. Appl. No. 14/498,036. cited by applicant
USPTO; Non-Final Office Action dated Apr. 10, 2015 in U.S. Appl. No. 14/505,290. cited by applicant
USPTO; Notice of Allowance dated Aug. 21, 2015 in U.S. Appl. No. 14/505,290. cited by applicant
USPTO; Non-Final Office Action dated Dec. 17, 2015 in U.S. Appl. No. 14/508,296. cited by applicant
USPTO; Final Office Action dated May 26, 2016 in U.S. Appl. No. 14/508,296. cited by applicant
USPTO; Advisory Action dated Aug. 17, 2016 in U.S. Appl. No. 14/508,296. cited by applicant
USPTO; Non-Final Office Action dated Sep. 8, 2016 in U.S. Appl. No. 14/508,296. cited by applicant
USPTO; Final Office Action dated Dec. 7, 2016 in U.S. Appl. No. 14/508,296. cited by applicant
USPTO; Notice of Allowance dated Jan. 27, 2017 in U.S. Appl. No. 14/508,296. cited by applicant
USPTO; Non-Final Office Action dated Apr. 6, 2017 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Final Office Action dated Oct. 4, 2017 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Non-Final Office Action dated May 15, 2018 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Final Office Action dated Nov. 28, 2018 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Non-Final Office Action dated Apr. 4, 2019 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Final Office Action dated Aug. 20, 2019 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Advisory Action dated Oct. 28, 2019 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Non-Final Office Action dated Mar. 5, 2020 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Final Office Action dated Aug. 20, 2020 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Notice of Allowance dated Nov. 3, 2020 in U.S. Appl. No. 14/508,489. cited by applicant
USPTO; Non-Final Office Action dated Jan. 16, 2015 in U.S. Appl. No. 14/563,044. cited by applicant
USPTO; Final Office Action dated Jul. 16, 2015 in U.S. Appl. No. 14/563,044. cited by applicant
USPTO; Notice of Allowance dated Oct. 15, 2015 in U.S. Appl. No. 14/563,044. cited by applicant
USPTO; Notice of Allowance dated Dec. 2, 2015 in U.S. Appl. No. 14/563,044. cited by applicant
USPTO; Non-Final Office Action dated May 4, 2016 in U.S. Appl. No. 14/568,647. cited by applicant
USPTO; Final Office Action dated Sep. 29, 2016 in U.S. Appl. No. 14/568,647. cited by applicant
USPTO; Advisory Action dated Dec. 21, 2016 in U.S. Appl. No. 14/568,647. cited by applicant
USPTO; Non-Final Office Action dated Feb. 2, 2017 in U.S. Appl. No. 14/568,647. cited by applicant
USPTO; Final Office Action dated May 19, 2017 in U.S. Appl. No. 14/568,647. cited by applicant

USPTO; Non-Final Office Action dated Sep. 14, 2017 in U.S. Appl. No. 14/568,647. cited by applicant

USPTO; Final Office Action dated Jan. 23, 2018 in U.S. Appl. No. 14/568,647. cited by applicant

USPTO; Advisory Action dated Apr. 12, 2018 in U.S. Appl. No. 14/568,647. cited by applicant

USPTO; Non-Final Office Action dated May 25, 2018 in U.S. Appl. No. 14/568,647. cited by applicant

USPTO; Non-Final Office Action dated Oct. 1, 2015 in U.S. Appl. No. 14/571,126. cited by applicant

USPTO; Final Office Action dated Feb. 22, 2016 in U.S. Appl. No. 14/571,126. cited by applicant

USPTO; Notice of Allowance dated May 18, 2016 in U.S. Appl. No. 14/571,126. cited by applicant

USPTO; Notice of Allowance dated Jun. 2, 2016 in U.S. Appl. No. 14/571,126. cited by applicant

USPTO; Non-Final Office Action dated Nov. 25, 2015 in U.S. Appl. No. 14/598,532. cited by applicant

USPTO; Notice of Allowance dated May 16, 2016 in U.S. Appl. No. 14/598,532. cited by applicant

USPTO; Non-Final Office Action dated Jan. 15, 2016 in U.S. Appl. No. 14/606,364. cited by applicant

USPTO; Final Office Action dated Jun. 14, 2016 in U.S. Appl. No. 14/606,364. cited by applicant

USPTO; Advisory Action dated Aug. 25, 2016 in U.S. Appl. No. 14/606,364. cited by applicant

USPTO; Final Office Action dated Jan. 12, 2017 in U.S. Appl. No. 14/606,364. cited by applicant

USPTO; Non-Final Office Action dated May 10, 2017 in U.S. Appl. No. 14/606,364. cited by applicant

USPTO; Notice of Allowance dated Aug. 16, 2017 in U.S. Appl. No. 14/606,364. cited by applicant

USPTO; Non-Final Office Action dated Mar. 3, 2016 in U.S. Appl. No. 14/622,603. cited by applicant

USPTO; Notice of Allowance dated Aug. 2, 2016 in U.S. Appl. No. 14/622,603. cited by applicant

USPTO; Notice of Allowance dated Feb. 16, 2016 in U.S. Appl. No. 14/634,342. cited by applicant

USPTO; Non-Final Office Action dated Oct. 19, 2017 in U.S. Appl. No. 14/645,234. cited by applicant

USPTO; Non-Final Office Action dated May 16, 2018 in U.S. Appl. No. 14/645,234. cited by applicant

USPTO; Final Office Action dated Aug. 10, 2018 in U.S. Appl. No. 14/645,234. cited by applicant

USPTO; Notice of Allowance dated Aug. 15, 2019 in U.S. Appl. No. 14/645,234. cited by applicant

USPTO; Non-Final Office Action dated Jun. 7, 2017 in U.S. Appl. No. 14/656,588. cited by applicant

USPTO; Final Office Action dated Dec. 26, 2017 in U.S. Appl. No. 14/656,588. cited by applicant

USPTO; Non-Final Office Action dated Apr. 6, 2018 in U.S. Appl. No. 14/656,588. cited by applicant

USPTO; Notice of Allowance dated Nov. 19, 2018 in U.S. Appl. No. 14/656,588. cited by applicant

USPTO; Non-Final Office Action dated Mar. 21, 2016 in U.S. Appl. No. 14/659,152. cited by applicant

USPTO; Final Office Action dated Jul. 29, 2016 in U.S. Appl. No. 14/659,152. cited by applicant

USPTO; Notice of Allowance dated Nov. 22, 2016 in U.S. Appl. No. 14/659,152. cited by applicant

USPTO; Non-Final Office Action dated Nov. 19, 2015 in U.S. Appl. No. 14/659,437. cited by applicant

USPTO; Final Office Action dated Mar. 17, 2016 in U.S. Appl. No. 14/659,437. cited by applicant

USPTO; Notice of Allowance dated May 31, 2016 in U.S. Appl. No. 14/659,437. cited by applicant

USPTO; Non-Final Office Action dated Sep. 7, 2017 in U.S. Appl. No. 14/660,755. cited by applicant

USPTO; Notice of Allowance dated Oct. 2, 2017 in U.S. Appl. No. 14/660,755. cited by applicant

USPTO; Notice of Allowance dated Mar. 25, 2016 in U.S. Appl. No. 14/693,138. cited by applicant

USPTO; Non-Final Office Action dated Aug. 3, 2017 in U.S. Appl. No. 14/752,712. cited by applicant

USPTO; Final Office Action dated Nov. 29, 2017 in U.S. Appl. No. 14/752,712. cited by applicant

USPTO; Advisory Action dated Feb. 15, 2018 in U.S. Appl. No. 14/752,712. cited by applicant

USPTO; Non-Final Office Action dated Mar. 21, 2018 in U.S. Appl. No. 14/752,712. cited by applicant

USPTO; Final Office Action dated Sep. 5, 2018 in U.S. Appl. No. 14/752,712. cited by applicant

USPTO; Non-Final Office Action dated Dec. 28, 2018 in U.S. Appl. No. 14/752,712. cited by applicant

USPTO; Notice of Allowance dated Jun. 11, 2019 in U.S. Appl. No. 14/752,712. cited by applicant

USPTO; Non-Final Office Action dated Nov. 29, 2017 in U.S. Appl. No. 14/793,323. cited by applicant

USPTO; Final Office Action dated Mar. 29, 2018 in U.S. Appl. No. 14/793,323. cited by applicant

USPTO; Non-Final Office Action dated Aug. 10, 2018 in U.S. Appl. No. 14/793,323. cited by applicant

USPTO; Final Office Action dated Feb. 25, 2019 in U.S. Appl. No. 14/793,323. cited by applicant

USPTO; Non-Final Office Action dated Jun. 27, 2019 in U.S. Appl. No. 14/793,323. cited by applicant

USPTO; Notice of Allowance dated Nov. 14, 2019 in U.S. Appl. No. 14/793,323. cited by applicant

USPTO; Non-Final Office Action dated Jun. 16, 2017 in U.S. Appl. No. 14/798,136. cited by applicant

USPTO; Notice of Allowance dated Oct. 5, 2017 in U.S. Appl. No. 14/798,136. cited by applicant

USPTO; Non-Final Office Action dated Mar. 30, 2016 in U.S. Appl. No. 14/808,979. cited by applicant

USPTO; Final Office Action dated Sep. 30, 2016 in U.S. Appl. No. 14/808,979. cited by applicant

USPTO; Non-Final Office Action dated Dec. 20, 2016 in U.S. Appl. No. 14/808,979. cited by applicant

USPTO; Final Office Action dated Jun. 8, 2017 in U.S. Appl. No. 14/808,979. cited by applicant

USPTO; Non-Final Office Action dated Sep. 21, 2017 in U.S. Appl. No. 14/808,979. cited by applicant

USPTO; Final Office Action dated Mar. 14, 2018 in U.S. Appl. No. 14/808,979. cited by applicant

USPTO; Notice of Allowance dated Jun. 27, 2018 in U.S. Appl. No. 14/808,979. cited by applicant

USPTO; Non-Final Office Action dated Feb. 23, 2018 in U.S. Appl. No. 14/817,953. cited by applicant

USPTO; Notice of Allowance dated Jul. 11, 2018 in U.S. Appl. No. 14/817,953. cited by applicant

USPTO; Non-Final Office Action dated Sep. 1, 2016 in U.S. Appl. No. 14/827,177. cited by applicant

USPTO; Notice of Allowance dated Jan. 27, 2017 in U.S. Appl. No. 14/827,177. cited by applicant

USPTO; Non-Final Office Action dated Sep. 9, 2016 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Final Office Action dated Feb. 9, 2017 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Advisory Action dated Apr. 20, 2017 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Non-Final Office Action dated Sep. 19, 2017 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Final Office Action dated Mar. 5, 2018 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Advisory Action dated Aug. 10, 2018 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Non-Final Office Action dated Sep. 6, 2018 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Final Office Action dated Apr. 18, 2019 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Advisory Action dated Jul. 22, 2019 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Non-Final Office Action dated Sep. 19, 2019 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Final Office Action dated Mar. 3, 2020 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Advisory Action dated Jun. 22, 2020 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Non-Final Office Action dated Aug. 3, 2020 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Final Office Action dated Feb. 1, 2021 in U.S. Appl. No. 14/829,565. cited by applicant

USPTO; Non-Final Office Action dated Apr. 29, 2016 in U.S. Appl. No. 14/835,637. cited by applicant

USPTO; Final Office Action dated Nov. 25, 2016 in U.S. Appl. No. 14/835,637. cited by applicant

USPTO; Advisory Action dated Feb. 14, 2017 in U.S. Appl. No. 14/835,637. cited by applicant

USPTO; Notice of Allowance dated Apr. 25, 2017 in U.S. Appl. No. 14/835,637. cited by applicant

USPTO; Non-Final Office Action dated Jul. 29, 2016 in U.S. Appl. No. 14/884,695. cited by applicant

USPTO; Final Office Action dated Feb. 9, 2017 in U.S. Appl. No. 14/884,695. cited by applicant

USPTO; Advisory Action dated Apr. 20, 2017 in U.S. Appl. No. 14/884,695. cited by applicant

USPTO; Non-Final Office Action dated May 18, 2017 in U.S. Appl. No. 14/884,695. cited by applicant

USPTO; Notice of Allowance dated Oct. 20, 2017 in U.S. Appl. No. 14/884,695. cited by applicant

USPTO; Non-Final Office Action dated May 18, 2017 in U.S. Appl. No. 14/886,571. cited by applicant

USPTO; Final Office Action dated Sep. 21, 2017 in U.S. Appl. No. 14/886,571. cited by applicant

USPTO; Notice of Allowance dated Dec. 6, 2017 in U.S. Appl. No. 14/886,571. cited by applicant

USPTO; Non-Final Office Action dated Dec. 1, 2016 in U.S. Appl. No. 14/919,536. cited by applicant

USPTO; Final Office Action dated Mar. 28, 2017 in U.S. Appl. No. 14/919,536. cited by applicant

USPTO; Non-Final Office Action dated Aug. 29, 2017 in U.S. Appl. No. 14/919,536. cited by applicant

USPTO; Final Office Action dated May 11, 2018 in U.S. Appl. No. 14/919,536. cited by applicant

USPTO; Notice of Allowance dated Oct. 4, 2018 in U.S. Appl. No. 14/919,536. cited by applicant

USPTO; Notice of Allowance dated Nov. 19, 2018 in U.S. Appl. No. 14/919,536. cited by applicant

USPTO; Non-Final Office Action dated May 3, 2016 in U.S. Appl. No. 14/937,053. cited by applicant

USPTO; Notice of Allowance dated Jul. 26, 2016 in U.S. Appl. No. 14/937,053. cited by applicant

USPTO; Non-Final Office Action dated Dec. 15, 2016 in U.S. Appl. No. 14/938,180. cited by applicant

USPTO; Notice of Allowance dated Nov. 9, 2017 in U.S. Appl. No. 14/938,180. cited by applicant

USPTO; Non-Final Office Action dated Apr. 14, 2017 in U.S. Appl. No. 14/956,115. cited by applicant

USPTO; Final Office Action dated Jul. 21, 2017 in U.S. Appl. No. 14/956,115. cited by applicant

USPTO; Notice of Allowance dated Dec. 14, 2017 in U.S. Appl. No. 14/956,115. cited by applicant

USPTO; Notice of Allowance dated Feb. 3, 2017 in U.S. Appl. No. 14/977,291. cited by applicant

USPTO; Non-Final Office Action dated Aug. 12, 2016 in U.S. Appl. No. 14/981,434. cited by applicant

USPTO; Notice of Allowance dated Nov. 21, 2016 in U.S. Appl. No. 14/981,434. cited by applicant

USPTO; Non-Final Office Action dated Jan. 12, 2017 in U.S. Appl. No. 14/981,468. cited by applicant

USPTO; Notice of Allowance dated Jun. 7, 2017 in U.S. Appl. No. 14/981,468. cited by applicant

USPTO; Non-Final Office Action dated Mar. 22, 2016 in U.S. Appl. No. 14/987,420. cited by applicant

USPTO; Final Office Action dated Jun. 10, 2016 in U.S. Appl. No. 14/987,420. cited by applicant
USPTO; Non-Final Office Action dated Dec. 14, 2016 in U.S. Appl. No. 14/997,683. cited by applicant
USPTO; Final Office Action dated Apr. 14, 2017 in U.S. Appl. No. 14/997,683. cited by applicant
USPTO; Non-Final Office Action dated Sep. 1, 2017 in U.S. Appl. No. 14/997,683. cited by applicant
USPTO; Final Office Action dated Feb. 6, 2018 in U.S. Appl. No. 14/997,683. cited by applicant
USPTO; Advisory Action dated May 2, 2018 in U.S. Appl. No. 14/997,683. cited by applicant
USPTO; Non-Final Office Action dated Jun. 20, 2018 in U.S. Appl. No. 14/997,683. cited by applicant
USPTO; Final Office Action dated Dec. 10, 2018 in U.S. Appl. No. 14/997,683. cited by applicant
USPTO; Notice of Allowance dated Mar. 25, 2019 in U.S. Appl. No. 14/997,683. cited by applicant
USPTO; Non-Final Office Action dated Sep. 23, 2016 in U.S. Appl. No. 15/048,422. cited by applicant
USPTO; Notice of Allowance dated May 4, 2017 in U.S. Appl. No. 15/048,422. cited by applicant
USPTO; Non-Final Office Action dated Aug. 4, 2017 in U.S. Appl. No. 15/050,159. cited by applicant
USPTO; Notice of Allowance dated Feb. 7, 2018 in U.S. Appl. No. 15/050,159. cited by applicant
USPTO; Non-Final Office Action dated Apr. 22, 2016 in U.S. Appl. No. 15/055,122. cited by applicant
USPTO; Notice of Allowance dated Sep. 15, 2016 in U.S. Appl. No. 15/055,122. cited by applicant
USPTO; Non-Final Office Action dated Feb. 20, 2018 in U.S. Appl. No. 15/060,412. cited by applicant
USPTO; Final Office Action dated Oct. 19, 2018 in U.S. Appl. No. 15/060,412. cited by applicant
USPTO; Non-Final Office Action dated Jun. 3, 2019 in U.S. Appl. No. 15/060,412. cited by applicant
USPTO; Final Office Action dated Dec. 26, 2019 in U.S. Appl. No. 15/060,412. cited by applicant
USPTO; Notice of Allowance dated Mar. 5, 2020 in U.S. Appl. No. 15/060,412. cited by applicant
USPTO; Non-Final Office Action dated Aug. 27, 2018 in U.S. Appl. No. 15/067,028. cited by applicant
USPTO; Notice of Allowance dated Dec. 21, 2018 in U.S. Appl. No. 15/067,028. cited by applicant
USPTO; Non-Final Office Action dated Sep. 26, 2018 in U.S. Appl. No. 15/074,813. cited by applicant
USPTO; Notice of Allowance dated Feb. 25, 2019 in U.S. Appl. No. 15/074,813. cited by applicant
USPTO; Non-Final Office Action dated Jan. 9, 2018 in U.S. Appl. No. 15/135,224. cited by applicant
USPTO; Notice of Allowance dated Jun. 29, 2018 in U.S. Appl. No. 15/135,224. cited by applicant
USPTO; Non-Final Office Action dated Jan. 9, 2018 in U.S. Appl. No. 15/135,258. cited by applicant
USPTO; Final Office Action dated Jul. 6, 2018 in U.S. Appl. No. 15/135,258. cited by applicant
USPTO; Non-Final Office Action dated Nov. 23, 2018 in U.S. Appl. No. 15/135,258. cited by applicant
USPTO; Final Office Action dated Mar. 14, 2019 in U.S. Appl. No. 15/135,258. cited by applicant
USPTO; Non-Final Office Action dated Jul. 19, 2019 in U.S. Appl. No. 15/135,258. cited by applicant
USPTO; Final Office Action dated Oct. 24, 2019 in U.S. Appl. No. 15/135,258. cited by applicant
USPTO; Advisory Action dated Jan. 3, 2020 in U.S. Appl. No. 15/135,258. cited by applicant
USPTO; Non-Final Office Action dated Feb. 13, 2020 in U.S. Appl. No. 15/135,258. cited by applicant
USPTO; Final Office Action dated Jun. 1, 2020 in U.S. Appl. No. 15/135,258. cited by applicant

USPTO; Non-Final Office Action dated Jan. 9, 2018 in U.S. Appl. No. 15/135,333. cited by applicant

USPTO; Notice of Allowance dated Aug. 18, 2020 in U.S. Appl. No. 15/135,258. cited by applicant

USPTO; Notice of Allowance dated Sep. 14, 2018 in U.S. Appl. No. 15/135,333. cited by applicant

USPTO; Non-Final Office Action dated Nov. 21, 2016 in U.S. Appl. No. 15/144,481. cited by applicant

USPTO; Final Office Action dated May 26, 2017 in U.S. Appl. No. 15/144,481. cited by applicant

USPTO; Non-Final Office Action dated Sep. 21, 2017 in U.S. Appl. No. 15/144,481. cited by applicant

USPTO; Notice of Allowance dated Mar. 26, 2018 in U.S. Appl. No. 15/144,481. cited by applicant

USPTO; Notice of Allowance dated Apr. 11, 2018 in U.S. Appl. No. 15/144,481. cited by applicant

USPTO; Non-Final Office Action dated Apr. 13, 2017 in U.S. Appl. No. 15/144,506. cited by applicant

USPTO; Final Office Action dated Oct. 10, 2017 in U.S. Appl. No. 15/144,506. cited by applicant

USPTO; Final Office Action dated Jul. 26, 2018 in U.S. Appl. No. 15/144,506. cited by applicant

USPTO; Notice of Allowance dated Mar. 13, 2019 in U.S. Appl. No. 15/144,506. cited by applicant

USPTO; Non-Final Office Action dated Oct. 9, 2018 in U.S. Appl. No. 15/182,504. cited by applicant

USPTO; Final Office Action dated Mar. 28, 2019 in U.S. Appl. No. 15/182,504. cited by applicant

USPTO; Notice of Allowance dated Jul. 17, 2019 in U.S. Appl. No. 15/182,504. cited by applicant

USPTO; Non-Final Office Action dated Nov. 28, 2016 in U.S. Appl. No. 15/203,632. cited by applicant

USPTO; Final Office Action dated Jun. 7, 2017 in U.S. Appl. No. 15/203,632. cited by applicant

USPTO; Advisory Action dated Aug. 23, 2017 in U.S. Appl. No. 15/203,632. cited by applicant

USPTO; Notice of Allowance dated Sep. 20, 2017 in U.S. Appl. No. 15/203,632. cited by applicant

USPTO; Non-Final Office Action dated Nov. 29, 2016 in U.S. Appl. No. 15/203,642. cited by applicant

USPTO; Final Office Action dated Apr. 13, 2017 in U.S. Appl. No. 15/203,642. cited by applicant

USPTO; Advisory Action dated Jun. 22, 2017 in U.S. Appl. No. 15/203,642. cited by applicant

USPTO; Notice of Allowance dated Aug. 7, 2017 in U.S. Appl. No. 15/203,642. cited by applicant

USPTO; Non-Final Office Action dated Jun. 1, 2017 in U.S. Appl. No. 15/205,827. cited by applicant

USPTO; Final Office Action dated Oct. 16, 2017 in U.S. Appl. No. 15/205,827. cited by applicant

USPTO; Non-Final Office Action dated May 14, 2018 in U.S. Appl. No. 15/205,827. cited by applicant

USPTO; Final Office Action dated Oct. 9, 2018 in U.S. Appl. No. 15/205,827. cited by applicant

USPTO; Non-Final Office Action dated Mar. 28, 2019 in U.S. Appl. No. 15/205,827. cited by applicant

USPTO; Final Office Action dated Aug. 9, 2019 in U.S. Appl. No. 15/205,827. cited by applicant

USPTO; Advisory Action dated Oct. 22, 2019 in U.S. Appl. No. 15/205,827. cited by applicant

USPTO; Notice of Allowance dated Dec. 3, 2019 in U.S. Appl. No. 15/205,827. cited by applicant

USPTO; Non-Final Office Action dated Mar. 31, 2017 in U.S. Appl. No. 15/205,890. cited by applicant

USPTO; Notice of Allowance dated Oct. 16, 2017 in U.S. Appl. No. 15/205,890. cited by applicant

USPTO; Non-Final Office Action dated Jan. 20, 2017 in U.S. Appl. No. 15/210,256. cited by applicant

USPTO; Notice of Allowance dated May 18, 2017 in U.S. Appl. No. 15/210,256. cited by applicant

USPTO; Notice of Allowance dated Jul. 24, 2017 in U.S. Appl. No. 15/210,256. cited by applicant

USPTO; Non Final Office Action dated Apr. 21, 2017 in U.S. Appl. No. 15/222,715. cited by applicant

USPTO; Notice of Allowance dated Jul. 14, 2017 in U.S. Appl. No. 15/222,715. cited by applicant
USPTO; Notice of Allowance dated Sep. 27, 2017 in U.S. Appl. No. 15/222,715. cited by applicant
USPTO; Non-Final Office Action dated Feb. 3, 2017 in U.S. Appl. No. 15/222,738. cited by applicant
USPTO; Notice of Allowance dated May 22, 2017 in U.S. Appl. No. 15/222,738. cited by applicant
USPTO; Notice of Allowance dated Aug. 23, 2017 in U.S. Appl. No. 15/222,738. cited by applicant
USPTO; Non-Final Office Action dated Jan. 17, 2017 in U.S. Appl. No. 15/222,749. cited by applicant
USPTO; Final Office Action dated May 5, 2017 in U.S. Appl. No. 15/222,749. cited by applicant
USPTO; Non-Final Office Action dated Sep. 7, 2017 in U.S. Appl. No. 15/222,749. cited by applicant
USPTO; Final Office Action dated Jun. 4, 2018 in U.S. Appl. No. 15/222,749. cited by applicant
USPTO; Notice of Allowance dated Aug. 30, 2018 in U.S. Appl. No. 15/222,749. cited by applicant
USPTO; Non-Final Office Action dated Jan. 3, 2017 in U.S. Appl. No. 15/222,780. cited by applicant
USPTO; Final Office Action dated May 5, 2017 in U.S. Appl. No. 15/222,780. cited by applicant
USPTO; Non-Final Office Action dated Sep. 7, 2017 in U.S. Appl. No. 15/222,780. cited by applicant
USPTO; Final Office Action dated May 17, 2018 in U.S. Appl. No. 15/222,780. cited by applicant
USPTO; Non-Final Office Action dated Oct. 1, 2018 in U.S. Appl. No. 15/222,780. cited by applicant
USPTO; Notice of Allowance dated Apr. 19, 2019 in U.S. Appl. No. 15/222,780. cited by applicant
USPTO; Notice of Allowance dated Jul. 12, 2018 in U.S. Appl. No. 15/254,605. cited by applicant
USPTO; Non-Final Office Action dated Aug. 28, 2017 in U.S. Appl. No. 15/254,724. cited by applicant
USPTO; Notice of Allowance dated Jan. 17, 2018 in U.S. Appl. No. 15/254,724. cited by applicant
USPTO; Notice of Allowance dated Apr. 2, 2018 in U.S. Appl. No. 15/254,724. cited by applicant
USPTO; Non-Final Office Action dated May 22, 2018 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Non-Final Office Action dated Sep. 13, 2018 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Non-Final Office Action dated Jan. 30, 2019 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Final Office Action dated May 13, 2019 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Advisory Action dated Jul. 22, 2019 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Non-Final Office Action dated Aug. 5, 2019 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Final Office Action dated Nov. 19, 2019 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Advisory Action dated Jan. 30, 2020 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Non-Final Office Action dated Oct. 6, 2020 in U.S. Appl. No. 15/262,990. cited by applicant
USPTO; Non-Final Office Action dated Aug. 3, 2018 in U.S. Appl. No. 15/273,488. cited by applicant
USPTO; Final Office Action dated Jan. 11, 2019 in U.S. Appl. No. 15/273,488. cited by applicant
USPTO; Notice of Allowance dated Apr. 19, 2019 in U.S. Appl. No. 15/273,488. cited by applicant
USPTO; Non-Final Office Action dated Jul. 2, 2018 in U.S. Appl. No. 15/286,503. cited by applicant
USPTO; Final Office Action dated Feb. 7, 2019 in U.S. Appl. No. 15/286,503. cited by applicant
USPTO; Non-Final Office Action dated Jun. 27, 2019 in U.S. Appl. No. 15/286,503. cited by applicant

USPTO; Final Office Action dated Jan. 6, 2020 in U.S. Appl. No. 15/286,503. cited by applicant
USPTO; Advisory Action dated Mar. 31, 2020 in U.S. Appl. No. 15/286,503. cited by applicant
USPTO; Non-Final Office Action dated Apr. 16, 2020 in U.S. Appl. No. 15/286,503. cited by applicant
USPTO; Final Office Action dated Oct. 2, 2020 in U.S. Appl. No. 15/286,503. cited by applicant
USPTO; Non-Final Office Action dated Dec. 24, 2020 in U.S. Appl. No. 15/286,503. cited by applicant
USPTO; Non-Final Office Action dated Dec. 14, 2018 in U.S. Appl. No. 15/340,512. cited by applicant
USPTO; Notice of Allowance dated May 24, 2019 in U.S. Appl. No. 15/340,512. cited by applicant
USPTO; Non-Final Office Action dated Oct. 23, 2017 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Final Office Action dated Apr. 16, 2018 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Advisory Action dated Aug. 8, 2018 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Non-Final Office Action dated Nov. 14, 2018 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Final Office Action dated Jun. 25, 2019 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Non-Final Office Action dated Nov. 7, 2019 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Final Office Action dated May 13, 2020 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Non-Final Office Action dated Sep. 2, 2020 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Final Office Action dated Feb. 12, 2021 in U.S. Appl. No. 15/377,439. cited by applicant
USPTO; Notice of Allowance dated Aug. 8, 2017 in U.S. Appl. No. 15/380,895. cited by applicant
USPTO; Notice of Allowance dated Oct. 11, 2017 in U.S. Appl. No. 15/380,895. cited by applicant
USPTO; Non-Final Office Action dated May 31, 2019 in U.S. Appl. No. 15/380,909. cited by applicant
USPTO; Final Office Action dated Dec. 12, 2019 in U.S. Appl. No. 15/380,909. cited by applicant
USPTO; Non-Final Office Action dated Jul. 1, 2020 in U.S. Appl. No. 15/380,909. cited by applicant
USPTO; Non-Final Office Action dated Jan. 4, 2018 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO; Final Office Action dated Jun. 28, 2018 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO; Non-Final Office Action dated Feb. 25, 2019 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO; Final Office Action dated Sep. 18, 2019 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO; Non-Final Office Action dated Jan. 15, 2020 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO; Final Office Action dated Jul. 23, 2020 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO Non-Final Office Action dated Dec. 10, 2020 in U.S. Appl. No. 15/380,921. cited by applicant
USPTO; Non-Final Office Action dated Oct. 3, 2017 in U.S. Appl. No. 15/388,410. cited by applicant
USPTO; Final Office Action dated May 15, 2018 in U.S. Appl. No. 15/388,410. cited by applicant
USPTO; Notice of Allowance dated Nov. 14, 2018 in U.S. Appl. No. 15/388,410. cited by applicant
USPTO; Notice of Allowance dated Dec. 28, 2018 in U.S. Appl. No. 15/388,410. cited by applicant
USPTO; Non-Final Office Action dated Aug. 11, 2017 in U.S. Appl. No. 15/397,237. cited by applicant
USPTO; Notice of Allowance dated Dec. 22, 2017 in U.S. Appl. No. 15/397,237. cited by applicant
USPTO; Non-Final Office Action dated Apr. 12, 2017 in U.S. Appl. No. 15/397,319. cited by

applicant
USPTO; Final Office Action dated Jul. 12, 2017 in U.S. Appl. No. 15/397,319. cited by applicant
USPTO; Notice of Allowance dated Dec. 15, 2017 in U.S. Appl. No. 15/397,319. cited by applicant
USPTO; Non-Final Office Action dated Feb. 5, 2019 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Final Office Action dated May 21, 2019 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Advisory Action dated Jul. 29, 2019 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Non-Final Office Action dated Oct. 24, 2019 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Final Office Action dated Feb. 19, 2020 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Advisory Action dated Apr. 2, 2020 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Non-Final Office Action dated May 27, 2020 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Final Office Action dated Sep. 8, 2020 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Advisory Action dated Nov. 9, 2020 in U.S. Appl. No. 15/402,993. cited by applicant
USPTO; Non-Final Office Action dated Sep. 20, 2018 in U.S. Appl. No. 15/410,503. cited by applicant
USPTO; Final Office Action dated Feb. 4, 2019 in U.S. Appl. No. 15/410,503. cited by applicant
USPTO; Non-Final Office Action dated Apr. 25, 2019 in U.S. Appl. No. 15/410,503. cited by applicant
USPTO; Notice of Allowance dated Aug. 14, 2019 in U.S. Appl. No. 15/410,503. cited by applicant
USPTO; Non-Final Office Action dated Aug. 7, 2018 in U.S. Appl. No. 15/428,808. cited by applicant
USPTO; Final Office Action dated Jan. 11, 2019 in U.S. Appl. No. 15/428,808. cited by applicant
USPTO; Notice of Allowance dated Apr. 25, 2019 in U.S. Appl. No. 15/428,808. cited by applicant
USPTO; Non-Final Office Action dated Apr. 6, 2018 in U.S. Appl. No. 15/434,051. cited by applicant
USPTO; Final Office Action dated Aug. 29, 2018 in U.S. Appl. No. 15/434,051. cited by applicant
USPTO; Advisory Action dated Dec. 4, 2018 in U.S. Appl. No. 15/434,051. cited by applicant
USPTO; Non-Final Office Action dated Jan. 25, 2019 in U.S. Appl. No. 15/434,051. cited by applicant
USPTO; Notice of Allowance dated Jun. 3, 2019 in U.S. Appl. No. 15/434,051. cited by applicant
USPTO; Notice of Allowance dated Oct. 6, 2017 in U.S. Appl. No. 15/450,199. cited by applicant
USPTO; Non-Final Office Action dated Dec. 15, 2017 in U.S. Appl. No. 15/466,149. cited by applicant
USPTO; Notice of Allowance dated Apr. 20, 2018 in U.S. Appl. No. 15/466,149. cited by applicant
USPTO; Non-Final Office Action dated Apr. 6, 2018 in U.S. Appl. No. 15/472,750. cited by applicant
USPTO; Notice of Allowance dated Nov. 30, 2018 in U.S. Appl. No. 15/472,750. cited by applicant
USPTO; Non-Final Office Action dated Dec. 6, 2017 in U.S. Appl. No. 15/476,035. cited by applicant
USPTO; Notice of Allowance dated Mar. 21, 2018 in U.S. Appl. No. 15/476,035. cited by applicant
USPTO; Notice of Allowance dated Aug. 14, 2018 in U.S. Appl. No. 15/476,035. cited by applicant
USPTO; Non-Final Office Action dated Oct. 4, 2017 in U.S. Appl. No. 15/489,453. cited by applicant
USPTO; Final Office Action dated Apr. 19, 2018 in U.S. Appl. No. 15/489,453. cited by applicant
USPTO; Non-Final Office Action dated Sep. 10, 2018 in U.S. Appl. No. 15/489,453. cited by applicant
USPTO; Final Office Action dated Feb. 27, 2019 in U.S. Appl. No. 15/489,453. cited by applicant
USPTO; Non-Final Office Action dated Jun. 5, 2019 in U.S. Appl. No. 15/489,453. cited by

applicant

USPTO; Notice of Allowance dated Oct. 7, 2019 in U.S. Appl. No. 15/489,453. cited by applicant

USPTO; Notice of Allowance dated Dec. 19, 2017 in U.S. Appl. No. 15/489,660. cited by applicant

USPTO; Final Office Action dated May 1, 2019 in U.S. Appl. No. 15/491,726. cited by applicant

USPTO; Non-Final Office Action dated May 31, 2018 in U.S. Appl. No. 15/491,726. cited by applicant

USPTO; Non-Final Office Action dated Oct. 3, 2019 in U.S. Appl. No. 15/491,726. cited by applicant

USPTO; Final Office Action dated Apr. 17, 2020 in U.S. Appl. No. 15/491,726. cited by applicant

USPTO; Non-Final Office Action dated Sep. 30, 2020 in U.S. Appl. No. 15/491,726. cited by applicant

USPTO; Non-Final Office Action dated Jan. 16, 2018 in U.S. Appl. No. 15/499,647. cited by applicant

USPTO; Notice of Allowance dated May 23, 2018 in U.S. Appl. No. 15/499,647. cited by applicant

USPTO; Non-Final Office Action dated Jun. 21, 2018 in U.S. Appl. No. 15/499,647. cited by applicant

USPTO; Notice of Allowance dated Nov. 1, 2018 in U.S. Appl. No. 15/499,647. cited by applicant

USPTO; Notice of Allowance dated Nov. 15, 2018 in U.S. Appl. No. 15/499,647. cited by applicant

USPTO; Office Action dated Aug. 30, 2018 in U.S. Appl. No. 15/589,849. cited by applicant

USPTO; Final Office Action dated Mar. 6, 2019 in U.S. Appl. No. 15/589,849. cited by applicant

USPTO; Non-Final Office Action dated Jun. 28, 2019 in U.S. Appl. No. 15/589,849. cited by applicant

USPTO; Final Office Action dated Jan. 8, 2020 in U.S. Appl. No. 15/589,849. cited by applicant

USPTO; Notice of Allowance dated Feb. 28, 2020 in U.S. Appl. No. 15/589,849. cited by applicant

USPTO; Office Action dated May 3, 2018 in U.S. Appl. No. 15/589,861. cited by applicant

USPTO; Non-Final Office Action dated Dec. 21, 2018 in U.S. Appl. No. 15/589,861. cited by applicant

USPTO; Final Office Action dated Jun. 26, 2019 in U.S. Appl. No. 15/589,861. cited by applicant

USPTO; Advisory Action dated Sep. 20, 2019 in U.S. Appl. No. 15/589,861. cited by applicant

USPTO; Non-Final Office Action dated Jan. 13, 2020 in U.S. Appl. No. 15/589,861. cited by applicant

USPTO; Final Office Action dated Jul. 8, 2020 in U.S. Appl. No. 15/589,861. cited by applicant

USPTO; Notice of Allowance dated Oct. 16, 2020 in U.S. Appl. No. 15/589,861. cited by applicant

USPTO; Non-Final Office Action dated Apr. 4, 2018 in U.S. Appl. No. 15/592,730. cited by applicant

USPTO; Final Office Action dated Nov. 16, 2018 in U.S. Appl. No. 15/592,730. cited by applicant

USPTO; Advisory Action dated Mar. 15, 2019 in U.S. Appl. No. 15/592,730. cited by applicant

USPTO; Notice of Allowance dated Aug. 21, 2019 in U.S. Appl. No. 15/592,730. cited by applicant

USPTO; Non-Final Office Action dated Mar. 7, 2019 in U.S. Appl. No. 15/598,169. cited by applicant

USPTO; Final Office Action dated Jun. 25, 2019 in U.S. Appl. No. 15/598,169. cited by applicant

USPTO; Notice of Allowance dated Sep. 11, 2019 in U.S. Appl. No. 15/598,169. cited by applicant

USPTO; Non-Final Office Action dated Mar. 10, 2020 in U.S. Appl. No. 15/611,707. cited by applicant

USPTO; Final Office Action dated Sep. 16, 2020 in U.S. Appl. No. 15/611,707. cited by applicant

USPTO; Advisory Action dated Nov. 20, 2020 in U.S. Appl. No. 15/611,707. cited by applicant

USPTO; Non-Final Office Action dated Feb. 3, 2021 in U.S. Appl. No. 15/611,707. cited by applicant

USPTO; Ex Parte Quayle Action dated Mar. 21, 2019 in U.S. Appl. No. 15/615,489. cited by applicant

USPTO; Non-Final Office Action dated Sep. 4, 2019 in U.S. Appl. No. 15/615,489. cited by applicant

USPTO; Notice of Allowance dated Feb. 28, 2020 in U.S. Appl. No. 15/615,489. cited by applicant

USPTO; Non-Final Office Action dated Feb. 1, 2019 in U.S. Appl. No. 15/627,189. cited by applicant

USPTO; Notice of Allowance dated May 21, 2019 in U.S. Appl. No. 15/627,189. cited by applicant

USPTO; Non-Final Office Action dated Nov. 9, 2018 in U.S. Appl. No. 15/636,307. cited by applicant

USPTO; Final Office Action dated Mar. 6, 2019 in U.S. Appl. No. 15/636,307. cited by applicant

USPTO; Non-Final Office Action dated Jul. 16, 2019 in U.S. Appl. No. 15/636,307. cited by applicant

USPTO; Final Office Action dated Nov. 12, 2019 in U.S. Appl. No. 15/636,307. cited by applicant

USPTO; Advisory Action dated Jan. 17, 2020 in U.S. Appl. No. 15/636,307. cited by applicant

USPTO; Non-Final Office Action dated Mar. 11, 2020 in U.S. Appl. No. 15/636,307. cited by applicant

USPTO; Final Office Action dated Jun. 16, 2020 in U.S. Appl. No. 15/636,307. cited by applicant

USPTO; Non-Final Office Action dated Sep. 3, 2020 in U.S. Appl. No. 15/636,307. cited by applicant

USPTO; Final Office Action dated Jan. 19, 2021 in U.S. Appl. No. 15/636,307. cited by applicant

USPTO; Notice of Allowance dated Jul. 18, 2018 in U.S. Appl. No. 15/640,239. cited by applicant

USPTO; Notice of Allowance dated Aug. 30, 2018 in U.S. Appl. No. 15/640,239. cited by applicant

USPTO; Non-Final Office Action dated Jun. 5, 2018 in U.S. Appl. No. 15/650,686. cited by applicant

USPTO; Final Office Action dated Nov. 20, 2018 in U.S. Appl. No. 15/650,686. cited by applicant

USPTO; Notice of Allowance dated Jun. 24, 2019 in U.S. Appl. No. 15/650,686. cited by applicant

USPTO; Non-Final Office Action dated Sep. 21, 2018 in U.S. Appl. No. 15/659,631. cited by applicant

USPTO; Notice of Allowance dated Feb. 21, 2019 in U.S. Appl. No. 15/659,631. cited by applicant

USPTO; Non-Final Office Action dated Jul. 29, 2019 in U.S. Appl. No. 15/660,797. cited by applicant

USPTO; Notice of Allowance dated Nov. 7, 2019 in U.S. Appl. No. 15/660,797. cited by applicant

USPTO; Non-Final Office Action dated Aug. 9, 2018 in U.S. Appl. No. 15/660,805. cited by applicant

USPTO; Non-Final Office Action dated Mar. 1, 2019 in U.S. Appl. No. 15/660,805. cited by applicant

USPTO; Notice of Allowance dated Aug. 22, 2019 in U.S. Appl. No. 15/660,805. cited by applicant

USPTO; Non-Final Office Action dated Aug. 27, 2018 in U.S. Appl. No. 15/662,107. cited by applicant

USPTO; Notice of Allowance dated Feb. 21, 2019 in U.S. Appl. No. 15/662,107. cited by applicant

USPTO; Non-Final Office Action dated Dec. 4, 2018 in U.S. Appl. No. 15/672,063. cited by applicant

USPTO; Notice of Allowance dated Mar. 20, 2019 in U.S. Appl. No. 15/672,063. cited by applicant

USPTO; Non-Final Office Action dated Oct. 10, 2019 in U.S. Appl. No. 15/672,096. cited by applicant

USPTO; Final Office Action dated Mar. 27, 2020 in U.S. Appl. No. 15/672,096. cited by applicant

USPTO; Notice of Allowance dated May 20, 2020 in U.S. Appl. No. 15/672,096. cited by applicant

USPTO; Notice of Allowance dated Jul. 2, 2020 in U.S. Appl. No. 15/672,096. cited by applicant

USPTO; Non-Final Office Action dated Feb. 8, 2019 in U.S. Appl. No. 15/672,119. cited by applicant

USPTO; Final Office Action dated Jul. 16, 2019 in U.S. Appl. No. 15/672,119. cited by applicant

USPTO; Advisory Action dated Sep. 23, 2019 in U.S. Appl. No. 15/672,119. cited by applicant
USPTO; Notice of Allowance dated Feb. 7, 2020 in U.S. Appl. No. 15/672,119. cited by applicant
USPTO; Non-Final Office Action dated Jul. 27, 2018 in U.S. Appl. No. 15/673,110. cited by applicant
USPTO; Notice of Allowance dated Jan. 9, 2019 in U.S. Appl. No. 15/673,110. cited by applicant
USPTO; Non-Final Office Action dated Apr. 25, 2018 in U.S. Appl. No. 15/673,278. cited by applicant
USPTO; Notice of Allowance dated May 6, 2019 in U.S. Appl. No. 15/673,278. cited by applicant
USPTO; Non-Final Office Action dated Jan. 18, 2018 in U.S. Appl. No. 15/683,701. cited by applicant
USPTO; Notice of Allowance dated Jan. 9, 2019 in U.S. Appl. No. 15/683,701. cited by applicant
USPTO; Final Office Action dated Aug. 24, 2018 in U.S. Appl. No. 15/683,701. cited by applicant
USPTO; Advisory Action dated Nov. 26, 2018 in U.S. Appl. No. 15/683,701. cited by applicant
USPTO; Non-Final Office Action dated Dec. 18, 2018 in U.S. Appl. No. 15/690,017. cited by applicant
USPTO; Final Office Action dated Jul. 26, 2019 in U.S. Appl. No. 15/690,017. cited by applicant
USPTO; Non-Final Office Action dated Nov. 20, 2019 in U.S. Appl. No. 15/690,017. cited by applicant
USPTO; Final Office Action dated May 27, 2020 in U.S. Appl. No. 15/690,017. cited by applicant
USPTO; Advisory Action dated Aug. 6, 2020 in U.S. Appl. No. 15/690,017. cited by applicant
USPTO; Non-Final Office Action dated Aug. 24, 2020 in U.S. Appl. No. 15/690,017. cited by applicant
USPTO; Non-Final Office Action dated Aug. 9, 2018 in U.S. Appl. No. 15/691,241. cited by applicant
USPTO; Final Office Action dated Jan. 11, 2019 in U.S. Appl. No. 15/691,241. cited by applicant
USPTO; Non-Final Office Action dated Mar. 19, 2019 in U.S. Appl. No. 15/691,241. cited by applicant
USPTO; Final Office Action dated Jan. 24, 2020 in U.S. Appl. No. 15/691,241. cited by applicant
USPTO; Advisory Action dated Jun. 19, 2020 in U.S. Appl. No. 15/691,241. cited by applicant
USPTO; Non-Final Office Action dated Oct. 6, 2020 in U.S. Appl. No. 15/691,241. cited by applicant
USPTO; Non-Final Office Action dated Dec. 6, 2018 in U.S. Appl. No. 15/705,955. cited by applicant
USPTO; Notice of Allowance dated Apr. 16, 2019 in U.S. Appl. No. 15/705,955. cited by applicant
USPTO; Non-Final Office Action dated Feb. 11, 2019 in U.S. Appl. No. 15/707,786. cited by applicant
USPTO; Final Office Action dated Aug. 12, 2019 in U.S. Appl. No. 15/707,786. cited by applicant
USPTO; Notice of Allowance dated Nov. 7, 2019 in U.S. Appl. No. 15/707,786. cited by applicant
USPTO; Non-Final Office Action dated Jun. 14, 2018 in U.S. Appl. No. 15/711,989. cited by applicant
USPTO; Notice of Allowance dated Dec. 6, 2018 in U.S. Appl. No. 15/711,989. cited by applicant
USPTO; Non-Final Office Action dated May 29, 2018 in U.S. Appl. No. 15/719,208. cited by applicant
USPTO; Final Office Action dated Dec. 13, 2018 in U.S. Appl. No. 15/719,208. cited by applicant
USPTO; Non-Final Office Action dated Jun. 25, 2019 in U.S. Appl. No. 15/719,208. cited by applicant
USPTO; Notice of Allowance dated Jan. 29, 2020 in U.S. Appl. No. 15/719,208. cited by applicant
USPTO; Non-Final Office Action dated Oct. 4, 2018 in U.S. Appl. No. 15/726,222. cited by applicant
USPTO; Notice of Allowance dated Apr. 19, 2019 in U.S. Appl. No. 15/726,222. cited by applicant

USPTO; Non-Final Office Action dated Apr. 19, 2018 in U.S. Appl. No. 15/726,959. cited by applicant

USPTO; Final Office Action dated Nov. 14, 2018 in U.S. Appl. No. 15/726,959. cited by applicant

USPTO; Non-Final Office Action dated Jul. 8, 2019 in U.S. Appl. No. 15/726,959. cited by applicant

USPTO; Final Office Action dated Jan. 7, 2020 in U.S. Appl. No. 15/726,959. cited by applicant

USPTO; Advisory Action dated Mar. 16, 2020 in U.S. Appl. No. 15/726,959. cited by applicant

USPTO; Non-Final Office Action dated Apr. 3, 2020 in U.S. Appl. No. 15/726,959. cited by applicant

USPTO; Final Office Action dated Sep. 24, 2020 in U.S. Appl. No. 15/726,959. cited by applicant

USPTO; Non-Final Office Action dated Jan. 6, 2021 in U.S. Appl. No. 15/726,959. cited by applicant

USPTO; Non-Final Office Action dated Sep. 26, 2019 in U.S. Appl. No. 15/727,432. cited by applicant

USPTO; Notice of Allowance dated Jan. 6, 2020 in U.S. Appl. No. 15/727,432. cited by applicant

USPTO; Non-Final Office Action dated May 17, 2018 in U.S. Appl. No. 15/729,485. cited by applicant

USPTO; Notice of Allowance dated Jan. 23, 2019 in U.S. Appl. No. 15/729,485. cited by applicant

USPTO; Non-Final Office Action dated Nov. 28, 2018 in U.S. Appl. No. 15/795,056. cited by applicant

USPTO; Final Office Action dated Apr. 19, 2019 in U.S. Appl. No. 15/795,056. cited by applicant

USPTO; Final Office Action dated Jul. 31, 2019 in U.S. Appl. No. 15/795,056. cited by applicant

USPTO; Non-Final Office Action dated Nov. 20, 2019 in U.S. Appl. No. 15/795,056. cited by applicant

USPTO; Notice of Allowance dated Mar. 12, 2020 in U.S. Appl. No. 15/795,056. cited by applicant

USPTO; Non-Final Office Action dated Jun. 26, 2018 in U.S. Appl. No. 15/796,593. cited by applicant

USPTO; Final Office Action dated Feb. 21, 2019 in U.S. Appl. No. 15/796,593. cited by applicant

USPTO; Non-Final Office Action dated Jun. 14, 2019 in U.S. Appl. No. 15/796,593. cited by applicant

USPTO; Notice of Allowance dated Dec. 31, 2019 in U.S. Appl. No. 15/796,593. cited by applicant

USPTO; Non-Final Office Action dated Dec. 26, 2017 in U.S. Appl. No. 15/798,120. cited by applicant

USPTO; Notice of Allowance dated Jun. 13, 2018 in U.S. Appl. No. 15/798,120. cited by applicant

USPTO; Non-Final Office Action dated Dec. 21, 2018 in U.S. Appl. No. 15/798,150. cited by applicant

USPTO; Notice of Allowance dated May 14, 2019 in U.S. Appl. No. 15/798,150. cited by applicant

USPTO; Non-Final Office Action dated Aug. 9, 2018 in U.S. Appl. No. 15/798,201. cited by applicant

USPTO; Final Office Action dated Dec. 14, 2018 in U.S. Appl. No. 15/798,201. cited by applicant

USPTO; Non-Final Office Action dated Oct. 2, 2019 in U.S. Appl. No. 15/798,201. cited by applicant

USPTO; Final Office Action dated Mar. 19, 2020 in U.S. Appl. No. 15/798,201. cited by applicant

USPTO; Advisory Action dated May 26, 2020 in U.S. Appl. No. 15/798,201. cited by applicant

USPTO; Non-Final Office Action dated Jul. 10, 2020 in U.S. Appl. No. 15/798,201. cited by applicant

USPTO; Notice of Allowance dated Oct. 21, 2020 in U.S. Appl. No. 15/798,201. cited by applicant

USPTO; Non-Final Office Action dated Nov. 15, 2019 in U.S. Appl. No. 15/802,154. cited by applicant

USPTO; Final Office Action dated Mar. 11, 2020 in U.S. Appl. No. 15/802,154. cited by applicant

USPTO; Advisory Action dated May 19, 2020 in U.S. Appl. No. 15/802,154. cited by applicant
USPTO; Non-Final Office Action dated Jul. 2, 2020 in U.S. Appl. No. 15/802,154. cited by applicant
USPTO; Notice of Allowance dated Oct. 26, 2020 in U.S. Appl. No. 15/802,154. cited by applicant
USPTO; Non-Final Office Action dated Jul. 2, 2018 in U.S. Appl. No. 15/815,483. cited by applicant
USPTO; Final Office Action dated Mar. 7, 2019 in U.S. Appl. No. 15/815,483. cited by applicant
USPTO; Non-Final Office Action dated Aug. 20, 2019 in U.S. Appl. No. 15/815,483. cited by applicant
USPTO; Final Office Action dated Jan. 13, 2020 in U.S. Appl. No. 15/815,483. cited by applicant
USPTO; Advisory Action dated Mar. 25, 2020 in U.S. Appl. No. 15/815,483. cited by applicant
USPTO; Non-Final Office Action dated Apr. 24, 2020 in U.S. Appl. No. 15/815,483. cited by applicant
USPTO; Notice of Allowance dated Sep. 30, 2020 in U.S. Appl. No. 15/815,483. cited by applicant
USPTO; Non-Final Office Action dated Sep. 26, 2018 in U.S. Appl. No. 15/832,188. cited by applicant
USPTO; Notice of Allowance dated Dec. 5, 2017 in U.S. Appl. No. 15/832,188. cited by applicant
USPTO; Non-Final Office Action dated Mar. 17, 2020 in U.S. Appl. No. 15/835,328. cited by applicant
USPTO; Final Office Action dated Sep. 17, 2020 in U.S. Appl. No. 15/835,328. cited by applicant
USPTO; Advisory Action dated Dec. 15, 2020 in U.S. Appl. No. 15/835,328. cited by applicant
USPTO; Non-Final Office Action dated Feb. 20, 2020 in U.S. Appl. No. 15/835,352. cited by applicant
USPTO; Final Office Action dated Aug. 6, 2020 in U.S. Appl. No. 15/835,352. cited by applicant
USPTO; Notice of Allowance dated Jan. 14, 2021 in U.S. Appl. No. 15/835,352. cited by applicant
USPTO; Non-Final Office Action dated Sep. 10, 2018 in U.S. Appl. No. 15/836,547. cited by applicant
USPTO; Non-Final Office Action dated Mar. 13, 2019 in U.S. Appl. No. 15/836,547. cited by applicant
USPTO; Notice of Allowance dated Aug. 16, 2019 in U.S. Appl. No. 15/836,547. cited by applicant
USPTO; Notice of Allowance dated Aug. 17, 2020 in U.S. Appl. No. 15/843,899. cited by applicant
USPTO; Non-Final Office Action dated Jul. 31, 2019 in U.S. Appl. No. 15/860,564. cited by applicant
USPTO; Final Office Action dated Nov. 13, 2019 in U.S. Appl. No. 15/860,564. cited by applicant
USPTO; Advisory Action dated Jan. 21, 2020 in U.S. Appl. No. 15/860,564. cited by applicant
USPTO; Non-Final Office Action dated Mar. 30, 2020 in U.S. Appl. No. 15/860,564. cited by applicant
USPTO; Notice of Allowance dated Jun. 30, 2020 in U.S. Appl. No. 15/860,564. cited by applicant
USPTO; Non-Final Office Action dated Mar. 20, 2020 in U.S. Appl. No. 15/861,418. cited by applicant
USPTO; Final Office Action dated Jun. 11, 2020 in U.S. Appl. No. 15/861,418. cited by applicant
USPTO; Notice of Allowance dated Aug. 19, 2020 in U.S. Appl. No. 15/861,418. cited by applicant
USPTO; Non-Final Office Action dated Jul. 23, 2018 in U.S. Appl. No. 15/863,340. cited by applicant
USPTO; Notice of Allowance dated Dec. 10, 2018 in U.S. Appl. No. 15/863,340. cited by applicant
USPTO; Non-Final Office Action dated Jan. 11, 2019 in U.S. Appl. No. 15/879,209. cited by applicant
USPTO; Non-Final Office Action dated Jan. 22, 2019 in U.S. Appl. No. 15/879,209. cited by applicant
USPTO; Final Office Action dated Aug. 21, 2019 in U.S. Appl. No. 15/879,209. cited by applicant

USPTO; Advisory Action dated Nov. 5, 2019 in U.S. Appl. No. 15/879,209. cited by applicant
USPTO; Notice of Allowance dated Jan. 16, 2020 in U.S. Appl. No. 15/879,209. cited by applicant
USPTO; Non-Final Office Action dated Apr. 17, 2019 in U.S. Appl. No. 15/886,225. cited by applicant
USPTO; Notice of Allowance dated Sep. 23, 2019 in U.S. Appl. No. 15/886,225. cited by applicant
USPTO; Non-Final Office Action dated Nov. 15, 2018 in U.S. Appl. No. 15/890,037. cited by applicant
USPTO; Final Office Action dated May 2, 2019 in U.S. Appl. No. 15/890,037. cited by applicant
USPTO; Non-Final Office Action dated Oct. 24, 2019 in U.S. Appl. No. 15/890,037. cited by applicant
USPTO; Final Office Action dated Feb. 26, 2020 in U.S. Appl. No. 15/890,037. cited by applicant
USPTO; Advisory Action dated Jun. 5, 2020 in U.S. Appl. No. 15/890,037. cited by applicant
USPTO; Non-Final Office Action dated Sep. 2, 2020 in U.S. Appl. No. 15/890,037. cited by applicant
USPTO; Final Office Action dated Jan. 11, 2021 in U.S. Appl. No. 15/890,037. cited by applicant
USPTO; Notice of Allowance dated Feb. 8, 2019 in U.S. Appl. No. 15/892,756. cited by applicant
USPTO; Non-Final Office Action dated Apr. 24, 2019 in U.S. Appl. No. 15/896,986. cited by applicant
USPTO; Final Office Action dated Nov. 19, 2019 in U.S. Appl. No. 15/896,986. cited by applicant
USPTO; Advisory Action dated Feb. 7, 2020 in U.S. Appl. No. 15/896,986. cited by applicant
USPTO; Non-Final Office Action dated Mar. 24, 2020 in U.S. Appl. No. 15/896,986. cited by applicant
USPTO; Final Office Action dated Jun. 15, 2020 in U.S. Appl. No. 15/896,986. cited by applicant
USPTO; Notice of Allowance dated Aug. 26, 2020 in U.S. Appl. No. 15/896,986. cited by applicant
USPTO; Non-Final Office Action dated Sep. 19, 2019 in U.S. Appl. No. 15/897,578. cited by applicant
USPTO; Notice of Allowance dated Mar. 27, 2020 in U.S. Appl. No. 15/897,578. cited by applicant
USPTO; Non-Final Office Action dated May 30, 2019 in U.S. Appl. No. 15/900,425. cited by applicant
USPTO; Notice of Allowance dated Dec. 17, 2019 in U.S. Appl. No. 15/900,425. cited by applicant
USPTO; Non-Final Office Action dated Sep. 9, 2020 in U.S. Appl. No. 15/909,705. cited by applicant
USPTO; Non-Final Office Action dated Mar. 8, 2019 in U.S. Appl. No. 15/917,224. cited by applicant
USPTO; Final Office Action dated Aug. 28, 2019 in U.S. Appl. No. 15/917,224. cited by applicant
USPTO; Advisory Action dated Nov. 27, 2019 in U.S. Appl. No. 15/917,224. cited by applicant
USPTO; Non-Final Office Action dated May 1, 2020 in U.S. Appl. No. 15/917,224. cited by applicant
USPTO; Final Office Action dated Sep. 24, 2020 in U.S. Appl. No. 15/917,224. cited by applicant
USPTO; Non-Final Office Action dated Feb. 8, 2019 in U.S. Appl. No. 15/917,262. cited by applicant
USPTO; Final Office Action dated Jun. 14, 2019 in U.S. Appl. No. 15/917,262. cited by applicant
USPTO; Notice of Allowance dated Aug. 30, 2019 in U.S. Appl. No. 15/917,262. cited by applicant
USPTO; Non-Final Office Action dated Dec. 23, 2019 in U.S. Appl. No. 15/923,834. cited by applicant
USPTO; Final Office Action dated Mar. 12, 2020 in U.S. Appl. No. 15/923,834. cited by applicant
USPTO; Advisory Action dated May 15, 2020 in U.S. Appl. No. 15/923,834. cited by applicant
USPTO; Non-Final Office Action dated Jul. 6, 2020 in U.S. Appl. No. 15/923,834. cited by applicant
USPTO; Final Office Action dated Oct. 19, 2020 in U.S. Appl. No. 15/923,834. cited by applicant

USPTO; Advisory Action dated Jan. 8, 2021 in U.S. Appl. No. 15/923,834. cited by applicant
USPTO; Non-Final Office Action dated May 8, 2019 in U.S. Appl. No. 15/925,532. cited by applicant
USPTO; Final Office Action dated Nov. 27, 2019 in U.S. Appl. No. 15/925,532. cited by applicant
USPTO; Non-Final Office Action dated May 1, 2020 in U.S. Appl. No. 15/925,532. cited by applicant
USPTO; Final Office Action dated Oct. 15, 2020 in U.S. Appl. No. 15/925,532. cited by applicant
USPTO; Advisory Action dated Dec. 24, 2020 in U.S. Appl. No. 15/925,532. cited by applicant
USPTO; Non-Final Office Action dated Jan. 25, 2021 in U.S. Appl. No. 15/925,532. cited by applicant
USPTO; Non-Final Office Action dated Oct. 6, 2020 in U.S. Appl. No. 15/940,729. cited by applicant
USPTO; Final Office Action dated Feb. 18, 2021 in U.S. Appl. No. 15/940,729. cited by applicant
USPTO; Non-Final Office Action dated Jul. 22, 2019 in U.S. Appl. No. 15/940,759. cited by applicant
USPTO; Final Office Action dated Feb. 19, 2020 in U.S. Appl. No. 15/940,759. cited by applicant
USPTO; Advisory Action dated Apr. 23, 2020 in U.S. Appl. No. 15/940,759. cited by applicant
USPTO; Non-Final Office Action dated Aug. 3, 2020 in U.S. Appl. No. 15/940,759. cited by applicant
USPTO; Final Office Action dated Jan. 12, 2021 in U.S. Appl. No. 15/940,759. cited by applicant
USPTO; Non-Final Office Action dated Mar. 29, 2019 in U.S. Appl. No. 15/940,801. cited by applicant
USPTO; Notice of Allowance dated Aug. 26, 2019 in U.S. Appl. No. 15/940,801. cited by applicant
USPTO; Non-Final Rejection dated Nov. 29, 2019 in U.S. Appl. No. 15/949,990. cited by applicant
USPTO; Notice of Allowance dated Mar. 12, 2020 in U.S. Appl. No. 15/949,990. cited by applicant
USPTO; Notice of Allowance dated May 31, 2019 in U.S. Appl. No. 15/957,565. cited by applicant
USPTO; Non-Final Office Action dated Jan. 31, 2020 in U.S. Appl. No. 15/962,980. cited by applicant
USPTO; Final Office Action dated Apr. 30, 2020 in U.S. Appl. No. 15/962,980. cited by applicant
USPTO; Advisory Action dated Jul. 2, 2020 in U.S. Appl. No. 15/962,980. cited by applicant
USPTO; Non-Final Office Action dated Sep. 14, 2020 in U.S. Appl. No. 15/962,980. cited by applicant
USPTO; Final Office Action dated Dec. 15, 2020 in U.S. Appl. No. 15/962,980. cited by applicant
USPTO; Non-Final Office Action dated Oct. 6, 2020 in U.S. Appl. No. 15/967,146. cited by applicant
USPTO; Non-Final Office Action dated Apr. 2, 2020 in U.S. Appl. No. 15/974,948. cited by applicant
USPTO; Final Office Action dated Sep. 3, 2020 in U.S. Appl. No. 15/974,948. cited by applicant
USPTO; Advisory Action dated Oct. 27, 2020 in U.S. Appl. No. 15/974,948. cited by applicant
USPTO; Non-Final Office Action dated May 15, 2020 in U.S. Appl. No. 15/974,988. cited by applicant
USPTO; Notice of Allowance dated Oct. 14, 2020 in U.S. Appl. No. 15/974,988. cited by applicant
USPTO; Non-Final Office Action dated Jan. 16, 2020 in U.S. Appl. No. 15/985,261. cited by applicant
USPTO; Notice of Allowance dated Apr. 3, 2020 in U.S. Appl. No. 15/985,261. cited by applicant
USPTO; Non-Final Office Action dated Apr. 19, 2019 in U.S. Appl. No. 15/985,298. cited by applicant
USPTO; Notice of Allowance dated Aug. 22, 2019 in U.S. Appl. No. 15/985,298. cited by applicant
USPTO; Non-Final Office Action dated Jun. 12, 2020 in U.S. Appl. No. 15/985,539. cited by applicant

USPTO; Final Office Action dated Nov. 25, 2020 in U.S. Appl. No. 15/985,539. cited by applicant
USPTO; Non-Final Office Action dated Feb. 21, 2019 in U.S. Appl. No. 15/987,755. cited by applicant
USPTO; Notice of Allowance dated Jul. 31, 2019 in U.S. Appl. No. 15/987,755. cited by applicant
USPTO; Non-Final Office Action dated Feb. 20, 2020 in U.S. Appl. No. 15/996,286. cited by applicant
USPTO; Final Office Action dated Jul. 24, 2020 in U.S. Appl. No. 15/996,286. cited by applicant
USPTO; Advisory Action dated Oct. 7, 2020 in U.S. Appl. No. 15/996,286. cited by applicant
USPTO; Non-Final Office Action dated Apr. 1, 2020 in U.S. Appl. No. 15/997,445. cited by applicant
USPTO; Final Office Action dated Oct. 2, 2020 in U.S. Appl. No. 15/997,445. cited by applicant
USPTO; Notice of Allowance dated Jul. 10, 2020 in U.S. Appl. No. 15/998,775. cited by applicant
USPTO; Non-Final Office Action dated Mar. 3, 2020 in U.S. Appl. No. 16/000,109. cited by applicant
USPTO; Final Office Action dated Sep. 14, 2020 in U.S. Appl. No. 16/000,109. cited by applicant
USPTO; Advisory Action dated Dec. 10, 2020 in U.S. Appl. No. 16/000,109. cited by applicant
USPTO; Non-Final Office Action dated Jul. 29, 2019 in U.S. Appl. No. 16/000,125. cited by applicant
USPTO; Final Office Action dated Jan. 13, 2020 in U.S. Appl. No. 16/000,125. cited by applicant
USPTO; Advisory Action dated Apr. 3, 2020 in U.S. Appl. No. 16/000,125. cited by applicant
USPTO; Non-Final Office Action dated Jun. 5, 2020 in U.S. Appl. No. 16/000,125. cited by applicant
USPTO; Final Office Action dated Oct. 6, 2020 in U.S. Appl. No. 16/000,125. cited by applicant
USPTO; Advisory Action dated Dec. 15, 2020 in U.S. Appl. No. 16/000,125. cited by applicant
USPTO; Non-Final Office Action dated Apr. 22, 2020 in U.S. Appl. No. 16/000,156. cited by applicant
USPTO; Final Office Action dated Oct. 22, 2020 in U.S. Appl. No. 16/000,156. cited by applicant
USPTO; Advisory Action dated Dec. 14, 2020 in U.S. Appl. No. 16/000,156. cited by applicant
USPTO; Non-Final Office Action dated May 7, 2020 in U.S. Appl. No. 16/004,041. cited by applicant
USPTO; Final Office Action dated Oct. 15, 2020 in U.S. Appl. No. 16/004,041. cited by applicant
USPTO; Advisory Action dated Dec. 22, 2020 in U.S. Appl. No. 16/004,041. cited by applicant
USPTO; Non-Final Office Action dated Jul. 16, 2019 in U.S. Appl. No. 16/014,981. cited by applicant
USPTO; Non-Final Office Action dated Jan. 16, 2020 in U.S. Appl. No. 16/014,981. cited by applicant
USPTO; Notice of Allowance dated Jun. 24, 2020 in U.S. Appl. No. 16/014,981. cited by applicant
USPTO; Non-Final Office Action dated Jan. 24, 2019 in U.S. Appl. No. 16/018,692. cited by applicant
USPTO; Final Office Action dated Aug. 23, 2019 in U.S. Appl. No. 16/018,692. cited by applicant
USPTO; Non-Final Office Action dated Dec. 11, 2019 in U.S. Appl. No. 16/018,692. cited by applicant
USPTO; Notice of Allowance dated Jan. 30, 2020 in U.S. Appl. No. 16/018,692. cited by applicant
USPTO; Non-Final Office Action dated Aug. 7, 2019 in U.S. Appl. No. 16/024,390. cited by applicant
USPTO; Notice of Allowance dated Nov. 26, 2019 in U.S. Appl. No. 16/024,390. cited by applicant
USPTO; Notice of Allowance dated Apr. 9, 2019 in U.S. Appl. No. 16/026,711. cited by applicant
USPTO; Non-Final Office Action dated Mar. 4, 2020 in U.S. Appl. No. 16/036,692. cited by applicant
USPTO; Notice of Allowance dated May 26, 2020 in U.S. Appl. No. 16/036,692. cited by applicant

USPTO; Non-Final Office Action dated Apr. 25, 2019 in U.S. Appl. No. 16/038,024. cited by applicant

USPTO; Final Office Action dated Nov. 29, 2019 in U.S. Appl. No. 16/038,024. cited by applicant

USPTO; Advisory Action dated Feb. 28, 2020 in U.S. Appl. No. 16/038,024. cited by applicant

USPTO; Notice of Allowance dated Mar. 23, 2020 in U.S. Appl. No. 16/038,024. cited by applicant

USPTO; Non-Final Office Action dated Sep. 8, 2020 in U.S. Appl. No. 16/039,817. cited by applicant

USPTO; Non-Final Office Action dated Jul. 25, 2019 in U.S. Appl. No. 16/039,867. cited by applicant

USPTO; Final Office Action dated Jan. 28, 2020 in U.S. Appl. No. 16/039,867. cited by applicant

USPTO; Notice of Allowance dated Apr. 3, 2020 in U.S. Appl. No. 16/039,867. cited by applicant

USPTO; Non-Final Office Action dated Nov. 24, 2020 in U.S. Appl. No. 16/039,938. cited by applicant

USPTO; Non-Final Office Action dated Aug. 4, 2020 in U.S. Appl. No. 16/042,791. cited by applicant

USPTO; Notice of Allowance dated Jul. 10, 2019 in U.S. Appl. No. 16/046,218. cited by applicant

USPTO; Non-Final Office Action dated Apr. 30, 2020 in U.S. Appl. No. 16/055,532. cited by applicant

USPTO; Final Office Action dated Jul. 30, 2020 in U.S. Appl. No. 16/055,532. cited by applicant

USPTO; Final Office Action dated Nov. 3, 2020 in U.S. Appl. No. 16/055,532. cited by applicant

USPTO; Advisory Action dated Jan. 14, 2021 in U.S. Appl. No. 16/055,532. cited by applicant

USPTO; Non-Final Office Action dated May 27, 2020 in U.S. Appl. No. 16/100,012. cited by applicant

USPTO; Notice of Allowance dated Sep. 14, 2020 in U.S. Appl. No. 16/100,012. cited by applicant

USPTO; Non-Final Office Action dated Feb. 20, 2020 in U.S. Appl. No. 16/105,745. cited by applicant

USPTO; Final Office Action dated Jun. 23, 2020 in U.S. Appl. No. 16/105,745. cited by applicant

USPTO; Advisory Action dated Aug. 25, 2020 in U.S. Appl. No. 16/105,745. cited by applicant

USPTO; Non-Final Office Action dated Nov. 25, 2020 in U.S. Appl. No. 16/105,745. cited by applicant

USPTO; Non-Final Office Action dated Feb. 24, 2020 in U.S. Appl. No. 16/105,761. cited by applicant

USPTO; Final Office Action dated Apr. 21, 2020 in U.S. Appl. No. 16/105,761. cited by applicant

USPTO; Advisory Action dated Jul. 8, 2020 in U.S. Appl. No. 16/105,761. cited by applicant

USPTO; Non-Final Office Action dated Nov. 25, 2020 in U.S. Appl. No. 16/105,761. cited by applicant

USPTO; Non-Final Office Action dated Mar. 19, 2020 in U.S. Appl. No. 16/105,802. cited by applicant

USPTO; Final Office Action dated Jun. 26, 2020 in U.S. Appl. No. 16/105,802. cited by applicant

USPTO; Final Office Action dated Nov. 17, 2020 in U.S. Appl. No. 16/105,802. cited by applicant

USPTO; Non-Final Office Action dated Sep. 8, 2020 in U.S. Appl. No. 16/108,950. cited by applicant

USPTO; Final Office Action dated Dec. 15, 2020 in U.S. Appl. No. 16/108,950. cited by applicant

USPTO; Non-Final Office Action dated Mar. 16, 2020 in U.S. Appl. No. 16/116,708. cited by applicant

USPTO; Final Office Action dated Sep. 2, 2020 in U.S. Appl. No. 16/116,708. cited by applicant

USPTO; Advisory Action dated Nov. 17, 2020 in U.S. Appl. No. 16/116,708. cited by applicant

USPTO; Non-Final Office Action dated Feb. 6, 2020 in U.S. Appl. No. 16/117,530. cited by applicant

USPTO; Final Office Action dated Jan. 6, 2021 in U.S. Appl. No. 16/117,530. cited by applicant

USPTO; Non-Final Office Action dated Jun. 22, 2020 in U.S. Appl. No. 16/128,282. cited by applicant

USPTO; Final Office Action dated Nov. 19, 2020 in U.S. Appl. No. 16/128,282. cited by applicant

USPTO; Notice of Allowance dated Feb. 12, 2021 in U.S. Appl. No. 16/128,282. cited by applicant

USPTO; Non-Final Office Action dated Apr. 21, 2020 in U.S. Appl. No. 16/130,798. cited by applicant

USPTO; Notice of Allowance dated Aug. 5, 2020 in U.S. Appl. No. 16/130,798. cited by applicant

USPTO; Non-Final Office Action dated Jan. 6, 2020 in U.S. Appl. No. 16/132,142. cited by applicant

USPTO; Final Office Action dated Jun. 5, 2020 in U.S. Appl. No. 16/132,142. cited by applicant

USPTO; Advisory Action dated Aug. 24, 2020 in U.S. Appl. No. 16/132,142. cited by applicant

USPTO; Non-Final Office Action dated Oct. 16, 2020 in U.S. Appl. No. 16/132,142. cited by applicant

USPTO; Notice of Allowance dated Feb. 8, 2021 in U.S. Appl. No. 16/132,142. cited by applicant

USPTO; Non-Final Office Action dated Feb. 28, 2020 in U.S. Appl. No. 16/137,974. cited by applicant

USPTO; Final Office Action dated Aug. 20, 2020 in U.S. Appl. No. 16/137,974. cited by applicant

USPTO; Notice of Allowance dated Nov. 3, 2020 in U.S. Appl. No. 16/137,974. cited by applicant

USPTO; Non-Final Office Action dated Apr. 2, 2019 in U.S. Appl. No. 16/147,047. cited by applicant

USPTO; Final Office Action dated Sep. 25, 2019 in U.S. Appl. No. 16/147,047. cited by applicant

USPTO; Notice of Allowance dated Jan. 3, 2020 in U.S. Appl. No. 16/147,047. cited by applicant

USPTO; Non-Final Office Action dated Jul. 30, 2020 in U.S. Appl. No. 16/151,074. cited by applicant

USPTO; Final Office Action dated Dec. 14, 2020 in U.S. Appl. No. 16/151,074. cited by applicant

USPTO; Advisory Action dated Feb. 16, 2021 in U.S. Appl. No. 16/151,074. cited by applicant

USPTO; Non-Final Office Action dated Nov. 5, 2019 in U.S. Appl. No. 16/152,260. cited by applicant

USPTO; Advisory Action dated Sep. 16, 2020 in U.S. Appl. No. 16/152,260. cited by applicant

USPTO; Final Office Action dated Jul. 15, 2020 in U.S. Appl. No. 16/152,260. cited by applicant

USPTO; Non-Final Office Action dated Nov. 9, 2020 in U.S. Appl. No. 16/152,260. cited by applicant

USPTO; Non-Final Office Action dated Aug. 6, 2019 in U.S. Appl. No. 16/158,077. cited by applicant

USPTO; Final Office Action dated Mar. 2, 2020 in U.S. Appl. No. 16/158,077. cited by applicant

USPTO; Advisory Action dated Jun. 9, 2020 in U.S. Appl. No. 16/158,077. cited by applicant

USPTO; Notice of Allowance dated Jul. 24, 2020 in U.S. Appl. No. 16/158,077. cited by applicant

USPTO; Non-Final Office Action dated Aug. 29, 2019 in U.S. Appl. No. 16/161,744. cited by applicant

USPTO; Final Office Action dated Mar. 17, 2020 in U.S. Appl. No. 16/161,744. cited by applicant

USPTO; Notice of Allowance dated Jun. 19, 2020 in U.S. Appl. No. 16/161,744. cited by applicant

USPTO; Non-Final Office Action dated Jul. 23, 2020 in U.S. Appl. No. 16/167,164. cited by applicant

USPTO; Final Office Action dated Dec. 18, 2020 in U.S. Appl. No. 16/167,164. cited by applicant

USPTO; Non-Final Office Action dated Aug. 16, 2019 in U.S. Appl. No. 16/167,225. cited by applicant

USPTO; Final Office Action dated Dec. 16, 2019 in U.S. Appl. No. 16/167,225. cited by applicant

USPTO; Notice of Allowance dated Mar. 5, 2020 in U.S. Appl. No. 16/167,225. cited by applicant

USPTO; Notice of Allowance dated Apr. 17, 2019 in U.S. Appl. No. 16/171,098. cited by applicant

USPTO; Notice of Allowance dated May 1, 2019 in U.S. Appl. No. 16/171,098. cited by applicant

USPTO; Non-Final Office Action dated Aug. 7, 2020 in U.S. Appl. No. 16/172,535. cited by applicant

USPTO; Final Office Action dated Oct. 16, 2020 in U.S. Appl. No. 16/172,535. cited by applicant

USPTO; Advisory Action dated Dec. 21, 2020 in U.S. Appl. No. 16/172,535. cited by applicant

USPTO; Non-Final Office Action dated Nov. 4, 2020 in U.S. Appl. No. 16/176,517. cited by applicant

USPTO; Final Office Action dated Jan. 29, 2021 in U.S. Appl. No. 16/176,517. cited by applicant

USPTO; Non-Final Office Action dated Oct. 29, 2020 in U.S. Appl. No. 16/183,258. cited by applicant

USPTO; Non-Final Office Action dated Apr. 2, 2019 in U.S. Appl. No. 16/188,690. cited by applicant

USPTO; Final Office Action dated Sep. 26, 2019 in U.S. Appl. No. 16/188,690. cited by applicant

USPTO; Notice of Allowance dated Dec. 16, 2019 in U.S. Appl. No. 16/188,690. cited by applicant

USPTO; Notice of Allowance dated Jul. 28, 2020 in U.S. Appl. No. 16/193,789. cited by applicant

USPTO; Notice of Allowance dated Jul. 7, 2020 in U.S. Appl. No. 16/194,041. cited by applicant

USPTO; Notice of Allowance dated Oct. 3, 2019 in U.S. Appl. No. 16/200,100. cited by applicant

USPTO; Non-Final Office Action dated Dec. 11, 2021 in U.S. Appl. No. 16/202,941. cited by applicant

USPTO; Non-Final Office Action dated May 29, 2020 in U.S. Appl. No. 16/205,899. cited by applicant

USPTO; Final Office Action dated Nov. 27, 2020 in U.S. Appl. No. 16/205,899. cited by applicant

USPTO; Non-Final Office Action dated Aug. 19, 2019 in U.S. Appl. No. 16/208,062. cited by applicant

USPTO; Notice of Allowance dated Jan. 28, 2020 in U.S. Appl. No. 16/208,062. cited by applicant

USPTO; Non-Final Office Action dated Jul. 15, 2020 in U.S. Appl. No. 16/210,922. cited by applicant

USPTO; Final Office Action dated Dec. 9, 2020 in U.S. Appl. No. 16/210,922. cited by applicant

USPTO; Non-Final Office Action dated Sep. 16, 2019 in U.S. Appl. No. 16/213,702. cited by applicant

USPTO; Notice of Allowance dated Dec. 9, 2019 in U.S. Appl. No. 16/213,702. cited by applicant

USPTO; Non-Final Office Action dated Jan. 8, 2020 in U.S. Appl. No. 16/219,555. cited by applicant

USPTO; Final Office Action dated Jun. 19, 2020 in U.S. Appl. No. 16/219,555. cited by applicant

USPTO; Advisory Action dated Sep. 11, 2020 in U.S. Appl. No. 16/219,555. cited by applicant

USPTO; Non-Final Office Action dated Sep. 30, 2020 in U.S. Appl. No. 16/219,555. cited by applicant

USPTO; Non-Final Office Action dated Feb. 19, 2020 in U.S. Appl. No. 16/242,829. cited by applicant

USPTO; Final Office Action dated Jun. 15, 2020 in U.S. Appl. No. 16/242,829. cited by applicant

USPTO; Advisory Action dated Aug. 11, 2020 in U.S. Appl. No. 16/242,829. cited by applicant

USPTO; Notice of Allowance dated Sep. 23, 2020 in U.S. Appl. No. 16/242,829. cited by applicant

USPTO; Non-Final Office Action dated Sep. 3, 2020 in U.S. Appl. No. 16/242,852. cited by applicant

USPTO; Notice of Allowance dated Dec. 15, 2020 in U.S. Appl. No. 16/242,852. cited by applicant

USPTO; Non-Final Office Action dated Jan. 30, 2020 in U.S. Appl. No. 16/245,006. cited by applicant

USPTO; Notice of Allowance dated May 29, 2020 in U.S. Appl. No. 16/245,006. cited by applicant

USPTO; Non-Final Office Action dated Sep. 24, 2019 in U.S. Appl. No. 16/251,534. cited by applicant

USPTO; Final Office Action dated Jan. 10, 2020 in U.S. Appl. No. 16/251,534. cited by applicant

USPTO; Advisory Action dated Apr. 3, 2020 in U.S. Appl. No. 16/251,534. cited by applicant
USPTO; Non-Final Office Action dated Apr. 24, 2020 in U.S. Appl. No. 16/251,534. cited by applicant
USPTO; Final Office Action dated Oct. 14, 2020 in U.S. Appl. No. 16/251,534. cited by applicant
USPTO; Non-Final Office Action dated May 27, 2020 in U.S. Appl. No. 16/252,567. cited by applicant
USPTO; Final Office Action dated Nov. 18, 2020 in U.S. Appl. No. 16/252,567. cited by applicant
USPTO; Advisory Action dated Jan. 22, 2021 in U.S. Appl. No. 16/252,567. cited by applicant
USPTO; Non-Final Office Action dated Apr. 16, 2020 in U.S. Appl. No. 16/255,639. cited by applicant
USPTO; Notice of Allowance dated Aug. 6, 2020 in U.S. Appl. No. 16/255,639. cited by applicant
USPTO; Non-Final Office Action dated Jun. 30, 2020 in U.S. Appl. No. 16/258,187. cited by applicant
USPTO; Notice of Allowance dated Aug. 5, 2020 in U.S. Appl. No. 16/258,187. cited by applicant
USPTO; Non-Final Office Action dated Dec. 18, 2019 in U.S. Appl. No. 16/280,964. cited by applicant
USPTO; Notice of Allowance dated May 21, 2020 in U.S. Appl. No. 16/280,964. cited by applicant
USPTO; Non-Final Office Action dated Jan. 9, 2020 in U.S. Appl. No. 16/317,774. cited by applicant
USPTO; Notice of Allowance dated Apr. 15, 2020 in U.S. Appl. No. 16/317,774. cited by applicant
USPTO; Non-Final Office Action dated Mar. 2, 2020 in U.S. Appl. No. 16/356,394. cited by applicant
USPTO; Final Office Action dated Jun. 30, 2020 in U.S. Appl. No. 16/356,394. cited by applicant
USPTO; Notice of Allowance dated Aug. 31, 2020 in U.S. Appl. No. 16/356,394. cited by applicant
USPTO; Notice of Allowance dated Jun. 13, 2019 in U.S. Appl. No. 16/396,475. cited by applicant
USPTO; Non-Final Office Action dated Aug. 5, 2020 in U.S. Appl. No. 16/397,045. cited by applicant
USPTO; Final Office Action dated Dec. 8, 2020 in U.S. Appl. No. 16/397,045. cited by applicant
USPTO; Advisory Action dated Jan. 27, 2021 in U.S. Appl. No. 16/397,045. cited by applicant
USPTO; Non-Final Office Action dated Aug. 19, 2020 in U.S. Appl. No. 16/400,814. cited by applicant
USPTO; Final Office Action dated Dec. 14, 2020 in U.S. Appl. No. 16/400,814. cited by applicant
USPTO; Non-Final Office Action dated Dec. 26, 2019 in U.S. Appl. No. 16/417,938. cited by applicant
USPTO; Notice of Allowance dated Mar. 30, 2020 in U.S. Appl. No. 16/417,938. cited by applicant
USPTO; Non-Final Office Action dated Dec. 31, 2019 in U.S. Appl. No. 16/427,288. cited by applicant
USPTO; Notice of Allowance dated Apr. 16, 2020 in U.S. Appl. No. 16/427,288. cited by applicant
USPTO; Non-Final Office Action dated Jan. 12, 2021 in U.S. Appl. No. 16/453,249. cited by applicant
USPTO; Non-Final Office Action dated May 12, 2020 in U.S. Appl. No. 16/454,063. cited by applicant
USPTO; Notice of Allowance dated Sep. 18, 2020 in U.S. Appl. No. 16/454,063. cited by applicant
USPTO; Non-Final Office Action dated Dec. 31, 2019 in U.S. Appl. No. 16/455,406. cited by applicant
USPTO; Notice of Allowance dated Apr. 17, 2020 in U.S. Appl. No. 16/455,406. cited by applicant
USPTO; Non-Final Office Action dated Nov. 27, 2020 in U.S. Appl. No. 16/468,258. cited by applicant
USPTO; Non-Final Office Action dated Dec. 28, 2020 in U.S. Appl. No. 16/517,122. cited by applicant

USPTO; Non-Final Office Action dated Nov. 5, 2020 in U.S. Appl. No. 16/546,543. cited by applicant

USPTO; Non-Final Office Action dated Feb. 11, 2021 in U.S. Appl. No. 16/565,460. cited by applicant

USPTO; Non-Final Office Action dated Jun. 11, 2020 in U.S. Appl. No. 16/598,768. cited by applicant

USPTO; Notice of Allowance dated Sep. 16, 2020 in U.S. Appl. No. 16/598,768. cited by applicant

USPTO; Final Office Action dated Jan. 10, 2020 in U.S. Appl. No. 16/674,894. cited by applicant

USPTO; Notice of Allowance dated Apr. 21, 2020 in U.S. Appl. No. 16/674,894. cited by applicant

USPTO; Non-Final Office Action dated Oct. 16, 2020 in U.S. Appl. No. 16/679,885. cited by applicant

USPTO; Notice of Allowance dated Jan. 25, 2021 in U.S. Appl. No. 16/679,885. cited by applicant

USPTO; Non-Final Office Action dated Feb. 8, 2021 in U.S. Appl. No. 16/685,787. cited by applicant

USPTO; Non-Final Office Action dated Nov. 16, 2020 in U.S. Appl. No. 16/713,311. cited by applicant

USPTO; Non-Final Office Action dated Jan. 8, 2021 in U.S. Appl. No. 16/752,514. cited by applicant

USPTO; Non-Final Office Action dated Apr. 13, 2020 in U.S. Appl. No. 16/833,279. cited by applicant

USPTO; Final Office Action dated Oct. 2, 2020 in U.S. Appl. No. 16/833,279. cited by applicant

USPTO; Notice of Allowance dated Dec. 28, 2020 in U.S. Appl. No. 16/833,279. cited by applicant

USPTO; Non-Final Office Action dated Jul. 9, 2020 in U.S. Appl. No. 16/897,158. cited by applicant

USPTO; Notice of Allowance dated Nov. 12, 2020 in U.S. Appl. No. 16/897,158. cited by applicant

USPTO; Non-Final Office Action dated Aug. 4, 2020 in U.S. Appl. No. 16/926,192. cited by applicant

USPTO; Notice of Allowance dated May 14, 2012 in U.S. Appl. No. 29/411,637. cited by applicant

USPTO; Notice of Allowance dated Oct. 2, 2013 in U.S. Appl. No. 29/412,887. cited by applicant

USPTO; Non-Final Office Action dated Mar. 16, 2015 in U.S. Appl. No. 29/447,298. cited by applicant

USPTO; Notice of Allowance dated Jul. 6, 2015 in U.S. Appl. No. 29/447,298. cited by applicant

USPTO; Notice of Allowance dated Dec. 19, 2013 in U.S. Appl. No. 29/448,094. cited by applicant

USPTO; Notice of Allowance dated Nov. 26, 2014 in U.S. Appl. No. 29/481,301. cited by applicant

USPTO; Notice of Allowance dated Feb. 17, 2015 in U.S. Appl. No. 29/481,308. cited by applicant

USPTO; Notice of Allowance dated Jan. 12, 2015 in U.S. Appl. No. 29/481,312. cited by applicant

USPTO; Notice of Allowance dated Apr. 30, 2015 in U.S. Appl. No. 29/481,315. cited by applicant

USPTO; Notice of Allowance dated May 11, 2015 in U.S. Appl. No. 29/511,011. cited by applicant

USPTO; Notice of Allowance dated May 11, 2015 in U.S. Appl. No. 29/514,153. cited by applicant

USPTO; Notice of Allowance dated Dec. 14, 2015 in U.S. Appl. No. 29/514,264. cited by applicant

USPTO; Notice of Allowance dated Jun. 16, 2017 in U.S. Appl. No. 29/570,711. cited by applicant

USPTO; Non-Final Office Action dated Apr. 16, 2019 in U.S. Appl. No. 29/604,101. cited by applicant

USPTO; Notice of Allowance dated Oct. 30, 2019 in U.S. Appl. No. 29/604,101. cited by applicant

USPTO; Notice of Allowance dated Jun. 26, 2018 in U.S. Appl. No. 29/604,288. cited by applicant

USPTO; Ex Parte Quayle Action dated Dec. 17, 2019 in U.S. Appl. No. 29/615,000. cited by applicant

USPTO; Notice of Allowance dated Jun. 10, 2020 in U.S. Appl. No. 29/615,000. cited by applicant

USPTO; Non-Final Office Action dated Aug. 21, 2019 in U.S. Appl. No. 29/634,768. cited by applicant

USPTO; Notice of Allowance dated Jan. 23, 2020 in U.S. Appl. No. 29/634,768. cited by applicant
USPTO; Non-Final Office Action dated Feb. 20, 2019 in U.S. Appl. No. 29/646,377. cited by applicant
USPTO; Final Office Action dated Jul. 15, 2019 in U.S. Appl. No. 29/646,377. cited by applicant
USPTO; Notice of Allowance dated Nov. 14, 2019 in U.S. Appl. No. 29/646,377. cited by applicant
USPTO; Non-Final Office Action dated Apr. 24, 2020 in U.S. Appl. No. 29/726,623. cited by applicant
USPTO; Final Office Action dated Sep. 9, 2020 in U.S. Appl. No. 29/726,623. cited by applicant
USPTO; Notice of Allowance dated Nov. 23, 2020 in U.S. Appl. No. 29/726,623. cited by applicant
WIPO; International Search Report dated Oct. 13, 1998 in Application No. PCT/FI1998/000571. cited by applicant
WIPO; International Search Report and Written Opinion dated Nov. 16, 2017 in Application No. PCT/IB2017/001015. cited by applicant
WIPO; International Search Report and Written Opinion dated Nov. 13, 2017 in Application No. PCT/IB2017/001050. cited by applicant
WIPO; International Search Report and Written Opinion dated Nov. 30, 2017 in Application No. PCT/IB2017/001070. cited by applicant
WIPO; International Search Report and Written Opinion dated Jan. 25, 2018 in Application No. PCT/IB2017/001262. cited by applicant
WIPO; International Search Report and Written Opinion dated Sep. 14, 2018 in Application No. PCT/IB2017/001640. cited by applicant
WIPO; International Search Report and Written Opinion dated Jun. 1, 2018 in Application No. PCT/IB2017/001644. cited by applicant
WIPO; International Search Report and Written Opinion dated Nov. 6, 2018 in Application No. PCT/IB2017/001652. cited by applicant
WIPO; International Search Report and Written Opinion dated Jun. 1, 2018 in Application No. PCT/IB2017/001656. cited by applicant
WIPO; International Search Report and Written Opinion dated Jan. 25, 2019 in Application No. PCT/IB2018/000192. cited by applicant
WIPO; International Search Report and Written Opinion dated Jul. 9, 2018 in Application No. PCT/IB2018/000419. cited by applicant
WIPO; International Search Report and Written Opinion dated Jan. 4, 2019 in Application No. PCT/IB2018/000936. cited by applicant
WIPO; International Search Report and Written Opinion dated Dec. 20, 2018 in Application No. PCT/IB2018/001003. cited by applicant
WIPO; International Search Report and Written Opinion dated Dec. 20, 2018 in Application No. PCT/IB2018/001022. cited by applicant
WIPO; International Search Report and Written Opinion dated Jun. 28, 2019 in Application No. PCT/IB2019/000084. cited by applicant
WIPO; International Search Report and Written Opinion dated Nov. 19, 2019 in Application No. PCT/IB2019/000127. cited by applicant
WIPO; International Search Report and Written Opinion dated Jan. 10, 2020 in Application No. PCT/IB2019/000729. cited by applicant
WIPO; International Search Report and Written Opinion dated Dec. 20, 2019 in Application No. PCT/IB2019/000805. cited by applicant
WIPO; International Search Report and Written Opinion dated Dec. 20, 2019 in Application No. PCT/IB2019/000817. cited by applicant
WIPO; International Search Report and Written Opinion dated May 23, 2019 in Application No. PCT/IB2019/050974. cited by applicant
WIPO; International Search Report dated Jul. 13, 1990 in Application No. PCT/NL1990/000027.

cited by applicant
WIPO; International Search Report dated Jan. 11, 1990 in Application No. PCT/NL1990/000042.
cited by applicant
WIPO; International Search Report dated Feb. 26, 1991 in Application No. PCT/NL1990/000166.
cited by applicant
WIPO; International Search Report dated Oct. 25, 1991 in Application No. PCT/NL1991/000108.
cited by applicant
WIPO; International Search Report dated Oct. 17, 1997 in Application No. PCT/NL1997/000398.
cited by applicant
WIPO; International Preliminary Examination Report dated Oct. 2, 1998 in Application No. PCT/NL1997/000398. cited by applicant
WIPO; International Search Report dated Apr. 24, 1998 in Application No. PCT/NL1998/000055.
cited by applicant
WIPO; International Preliminary Examination Report dated Feb. 24, 1999 in Application No. PCT/NL1998/000055. cited by applicant
WIPO; International Search Report dated May 27, 1998 in Application No. PCT/NL1998/000105.
cited by applicant
WIPO; International Search Report dated May 8, 1998 in Application No. PCT/NL1998/000151.
cited by applicant
WIPO; International Preliminary Examination Report dated Mar. 23, 1999 in Application No. PCT/NL1998/000151. cited by applicant
WIPO; International Search Report dated May 27, 1998 in Application No. PCT/NL1998/000167.
cited by applicant
WIPO; International Preliminary Examination Report dated Jul. 13, 1999 in Application No. PCT/NL1998/000167. cited by applicant
WIPO; International Search Report dated Jul. 8, 1998 in Application No. PCT/NL1998/000204.
cited by applicant
WIPO; International Search Report dated Aug. 21, 1998 in Application No. PCT/NL1998/000246.
cited by applicant
WIPO; International Search Report dated Oct. 16, 1998 in Application No. PCT/NL1998/000383.
cited by applicant
WIPO; International Preliminary Examination Report dated Sep. 16, 1999 in Application No. PCT/NL1998/000383. cited by applicant
WIPO; International Search Report dated Apr. 15, 1999 in Application No. PCT/NL1999/000047.
cited by applicant
WIPO; International Search Report dated Oct. 12, 1999 in Application No. PCT/NL1999/000487.
cited by applicant
WIPO; International Search Report dated Dec. 16, 1999 in Application No. PCT/NL1999/000583.
cited by applicant
WIPO; Written Opinion dated Jul. 24, 2000 in Application No. PCT/NL1999/000583. cited by applicant
WIPO; International Preliminary Examination Report dated Jan. 22, 2001 in Application No. PCT/NL1999/000583. cited by applicant
WIPO; International Search Report dated Apr. 19, 2000 in Application No. PCT/NL2000/000020.
cited by applicant
WIPO; Written Opinion dated Oct. 17, 2000 in Application No. PCT/NL2000/000020. cited by applicant
WIPO; International Search Report and Written Opinion dated Apr. 17, 2019 in Application No. PCT/NL2018/050787. cited by applicant
WIPO; International Search Report and Written Opinion dated Apr. 29, 2019 in Application No.

PCT/NL2018/050791. cited by applicant

WIPO; International Search Report dated Dec. 16, 1999 in Application No. PCT/RU1999/000328. cited by applicant

WIPO; International Preliminary Examination Report dated Jun. 29, 2001 in Application No. PCT/RU1999/000328. cited by applicant

WIPO; International Preliminary Report on Patentability dated Nov. 24, 2009 and International Search Report dated Jul. 31, 2008 in Application No. PCT/US2008/063919. cited by applicant

WIPO; International Preliminary Report on Patentability dated Feb. 24, 2010 in Application No. PCT/US2008/074063. cited by applicant

WIPO; International Preliminary Report on Patentability dated Nov. 26, 2009 in Application No. PCT/US2009/043454. cited by applicant

WIPO; International Preliminary Report on Patentability dated Jun. 14, 2011 in Application No. PCT/US2009/066377. cited by applicant

WIPO; International Search report and Written Opinion dated Nov. 12, 2010 in Application No. PCT/US2010/030126. cited by applicant

WIPO; International Preliminary Report on Patentability dated Oct. 11, 2011 in Application No. PCT/US2010/030126. cited by applicant

WIPO; International Preliminary Report on Patentability dated Nov. 9, 2011 in Application No. PCT/US2010/033244. cited by applicant

WIPO; International Preliminary Report on Patentability dated Nov. 9, 2011 in Application No. PCT/US2010/033248. cited by applicant

WIPO; International Preliminary Report on Patentability dated Nov. 9, 2011 in Application No. PCT/US2010/033252. cited by applicant

WIPO; International Search report and Written Opinion dated Jan. 20, 2011 in Application No. PCT/US2010/045368. cited by applicant

WIPO; International Search report and Written Opinion dated Feb. 6, 2013 in Application No. PCT/US2012/065343. cited by applicant

WIPO; International Search report and Written Opinion dated Feb. 13, 2013 in Application No. PCT/US2012/065347. cited by applicant

Akiyama et al. "Macro- and micro-scale simulation of growth rate and composition in MOCVD of yttria-stabilized zirconia" J. Crystal Growth 241, pp. 352-362 (2002). cited by applicant

Alen, "Atomic layer deposition of TaN, NbN and MoN films for Cu Metallizations," University of Helsinki Finland, 72 pages, (2005). cited by applicant

Altieri et al. "Review Article: Plasma-surface interactions at the atomic scale for patterning metals" J. Vac. Sci. Technol. A 35, 05C203, 13 pages (2017). cited by applicant

Ande et al. "Role of Surface Termination in Atomic Layer Deposition of Silicon Nitride" J. Phys. Chem. Lett., 6(18), 16 pages (2015). cited by applicant

Aoyagi et al. "Laser-assisted atomic layer epitaxy" Materials Science and Engineering, pp. 121-132 (1991). cited by applicant

Arita et al. "Electrical and optical properties of germanium-doped zinc oxide thin films" Materials Transactions, vol. 45, No. 11, pp. 3180-3183 (2004). cited by applicant

Arnold et al., "Novel single-layer vanadium sulphide phases" 2D Materials, 5, 045009, 11 pages (2018). cited by applicant

Athavale et al., "Realization of Atomic Layer Etching of Silicon", Journal of Vacuum Science and Technology B, vol. 14, pp. 3702-3705 (1996). cited by applicant

Atta et al. "The Catalytic Activity of Ruthenates ARuO₃ (A=Ca, Sr or Ba) for the Hydrogen Evolution Reaction in Acidic Medium" Int. J. Electrochem. Sci. 7, pp. 725-746 (2012). cited by applicant

Ballal "Mass Transfer in Porous Solids under Pulsating Pressure Conditions" SLJ International, vol. 35, Nov. 4, pp. 446-448 (1995). cited by applicant

Bark et al. "Large-area niobium disulfide thin films as transparent electrodes for devices based on two-dimensional materials," *Nanoscale*, published online, 7 pages (2012). cited by applicant

Barreca et al. "Cobalt oxide nanomaterials by vapor phase synthesis for fast and reversible lithium storage" *J Phys Chem C*, 114, 10054-10060 (2010). cited by applicant

Basuvalingam et al. "NS-WeA6—Low Temperature ALD for Phase-controlled Synthesis of 2D Transition Metal (M=Ti, Nb) di- (MX₂) and Tri- (MX₃) Sulfides," AVS 19th International Conference on Atomic Layer Deposition (ALD 2019)), Jul. 22, 2019, Abstract, 1 page (2019). cited by applicant

Bearzotti, et al., "Fast Humidity Response of a Metal Halide-Doped Novel Polymer," *Sensors and Actuators B*, 7, pp. 451-454, (1992). cited by applicant

Becker et al., "Atomic Layer Deposition of Insulating Hafnium and Zirconium Nitrides," *Chem. Mater.*, 16, 3497-3501 (2004). cited by applicant

Beynet et al. "Low temperature plasma-enhanced ALD enables cost-effective spacer defined double patterning," *Proceedings of SPIE*, 7520, (2009). cited by applicant

Bhatnagar et al., "Copper Interconnect Advances to Meet Moore's Law Milestones," *Solid State Technology*, 52, 10 (2009). cited by applicant

Boscher et al., "Atmosphere Pressure Chemical Vapour Deposition of NbSe₂ Thin Films on Glass" *Eur. J. Inorg. Chem.*, pp. 1255-1259 (2006). cited by applicant

Brewin et al. "UPS to deploy Bluetooth, wireless LAN network" *Internet*, pp. 1-5 (2001). cited by applicant

Bulletin Plasma Source—Dec. 2002: Type AX7610—Downstream Plasma Source. Source location: MKS Instruments, Inc. <http://www.mksinst.com/>. Accessed: Aug. 6, 2015. pp. 1-4 (2002). cited by applicant

Bulletin ASTRONe—Apr. 2002: Type AX7680—Reactive Gas Generator. Source location: MKS Instruments, Inc. <http://www.mksinst.com/>. Accessed: Aug. 6, 2015. pp. 1-2 (2002). cited by applicant

Bulletin ASTRONex—Jul. 2009: Type AX7685—Reactive Gas Generator. Source location: MKS Instruments, Inc. <http://www.mksinst.com/>. Accessed: Aug. 6, 2015. pp. 1-2 (2002). cited by applicant

Bulletin ASTRON hf-s—Mar. 2005: Type AX7645—Reactive Gas Generator. Source location: MKS Instruments, Inc. <http://www.mksinst.com/>. Accessed: Aug. 6, 2015. pp. 1-2 (2005). cited by applicant

Bulletin ENG-Xstream-340-09 OM Aug. 2006: Xstream with Active Matching Network Remote Plasma Source. Advanced Energy Industries, Inc. <http://www.advanced-energy.net.cn/upload/File/Sources/ENG-Xstream-340-09.pdf>. Accessed: Jul. 19, 2016. pp. 1-4 (2006). cited by applicant

Bulletin R*evolution—Sep. 2006: Type AX7690—Remote Plasma Source. Source location: MKS Instruments, Inc. <http://www.mksinst.com/>. Accessed: Aug. 6, 2015. pp. 1-2 (2006). cited by applicant

Buriak, "Organometallic Chemistry on Silicon and Germanium Surfaces," *Chemical Reviews*, 102, 5 (2002). cited by applicant

Cahiez et al. "Cobalt-Catalyzed cross coupling reaction between functionalized primary and secondary alkyl halides and aliphatic grignard reagents" *Adv. Synth. Catal*, 350, 1484-1488 (2008). cited by applicant

Callaghan et al. "Magnetic Interactions in Ternary Ruthenium Oxides" *Inorganic Chemistry*, vol. 5, No. 9, pp. 1572-1576 (1966). cited by applicant

Cant et al., "Chemisorption Sites on Porous Silica Glass and on Mixed-Oxide Catalysis," *Can. J. Chem.* 46, 1373 (1968). cited by applicant

Carmalt et al., "Chemical Vapor Deposition of Niobium Disulfide Thin Films" *Eur. J. Inorg. Chem.*, pp. 4470-4476 (2004). cited by applicant

Casey et al. "Chemical Vapor Deposition of Mo onto Si" J. Electrochem. Soc.: Solid State Science, 114 (2), pp. 201-204 (1967). cited by applicant

Chang et al. "Small-Subthreshold-Swing and Low-Voltage Flexible Organic Thin-Film Transistors Which Use HfLaO as the Gate Dielectric," IEEE Electron Device Letters, Feb. 2009, pp. 133-135; vol. 30, No. 2; IEEE Electron Device Society. cited by applicant

Chatterjee et al. "Chemical Vapor Deposition of Boron Nitride Nanosheets on Metallic Substrates via Decaborane/Ammonia Reactions", Chem. Mater. 23, pp. 4414-4416 (2011). cited by applicant

Chatterjee et al., "Sub-100nm Gate Length Metal Gate NMOS Transistors Fabricated by a Replacement by a Replacement Gate Process," IEEE Semiconductor Process and Device Center, 821-824 (1997). cited by applicant

Chen et al., "A Self-Aligned Airgap Interconnect Scheme," IEEE International Interconnect Technology Conference, vol. 1-3, 146-148 (2009). cited by applicant

Chen et al., "Develop Gap-fill Process of Shallow Trench Isolation in 450mm Wafer by Advanced Flowable CVD Technology for Sub-20nm Node," 2016 27th Annual Semi Advanced Semiconductor. cited by applicant

Cheng et al., "Effect of carrier gas on the structure and electric properties of low dielectric constant SiCOH film using trimethylsilane prepared by plasma enhanced chemical vapor deposition," Thin Solid Films vol. 469-470, pp. 178-183 (2004). cited by applicant

Choi et al., "Improvement of Silicon Direct Bonding using Surfaces Activated by Hydrogen Plasma Treatment," Journal of the Korean Physical Society, 37, 6, 878-881 (2000). cited by applicant

Choi et al., "Low Temperature Formation of Silicon Oxide Thin Films by Atomic Layer Deposition Using NH₃/O₂ Plasma," ECS Solid State Letters, 2(12) pp. 114-116 (2013). cited by applicant

Choi et al. "UV irradiation effects on the bonding structure and electrical properties of ultra low-k SiOC(-H) thin films for 45 nm technology node" Current Applied Physics, 11, S109-S113 (2011). cited by applicant

Closser et al., "Molecular Layer Deposition of a Highly Stable Silicon Oxycarbide Thin Film Using An Organic Chlorosilane And Water," ACS Applied Materials & Interfaces 10, pp. 24266-24274 (2018). cited by applicant

Coates, "Process Analytical Technology: Spectroscopic Tools and Implementation Strategies for the Chemical and Pharmaceutical Industries." Blackwell Publishing Ltd, 91-132, (2005). cited by applicant

Conroy et al., "The Preparation and Properties of Single Crystals of the 1S and 2S Polymorphs of Tantalum Disulfide" J. Solid State Chemistry, 4, pp. 345-350 (1972). cited by applicant

Crowell, "Chemical methods of thin film deposition: Chemical vapor deposition, atomic layer deposition, and related technologies," Journal of Vacuum Science & Technology A 21.5, (2003): S88-S95. cited by applicant

Cui et al., "Impact of Reductive N₂/H₂ Plasma on Porous Low-Dielectric Constant SiCOH Thin Films," Journal of Applied Physics 97, 113302, 1-8 (2005). cited by applicant

Crystal IS "Application Note: Using UV Reflective Materials to Maximize Disinfection"; AN011; Jun. 16, 2016, found at <https://www.klaran.com/using-uv-reflective-materials-to-maximize-disinfection>. cited by applicant

De Silva et al., "Inorganic Hardmask Development for Extreme Ultraviolet Patterning," Journal of Micro/Nanolithography, MEMS, and MOEMS 18(1) (2018). cited by applicant

Dingemans et al., "Comparison Between Aluminum Oxide Surface Passivation Films Deposited with Thermal Aid," Plasma Aid and Pecvd, 35th IEEE PVCS, June (2010). cited by applicant

Drummond et al., "Hydrophobic Radiofrequency Plasma-Deposited Polymer Films: Dielectric Properties and Surface Forces," Colloids and Surfaces A, 129-130, 117-129 (2006). cited by applicant

Duffey et al., "Raman Scattering from 1T-TaS₂" Solid State Communications 20, pp. 617-621 (1976). cited by applicant

Dufond et al. "Quantifying the Extent of Ligand Incorporation and the Effect on Properties of TiO₂ Thin Films Grown by Atomic Layer Deposition Using an Alkoxide or an Alkylamide" *Chemistry of Material*, American Chemical Society 32(4), pp. 1393-1407 (2020). cited by applicant

Dultsev et al. "Effect of UV radiation on the reactivity of porous methyl-doped SiO₂ layers" *Phys. Status Solidi C*, 8, No. 6, pp. 1946-1949 (2011). cited by applicant

Easley et al., "Thermal Isolation of Microchip Reaction Chambers for Rapid Non-Contact DNA Amplification," *J. Micromech. Microeng.* 17, 1758-1766 (2007). cited by applicant

Elam et al., "New Insights into Sequential Infiltration Synthesis", *ECS Transactions*, vol. 69, pp. 147-157 (2015). cited by applicant

Elers et al. "Film Uniformity in Atomic Layer Deposition," *Chemical Vapor Deposition*, 12, pp. 13-24 (2006). cited by applicant

Ellis et al. "Nitrous Oxide (N₂O) Processing for Silicon Oxynitride Gate Dielectrics." *IBM Journal of Research and Development*. 1999. 43[3]. pp. 287-300. (1999). cited by applicant

Esposito et al. "Preparation and characterization of lead ruthenate based composite cathodes for SOFC applications" *Mater. Res. Soc. Symp. Proc.*, vol. 835, 6 pages (2005). cited by applicant

Fallica et al., "Dynamic Absorption Coefficients Of Chemically Amplified Resists And Nonchemically Amplified Resists At Extreme Ultraviolet", *J. Micro/Nanolith. MEMS MOEMS* 15(3), pp. 033506-1-033506-7 (2016). cited by applicant

"Fiji F200 200mm Thermal/Plasma ALD Systems: Installation and Use Manual." CAW-02635 Rev. 0.6 (Mar. 13, 2012). Cambridge NanoTech Inc. pp 1-164 (2012). cited by applicant

Fray et al. "Aspects of the Application of Electrochemistry to the Extraction of Titanium and Its Applications" *Materials Transactions*, vol. 58, No. 3, pp. 306-312 (2017). cited by applicant

Freund et al. "Single crystal growth from separated educts and its application to lithium transition-metal oxides" *Sci. Rep.* 6, 35362, pp. 1-5 (2016). cited by applicant

Frequently Asked Question, "What is a load lock chamber?", Kurt J. Lesker Company, www.lesker.com, p. 1 (2020). cited by applicant

Fu et al., "Controlled Synthesis of Atomically Thin 1T-TaS₂ for Tunable Charge Density Wave Phase Transitions" *Chem. Mater.* 28, pp. 7613-7618 (2016). cited by applicant

Galesic et al. "Formation of vanadium nitride by rapid thermal processing" *Thin Solid Films* 349(1), 7 pages (1999). cited by applicant

Ge et al., "Carbon Nanotube-Based Synthetic Gecko Tapes," *Department of Polymer Science, PNAS*, 10792-10795 (2007). cited by applicant

Ge et al. "Large-scale synthesis of NbS₂ nanosheets with controlled orientation on graphene by ambient pressure CVD," *Nanoscale*, vol. 5, 5773-5778 (2013). cited by applicant

George et al., "Atomic Layer Deposition: An Overview," *Chem. Rev.* 110, 111-131 (2010). cited by applicant

Gesheva et al. "Composition and Microstructure of Black Molybdenum Photothermal Converter Layers Deposited by the Pyrolytic Hydrogen Reduction of MoO₃.SUB.2.Cl.SUB.2." *Thin Solid Films*, 79, pp. 39-49 (1981). cited by applicant

Glass et al. "The Chemical Vapor Deposition of Metal Boride Thin Films from Polyhedral Cluster Species", Winter and Hoffman, *Inorganic Materials Synthesis. ACS Symposium Series*; American Chemical Society: Washington DC, Chapter 10, pp. 130-143 (1999). cited by applicant

Gole et al. "Preparation of Nickel Sulfide Thin Films and Nanocrystallites Using Nickel Furfuraldehyde Thiosemicarbazone as Single-source Precursor," *Advanced Materials Research*, vols. 383-390, pp. 3828-3834 (2012). cited by applicant

Gordon et al. "A Kinetic Model for Step Coverage by Atomic Layer Deposition in Narrow Holes or Trenches." *Chemical Vapor Deposition*. 9 [2]. pp 73-78 (2003). cited by applicant

Grill et al., "The Effect of Plasma Chemistry on the Damage Induced Porous SiCOH Dielectrics," *IBM Research Division, RC23683 (W0508-008), Materials Science*, 1-19 (2005). cited by applicant

Guan et al., "Voltage gated ion and molecule transport in engineered nanochannels: theory, fabrication and applications," *Nanotechnology* 25 (2014) 122001. cited by applicant

Gupta et al., "Charge carrier transport and electroluminescence in atomic layer deposited poly-GaN/c-Si heterojunction diodes," *Journal of Applied Physics*, 124, 084503 (2018). cited by applicant

Gupta et al., "Conversion of Metal Carbides to Carbide Derived Carbon by Reactive Ion Etching in Halogen Gas," *Proceedings of SPIE—The International Society for Optical Engineering and Nanotechnologies for Space Applications*, ISSN: 0277-786X (2006). cited by applicant

Habib et al. "Atmospheric oxygen plasma activation of silicon (100) surfaces," *American Vacuum Society*, 28(3), pp. 476-485 (2010). cited by applicant

Hamalainen et al., "Atomic Layer Deposition of Rhenium Disulfide," *Adv. Mater.* 30.24, 6 pages (2018). cited by applicant

Han et al., "van der Waals Metallic Transition Metal Dichalcogenides" *Chem. Rev.* 118, pp. 6297-6336 (2018). cited by applicant

Hansen. "A Primer on Vacuum Pressure Measurement." *Vacuum Technology & Coating*. Jun. 2009. pp. 36-42 (2009). cited by applicant

Hansen. "Speed, Pressure and Throughput: Part 1 System Diagnostics." *Vacuum Technology & Coating*. Sep. 2011. pp. 14-17 (2011). cited by applicant

Hansen. "Speed, Pressure and Throughput: Part 2 Managing Gas Flow in High Vacuum Systems." *Vacuum Technology & Coating*. Oct. 2011. pp. 19-22 (2011). cited by applicant

Hansen. "Speed, Pressure and Throughput: Part 3 Automating the Pressure Control Process." *Vacuum Technology & Coating*. Nov. 2011. pp. 22-25 (2011). cited by applicant

Hansen. "Speed, Pressure and Throughput: Part 4 Outgassing and Base Pressure." *Vacuum Technology & Coating*. Dec. 2011. pp. 22-25 (2011). cited by applicant

Hansen. "Speed, Pressure and Throughput: Part 5 Leaks and Gas Flow in Leak Detection." *Vacuum Technology & Coating*. Jan. 2012. pp. 18-21 (2011). cited by applicant

Hargreaves et al., "New Fluorides and Oxyfluorides of Rhenium," *J. Chem. Soc.*, pp. 1099-1103 (1960). cited by applicant

Harrison et al., "Poly-gate Replacement Through Contact Hole (PRETCH): A New Method for High-K/ Metal Gate and Multi-Oxide Implementation on Chip," *IEEE* (2004). cited by applicant

Hayashi et al. "Spectroscopic properties of nitrogen doped hydrogenated amorphous carbon films grown by radio frequency plasma-enhanced chemical vapor deposition," *Journal of Applied Physics*. vol. 89, No. 12, pp. 7924-7931 (2001). cited by applicant

Heo et al., "Structural Characterization of Nanoporous Low-Dielectric Constant SiCOH Films Using Organosilane Precursors," *NSTI-Nanotech*, vol. 4, 122-123 (2007). cited by applicant

Henke et al., "X-Ray Interactions: Photo absorption, Scattering, Transmission, and Reflection at E=50-30,000 eV, Z=1-92," *Atomic Data and Nuclear Data Tables*, 54, 181-342 (1993). cited by applicant

Heyne et al., "The conversion mechanism of amorphous silicon to stoichiometric WS₂" *J. Materials Chemistry C*, 6, pp. 4122-4130 (2018). cited by applicant

Hong et al. "Ultralow-dielectric-constant amorphous boron nitride" *Nature*, vol. 582, 16 pages (2020). cited by applicant

Hossain et al., "Recent Advances in Two-Dimensional Materials with Charge Density Waves: Synthesis, Characterization and Applications" *Crystals* 7, 298, 19 pages (2017). cited by applicant

Hubert et al., "A Stacked SONOS Technology, up to 4 Levels and 6nm Crystalline Nanowires, With Gate-All-Around or Independent Gates (-Flash), Suitable for Full 3D Integration," *Minattec, IEDM09-637-640* (2009). cited by applicant

Hudis, "Surface Crosslinking of Polyethylene Using a Hydrogen Glow Discharge," *J. Appl. Polym. Sci.*, 16 (1972) 2397. cited by applicant

Ivanova et al. "ALD of Sc₂O₃ using Sc(Cp)₃ and a Novel Heteroleptic Precursors" Presented on

the AVS 19th International Conference on Atomic Layer Deposition, Jul. 23, 2019, Abstract, 1 page (2019). cited by applicant

Jensen et al. "Titanium, Zirconium, and Hafnium Tetrahydroborates as "Tailored" CVD Precursors for Metal Diboride Thin Films", J. Am. Chem. Soc. 110, pp. 1643-1644 (1988). cited by applicant

Johansson et al. "Towards absolute asymmetric synthesis. Synthesis and crystal structure of stereochemically labile MCl_2 ($M=CO$, Ni, Cu, Zn) complexes with diamine ligands," *Inorganica Chimica Acta* 358, pp. 3293-3302 (2005). cited by applicant

Johnson et al. "A brief review of atomic layer deposition: from fundamentals to applications" *Materials Today*, vol. 17, Issue 5, pp. 236-246 (2014). cited by applicant

Jones et al., "Growth of Aluminum Films by Low Pressure Chemical Vapour Deposition Using Tritertiarybutylaluminium," *Journal of Crystal Growth* 135, pp. 285-289, Elsevier Science B.V. (1994). cited by applicant

Jones et al., "Recent Developments in Metalorganic Precursors for Metalorganic Chemical Vapour Deposition," *Journal of Crystal Growth* 146, pp. 503-510, Elsevier Science B.V. (1995). cited by applicant

Jung et al., "Double Patterning of Contact Array with Carbon Polymer," *Proc. Of SPIE*, 6924, 69240C, 1-10 (2008). cited by applicant

Jung et al. "New Mechanisms for Ozone-Based ALO Growth of High-k Dielectrics via Nitrogen-Oxygen Species" *ECS Transactions*, 33(2), pp. 91-99 (2010). cited by applicant

Juppo et al. "Deposition of molybdenum thin films by an alternate supply of $MoCl_5$ and Zn" *J of Vacuum Science & Technology A*, 16, pp. 2845-2850 (1998). cited by applicant

Katamreddy et al., "ALD and Characterization of Aluminum Oxide Deposited on Si(100) using Tris(diethylamino) Aluminum and Water Vapor," *Journal of The Electrochemical Society*, 153 (10) C701-C706 (2006). cited by applicant

Kern et al., "Chemically Vapor-Deposited Borophosphosilicate Glasses for Silicon Device Applications" *RCE Review*, 43, 3, pp. 423-457 (1982). cited by applicant

Kerrigan et al. "Low Temperature, Selective Atomic Layer Deposition of Cobalt Metal Films Using Bis(1,4-di-tert-butyl-1,3-diazadienyl)cobalt and Alkylamine Precursors," *Chem. Materials*, 29, pp. 7458-7466 (2017). cited by applicant

Khandelwal et al. "Low-temperature Ar/ N_2 remote plasma nitridation of SiO_2 thin films," *J. Vacuum Science & Technology A*, 20(6), pp. 1989-1996 (2002). cited by applicant

Kim et al. "Characteristics of Low Temperature High Quality Silicon Oxide by Plasma Enhanced Atomic Layer Deposition with In-Situ Plasma Densification Process," *The Electrochemical Society, ECS Transactions*, College of Information and Communication Engineering, Sungkyunkwan University, 53(1), 321-329 (2013). cited by applicant

Kim et al. "High Growth Rate in Atomic Layer Deposition of TiO_2 Thin Films by UV Irradiation" *Electrochemical and Solid-State Letters*, 14(4), pp. H146-H148 (2011). cited by applicant

Kim et al. "Novel Flowable CVD Process Technology for sub-20nm Interlayer Dielectrics," *IEEE International Interconnect Technology Conference (IITC 2012)*, San Jose, California, USA, Jun. 4-6, 2012, pp. 1-3 (2012). cited by applicant

Kim et al. "Passivation Effect on Low-k S/OC Dielectrics by H_2 Plasma Treatment," *Journal of the Korean Physical Society*, 40, 1, 94-98 (2002). cited by applicant

Kim et al. "Thermal Decomposition of Tetrakis(ethylmethyamido) Titanium for Chemical Vapor Deposition of Titanium Nitride" *Bull. Korean Chem. Soc.*, vol. 27, No. 2, 5 pages (2006). cited by applicant

King, Plasma Enhanced Atomic Layer Deposition of SiN_x : H and SiO_2 , *J. Vac. Sci. Technol.*, A29(4) (2011). cited by applicant

Kirsch et al. "Dipole model explaining high-k/metal gate field effect transistor threshold voltage tuning" *Applied Physics Letters* 92, 092901, 3 pages (2008). cited by applicant

Kjesbu, ABB Corporate Research, Internet, pp. 1-37 (2002). cited by applicant

Kliem et al. "Silyl Group Migrations Between Oxygen and Nitrogen in Aminosiloxanes"; *Organosilicon Chemistry VI: from Molecules to Materials*; pp. 216-221 (2005). cited by applicant

Klug et al., "Atomic Layer Deposition of Amorphous Niobium Carbide-Based Thin Film Superconductors," *The Journal of Physical Chemistry C*, vol. 115, pp. 25063-25071, (2011). cited by applicant

Kobayshi, et al., "Temperature Dependence of SiO₂ Film Growth with Plasma-Enhanced Atomic Layer Deposition," regarding Thin Solid Films, published by Elsevier in the *International Journal on the Science and Technology of Condensed Matter*, in vol. 520, No. 11, 3994-3998 (2012). cited by applicant

Kofuji et al. "Line-edge roughness increase due to wiggling enhanced by initial pattern waviness" *Jpn. J. Appl. Phys.* 53, 03DE01, pp. 1-7 (2014). cited by applicant

Kofuji et al. "Mechanism of wiggling enhancement due to HBr gas addition during amorphous carbon etching" *Jpn. J. Appl. Phys.* 54, 06FH04, pp. 107 (2015). cited by applicant

Kogelschatz et al. "Ozone Generation from Oxygen and Air: Discharge Physics And Reaction Mechanisms" *Ozone Science & Engineering*, 10, pp. 367-378 (1998). cited by applicant

Koo et al., "Characteristics of Al₂O₃ Thin Films Deposited Using Dimethylaluminum Isopropoxide and Trimethylaluminum Precursors by the Plasma-Enhanced Atomic-Layer Deposition Method," *Journal of Physical Society*, 48, 1, 131-136 (2006). cited by applicant

Koutsokeras et al. "Texture and Microstructure Evolution in Single-Phase Ti_xTa_{1-x}N Alloys of Rocksalt Structure," *Journal of Applied Physics*, 110, pp. 043535-1-043535-6, (2011). cited by applicant

Kovalgin et al. "Hot-Wire Assisted ALD: A Study Powered by In Situ Spectroscopic Ellipsometry" *Adv. Mater. Interfaces* 4, 1700058, 11 pages (2017). cited by applicant

Knoops et al., "Atomic Layer Deposition of Silicon Nitride from Bis(tert-butyloamino) silane and N₂ Plasma," *Applied Materials & Interfaces*, American Chemical Society, A-E (2015). cited by applicant

Kreněk et al. "IR Laser CVD of Nanodisperse Ge—Si—Sn Alloys Obtained by Dielectric Breakdown of GeH₄/SiH₄/SnH₄ Mixtures", *NanoCon*, Nov. 5-7, 2014, Brno, Czech Republic, EU. cited by applicant

Krumdieck et al. "Experimental Characterization and Modeling of Pulsed MOCVD with Ultrasonic Atomization of Liquid Precursor" *Chem. Vap. Deposition*, 7 (2), pp. 85-90 (2001). cited by applicant

Krumdieck "Kinetic Model of Low Pressure Film Deposition from Single Precursor Vapor in a Well-Mixed, Cold-Wall Reactor" *Acta mater.*, 49, pp. 583-588 (2001). cited by applicant

Kucheyev et al. "Mechanisms of Atomic Layer Deposition on Substrates with Ultrahigh Aspect Ratios." *Langmuir*. 24 [3]. pp 943-948 (2008). cited by applicant

Kuchumov et al. "Pulsed MO CVD Processes of MgO Layer Deposition from Mg(thd)₂" *ECS Trans.* 25, pp. 927-934 (2009). cited by applicant

Kukli et al. "Atomic Layer Deposition and Chemical Vapor Deposition of Tantalum Oxide by Successive and Simultaneous Pulsing of Tantalum Ethoxide and Tantalum Chloride" *Chem. Mater.* 12, pp. 1914-1920 (2000). cited by applicant

Kukli et al. "Atomic Layer Epitaxy Growth of Tantalum Oxide Thin Films from Ta(OC₂H₅)₅ and H₂O" *The Electrochemical Society*, vol. 142, No. 5, pp. 1670-1674 (1995). cited by applicant

Kukli et al., "Influence of atomic layer deposition parameters on the phase content of Ta₂O₅ films" *J. Crystal Growth*, 212, pp. 459-468 (2000). cited by applicant

Kukli et al., "Properties of hafnium oxide films grown by atomic layer deposition from hafnium tetraiodide and oxygen". *Journal of Applied Physics*, vol. 92, No. 10, Nov. 15, 2002, pp. 5698-5703 (2002). cited by applicant

Kukli et al. "Properties of tantalum oxide thin films grown by atomic layer deposition" *Thin Solid Films*, 260, pp. 135-142 (1995). cited by applicant

Kumar et al. "Bringing the benefits of Java to Bluetooth" <http://www.allembedded.com/story/OEG20020519S001>, pp. 1-3 (2002). cited by applicant

Kurosawa et al. "Synthesis and Characterization of Plasma-Polymerized Hexamethyldisiloxane Films," *Thin Solid Films*, 506-507, 176-179 (2006). cited by applicant

Kwon et al. "Substrate Selectivity of (tBu-Allyl)Co(CO)₃ during Thermal Atomic Layer Deposition of Cobalt," *Chem. Materials*, 24, pp. 1025-1030 (2012). cited by applicant

Lanford et al., "The Hydrogen Content of Plasmadeposited Silicon Nitride," *J. Appl. Phys.*, 49, 2473 (1978). cited by applicant

Lee et al. "Characteristics Of Low-K Sioc Films Deposited Via Atomic Layer Deposition," *Thin Solid Films* 645, pp. 334-339 (2018). cited by applicant

Lee et al. "Enhancement of Iodine Adsorption on Ruthenium Glue Layer for Seedless CECVD of Cu" *Electrochemical and Solid-State Letters*, 8(2) C39-C42 (2005). cited by applicant

Lee et al. "Growth without Postannealing of Monoclinic VO₂ Thin Film by Atomic Layer Deposition Using VC14 as Precursor" *Coatings*, 8, 431, pp. 1-11 (2018). cited by applicant

Lee et al. "Layer Selection by Multi-Level Permutation in 3-D Stacked NAND Flash Memory," *IEEE Electron Device Letters*, vol. 37, No. 7, 866-869 (2016). cited by applicant

Lee et al. "Photo-Induced Atomic Layer Deposition of Tantalum Oxide Thin Films from Ta(OC₂H₅)₅ and O₂" *J of the Electrochemical Society*. 151(1) pp. C52-C55 (2004). cited by applicant

Levy et al., "Reflow Mechanisms of Contact Vias in VLSI Processing" *J. Electrochem. Soc.: Solid-State Science and Technology*, 133, 7, pp. 1417-1424 (1986). cited by applicant

Lewis et al. "Hawley's Condensed Chemical Dictionary" 12th Edition; excerpt pp. 1027-1038 (2005). cited by applicant

Li et al. "Metallic Transition-Metal Dichalcogenide Nanocatalysts for Energy Conversion" *Chem.* 4, pp. 1510-1537 (2018). cited by applicant

Liang et al. "Conversion of Metal Carbides to Carbide Derived Carbon by Reactive Ion Etching in Halogen Gas" *Micro (MEMS) and Nanotechnologies for Space Applications*, Thomas George et al. vol. 6223, 2006 p. 62230J-I to 62230J-11 lines 3-14 in the "Abstract" section and lines 7-9 in the "Introduction" section of p. 1, lines 3-4 in the "Introduction" section and lines 3-4 in the "Experimental Procedure" section of p. 2 (2006). cited by applicant

Lieberman, et al., "Principles of Plasma Discharges and Materials Processing," Second Edition, 368-381 (2005). cited by applicant

Lim et al., "Low-Temperature Growth of SiO₂ Films by Plasma-Enhanced Atomic Layer Deposition," *ETRI Journal*, 27 (1), 118-121 (2005). cited by applicant

Lim et al. "Synthesis and Characterization of Volatile, Thermally Stable, Reactive Transition Metal Amidinates," *Inorg. Chem.*, 42, pp. 7951-7958 (2003). cited by applicant

Liu et al., "Research, Design, and Experiment of End Effector for Wafer Transfer Robot," *Industrial Robot: An International Journal*, 79-91 (2012). cited by applicant

Liu et al., "Van der Waals metal-semiconductor junction: Weak Fermi level pinning enables effective tuning of Schottky barrier" *Sci. Adv.* 2: e1600069, 7 pages (2016). cited by applicant

Londergan et al. "Engineered Low Resistivity Titanium-Tantalum Nitride Films by Atomic Layer Deposition," *Mat. Res. Soc. Symp. Proc.*, vol. 714E, pp. L5.3.1-L5.3.6 (2001). cited by applicant

Longrie et al., "Plasma-Enhanced ALD of Platinum with O₂, N₂ and NH₃ Plasmas", *ECS Journal of Solid State Science and Technology*, vol. 1, pp. Q123-Q129 (2012). cited by applicant

Mackenzie et al. "Stress Control of Si-Based PEVCD Dielectrics," *Proc. Symp. Silicon Nitride and Silicon Dioxide Thin Insulating Films & Other Emerging Dielectrics VIII*, 148-159 (2005). cited by applicant

Mackus et al., "Optical Emission Spectroscopy as a Tool for Studying Optimizing and Monitoring Plasma-Assisted Atomic Layer Deposition Processes," *Journal of Vacuum Science and Technology*, 77-87 (2010). cited by applicant

Maeng et al. Electrical properties of atomic layer deposition HfO₂ and HfO_xNy on Si substrates with various crystal orientations, Journal of the Electrochemical Society, Apr. 2008, p. H267-H271, vol. 155, No. 4, Department of Materials Science and Engineering, Pohang University of Science and Technology, Pohang, Korea (2008). cited by applicant

Maeno, "Gecko Tape Using Carbon Nanotubes," Nitto Denko Gihou, 47, 48-51 (2009). cited by applicant

Makela et al. "Thermal Atomic Layer Deposition of Continuous and Highly Conducting Gold Thin Films," Chem. Materials, 29, pp. 6130-6136 (2017). cited by applicant

Mameli et al. "Area-Selective Atomic Layer Deposition of SiO₂ Using Acetylacetone as a Chemoselective Inhibitor in an ABC-Type Cycle" ACS Nano 11, pp. 9303-9311 (2017). cited by applicant

Mameli et al. "Isotropic Atomic Layer Etching of ZnO Using Acetylacetone and O₂ Plasma" ACS Appl. Mater. Interfaces 10, pp. 38588-38595 (2018). cited by applicant

Marsik et al., "Effect of Ultraviolet Curing Wavelength on Low-k Dielectric Material Properties and Plasma Damage Resistance," Sciencedirect.com, 519, 11, 3619-3626 (2011). cited by applicant

Mason et al., "Hydrolysis of Tri-tert-butylaluminum: The First Structural Characterization of Alkylalumoxanes [(R₂Al)₂O]_n and (RAIO)_n," J. American Chemical Society, vol. 115, No. 12, pp. 4971-4984 (1993). cited by applicant

Massachusetts Institute of Technology Lincoln Laboratory, "Solid State Research," Quarterly Technical Report (1995). cited by applicant

Mattinen et al., "Crystalline tungsten sulfide thin films by atomic layer deposition and mild annealing" J. Vac. Sci. Tech. 37, 020921, 35 pages (2019). cited by applicant

Maydannik et al., "Spatial atomic layer deposition: Performance of low temperature H₂O and O₃ oxidant chemistry for flexible electronics encapsulation", Journal of Vacuum Science and Technology: Part A AVS/ AIP, vol. 33 (1991). cited by applicant

Meng et al., "Atomic Layer of Deposition of Silicon Nitride Thin Films: A Review of Recent Progress, Challenges, and Outlooks," Materials, 9, 1007 (2016). cited by applicant

Merdrignac-Conanec et al. "Nitridation under ammonia of high surface area vanadium aerogels" Journal of Solid State Chemistry 178(1), 8 pages (2005). cited by applicant

Miller et al. "Carbon nitrides: synthesis and characterization of a new class of functional materials," Phys.Chem.Chem.Phys., 19, pp. 15613-15638 (2017). cited by applicant

Mix et al., "Characterization of plasma-polymerized allyl alcohol polymers and copolymers with styrene," Adhes. Sci. Technol., 21 (2007), S. 487-507. cited by applicant

Moeen, "Design, Modelling and Characterization of Si/SiGe Structures for IR Bolometer Applications," KTH Royal Institute of Technology. Information and Communication Technology, Department of Integrated Devices and Circuits, Stockholm Sweden (2015). cited by applicant

Morishige et al., "Thermal Desorption and Infrared Studies of Ammonia Amines and Pyridines Chemisorbed on Chromic Oxide," J. Chem. Soc., Faraday Trans. 1, 78, 2947-2957 (1982). cited by applicant

Mosleh et al., "Enhancement of Material Quality of (Si)GeSn Films Grown by SnCl₄ Precursor," ECS Transactions, 69 (5), 279-285 (2015). cited by applicant

Mukai et al., "A Study of CD Budget in Spacer Patterning Technology," Proc. Of SPIE, 6924, 1-8 (2008). cited by applicant

Naito et al. "Electrical Transport Properties in 2H—NbS₂, —NbSe₂, —TaS₂ and —TaSe₂," J. of Physical Society of Japan, vol. 51, No. 1, 219-227 (1982). cited by applicant

Nakano et al., "Layer-by-Layer Epitaxial Growth of Scalable WSe₂ on Sapphire by Molecular Beam Epitaxy" Nano. Lett. 17, pp. 5595-5599 (2017). cited by applicant

Naslain et al. "Synthesis of highly tailored ceramic matrix composites by pressure-pulsed CVI" Solid State Ionics, 141-142, pp. 541-548 (2001). cited by applicant

Ngo et al. "Atomic layer deposition of photoactive CoO/SrTiO₃ and CoO/TiO₂ on Si(001) for

visible light driven photoelectrochemical water oxidation,” J. Applied Physics, 114, 9 pages (2013). cited by applicant

Niinisto et al. “Advanced electronic and optoelectronic materials by Atomic Layer Deposition: An overview with special emphasis on recent progress in processing of high-k dielectrics and other oxide materials” Pus. Stat. Sol. (a) 201, No. 7, pp. 1443-1452 (2004). cited by applicant

Nikolic et al. “The dependence of the work function of rare earth metals on their electron structure” Microelectronics Journal 27, 93-96 (1996). cited by applicant

Nogueira et al., “Production of Highly Hydrophobic Films Using Low Frequency and High Density Plasma,” Revista Brasileira de Aplicacoes de Vacuo, 25(1), 45-53 (2006). cited by applicant

Novaro et al. Theoretical Study on a Reaction Pathway of Ziegler-Natta-Type Catalysis, J. Chem. Phys. 68(5), Mar. 1, 1978 p. 2337-2351. cited by applicant

Ohchi et al. “Reducing damage to Si substrates during gate etching processes.” Japanese Journal of Applied Physics 47.7R 5324 (2008). cited by applicant

Ohtsu et al. “Influences of Gap Distance on Plasma Characteristics in Narrow Gap Capacitively Coupled Radio-Frequency Discharge,” vol. 43, No. 2, pp. 795-799 (2004). cited by applicant

Ohzawa et al. “Preparation of fibrous SiC shape using pressure-pulsed chemical vapour infiltration and its properties as a high-temperature filter” J. of Materials Processing Technology, 96, pp. 151-156 (1999). cited by applicant

Okamoto et al., “Luminescent Properties of Pr³⁺-sensitized LaPO₄: Gd³⁺ Ultraviolet-B Phosphor Under Vacuum-Ultraviolet Light Excitation,” J. App. Phys. 106, 013522 (2009). cited by applicant

O'Malley et al. “Structure and properties of ordered Li₂IrO₃ and Li₂PtO₃” Journal of Solid State Chemistry 181, pp. 1803-1809 (2008). cited by applicant

Park et al. “Atomic layer deposition of Y₂O₃ films using hetroleptic liquid (iPrCp)₂Y(iPr-amd) precursor” J. Mater. Chem. C, 2, 9 pages (2014). cited by applicant

Park, “Substituted Aluminum Metal Gate on High-K Dielectric for Low Work-Function and Fermi-Level Pinning Free,” 4 pages, IEEE 0-7803-8684-1/04 (2004). cited by applicant

Peters et al., “Aerosol-Assisted Chemical Vapor Deposition of NbS₂ and TaS₂ Thin Films from Pentakis(dimethylamido)metal Complexes and 2-Methylpropanethiol” Eur. J. Inorg. Chem., pp. 4179-4185 (2005). cited by applicant

Pichler. “Intrinsic Point Defects, Impurities and Their Diffusion in Silicon,” Springer-Verlag Wien, p. 367 (2004). cited by applicant

Portet et al., “Impact of Synthesis Conditions on Surface Chemistry and Structure of Carbide-Derived Carbons,” Thermochemica Acta, 497, 137-142 (2010). cited by applicant

Posseme et al. “Alternative process for thin layer etching: Application to nitride spacer etching stopping on silicon germanium” Apply. Phys. Lett. 105, 051604, pp. 1-4 (2014). cited by applicant

Potts et al., “Low Temperature Plasma-Enhanced Atomic Layer Deposition of metal Oxide Thin Films,” Journal of the Electrochemical Society, 157, 66-74 (2010). cited by applicant

Potts et al., “Room-Temperature ALD of Metal Oxide Thin Films by Energy-Enhanced ALD”, Chemical Vapor Deposition, vol. 19, pp. 125-133 (2013). cited by applicant

Presser, et al., “Effect of Pore Size on Carbon Dioxide Sorption by Carbide Derived Carbon,” Energy & Environmental Science 4.8, 3059-3066 (2011). cited by applicant

Provine et al. “Atomic Layer Deposition: Introduction to the Theory and Cambridge Nanotech Savannah & Fiji” NNIN ALD Roadshow Presentation, 49 pages (2012). cited by applicant

Provine et al., “Correlation of Film Density and Wet Etch Rate in Hydrofluoric Acid of Plasma Enhanced Atomic Layer Deposited Silicon Nitride,” AIP Advances, 6 (2016). cited by applicant

Putkonen et al. “Low-Temperature ALE Deposition of Y₂O₃ Thin Films from β-Diketonate Precursors” Chemical Vapor Deposition 7(1) pp. 44-50 (2001). cited by applicant

Qin et al., “Chemical Vapor Deposition Growth of Degenerate p-Type Mo-Doped ReS₂ Films and Their Homo Junction,” ACS Appl. Mater. Interfaces, 9(18), pp. 15583-15591 (2007). cited by applicant

Radamson et al. "Growth of Sn-alloyed Group IV Materials for Photonic and Electronic Applications" Chapter 5 pp. 129-144, Manufacturing Nano Structures (2014). cited by applicant

Roddy, "Transport Reactions and Vaporization Studies of Some Vanadium Halides", Iowa State university of Science and Technology dissertation, 207 pages (1962). cited by applicant

Rosemount 1199 Diaphragm Seal Systems, Product Data Sheet, 92 pages (2008). cited by applicant

Rossing et al. "Acoustics of Eastern and Western bells, Old and New" J of Acoustical Society of Japan; 10(5); pp. 241-252 (1989). cited by applicant

Ryu et al., "Persistent Charge-Density-Wave Order in Single-Layer TaSe₂" Nano. Lett. 18, pp. 689-694 (2018). cited by applicant

Saeki et al. "Reaction Process of Vanadium Tetrachloride with Ammonia in the Vapor Phase and Properties of the Vanadium Nitride Formed" Bull. Chem. Soc. Jpn., 55, pp. 3446-3449 (1982). cited by applicant

Sakuma et al., "Highly Scalable Horizontal Channel 3-D NAND Memory Excellent in Compatibility with Conventional Fabrication Technology," IEEE Electron Device Letters, vol. 34, No. 9, 1142-1144 (2013). cited by applicant

Salim, "In-situ Fourier Transform Infrared Spectroscopy of Chemistry and Growth in Chemical Vapor Deposition," Massachusetts Institute of Technology, 187 pages (1995). cited by applicant

Salim et al., "In Situ Concentration Monitoring in a Vertical OMVPE Reactor by Fiber-Optics-Based Fourier Transform Infrared Spectroscopy," Journal of Crystal Growth 169, pp. 443-449, Elsevier Science B.V. (1996). cited by applicant

Samal et al., "Low-Temperature (<200° C.) Plasma Enhanced Atomic Deposition of Dense Titanium Nitride Thin Films" (2012). cited by applicant

Sanders et al., "Crystalline and electronic structure of single-layer TaS₂." Phys. Rev. B. 94, 081404, 6 pages (2016). cited by applicant

Schindler, Dissertation, Next Generation High-k Dielectrics for DRAM Produced by Atomic Layer Deposition Studied by Transmission Electron Microscopy (2015). cited by applicant

Schmatz et al., "Unusual Isomerization Reactions in 1,3-Diaza-2-Silacyclopentanes," Organometallics, 23, 1180-1182 (2004). cited by applicant

Sellers, Making Your Own Timber Dogs, Paul Sellers blog, Published on Nov. 18, 2014, [online], [site visted Jun. 10, 2017]. Available from Internet, <URL: <https://paulsellers.com/2014/11/making-your-own-timber-dogs/>>. cited by applicant

Selvaraj et al., "Selective Atomic Layer Deposition of Zirconia on Copper Patterned Silicon Substrates Using Ethanol as Oxygen Source as Well as Copper Reductant," J. Vac. Sci. Technol. A32(1), (2014). cited by applicant

Selvaraj et al., "Surface Selective Atomic Layer Deposition of Hafnium Oxide for Copper Diffusion Barrier Application Using Tetrakis (diethylamino) Hafnium and Ethanol," 225th ECS Meeting, Meeting Abstract, (May 12, 2014). cited by applicant

Senateur et al. "Pulsed Injection MOCVD of Functional Electronic Oxides" Adv. Mater. Opt. Electron, 10, pp. 155-161 (2000). cited by applicant

Seshadri et al., "Ultrathin Extreme Ultraviolet Patterning Stack Using Polymer Brush As An Adhesion Promotion Layer," Journal of Micro/Nanolithography, MEMS, and MOEMS 16(3) (2017). cited by applicant

Shamma et al., "PDL Oxide Enabled Doubling," Proc. Of SPIE, 6924, 69240D, 1-10 (2008). cited by applicant

Shevtsov et al. "An Apparatus for Pulse Chemical Vapor Deposition of Layers" Instruments and Experimental Techniques, vol. 56, No. 3, pp. 353-357 (2013). cited by applicant

Shevtsov et al. "Effect of Spatial Image Transfer in a Pulse MOCVD Process" Physics Procedia 46, pp. 27-32 (2013). cited by applicant

Simchi et al., "Sulfidation of 2D transition metals (Mo, W, Re, Nb, Ta): thermodynamics, processing, and characterization" J. Materials Science 52: 17, 9 pages (2017). cited by applicant

Spear et al., "Chemical Transport Reactions in the Vanadium-Silicon-Oxygen System and the Ternary Phase Diagram", *Journal of the Less-Common Metals*, 14, pp. 69-75 (1968). cited by applicant

Stanley et al. "Feedgas for Modern High-Performance Ozone Generators" Ozonia Ltd., Duebendorf, Switzerland. 7 pages. Available Jul. 14, 2017 online at: http://www.degremont-technologies.com/cms_medias/pdf/tech_ozonia_feedgas.pdf (1999). cited by applicant

Stannowski et al. "Growth process and properties of silicon nitride deposited by hot-wire chemical vapor deposition" *J Applied Physics*, vol. 93 No. 5, 9 pages (2013). cited by applicant

Su et al. "A Polymer Precursor Route to Metal Borides", *Chem. Mater.*, 5, pp. 1659-1668 (1993). cited by applicant

Svetin et al., "Three-dimensional resistivity and switching between correlated electronic states in 1T-TaS₂." *Nature, Scientific Reports* Apr. 12, 2017, 7:46048, 10 pages (2017). cited by applicant

Tatehaba et al., "Adhesion Energy of Polystyrene and Substrate in Function Water," 5th International Symposium of Cleaning Technology in Semiconductor Device Manufacturing, pp. 560-565 (1998). cited by applicant

Tidman et al. "Resistivity of thin TaS₂ crystals," *Can. J. Phys.*, vol. 54, 2306-2309 (1976). cited by applicant

Todi et al., "Characterization of Pt—Ru Binary Alloy Thin Films for Work Function Tuning," *IEEE Electron Device Letters*, vol. 27, No. 7, pp. 542-545 (2006). cited by applicant

Tomozeiu et al. "Effects of UV photon irradiation on SiO_x (0<x<2) structural properties" *Applied Surface Science* 253, pp. 376-380 (2006). cited by applicant

Trumbore et al. "Solid solubilities of aluminum and gallium in germanium," *J. of Physics and Chemistry of Solids*, vol. 11, Issues 3-4, 239-240 (1959). cited by applicant

Tseng et al., "Etch Properties of Resists Modified by Sequential Infiltration Synthesis," *American Vacuum Society* (2011). cited by applicant

Tseng et al., "Enhanced Block Copolymer Lithography Using Sequential Infiltration Synthesis," *Journal of Physical Chemistry*, vol. 5, 17725-17729 (2011). cited by applicant

Ueda et al. "Enhanced Sidewall Grown (ESG) process: towards PEALD with conformality above 100%," *Extended Abstracts of the 2011 International Conference on Solid State Devices and Materials*, Nagoya, pp. 34-35 (2011). cited by applicant

U.S. Appl. No. 60/394,086, filed Jul. 3, 2002 in the name of Jacques Schmitt, and entitled "ALD on a Rotary Susceptor" pp. 1-23 (2002). cited by applicant

U.S. Appl. No. 62/274,238, filed Jan. 1, 2016 in the names of Alexander S. Polyak et al., and entitled "Non-Metallic Thermal CVD/ALD Gas Injector and Purge System" pp. 1-63 (2016). cited by applicant

U.S. Appl. No. 62/293,897, filed Feb. 11, 2016 in the names of Bernardo Donoso et al., and entitled "Vapor Based Site-Isolated Processing Systems and Methods" pp. 1-51 (2016). cited by applicant

Vallat et al. "Area selective deposition of TiO₂ by intercalation of plasma etching cycles in PEALD process: A bottom up approach for the simplification of 3D integration scheme" *Journal of Vacuum Science & Technology A* 37(2), 12 pages (2019). cited by applicant

Vallat et al. "Selective deposition of Ta₂O₅ by adding plasma etching super-cycles in plasma enhanced atomic layer deposition steps" *Journal of Vacuum Science & Technology A* 35(1), 7 pages (2016). cited by applicant

Varma, et al., "Effect of Metal Halides on Thermal, Mechanical, and Electrical Properties of Polypyromelitimide Films," *Journal of Applied Polymer Science*, vol. 32, pp. 3987-4000, (1986). cited by applicant

Vasilev, "Borophosphosilicate Glass Films in Silicon Microelectronics, Part 1: Chemical Vapor Deposition, Composition, and Properties" *Russian Microelectronics*, vol. 33, No. 5, pp. 271-284 (2004). cited by applicant

Vayrynen et al. "Photo-Assisted Atomic Layer Deposition and Chemical Vapor Deposition of Metal and Metal Oxide Thin Films" Thesis, Oct. 2015, 146 pages (2005). cited by applicant

Venkatesan et al. "A rugged lead-ruthenate pyrochlore membrane catalyst for highly selective oxidation of alcohols" *J. Molecular Catalysis A: Chemical* 250, pp. 87-93 (2006). cited by applicant

Voltaix, "Material Safety Data Sheet for: Trisilylamine", pp. 1-8, (2014). cited by applicant

Wang et al., "Tertiarybutylaluminum as an Organometallic Source for Epitaxial Growth of AlGaSb," *Appl. Phys. Lett.* 67 (10), Sep. 4, pp. 1384-1386, American Institute of Physics (1995). cited by applicant

Wirths, et al., "SiGeSn Growth Studies Using Reduced Pressure Chemical Vapor Deposition Towards Optoelectronic Applications," *Thin Solid Films*, 557, 183-187 (2014). cited by applicant

Xiao "Introduction to Semiconductor Manufacturing Technology" SPIE Press, ISBN 978-0-08194-9092-6, pp. 237-245 (2012). cited by applicant

Xing et al., "Ising Superconductivity and Quantum Phase Transition in Macro-Size Monolayer NbSe₂" *Nano. Lett.* 17, pp. 6802-6807 (2017). cited by applicant

Xu et al., "14NM Metal Gate Film Stack Development and Challenges," Smic et al. (2016). cited by applicant

Xu et al., "Contacts between Two- and Three-Dimensional Materials: Ohmic, Schottky, and p-n Heterojunctions" *ACS Nano* 10, pp. 4895-4919 (2016). cited by applicant

Yoshida, et al., Threshold Voltage Tuning for 10NM and Beyond CMOS Integration, *Solid State Technology*, 57(7): 23-25 (2014). cited by applicant

Yu et al., "Modulation of the Ni Workfunction by Yb Doping: from Midgap to N-Type Band-Edge," 4 pages, *IEEE* 0-7803-9269-8/05 (2005). cited by applicant

Yuan et al. "A Bifunctional Air Electrode Catalyzed by Lead Ruthenate for Li-Air Batteries" *ECS Transactions* 69(19), pp. 23-32 (2015). cited by applicant

Yuan et al., "Facile Synthesis of Single Crystal Vanadium Disulfide Nanosheets by Chemical Vapor Deposition for Efficient Hydrogen Evolution Reaction" *Adv. Mater.* 27, pp. 5605-5609 (2015). cited by applicant

Yun et al., "Behavior of Various Organosilicon Molecules in PECVD Processes for Hydrocarbon-Doped Silicon Oxide Films," *Solid State Phenomena*, vol. 124-126, 347-350 (2007). cited by applicant

Yun et al., "Comparison of Atomic Scale Etching of Poly-Si in Inductively Coupled Ar and He Plasmas", *Korean Journal of Chemical Engineering*, vol. 24, 670-673 (2007). cited by applicant

Yun et al., "Single-Crystalline Si Stacked Array (STAR) NAND Flash Memory," *IEEE Transactions on Electron Devices*, vol. 58, No. 4, 1006-1014 (2011). cited by applicant

Yun et al., "Effect of Plasma on Characteristics of Zirconium Oxide Films Deposited by Plasma-Enhanced Atomic Layer Deposition," *Electrochemical and Solid State Letters*, 8(11) F47-F50 (2005). cited by applicant

Yushin et al., "Carbon-Derived Carbon," Department of Materials Science and Engineering, Taylor & Francis Group, LLC (2006). cited by applicant

Zhao et al. "Surface Chemistry of Thermal Dry Etching of Cobalt Thin Films Using Hexafluoroacetylacetone (hfacH)" *Appl Surf Sci.* 455, pp. 438-445 (2018). cited by applicant

Zhao et al. "Thermal and Plasma-Enhanced Atomic Layer Deposition of Yttrium Oxide Films and the Properties of Water Wettability" *Appl. Mater. Interfac.* 12(2), 9 pages (2019). cited by applicant

Zhou et al., "A library of atomically thin metal chalcogenides" *Nature* 556, pp. 355-361 (2018). cited by applicant

Chemistry Stack Exchange, "Why is CF₄ Non-Polar and CHF₃ Polar," <https://chemistry.stackexchange.com/questions/31604/why-is-cf4-non-polar-and-chf3-polar>, (2015). cited by applicant

Crystal IS "Application Note: Using UV Reflective Materials to Maximize Disinfection"; AN011;

Jun. 16, 2016. cited by applicant
IPS Water Heater Pan Adapter Kit, Nov. 1, 2015, [online], [site visited Dec. 4, 2019]; URL: <http://es.ipscorp.com/watertite/protectivesystem/whpanadapter> (2015). cited by applicant
“Polyurethane_HF”; webpage; no date. Cited in Notice of References dated May 18, 2017 in U.S. Appl. No. 14/884,695. cited by applicant
Rhenium trioxide; https://en.wikipedia.org/wiki/Rhenium_trioxide [online]; last edited on Feb. 18, 2017. cited by applicant
Scientific and Technical Information Center EIC 2800 Search Report dated Feb. 16, 2012. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/030,174 filed May 26, 2020 titled METHOD FOR DEPOSITING BORON AND GALLIUM CONTAINING SILICON GERMANIUM LAYERS, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD OF INVENTION

(1) The present disclosure generally relates to methods and systems suitable for forming electronic devices. More particularly, the disclosure relates to methods and systems that can be used for depositing material, for example for selectively depositing material, such as doped semiconductor material, on a surface of a substrate.

BACKGROUND OF THE DISCLOSURE

(2) The scaling of semiconductor devices, such as, for example, complementary metal-oxide-semiconductor (CMOS) devices, has led to significant improvements in speed and density of integrated circuits. However, conventional device scaling techniques face significant challenges for future technology nodes.

(3) One particular challenge relates to reduction in contact resistance to the active regions of a semiconductor device structure. Furthermore, in many applications, it may be desirable to selectively deposit semiconductor material (e.g., highly-doped Group IV semiconductor material) that incorporates a dopant. However, such techniques may not be well developed. Accordingly, improved methods and systems for depositing doped semiconductor material are desired.

(4) In addition, there is a particular need for depositing semiconductor material at ever lower temperatures because the thermal budget that many advanced electronic devices can withstand is limited.

(5) In addition, there is a particular need for depositing semiconductor material on which contacts with a very low contact resistance can be made. There is a specific need for methods which allow selectively depositing such materials.

(6) U.S. Publication No. 2014120678 A1 discloses methods for selective and conformal epitaxy of highly doped Si-containing materials for three dimensional structures.

(7) Any discussion, including discussion of problems and solutions, set forth in this section has been included in this disclosure solely for the purpose of providing a context for the present disclosure. Such discussion should not be taken as an admission that any or all of the information was known at the time the invention was made or otherwise constitutes prior art.

SUMMARY OF THE DISCLOSURE

(8) Various embodiments of the present disclosure relate to deposition methods, e.g., selective deposition methods, to structures and devices formed using such methods, and to apparatus for performing the methods and/or for forming the structure and/or devices. While the ways in which various embodiments of the present disclosure address drawbacks of prior methods and systems are discussed in more detail below, in general, various embodiments of the disclosure provide improved methods of selectively depositing doped semiconductor layers. The doped semiconductor layers can exhibit relatively low contact resistance.

(9) Described herein is a method for epitaxially growing a boron- and gallium-doped silicon germanium layer. The method comprises providing a substrate comprising a monocrystalline surface in a reactor chamber. The method further comprises introducing a silicon precursor, a germanium precursor, a boron precursor, a gallium precursor. Thus, a boron and gallium doped silicon germanium layer is grown on the monocrystalline surface. The silicon precursor is selected from the list consisting of silanes, cyclosilanes, alkylsilanes, and alkynylsilanes. The germanium precursor is selected from the list consisting of germanes, cyclogermanes, alkylgermanes, and alkynylgermanes. The boron precursor is selected from the list consisting of boranes, and organic borohydrides. The organic borohydrides have the general formula $R_{x-3}M(BH_4)_3$, wherein R is independently chosen from H, CH_3 , C_2H_5 , C_6H_5 , and NH_2 ; M is a Group IIIA metal independently chosen from gallium, aluminum and indium; and x is an integer from 1-3. The gallium precursor is selected from the list consisting of gallium alkyls, $Ga(BH_4)_3$, and GaH_3 .

(10) Further described herein is a method for epitaxially growing a boron- and gallium-doped silicon germanium layer. The method comprises providing a substrate comprising a monocrystalline surface in a reactor chamber. The method further comprises introducing a silicon precursor, a germanium precursor, a boron precursor, and a gallium precursor into the reactor chamber. Thus, a boron and gallium-doped silicon germanium layer is grown on the monocrystalline surface. The silicon precursor, the germanium precursor, the boron precursor, and the gallium precursor, are substantially free of chlorine.

(11) In some embodiments, the silicon precursor, the germanium precursor, the boron precursor, and the gallium precursor are substantially free of halogens.

(12) In some embodiments, the method further comprises introducing a carrier gas into the reactor chamber.

(13) In some embodiments, the carrier gas does not contain chlorine.

(14) In some embodiments, the carrier gas does not contain a halogen.

(15) In some embodiments, the carrier gas is selected from the list consisting of hydrogen, a noble gas, and nitrogen.

(16) In some embodiments, the carrier gas comprises a noble gas selected from the list consisting of helium, neon, krypton, argon, and xenon.

(17) In some embodiments, the silicon precursor comprises a silane and is selected from the list consisting of silane (SiH_4), disilane (Si_2H_6), trisilane (Si_3H_8), tetrasilane (Si_4H_{10}), pentasilane (Si_5H_{12}).

(18) In some embodiments, the silicon precursor comprises methylsilane (CH_3-SiH_3).

(19) In some embodiments, the germanium precursor is selected from the list consisting of germane (GeH_4), digermane (Ge_2H_6), trigermane (Ge_3H_8), or germysilane ($H_3Ge-SiH_3$).

(20) In some embodiments, the gallium precursor is a gallium alkyl.

(21) In some embodiments, the gallium precursor is selected from the list consisting of trimethylgallium (TMG) or triethylgallium (TEG), tritertiarybutylgallium (TTBGa), $Ga(BH_4)_3$, and GaH_3 .

(22) In some embodiments, the boron precursor comprises a borane.

(23) In some embodiments, the boron precursor comprises diborane (B_2H_6).

- (24) In some embodiments, the substrate comprises a first surface and a second surface. The first surface is a monocrystalline surface. The second surface is a dielectric surface. The boron- and gallium-doped silicon germanium layer is then selectively and epitaxially grown on the first surface.
- (25) In some embodiments, the method further comprises the step of introducing one or more cap layer precursors into the reactor chamber. Thus, an epitaxial cap layer overlying the boron- and gallium-doped silicon germanium layer is formed. Afterwards, an etch gas may be introduced into the reactor chamber. Thus, the epitaxial cap layer is etched.
- (26) In some embodiments, the following steps are repeated until the boron and gallium doped epitaxial silicon germanium layer on the first surface has reached a predetermined thickness: introducing the silicon precursor, the germanium precursor, the boron precursor, the gallium precursor, and the carrier gas into the reactor chamber; introducing one or more cap layer precursors into the reactor chamber; and, introducing an etch gas into the reactor chamber.
- (27) In some embodiments, the one or more cap layer precursors comprise a silicon precursor and a boron precursor, and the cap layer comprises silicon and boron.
- (28) In some embodiments, the etch gas comprises a halogen.
- (29) In some embodiments, the etch gas is selected from the list consisting of HCl, Cl₂, and HBr.
- (30) In some embodiments, the second surface is selected from the list consisting of a silicon oxide surface, a silicon nitride surface, a silicon oxycarbide surface, a silicon oxynitride surface, a hafnium oxide surface, a zirconium oxide surface, and an aluminum oxide surface.
- (31) In some embodiments, the reactor chamber is maintained at a temperature of at least 300° C. to at most 600° C.
- (32) In some embodiments, the reactor chamber is maintained at a pressure of at least 10 Torr to at most 150 Torr.
- (33) In some embodiments, the silicon precursor is a silane, the germanium precursor is a germane, the boron precursor is a borane, and the gallium precursor is a gallium alkyl.
- (34) In some embodiments, the silicon precursor is SiH₄, the germanium precursor is GeH₄, the boron precursor is B₂H₆, and the gallium precursor is (CH₃CH₂)₃Ga.
- (35) In some embodiments, the monocrystalline surface comprises a monocrystalline silicon surface.
- (36) In some embodiments, the monocrystalline surface comprises a monocrystalline silicon germanium surface.
- (37) In some embodiments, the monocrystalline silicon germanium surface comprises a boron doped silicon germanium surface.
- (38) In some embodiments, the monocrystalline silicon germanium surface comprises a boron and gallium doped silicon germanium surface.
- (39) Further provided herein is a system comprising one or more reaction chambers, a gas injection system, and a controller configured for causing the system to perform a method as described herein.
- (40) Further provided herein is a field effect transistor comprising a boron and gallium doped silicon germanium layer as a source and/or drain contact. The boron and gallium doped silicon germanium layer is deposited by means of a method as described herein.
- (41) In accordance with yet additional examples of the disclosure, a system to perform a method as described herein and/or to form a structure, device, or portion of either is disclosed.
- (42) These and other embodiments will become readily apparent to those skilled in the art from the following detailed description of certain embodiments having reference to the attached figures. The invention is not being limited to any particular embodiments disclosed.
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Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

- (1) A more complete understanding of the embodiments of the present disclosure may be derived by referring to the detailed description and claims when considered in connection with the following illustrative figures.
- (2) FIG. 1 illustrates a method in accordance with exemplary embodiments of the disclosure.
- (3) FIG. 2 illustrates a substrate (**200**) on which boron and gallium-doped silicon germanium layer may be deposited in accordance with exemplary embodiments of the disclosure.
- (4) FIG. 3 illustrates a system (**300**) in accordance with additional exemplary embodiments of the disclosure.
- (5) FIG. 4, panel A, shows X-ray reflectivity (XRR) measurements of layers that were deposited by means of a method as disclosed herein.
- (6) FIG. 4, panel B, shows x-ray diffraction (XRD) patterns of layers that were deposited by means of a method as disclosed herein.
- (7) FIG. 5 shows the presence of nucleation delay on silicon nitride versus silicon oxide, and shows a lower growth rate for growth on silicon nitride compared to monocrystalline silicon.
- (8) FIG. 6 shows selective growth of gallium and boron doped silicon germanium on monocrystalline silicon versus silicon nitride and silicon oxide.
- (9) Throughout the drawings, the following numbering is used: **100**—method; **102**—substrate providing step; **104**—deposition step; **105**—cap step; **108**—etching step; **110**—cyclic loop/repetition of deposition steps **104** and **105** and etching step **108**; **112**—method end; **200**—substrate; **202**—monocrystalline material; **204**—non-monocrystalline material; **206**—first area; **208**—second area; **210**—monocrystalline surface; **212**—non-monocrystalline surface; **300**—system; **302**—substrate handling system; **304**—reaction chamber; **306**—injection system; **308**—wall; **310**—first gas source; **311**—second gas source; **312**—third gas source; **314**—fourth gas source; **316**—fifth gas source; **318**—line; **320**—line; **321**—line; **322**—line; **324**—line; **326**—exhaust; **328**—controller; **610**—boron and gallium-doped silicon germanium; **620**—monocrystalline silicon; **630**—silicon oxide; **640**—silicon nitride.
- (10) It will be appreciated that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of illustrated embodiments of the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

- (11) The description of exemplary embodiments of methods, structures, devices and systems provided below is merely exemplary and is intended for purposes of illustration only; the following description is not intended to limit the scope of the disclosure or the claims. Moreover, recitation of multiple embodiments having stated features is not intended to exclude other embodiments having additional features or other embodiments incorporating different combinations of the stated features. For example, various embodiments are set forth as exemplary embodiments and may be recited in the dependent claims. Unless otherwise noted, the exemplary embodiments or components thereof may be combined or may be applied separate from each other.
- (12) As set forth in more detail below, various embodiments of the disclosure provide methods for depositing boron and gallium doped silicon germanium on a surface of a substrate. Exemplary methods can be used to, for example, form source and/or drain regions of semiconductor devices that exhibit relatively high mobility, relatively low contact resistance, and that maintain the structure and composition of the deposited layers. In particular, the layers can be used as p-type source and/or drain regions in n-channel MOSFETs. Exemplary MOSFETs in which these layers can be used include FinFETs and GAA (Gate-All-Around) FETs. In addition, the present layers are

especially useful for the formation of shallow junctions because of a reduced channeling effect. In some embodiments, the present methods involve selectively depositing boron and gallium doped silicon germanium.

(13) In this disclosure, “gas” can include material that is a gas at normal temperature and pressure (NTP), a vaporized solid and/or a vaporized liquid, and can be constituted by a single gas or a mixture of gases, depending on the context. A gas other than the process gas, i.e., a gas introduced without passing through a gas distribution assembly, a multi-port injection system, other gas distribution device, or the like, can be used for, e.g., sealing the reaction space, and can include a seal gas, such as a noble gas. In some cases, the term “precursor” can refer to a compound that participates in the chemical reaction that produces another compound, and particularly to a compound that constitutes a film matrix or a main skeleton of a film; the term “reactant” can be used interchangeably with the term precursor.

(14) As used herein, the term “substrate” can refer to any underlying material or materials that can be used to form, or upon which, a device, a circuit, or a film can be formed. As set forth in more detail below, a surface of a substrate can include two or more areas, wherein each of the two or more areas comprise different material.

(15) As used herein, the term “epitaxial layer” can refer to a substantially single crystalline layer upon an underlying substantially single crystalline substrate or layer.

(16) As used herein, the term “chemical vapor deposition” can refer to any process wherein a substrate is exposed to one or more volatile precursors, which react and/or decompose on a substrate surface to produce a desired deposition.

(17) As used herein, the term “film” and/or “layer” can refer to any continuous or non-continuous structures and material, such as material deposited by the methods disclosed herein. For example, film and/or layer can include two-dimensional materials, three-dimensional materials, nanoparticles or even partial or full molecular layers or partial or full atomic layers or clusters of atoms and/or molecules. A film or layer may comprise material or a layer with pinholes, which may be at least partially continuous. Alternatively, a film or layer may consist entirely of isolated islands.

(18) Further, in this disclosure, any two numbers of a variable can constitute a workable range of the variable, and any ranges indicated may include or exclude the endpoints. Additionally, any values of variables indicated (regardless of whether they are indicated with “about” or not) may refer to precise values or approximate values and include equivalents, and may refer to average, median, representative, majority, or the like. Further, in this disclosure, the terms “including,” “constituted by” and “having” refer independently to “typically or broadly comprising,” “comprising,” “consisting essentially of,” or “consisting of” in some embodiments. It shall be understood that when a composition, method, device, etc. is said to comprise certain features, it means that it includes those features, and that it does not necessarily excludes the presence of other features, as long as they do not render the claim unworkable. This notwithstanding, the wording “comprises” includes the meaning of “consists of,” i.e., the case when the composition, method, device, etc. in question only includes the features, components, and/or steps that are listed, and does not contain any other features, components, steps, etc.

(19) In this disclosure, any defined meanings do not necessarily exclude ordinary and customary meanings in some embodiments.

(20) The term “carrier gas” as used herein may refer to a gas that is provided to a reactor chamber together with one or more precursors. For example, a carrier gas may be provided to the reactor chamber together with one or more of the precursors used herein. Exemplary carrier gasses include N₂, H₂, and noble gasses such as He, Ne, Kr, Ar, and Xe.

(21) As opposed to a carrier gas, a purge gas may be provided to a reactor chamber separately, i.e., not together with one or more precursors. This notwithstanding, gasses which are commonly used as carrier gas may also be used as a purge gas, even within the same process. For example, in a cyclic deposition-etch process, N₂ used as a carrier gas may be provided together with one or

more precursors during deposition pulses, and N.sub.2 used as a purge gas may be used to separate deposition and etch pulses. Hence, it is the manner of how a gas is provided to the reactor chamber that determines whether it serves as a purge gas or a carrier gas in a specific context. Thus, as used herein, the term “purge” may refer to a procedure in which an inert or substantially inert gas is provided to a reactor chamber in between two pulses of gasses which react with each other. For example, a purge, e.g., using nitrogen gas, may be provided between a precursor pulse and a reactant pulse, thus avoiding or at least minimizing gas phase interactions between the precursor and the reactant. It shall be understood that a purge can be effected either in time or in space, or both. For example, in the case of temporal purges, a purge step can be used, e.g., in the temporal sequence of providing a first precursor to a reactor chamber, providing a purge gas to the reactor chamber, and providing a second precursor to the reactor chamber, wherein the substrate on which a layer is deposited does not move. For example, in the case of spatial purges, a purge step can take the following form: moving a substrate from a first location to which a first precursor is continually supplied, through a purge gas curtain, to a second location to which a second precursor is continually supplied.

(22) As set forth in more detail below, various steps of exemplary methods described herein can be performed in the same reaction chamber or in different reaction chambers of, for example, the same cluster tool.

(23) The present disclosure relates to the epitaxial deposition of boron and gallium-doped silicon germanium. The presently disclosed methods and devices allow for selective or non-selective deposition of layers with low resistance, at low temperatures. The layers may be used, for example, as a p-type source and/or drain regions in field effect transistors.

(24) Thus, described herein is a method for epitaxially growing a boron- and gallium-doped silicon germanium layer. The method comprises providing a substrate in a reactor chamber. The substrate comprises a monocrystalline material. In other words, the substrate comprises a monocrystalline surface. Advantageously, the substrate comprises a monocrystalline silicon surface. Additionally or alternatively, the substrate may comprise a monocrystalline silicon germanium surface.

Additionally or alternatively, the substrate may comprise a boron doped silicon germanium surface. Additionally or alternatively, the substrate may comprise a boron and gallium doped silicon germanium surface.

(25) The method further comprises introducing a silicon precursor, a germanium precursor, a boron precursor, and a gallium precursor into the reaction chamber. Thus, a boron and gallium-doped silicon germanium layer is epitaxially grown on the surface of the substrate.

(26) In some embodiments, the boron precursor, the gallium precursor, the silicon precursor, and the germanium precursor are continually provided to the reaction chamber. Alternatively, the layer may be formed by means of a cyclical deposition process. Thus, in some embodiments, the boron precursor, the gallium precursor, the silicon precursor, and the germanium precursor are sequentially provided to the reactor chamber. Alternatively, any two precursors selected from the boron precursor, the gallium precursor, the silicon precursor, and the germanium precursor may be simultaneously provided to the reaction chamber in a combined precursor pulse, whereas the remaining precursors are provided to the reactor chamber in separate precursors. Alternatively, any three precursors selected from the boron precursor, the gallium precursor, the silicon precursor, and the germanium precursor may be simultaneously provided to the reaction chamber in a combined precursor pulse, whereas the remaining precursor is provided to the reactor chamber in separate precursors. In some embodiments, one or more precursors are provided continually to the reactor chamber, and the remaining precursors are provided to the reactor chamber in pulses. Optionally, any or all of the above-mentioned precursor pulses are separated by purge steps.

(27) It shall be understood that the present methods may be carried out after any suitable pre-clean. One possible pre-clean is a plasma clean that results in an H-terminated silicon surface. Another possible pre-clean uses wet chemistry. For example, the following sequence may be used: surface

oxidation in a mixture consisting of NH_3 , H_2O , and H_2 ; followed by a rinse; followed by an HF dip; followed by a rinse. A suitable HF dip comprises, for example, a dip in a mixture consisting of from at least 0.1 vol. % to at most 1.5 vol. % HF in water. Additionally or alternatively, a gas-phase pre-clean may be used.

(28) In some embodiments, the method further comprises introducing a carrier gas into the reactor chamber. This can be particularly useful when, for example, hard to volatilize precursors are used, in which case a carrier gas can help with bringing the precursors to the reaction chamber.

(29) In some embodiments, the silicon precursor, the germanium precursor, the boron precursor, and the gallium precursor are substantially free of chlorine. In some embodiments, they do not contain chlorine. In other words, in some embodiments, none of the precursors contain chlorine.

(30) In some embodiments, the silicon precursor, the germanium precursor, the boron precursor, and the gallium precursor are substantially free of halogens. In some embodiments, they do not contain any halogens. In other words, in some embodiments, none of the precursors contain a halogen.

(31) Without the invention being bound to any particular theory or mode of operation, it is believed that the use of chlorine-free or halogen-free precursors prevents preferential gallium etching during epitaxial growth. Thus, the total dopant concentration (i.e., the total of the gallium concentration plus the boron concentration) in the resulting silicon-germanium layer may be increased compared to processes using chlorine- or halogen-containing precursors.

(32) It shall be understood that, when a carrier gas is used, the carrier gas is preferably substantially chlorine-free and/or substantially halogen-free as well. Also, in case other gasses are introduced into the reactor apart from the above-mentioned precursors and the carrier gas, these further gasses preferably do not comprise any chlorine and/or halogens either.

(33) In some embodiments, the silicon precursor is selected from the list consisting of silanes, cyclosilanes, alkylsilanes, and alkynylsilanes.

(34) In some embodiments, the silicon precursor comprises a silane and is selected from the list consisting of silane (SiH_4), disilane (Si_2H_6), trisilane (Si_3H_8), tetrasilane (Si_4H_{10}), pentasilane (Si_5H_{12}).

(35) In some embodiments, the silicon precursor comprises methylsilane ($\text{CH}_3\text{—SiH}_3$).

(36) In some embodiments, the germanium precursor is selected from the list consisting of germanes, cyclogermanes, alkylgermanes, and alkynylgermanes.

(37) In some embodiments, the germanium precursor is selected from the list consisting of germane (GeH_4), digermane (Ge_2H_6), trigermane (Ge_3H_8), and germysilane ($\text{H}_3\text{Ge—SiH}_3$).

(38) In some embodiments, the boron precursor is selected from the list consisting of boranes, and organic borohydrides, the organic borohydrides having the general formula $\text{R}_x\text{M}(\text{BH}_4)_{3-x}$, wherein R is independently chosen from H, CH_3 , C_2H_5 , C_6H_5 , and NH_2 ; M is a Group IIIA metal independently chosen from gallium, aluminum and indium; and x is an integer from 1-3.

(39) In some embodiments, the boron precursor comprises a borane.

(40) In some embodiments, the boron precursor comprises diborane (B_2H_6).

(41) In some embodiments, the gallium precursor is selected from the list consisting of gallium alkyls, $\text{Ga}(\text{BH}_4)_3$, and GaH_3 .

(42) In some embodiments, the gallium precursor comprises a gallium alkyl. Suitable gallium alkyls include trimethylgallium (TMG) or triethylgallium (TEG), and tritertiarybutylgallium (TTBGa).

(43) In some embodiments, the gallium precursor comprises $\text{Ga}(\text{BH}_4)_3$.

(44) In some embodiments, the gallium precursor comprises GaH_3 .

(45) In some embodiments, the carrier gas is selected from the list consisting of hydrogen, a noble gas, and nitrogen. In some embodiments, the noble gas comprises helium. In some embodiments,

the noble gas comprises krypton. In some embodiments, the noble gas comprises neon. In some embodiments, the noble gas comprises argon. In some embodiments, the noble gas comprises xenon. In some embodiments, N.sub.2 is used as a carrier gas.

(46) In some embodiments, the carrier gas is provided to the reactor chamber at a flow rate from at least 100 sccm to at most 30000 sccm, or from at least 200 sccm to at most 20000 sccm, or from at least 300 sccm to at most 10000 sccm, or from at least 500 sccm to at most 5000 sccm, or from at least 750 sccm to at most 2500 sccm, or from at least 10000 sccm to at most 20000 sccm, or of 15000 sccm.

(47) In some embodiments, the silicon precursor is a silane, the germanium precursor is a germane, the boron precursor is a borane, and the gallium precursor is a gallium alkyl. In some embodiments, the silicon precursor is SiH.sub.4, the germanium precursor is GeH.sub.4, the boron precursor is B.sub.2H.sub.6, and the gallium precursor is (CH.sub.3CH.sub.2).sub.3Ga.

(48) In some embodiments, the silicon precursor is SiH.sub.4. SiH.sub.4 may be provided to the reactor chamber as 100% SiH.sub.4. Alternatively, SiH.sub.4 may be diluted, e.g., in H.sub.2, e.g., as from at least 1.0 to at most 2.0 vol. %, from at least 2.0 vol. % to at most 5.0 vol. %, from at least 5.0 vol. % to at most 10.0 vol. %, from at least 10.0 vol. % to at most 20.0 vol. %, from at least 20.0 vol. % to at most 50.0 vol. %, or from at least 50.0 vol. % to at most 99.9 vol. % SiH.sub.4 in H.sub.2.

(49) In some embodiments, the germanium precursor is GeH.sub.4. GeH.sub.4 may be provided to the reactor chamber as 100% GeH.sub.4. Alternatively, GeH.sub.4 may be diluted, e.g., in H.sub.2. E.g., GeH.sub.4 in H.sub.2 may be provided in a concentration from at least 1.0 to at most 2.0 vol. %, from at least 2.0 vol. % to at most 5.0 vol. %, from at least 3.0 vol. % to at most 7.0 vol. %, from at least 5.0 vol. % to at most 10.0 vol. %, from at least 10.0 vol. % to at most 20.0 vol. %, from at least 20.0 vol. % to at most 50.0 vol. %, or from at least 50.0 vol. % to at most 99.9 vol. % GeH.sub.4 in H.sub.2.

(50) In some embodiments, the boron precursor is diborane, and the diborane is provided to the reactor chamber together with H.sub.2 as a carrier gas. In some embodiments, a mixture of 0.1 vol. % to 10.0 vol. %, or 0.2 vol. % to 5.0 vol. %, or 0.4 vol. % to 2.5 vol. %, or 0.6 vol. % to 1.5 vol. %, or 0.8 vol. % to 1.2 vol. % diborane in H.sub.2 may be used.

(51) In some embodiments, the gallium precursor is (CH.sub.3CH.sub.2).sub.3Ga. (CH.sub.3CH.sub.2).sub.3Ga may be provided to the reactor chamber pure, i.e., at a concentration of 100 vol. % (CH.sub.3CH.sub.2).sub.3Ga. Alternatively, (CH.sub.3CH.sub.2).sub.3Ga may be provided in a diluted form with a carrier gas.

(52) In some embodiments, 100 vol. % SiH.sub.4 is provided to the reactor chamber at a flow rate of at least 20 sccm to at most 100 sccm. In some embodiments, from at least 3.0 to at most 7.0 vol. % GeH.sub.4 in H.sub.2 is applied to the reactor chamber at a flow rate of at least 200 sccm to at most 400 sccm. In some embodiments, from at least 0.5 to at most 2.0 vol. % B.sub.2H.sub.6 in H.sub.2 is provided to the reaction chamber at a flow rate of at least 2.0 to at most 12.0 sccm. In some embodiments, 100 vol. % (CH.sub.3CH.sub.2).sub.3Ga is provided to the reaction chamber at a flow rate of at least 1.0 sccm to at most 5.0 sccm.

(53) The present methods may allow for intrinsic selective growth of boron and gallium doped silicon germanium layers within a pre-determined selectivity window. In other words, the present methods may be used to selectively grow boron and gallium doped silicon germanium on one part of a substrate (e.g., a monocrystalline silicon surface), whereas no, or no substantial amount of, growth occurs on another part of that substrate (e.g., a silicon oxide surface). A selectivity window is a thickness range of a grown layer in which the layer can be growth solely, or substantially solely, on one part of a substrate and not on one or more other parts of the substrate. Exemplary selectivity windows are 20 nm, 10 nm, 8 nm, 6 nm, 5 nm, 4 nm, 3 nm, 2 nm, and 1 nm.

(54) In one embodiment that enables selective growth, the substrate comprises a first surface and a second surface. The first surface is a monocrystalline surface, e.g., a monocrystalline silicon

surface or a monocrystalline silicon germanium surface. Additionally or alternatively, the substrate may comprise a monocrystalline silicon germanium surface. Additionally or alternatively, the substrate may comprise a boron doped silicon germanium surface. Additionally or alternatively, the substrate may comprise a boron and gallium doped silicon germanium surface. The first surface preferably has a hydrogen termination. The first surface may be a surface of a doped layer, e.g., a boron-doped silicon layer. Alternatively, the first surface may be a surface of an undoped layer. The second surface is a dielectric surface, and the boron- and gallium-doped silicon germanium layer is selectively and epitaxially grown on the first surface. In other words, the boron and gallium doped silicon germanium layer is grown on the first surface and not, or not substantially, on the second surface. Without the invention being limited by any theory or mode of operation, it shall be understood that such selectivity may be obtained through nucleation delay on the second surface compared to the first surface.

(55) In some embodiments, the second surface is selected from the list consisting of a silicon oxide surface, a silicon nitride surface, a silicon oxycarbide surface, a silicon oxynitride surface, a hafnium oxide surface, a zirconium oxide surface, and an aluminum oxide surface. In some embodiments, material exposed on the second surface can include, for example, a dielectric material, such as an oxide, a nitride, an oxynitride, an oxycarbide, an oxycarbide nitride, and/or the like, such as silicon nitride, silicon oxide (SiO_2), silicon carbide and mixtures thereof, such as SiOC , SiOCN , SiON . In some embodiments, the second area has a silicon oxide surface. In other words, in some embodiments, the second material consists of silicon oxide (SiO_2).

(56) In some embodiments that enable selectively growing an epitaxial boron and gallium doped silicon germanium layer, the method comprises providing a substrate in a reactor chamber. The substrate comprises a first area comprising a first material a second area comprising a second material. The first area may comprise a monocrystalline silicon surface. Additionally or alternatively, the first area may comprise a monocrystalline silicon germanium surface. Additionally or alternatively, the first area may comprise a boron doped silicon germanium surface. Additionally or alternatively, the first area may comprise a boron and gallium doped silicon germanium surface.

(57) The method comprises selectively growing a boron and gallium doped epitaxial silicon germanium layer on a surface in the first area. Such methods allow growing relatively thin, i.e., with a thickness within a pre-determined selectivity window, epitaxial silicon germanium layers without an etch back. Exemplary selectivity windows include 5 nm, 10 nm, 15 nm, or 20 nm.

(58) When it is desirable to selectively grow a boron and gallium doped silicon germanium layer having a thickness that is higher than the selectivity window, a cap-and-etch approach may be used. This may be done, for example, by introducing one or more cap layer precursors into the reactor chamber, thereby forming an epitaxial cap layer overlying the boron- and gallium-doped silicon germanium layer, and then introducing an etch gas into the reactor chamber, thereby etching the epitaxial cap layer. Suitable etch gasses include halogen-containing compounds. Exemplary halogens include fluorine, chlorine, bromine, and iodine. In some embodiments, the etch gas comprises chlorine. Exemplary chlorine containing etch gasses include HCl and Cl_2 . An exemplary bromine containing etch gas includes HBr . A suitable cap-and-etch approach is described, for example, in U.S. Provisional Application No. 62/930,752, which is hereby incorporated by reference in its entirety.

(59) Without the invention being limited by any theory or particular mode of operation, it is believed that during epitaxial growth of boron and gallium doped silicon germanium on the first surface, parasitic nuclei of boron and gallium doped silicon germanium capped with cap layer may be formed on the second surface are etched as well. Introducing the etch gas into the reactor chamber, completely or substantially completely etches the nuclei in the second area while etching only the cap layer in the first area.

(60) In some embodiments, the step of depositing the boron and gallium doped silicon germanium

layer and the cap layer deposition step are separated by a purge step.

(61) The aforementioned cap-and-etch approach may be repeated in order to epitaxially grow layers of any desired thickness. Accordingly, in some embodiments, the following steps are repeated until the until the boron and gallium doped epitaxial silicon germanium layer on the first surface has reached a pre-determined thickness: introducing the silicon precursor, the germanium precursor, the boron precursor, the gallium precursor, and the carrier gas into the reactor chamber; introducing one or more cap layer precursors into the reactor chamber; and, introducing an etch gas into the reactor chamber. Optionally, these steps are separated by purge steps.

(62) In other words, in some embodiments, the sequence of the deposition step, capping step, and etching step are repeated as desired until a pre-determined thickness of the epitaxial boron and gallium doped silicon germanium layer is formed overlaying the first area. For example, the deposition step and the etching step can be repeated from at least 1 to at most 1000 times, from at least 2 to at most 100 times, from at least 2 to at most 50 times, from at least 2 to at most 30 times, from at least 2 to at most 20 times, or from at least 5 to at most 15 times, or from at least 8 to at most 12 times.

(63) In some embodiments, the one or more cap layer precursors comprise a silicon precursor and a boron precursor, and the cap layer comprises silicon and boron. For example, the silicon precursors and boron precursors mentioned above in the context of the boron and gallium doped silicon germanium layer growth may be used as the silicon and boron precursors for the cap layer.

(64) In other words, selective growth may be achieved by cyclically performing the following sequence of sub steps i., ii., and iii. Step i. comprises epitaxially growing a boron and gallium doped silicon germanium layer on the monocrystalline surface and growing a boron and gallium doped amorphous and/or polycrystalline silicon germanium layer on the second surface by introducing silane, germane, diborane, triethylgallium, and a carrier gas into the reactor chamber. Step ii. comprises growing a capping layer as described herein. Step iii. comprises etching the epitaxial boron and gallium doped silicon germanium layer and etching the amorphous and/or polycrystalline boron and gallium doped silicon germanium layer on the second surface by introducing an etchant into the reactor chamber. The sequence of sub steps i., ii., and iii. is then repeated until the epitaxial phosphorous doped silicon layer on the monocrystalline surface has reached a pre-determined thickness. Optionally, any or all of steps i., ii., and iii. may be separated by a purge step.

(65) The etch step (iii) can be performed in the same reaction chamber used during the deposition step (i) and the capping step (ii). Alternatively, the etch step can be performed in another reaction chamber, such as another reaction chamber in the same cluster tool as the reaction chamber used during the deposition step. The temperature and/or pressure which is maintained during the etching step can be the same or similar to the temperature and/or pressure described above in connection with the deposition step.

(66) Without the invention being bound by any theory or specific mode of operation, it shall be understood that selectivity may be obtained through any one or a combination of the following mechanisms: 1) amorphous boron and gallium doped silicon germanium grows at a slower rate in the second area than epitaxial boron and gallium doped silicon germanium in the first area, 2) amorphous boron and gallium doped silicon germanium growth on the second surface exhibits delayed growth with respect to epitaxial boron and gallium doped silicon germanium in the first area, and/or 3) amorphous boron and gallium doped silicon germanium in the second area is etched at a faster rate than epitaxial boron and gallium doped silicon germanium in the first area. Thus, an epitaxial boron and gallium doped silicon germanium layer may be grown in the first area whereas no deposition occurs in the second area. In other words, an epitaxial boron and gallium doped silicon germanium film is grown on a first surface in the first area whereas after deposition, no or no substantial amount of amorphous boron and gallium doped silicon germanium remains on a second surface in the second area. Without the invention being bound by any theory or particular

mode of operation, it is believed that nucleation delay (i.e., mechanism 2) plays a major role in obtaining selectivity in the methods that are described herein.

(67) In some embodiments, the etchant comprises an elementary halogen. In some embodiments, the etchant comprises HCl. In some embodiments, the etchant comprises chlorine. In other words, Cl.sub.2 is used in some embodiments as an etchant. In some embodiments, Cl.sub.2 is provided during the etch cycles to the reaction chamber at a flow rate from at least 5.0 sccm to at most 100.0 sccm, or from at least 10.0 sccm to at most 50.0 sccm, or from at least 15.0 sccm to at most 40.0 sccm, or from at least 20.0 sccm to at most 30.0 sccm.

(68) In some embodiments, the etch cycles may last from at least 1.0 s to at most 400.0 s, or from at least 2.0 s to at most 200.0 s, or from at least 4.0 s to at most 100.0 s, or from at least 8.0 s to at most 50.0 s, or from at least 10.0 s to at most 40.0 s, or from at least 20.0 s to at most 30.0 s.

(69) In some embodiments, temperature and pressure are kept constant throughout the deposition cycles and the etch cycles, i.e., throughout the boron and gallium doped silicon germanium deposition steps, the cap layer deposition steps, and the etch steps.

(70) In some embodiments, etch cycles and deposition cycles can be at different pressures. In other words, in some embodiments, the step of depositing the boron and gallium doped silicon germanium layer, the step of depositing the cap layer and/or the step etching may be at different pressures. Preferably, the aforementioned pressures differ by no more than 10%, or by no more than 20%, or by no more than 50%, or by no more than 100%, or by no more than 200%, or by no more than 500%, or by no more than 1000%, relative to the lowest pressure that occurs. Keeping the pressure differences limited in this way may speed up processing time by limiting the amount of time needed for pumping between the cycles.

(71) In some embodiments, the deposition cycles and the etch cycles are separated by purges. In some embodiments, N.sub.2 is used as a purge gas. In some embodiments, the purges last from at least 5.0 s to at most 80.0 s, or from at least 10.0 s to at most 40.0 s, or from at least 15.0 to at most 30.0 s, or for about 20.0 s. In some embodiments, the purge gas is provided to the reaction chamber during the purges at a flow rate of at least 5,000 to at most 100,000 sccm, or of at least 10,000 to at most 50,000 sccm, or of at least 20,000 to at most 30,000 sccm.

(72) Suitable purge gasses include nitrogen and the noble gasses. Suitable noble gasses may include He, Ne, Ar, Kr, and Xe. In some embodiments, the purge gas consists of N.sub.2.

(73) In some embodiments, whether selective or not, the reactor chamber is maintained at a temperature of at least 300° C. to at most 600° C. For example, the reactor chamber may be maintained at a temperature of at least 300° C. to at most 350° C., or at a temperature of at least 350° C. to at most 400° C., or at a temperature of at least 400° C. to at most 450° C., or at a temperature of at least 450° C. to at most 500° C., or at a temperature of at least 500° C. to at most 550° C., or at a temperature of at least 550° C. to at most 600° C., or at a temperature of at least 300° C. to at most 400° C., or at a temperature of at least 40° C. to at most 500° C., or at a temperature of at least 500° C. to at most 600° C., or at a temperature of at least 450° C. to at most 550° C.

(74) In other words, in some embodiments, the reaction chamber is maintained at the aforementioned temperatures during the epitaxial deposition of the boron and gallium doped silicon germanium. The temperatures mentioned herein may be measured by means of a thermocouple under the substrate's susceptor and/or by means of a pyrometer suspended in the reactor chamber and above the substrate. The aforementioned temperatures may also be maintained during any cap and/or etch steps, if present.

(75) In some embodiments, the reactor chamber is maintained at a pressure of at least 10 Torr to at most 150 Torr, or at a pressure of at least 10 Torr to at most 80 Torr, or at a pressure of at least 80 Torr to at most 150 Torr, or at a pressure of at least 10 Torr to at most 40 Torr, or at a pressure of at least 40 Torr to at most 80 Torr, or at a pressure of at least 80 Torr to at most 115 Torr, or at a pressure of at least 115 Torr to at most 150 Torr. In other words, in some embodiments, the reaction

chamber is maintained at the aforementioned pressures during the epitaxial deposition of the boron and gallium doped silicon germanium.

(76) The aforementioned pressures may also be maintained during any cap and/or etch steps, if present. Thus, during a selective process comprising a sequence of deposition cycles (i.e., deposition cycles of boron and gallium doped silicon germanium and deposition cycles of the cap layer), and etch cycles, the pressure during the etch cycles may be the same as the pressure used during the deposition cycles. This notwithstanding, and in some embodiments, the pressure during the etch cycles may be different than the pressure used during the deposition cycles. In some embodiments, the pressure during the etch cycles equals the pressure during the deposition cycles within a margin of error of 50%, or within a margin of error of 40%, or within a margin of error of 30%, or within a margin of error of 20%, or within a margin of error of 10%, or within a margin of error of 5%.

(77) In some embodiments, the pressure during the etch cycles is at most 12,000 Pa. This pressure range may improve process safety. In some embodiments, the pressure during the etch cycles is from at least 1,300 Pa to at most 10,700 Pa, or from at least 3,000 Pa to at most 8,000 Pa, or from at least 4,000 Pa to at most 6,000 Pa.

(78) Further described herein is a system that comprises one or more reaction chambers, a gas injection system, and a controller. The controller is configured for causing the system to perform a method for epitaxially depositing a boron and gallium doped silicon germanium layer, as described herein.

(79) In accordance with yet additional embodiments of the disclosure, a device or portion thereof can be formed using a method and/or a structure as described herein. The boron and gallium doped silicon germanium layer can be used to form a source or drain contact of the device. The device can be, for example, a field effect transistor (FET) (e.g., a FinFET, gate all around transistor, or a stack comprising multiple transistor devices).

(80) In a first example, reference is made to FIG. 1. FIG. 1 illustrates a (e.g., selective deposition) method (100) in accordance with exemplary embodiments of the disclosure. The method (100) includes the steps of providing a substrate within a reaction chamber (step 102), selectively or non-selectively depositing a boron and gallium doped silicon germanium layer (step 104), an optional capping layer step (105) in which a boron doped silicon layer may be deposited, and optional etching step (step 108), optionally repeating the deposition steps (104, 105) and the etching step (108) (loop 110), and ending (step 112). Note that in case the etching step is carried out, it is advantageously preceded by the capping layer step (105).

(81) With reference to FIG. 1 and FIG. 2, a substrate (or structure) (200) provided during step (102) can include a first area (206) comprising a first material (e.g., (mono)-crystalline bulk material (202)) and a second area (208) comprising a second material (e.g., non-monocrystalline material (204)). The first material can include a monocrystalline surface (210); second area (208) can include a non-monocrystalline surface (212), such as a polycrystalline surface or an amorphous surface. The monocrystalline surface (210) may be a monocrystalline silicon surface. Additionally or alternatively, the monocrystalline surface (210) may comprise a monocrystalline silicon germanium surface. Additionally or alternatively, the monocrystalline surface (210) may comprise a boron doped silicon germanium surface. Additionally or alternatively, the monocrystalline surface (210) may comprise a boron and gallium doped silicon germanium surface.

(82) The non-monocrystalline surface (212) may include, for example, dielectric materials, such as oxides, oxynitrides, nitrides, oxycarbides, or oxycarbide nitrides, including, for example, silicon oxides, silicon nitrides, silicon oxynitrides, silicon carbides and mixtures thereof, such as SiOC, SiOCN, and SiON.

(83) As a non-limiting example, the reaction chamber used during the step (102) of providing the substrate may comprise a reaction chamber of a chemical vapor deposition system. This notwithstanding, other reaction chambers and alternative chemical vapor deposition systems may

also be utilized to perform the embodiments of the present disclosure. The reaction chamber can be a stand-alone reaction chamber or part of a cluster tool.

(84) The substrate providing step (102) can include heating the substrate to a desired deposition temperature within the reaction chamber. In some embodiments of the disclosure, the substrate providing step (102) includes heating the substrate to a temperature of less than approximately 600° C., or even to a temperature of less than approximately 550° C. For example, in some embodiments of the disclosure, heating the substrate to a deposition temperature may comprise heating the substrate to a temperature of at least 450° C. to at most 550° C. The deposition temperature is maintained throughout the step of depositing the boron and gallium doped silicon germanium layer (104), the optional capping step (105), and the optional etching step (108).

(85) Exemplary process conditions are discussed which were observed to yield a selectivity window of at least 10 nanometers for epitaxial growth of boron and germanium doped silicon germanium on monocrystalline silicon versus dielectrics, e.g., silicon oxide or silicon nitride: substrate temperature: 500° C.; reactor pressure: 40 Torr; silicon precursor: silane; germanium precursor: germane; boron precursor: diborane; gallium precursor: triethylgallium. The substrate temperature may be measured by means of a thermocouple under the wafer's susceptor and/or by means of a pyrometer suspended above the wafer. The following flow-rates may be used for a reactor configured for processing 300 mm wafers, but the flow rates used are readily transferable to other wafer sizes: silane flow rate: 60 standard cubic centimeters per minute (sccm), germane flow rate: 330 sccm, diborane flow rate: 5 sccm, and triethylgallium flow rate: 3 sccm. Such layers have excellent crystallinity as shown by the x-ray diffraction (XRD) pattern shown in FIG. 4, panel B. In addition, the resulting layers offer low surface roughness (0.3 nm), as was observed using atomic force microscopy (AFM) on a 1×1 μm measurement area, haze measurements, and X-Ray Reflectivity (XRR) measurements (FIG. 4, panel A). The selectivity of the process for growth on monocrystalline silicon versus silicon nitride is evidenced by the thickness measurements in FIG. 5, which show the presence of nucleation delay on silicon nitride versus silicon oxide, and which show a lower growth rate, and delayed growth, on silicon nitride compared to monocrystalline silicon. This allows intrinsic selective growth on monocrystalline silicon versus silicon nitride and silicon oxide within a selectivity window of at least 10 nm, as experimentally shown in FIG. 6. In particular, FIG. 6 panels A, B, and C show close-up transmission electron microscope images of a monocrystalline silicon (620) substrate having a monocrystalline silicon surface on which boron and gallium doped silicon germanium (610) is selectively grown. Conversely, no silicon germanium is grown on exposed silicon oxide (630) and silicon nitride (640) surfaces. For layer thicknesses within the selectivity window, no etchback is needed. Selective growth of layers having a thickness which is higher than the selectivity window can be obtained by means of an etchback. The above-described process resulted in a p-type active dopant concentration, i.e., a total combined active concentration of boron and gallium, in silicon of $1.1 \times 10^{21} \text{ cm}^{-3}$, as characterized using micro Hall Effect measurements. Titanium contacts on the boron and gallium doped silicon germanium yielded a contact resistivity of $7.6 \pm 1.4 \times 10^{-10} \Omega \cdot \text{cm}^2$, and a resistivity of only 0.3 mΩ·cm, all without post-epi annealing.

(86) FIG. 3 illustrates a system (300) in accordance with yet additional exemplary embodiments of the disclosure. The system (300) can be used to perform a method as described herein and/or form a structure or device portion as described herein.

(87) In the illustrated example, the system (300) includes an optional substrate handling system (302), one or more reaction chambers (304), a gas injection system (306), and optionally a wall (308) disposed between reaction chamber(s) (304) and substrate handling system (302). The system (300) can also include a first gas source (310), a second gas source (312), a third gas source (314), a fourth gas source (316), a fifth gas source (311), an exhaust (326), and a controller (328). At least one of the first through fifth gas source includes a silicon precursor source. The silicon precursor may be, for example, silane. At least one of the first through fifth gas source includes a carrier gas

source, for example a H.sub.2 source. At least one of the first through the fifth gas source includes a germanium precursor source. The germanium precursor may be, for example, germane. At least one of the first through fifth gas source includes a boron precursor source. The boron precursor may be, for example, diborane. At least one of the first through fifth gas source includes a gallium precursor source. The gallium precursor may be, for example, triethylgallium.

(88) Although illustrated with five gas sources (310-316), the system (300) can include any suitable number of gas sources. The gas sources (310-316) can each include, for example, precursor gasses, e.g., the silicon, germanium, boron, and gallium precursors mentioned herein, including mixtures of such precursors and/or mixtures of one or more precursors with a carrier gas. Additionally, one of gas sources 310-316 or another gas source can include an etchant, such as an elementary halogen—e.g., chlorine. Gas sources (310)-(316) can be coupled to reaction chamber (304) via lines (318)-(324), which can each include flow controllers, valves, heaters, and the like.

(89) The system (300) can include any suitable number of reaction chambers 304 and substrate handling systems (302). Further, one or more reaction chambers 304 can be or include a cross-flow, cold wall epitaxial reaction chamber.

(90) Vacuum source (320) can include one or more vacuum pumps.

(91) The controller (328) can be configured to perform various functions and/or steps as described herein. For example, the controller (328) can be configured for causing the system (300) to perform a method for epitaxially growing a boron and gallium doped silicon germanium layer as described herein.

(92) A controller (328) can include one or more microprocessors, memory elements, and/or switching elements to perform the various functions. Although illustrated as a single unit, the controller (328) can alternatively comprise multiple devices. By way of examples, the controller (328) can be used to control gas flow (e.g., by monitoring flow rates of precursors and/or other gasses from the gas sources (310-316) and/or controlling valves, motors, heaters, and the like). Further, when the system (300) includes two or more reaction chambers, the two or more reaction chambers can be coupled to the same/shared controller.

(93) During operation of reactor system (300), substrates, such as semiconductor wafers (not illustrated), are transferred from, e.g., a substrate handling system (302), to a reaction chamber (304). Once substrate(s) are transferred to the reaction chamber (304), one or more gasses from gas sources (310-316), such as precursors, dopants, carrier gases, and/or purge gases, are introduced into the reaction chamber (304) via a gas injection system (306). A gas injection system (306) can be used to meter and control gas flow of one or more gasses (e.g., from one or more gas sources (310-316)) during substrate processing and to provide desired flows of such gas(es) to multiple sites within the reaction chamber (304).

(94) The example embodiments of the disclosure described above do not limit the scope of the invention, since these embodiments are merely examples of the embodiments of the invention, which is defined by the appended claims and their legal equivalents. Any equivalent embodiments are intended to be within the scope of this invention. Indeed, various modifications of the disclosure, in addition to those shown and described herein, such as alternative useful combinations of the elements described, may become apparent to those skilled in the art from the description. Such modifications and embodiments are also intended to fall within the scope of the appended claims.

Claims

1. A method for epitaxially growing a boron- and gallium-doped silicon germanium layer comprising: providing a substrate in a reactor chamber, wherein the substrate comprises a first surface and a second surface, wherein the first surface is a monocrystalline surface, wherein the first surface has hydrogen termination, wherein the second surface is a dielectric surface;

introducing a silicon precursor, a germanium precursor, a boron precursor, and a gallium precursor into the reactor chamber, thereby epitaxially growing a boron and gallium-doped silicon germanium layer on the monocrystalline surface; and introducing one or more cap layer precursors into the reactor chamber, thereby forming an epitaxial cap layer overlying the boron- and gallium-doped silicon germanium layer, wherein the epitaxial cap layer consists of silicon and boron, wherein the silicon precursor, the germanium precursor, the boron precursor, and the gallium precursor, are substantially free of chlorine, wherein the temperature during the step of introducing a silicon precursor, a germanium precursor, a boron precursor, and a gallium precursor is greater than 450° C. and less than 600° C., wherein the gallium precursor comprises a compound selected from the list consisting of Ga(BH.sub.4).sub.3 and GaH.sub.3, and wherein the boron- and gallium-doped silicon germanium layer is selectively and epitaxially grown on the first surface.

2. The method according to claim 1, wherein a pressure during the step of introducing a silicon precursor, a germanium precursor, a boron precursor, and a gallium precursor is between about 80 Torr and about 150 Torr.
3. The method according to claim 1, wherein the gallium precursor comprises GaH.sub.3.
4. The method according to claim 1, wherein the boron precursor has the general formula R.sub.xM (BH.sub.4).sub.3-x, wherein R is independently chosen from H, CH.sub.3, C.sub.2H.sub.5, C.sub.6H.sub.5, and NH.sub.2; M is a Group IIIA metal chosen from gallium, aluminum and indium; and x is an integer from 1-3.
5. The method according to claim 1, wherein the germanium precursor comprises an alkynylgermane.
6. The method according to claim 1, wherein the silicon precursor comprises an alkynylsilane.
7. The method according to claim 1, wherein the germanium precursor comprises germynsilane (H.sub.3Ge—SiH.sub.3).
8. The method according to claim 1, wherein the boron precursor and the gallium precursor are different.
9. The method according to claim 1, wherein the temperature during the step of introducing a silicon precursor, a germanium precursor, a boron precursor, and a gallium precursor is between 500° C. and 600° C.
10. The method according to claim 1, further comprising the steps of: introducing an etch gas into the reactor chamber, thereby etching the epitaxial cap layer.
11. The method according to claim 10 wherein the steps of: introducing the silicon precursor, the germanium precursor, the boron precursor, the gallium precursor, and the carrier gas into the reactor chamber; introducing one or more cap layer precursors into the reactor chamber; and, introducing an etch gas into the reactor chamber; are repeated until the boron and gallium doped epitaxial silicon germanium layer on the first surface has reached a pre-determined thickness.
12. The method according to claim 10, wherein the one or more cap layer precursors consist of a silicon precursor and a boron precursor.
13. The method according to claim 1, wherein the second surface is selected from the list consisting of a silicon oxycarbide surface, a silicon oxynitride surface, a hafnium oxide surface, a zirconium oxide surface, and an aluminum oxide surface.
14. The method according to claim 1, further comprising a pre-clean step before the step of introducing a silicon precursor, a germanium precursor, a boron precursor, and a gallium precursor, wherein the pre-clean step comprises exposing the monocrystalline surface to a mixture comprising NH.sub.4OH, H.sub.2O.sub.2, and H.sub.2O.
15. The method according to claim 1 wherein the monocrystalline surface comprises a boron and gallium doped silicon germanium surface.
16. A system comprising one or more reaction chambers, a gas injection system, and a controller configured for causing the system to perform a method according to claim 1.

17. A method for epitaxially growing a boron- and gallium-doped silicon germanium layer comprising: providing a substrate in a reactor chamber, wherein the substrate comprises a first surface and a second surface, wherein the first surface is a monocrystalline surface, wherein the first surface has hydrogen termination, wherein the second surface is a dielectric surface; introducing a silicon precursor, a germanium precursor, a boron precursor, a gallium precursor, thereby epitaxially growing a boron and gallium-doped silicon germanium layer on the monocrystalline surface; and introducing one or more cap layer precursors into the reactor chamber, thereby forming an epitaxial cap layer overlying the boron- and gallium-doped silicon germanium layer, wherein the epitaxial cap layer consists of silicon and boron, wherein the silicon precursor is selected from the list consisting of silanes, cyclosilanes, alkylsilanes, and alkynylsilanes; wherein the germanium precursor is selected from the list consisting of germanes, cyclogermanes, alkylgermanes, and alkynylgermanes; wherein the boron precursor is selected from the list consisting of boranes, and organic borohydrides, the organic borohydrides having the general formula $R_{x-3}M(BH_4)_3$, wherein R is independently chosen from H, CH₃, C₂H₅, C₆H₅, and NH₂; M is a Group IIIA metal chosen from gallium, aluminum and indium; and x is an integer from 1-3; wherein the temperature during the step of introducing a silicon precursor, a germanium precursor, a boron precursor, and a gallium precursor is greater than 450° C. and less than 600 C; and, wherein the gallium precursor is selected from the list consisting of Ga(BH₄)₃, and GaH₃, and wherein the boron- and gallium-doped silicon germanium layer is selectively and epitaxially grown on the first surface.
